eRAN

Carrier Aggregation Feature Parameter Description

Issue 11

Date 2026-06-30

HUAWEI TECHNOLOGIES CO., LTD.

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Contents

1 Change History

The following describes the specific changes in different issues.

eRAN21.1 11 (2026-06-30)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes

  Added descriptions of the impact relationship between "UL CoMP based on eNodeB coordination" and "inter-eNodeB CA based on eNodeB coordination". For details, see **17.3.3.3 Function Impacts**.

  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

eRAN21.1 10 (2026-05-30)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes

  Corrected descriptions of the frequency requirements of downlink 2CC aggregation. For details, see **5.3.5 Networking**.

  Added descriptions of the impact relationships of Zero Guard Band Between Contiguous Intra-Band Carriers with the following functions: downlink 2CC aggregation and downlink FDD+TDD CA. For details, see **5.3.3.3 Function Impacts** and **13.3.2.3 Function Impacts**.

eRAN21.1 09 (2026-05-11)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between downlink 2CC aggregation and iSDU-controlled RF module power-off super dormancy. For details, see **5.3.3.3 Function Impacts**.	None	FDD TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes

  Added descriptions of the impact on the actual available number of RRC_CONNECTED UEs supported by the main control board in the SCell's eNodeB when inter-eNodeB CA based on relaxed backhaul is in effect. For details, see **16.2.2 Impacts**.

eRAN21.1 08 (2026-01-15)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes

  Added descriptions of the impact relationships of "precise scheduling for CA UEs" with "ultra-low-latency scheduling for intra-eNodeB CA" and "energy saving based on proactive scheduling". For details, see **5.3.3.3 Function Impacts**.

  Added related descriptions in the event of base station resource insufficiency. For details, see **4.4 Configuration Modes**.

  Added descriptions of BBP specifications. For details, see **16.3.4 Hardware**.

  Added descriptions of the impact of "collaboration between CA and VoLTE" on the packet loss rates and block error rates of voice services. For details, see **19.7 VoLTE**.

eRAN21.1 07 (2025-12-14)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for optimized selection of TDD high-load massive MIMO cells as SCells. For details, see: • **5.4.1.1.2 Adaptive Configuration Mode** • **5.4.1.2.4 Adaptive Configuration Mode (TDD)** • **19.2.1 Adaptive Handling of CA for MU Beamforming**	Activated the **RSVD_SW_PARAM0_BIT28** option of the reserved parameter **EnodebRsvdParamExt**.*RsvdSwParam0*.	TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes
  None

eRAN21.1 06 (2025-11-07)

This issue includes the following changes.

- Technical changes
  None
- Editorial changes
  Added the description of the impact relationship between inter-eNodeB CA based on relaxed backhaul and precise AMC. For details, see **16.3.3.3 Function Impacts**.
  Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

eRAN21.1 05 (2025-09-26)

This issue includes the following changes.

- Technical changes
  None
- Editorial changes
  Revised the descriptions of "CA for UEs with VoLTE services ongoing". For details, see **19.7 VoLTE**.
  Revised the descriptions of the impact relationships of "beamforming in SCells" with 3D beamforming and MU beamforming. For details, see **15.3.2.3 Function Impacts**.
  Corrected the function switches of single-stream beamforming and dual-stream beamforming. For details, see **15.3.2.3 Function Impacts**.
  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

eRAN21.1 04 (2025-08-30)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes

  Added the requirement on the channel spacing between the center frequencies of two intra-band non-contiguous CCs. For details, see **5.3.5 Networking**.

  Corrected the switch for TM9 split SDMA optimization, and revised the descriptions of the principles of beamforming in SCells. For details, see **15.1 Principles**.

  Added the descriptions of additional data preparation and MML command examples for beamforming in SCells. For details, see **15.4.1.1 Data Preparation** and **15.4.1.2 Using MML Commands**.

  Deleted the descriptions of the mutually exclusive relationships of massive MIMO with downlink massive CA and downlink FDD+TDD CA from **9.3.2.2 Mutually Exclusive Functions** and **13.3.2.2 Mutually Exclusive Functions**.

  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

eRAN21.1 03 (2025-06-28)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes

  Revised the reference document for UL CoMP between massive MIMO cells. For details, see **10.3.3.3 Function Impacts**.

  Added the descriptions of the measurement timer in the PCC anchoring procedure. For details, see **4.6.1.2 CA-Group-based PCC Anchoring Procedure** and **4.6.1.3 Adaptive PCC Anchoring Procedure**.

  Added the description of the dependency relationship between beamforming in SCells and the SCC transmission mode adaptation switch. For details, see **15.3.2.1 Prerequisite Functions**.

  Corrected the single-stream beamforming switch. For details, see **15.3.2.1 Prerequisite Functions**.

  Deleted the description of the impact relationship between intelligent selection of CA combinations and SPID-specific CA PCC anchoring from **5.3.3.3 Function Impacts**.

  Deleted the description of the impact relationship between uplink FDD+TDD CA and NSA PCC anchoring from **14.3.2.3 Function Impacts**.

  Added the description of the dependency relationship between beamforming in SCells and Adaptive Transmission Mode. For details, see **15.3.2.1 Prerequisite Functions**.

  Revised the descriptions of the impacts of downlink 2CC aggregation on network performance. For details, see **5.2.2 Impacts**.

Revised the descriptions of channel quality–triggered SCell deactivation. For details, see **Channel Quality–triggered SCell Deactivation** in **4.6.3.5 SCell Deactivation**.

Revised the descriptions of the impact relationships of GSM and LTE Spectrum Concurrency with downlink 2CC aggregation, downlink FDD+TDD CA, and inter-eNodeB CA based on relaxed backhaul. For details, see **5.3.3.3 Function Impacts**, **13.3.2.3 Function Impacts**, and **16.3.3.3 Function Impacts**.

Revised the descriptions of the impact relationships of downlink massive CA with load-based SCell configuration and adaptive selection of CA or high-order MIMO. For details, see **9.3.2.3 Function Impacts**.

Deleted the descriptions of the impact relationships between downlink massive CA and CA UE transfer, multi-band optimal carrier selection, and GSM and LTE Spectrum Concurrency from **9.3.2.3 Function Impacts**.

Revised the descriptions of the impact relationships of downlink FDD+TDD massive CA with load-based SCell configuration and adaptive selection of CA or high-order MIMO. For details, see **13.3.2.3 Function Impacts**.

Deleted the descriptions of the impact relationships between downlink FDD +TDD massive CA and CA UE transfer, multi-band optimal carrier selection, and GSM and LTE Spectrum Concurrency from **13.3.2.3 Function Impacts**.

Added a statement that CA PCC anchoring policies and SCell management policies can be defined on a per SPID basis only when adaptive CA is enabled. For details, see **19.6.5 SPID-specific CA Policies**.

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

eRAN21.1 02 (2025-04-27)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes

  Revised descriptions of the collaboration between CA and multi-band compatibility enhancement, and added the rules for selecting PCCs and SCCs for CA UEs. For details, see **19.1 Multi-Band Compatibility Enhancement**.

  Added descriptions of the mutually exclusive relationship between the "intelligent selection of serving cell combinations" function and the "adaptive handling of CA for MU beamforming" function in FDD. For details, see **12.3.3.2 Mutually Exclusive Functions**.

  Added the description of the mutually exclusive relationship between intelligent selection of serving cell combinations and the **PRED_TRAFFIC_AWARENESS_SCS_SW** option. For details, see **12.3.3.2 Mutually Exclusive Functions**.

  Revised the descriptions of the dependency relationships of downlink massive CA with inter-eNodeB CA based on relaxed backhaul, downlink FDD+TDD CA, and inter-eNodeB CA based on eNodeB coordination. For details, see **16.3.3.1 Prerequisite Functions**, **13.3.2.1 Prerequisite Functions**, and **17.3.3.1 Prerequisite Functions**.

  Added the descriptions of the mutually exclusive relationships of PUCCH measurement with downlink 3CC aggregation, downlink 4CC aggregation, and downlink 5CC aggregation in FDD. For details, see **6.3.3.2 Mutually Exclusive Functions**, **7.3.3.2 Mutually Exclusive Functions**, and **8.3.3.2 Mutually Exclusive Functions**.

Deleted the description of the mutually exclusive relationship between uplink FDD+TDD CA and uplink SU-MIMO from **14.3.2.2 Mutually Exclusive Functions**.

Corrected the A2 threshold calculation in CA-group-based configuration mode. For details, see **19.4.1 Mobility Management**.

Corrected the parameter that specifies *TimeToTrig* for event A6. For details, see **4.3 CA-related Events**.

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

eRAN21.1 01 (2025-03-10)

This issue includes the following changes.

- Technical changes

  None

- Editorial changes

  Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

eRAN21.1 Draft B (2025-02-10)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added QCI-specific VoLTE and CA coexistence. For details, see **4.6.1.1 Triggering Conditions** and **19.7 VoLTE**.	Modified parameter: Added the **QCI_VOLTE_CA_COEXIST_SW** option to the **CellQciPara**.***QciAlgoSwitch*** parameter.	FDD TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added UE-number-based SCell configuration control. For details, see: • 4.6.3.1.1 Triggering Conditions • 5.4.1.1.3 Common Parameters • 5.4.1.2 Using MML Commands	Added parameters: • **CaMgtCfg**.***Forbid2ccUeRatioThld*** • **CaMgtCfg**.***Forbid3ccUeRatioThld*** • **CaMgtCfg**.***Forbid4ccUeRatioThld*** • **CaMgtCfg**.***Forbid5ccUeRatioThld***	FDD TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

  Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

eRAN21.1 Draft A (2024-12-31)

This issue introduces the following changes to eRAN20.1 06 (2024-11-29).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship of uplink 2CC aggregation with UL CoMP between massive MIMO cells. For details, see **10.3.3.3 Function Impacts**.	None	TDD	3900 and 5900 series base stations (macro base stations)
Added PCC anchoring prohibition for high-mobility UEs. For details, see: • 4.6.1.1 Triggering Conditions • 5.4.1.1.3 Common Parameters • 5.4.1.2 Using MML Commands	Added the **CellOpHoCfg**.***HighMobiUeHoForbidSw*** parameter.	FDD TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added compatibility of UMPTha boards with inter-eNodeB CA based on relaxed backhaul. For details, see **16.3.5 Networking**.	None	FDD TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between downlink 2CC aggregation and downlink-load-based scheduling priority optimization (FDD). For details, see **5.3.3.3 Function Impacts**.	None	FDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added impact relationships of downlink 2CC aggregation with downlink scheduling priority enhancement and downlink slot awareness scheduling. For details, see **5.3.3.3 Function Impacts**.	None	TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between precise scheduling for CA UEs and LTE-NR multi-dimensional scheduling optimization. For details, see **5.3.3.3 Function Impacts**.	None	FDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between downlink 2CC aggregation and SOC level-based low power consumption mode. For details, see **5.3.3.3 Function Impacts**.	None	FDD TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between uplink 2CC aggregation and UL CIAS. For details, see **10.3.3.3 Function Impacts**.	None	FDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between uplink FDD+TDD CA and UL CIAS. For details, see **14.3.2.3 Function Impacts**.	None	FDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Optimized measurement of the number of multi-CC-active CA UEs and the number of transmitted bits and transmission duration of such UEs. For details, see: • Downlink 2CC aggregation – **5.4.1.1.3 Common Parameters** – **5.4.1.2 Using MML Commands** – **5.4.2 Activation Verification** – **5.4.3 Network Monitoring** • Downlink 3CC aggregation – **6.4.3 Activation Verification** – **6.4.4 Network Monitoring** • Downlink 4CC aggregation – **7.4.2 Activation Verification** – **7.4.3 Network Monitoring** • Downlink 5CC aggregation – **8.4.2 Activation Verification** – **8.4.3 Network Monitoring** • Downlink massive CA (FDD) – **9.4.2 Activation Verification** – **9.4.3 Network Monitoring** • Downlink FDD+TDD CA	Modified parameter: Added the **CA_UE_MULTI_CC_STAT_OPT_SW** option to the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter.	FDD TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
– **13.4.2 Activation Verification** – **13.4.3 Network Monitoring** ● Inter-eNodeB CA based on relaxed backhaul – **16.4.2 Activation Verification**			

- Editorial changes

  Added descriptions of the impact relationship between uplink FDD+TDD CA and NSA networking based on EPC. For details, see **14.2.2 Impacts**.

  Corrected the value of threshold 1 for event A5 used in the CA-group-based PCC anchoring procedure. For details, see **4.6.1.2 CA-Group-based PCC Anchoring Procedure**.

  Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in the names of reference documents.

2 About This Document

2.1 General Statements

Purpose

This document is intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Feature Differences Between RATs

The feature difference section only describes differences in switches or principles.

Unless otherwise stated, descriptions in this document apply to all RATs. If a description does not apply to all RATs, the specific RAT that it does apply to will be stated.

For example, in the statement "TDD cells are compatible with enhanced MU-MIMO", "TDD cells" indicates that this function cannot be used in non-TDD cells.

2.2 Applicable RAT

This document applies to FDD/TDD.

2.3 Features in This Document

This document describes the following FDD features.

Feature ID	Feature Name	Chapter/Section
LAOFD-001001	LTE-A Introduction	See the corresponding sections of its subfeatures.
LAOFD-00100101	Carrier Aggregation for Downlink 2CC	**5 Downlink 2CC Aggregation**
LAOFD-001002	Carrier Aggregation for Downlink 2CC in 40MHz	
LAOFD-080207	Carrier Aggregation for Downlink 3CC in 40MHz	**6 Downlink 3CC Aggregation**
LAOFD-080208	Carrier Aggregation for Downlink 3CC in 60MHz	
LEOFD-110303	Carrier Aggregation for Downlink 4CC and 5CC	**7 Downlink 4CC Aggregation** **8 Downlink 5CC Aggregation**
LEOFD-151308	Downlink Massive CA	**9 Downlink Massive CA (FDD)**
LAOFD-080202	Carrier Aggregation for Uplink 2CC	**10 Uplink 2CC Aggregation**
LAOFD-070201	Flexible CA from Multiple Carriers	**11 Flexible CA** **12 Intelligent Selection of Serving Cell Combinations**
MRFD-101222	FDD+TDD Downlink Carrier Aggregation(LTE FDD)	**13 Downlink FDD+TDD CA**
MRFD-151309	FDD+TDD Downlink Massive CA(LTE FDD)	
MRFD-111222	FDD+TDD Uplink Carrier Aggregation (LTE FDD)	**14 Uplink FDD+TDD CA**

Feature ID	Feature Name	Chapter/Section
LAOFD-080201	Inter-eNodeB CA Based on Relaxed Backhaul	**16 Inter-eNodeB CA Based on Relaxed Backhaul**
LAOFD-070202	Inter-eNodeB CA based on Coordinated eNodeB	**17 Inter-eNodeB CA Based on eNodeB Coordination**

This document describes the following TDD features.

Feature ID	Feature Name	Chapter/Section
TDLAOFD-001001	LTE-A Introduction	See the corresponding sections of its subfeatures.
TDLAOFD-00100111	Carrier Aggregation for Downlink 2CC	**5 Downlink 2CC Aggregation**
TDLAOFD-001002	Carrier Aggregation for Downlink 2CC in 40MHz	
TDLAOFD-081405	Carrier Aggregation for Downlink 3CC	**6 Downlink 3CC Aggregation**
TDLEOFD-081504	Carrier Aggregation for Downlink 4CC and 5CC	**7 Downlink 4CC Aggregation** **8 Downlink 5CC Aggregation**
TDLAOFD-081407	Carrier Aggregation for Uplink 2CC	**10 Uplink 2CC Aggregation**
TDLAOFD-070201	Flexible CA from Multiple Carriers	**11 Flexible CA** **12 Intelligent Selection of Serving Cell Combinations**
MRFD-101231	FDD+TDD Downlink Carrier Aggregation(LTE TDD)	**13 Downlink FDD+TDD CA**
MRFD-151401	FDD+TDD Downlink Massive CA(LTE TDD)	
MRFD-111232	FDD+TDD Uplink Carrier Aggregation (LTE TDD)	**14 Uplink FDD+TDD CA**
TDLAOFD-00100115	Beamforming in Scell	**15 Beamforming in SCells (TDD)**

Feature ID	Feature Name	Chapter/Section
TDLAOFD-081402	Inter-eNodeB CA Based on Relaxed Backhaul	**16 Inter-eNodeB CA Based on Relaxed Backhaul**
TDLAOFD-110401	Inter-eNodeB CA based on Coordinated eNodeB	**17 Inter-eNodeB CA Based on eNodeB Coordination**

2.4 Feature Differences Between FDD and TDD

FDD Feature	TDD Feature	Difference	Chapter/Section
LAOFD-00100101 Carrier Aggregation for Downlink 2CC	TDLAOFD-001001 11 Carrier Aggregation for Downlink 2CC	Maximum aggregated bandwidth: 20 MHz for FDD vs. 30 MHz for TDD	**5 Downlink 2CC Aggregation**
LAOFD-001002 Carrier Aggregation for Downlink 2CC in 40MHz	TDLAOFD-001002 Carrier Aggregation for Downlink 2CC in 40MHz	None	
LAOFD-080207 Carrier Aggregation for Downlink 3CC in 40MHz	TDLAOFD-081405 Carrier Aggregation for Downlink 3CC	In adaptive configuration mode, the FDD feature "Carrier Aggregation for Downlink 3CC in 60MHz" is further controlled by the **CaDl3CCExtSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter, while the TDD feature "Carrier Aggregation for Downlink 3CC" is not affected by this option.	**6 Downlink 3CC Aggregation**
LAOFD-080208 Carrier Aggregation for Downlink 3CC in 60MHz			
LEOFD-110303 Carrier Aggregation for Downlink 4CC and 5CC	TDLEOFD-081504 Carrier Aggregation for Downlink 4CC and 5CC	None	**7 Downlink 4CC Aggregation** **8 Downlink 5CC Aggregation**

FDD Feature	TDD Feature	Difference	Chapter/ Section
LEOFD-151308 Downlink Massive CA	None	This feature is supported only by FDD.	**9 Downlink Massive CA (FDD)**
LAOFD-080202 Carrier Aggregation for Uplink 2CC	TDLAOFD-081407 Carrier Aggregation for Uplink 2CC	None	**10 Uplink 2CC Aggregation**
LAOFD-070201 Flexible CA from Multiple Carriers	TDLAOFD-070201 Flexible CA from Multiple Carriers	None	**11 Flexible CA** **12 Intelligent Selection of Serving Cell Combinations**
MRFD-101222 FDD+TDD Downlink Carrier Aggregation(LTE FDD)	MRFD-101231 FDD+TDD Downlink Carrier Aggregation(LTE TDD)	None	**13 Downlink FDD+TDD CA**
MRFD-151309 FDD+TDD Downlink Massive CA(LTE FDD)	MRFD-151401 FDD+TDD Downlink Massive CA(LTE TDD)	None	**13 Downlink FDD+TDD CA**
MRFD-111222 FDD+TDD Uplink Carrier Aggregation (LTE FDD)	MRFD-111232 FDD+TDD Uplink Carrier Aggregation (LTE TDD)	None	**14 Uplink FDD+TDD CA**
None	TDLAOFD-001001 15 Beamforming in Scell	This feature is supported only by TDD.	**15 Beamforming in SCells (TDD)**
LAOFD-080201 Inter-eNodeB CA Based on Relaxed Backhaul	TDLAOFD-081402 Inter-eNodeB CA Based on Relaxed Backhaul	• These two features are activated using different switches. • These two features have different requirements for inter-eNodeB transmission quality.	**16 Inter-eNodeB CA Based on Relaxed Backhaul**

FDD Feature	TDD Feature	Difference	Chapter/ Section
LAOFD-070202 Inter-eNodeB CA based on Coordinated eNodeB	TDLAOFD-110401 Inter-eNodeB CA based on Coordinated eNodeB	The centralized, distributed, and hybrid architectures of eNodeB coordination are supported in FDD, but only the centralized architecture is supported in TDD.	**17 Inter-eNodeB CA Based on eNodeB Coordination**

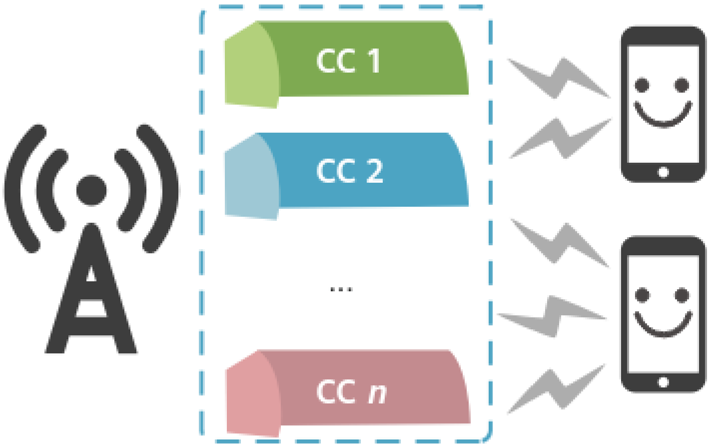
3 Overview

3.1 Definition

3GPP requires that LTE-Advanced networks provide a downlink peak data rate of 1 Gbit/s. **However, operators' spectrum resources may not be contiguous or may exceed the single-carrier bandwidth capability defined in 3GPP specifications. Against this backdrop, 3GPP introduced carrier aggregation (CA), which allows for aggregation of multiple contiguous or non-contiguous component carriers (CCs) to a larger bandwidth to meet 3GPP's 1 Gbit/s downlink rate requirement for LTE UEs and increase users' peak data rates in the uplink and downlink.**

Figure 3-1 illustrates an example of CA.

Figure 3-1 CA



3.2 Classification

CA can be classified from the following dimensions:

- By service direction: downlink CA and uplink CA
  - Downlink CA: The eNodeB aggregates intra- or inter-band carriers to widen the bandwidth for downlink data transmission to a CA UE.

  – Uplink CA: The eNodeB aggregates intra- or inter-band carriers to widen the bandwidth for uplink data transmission from a CA UE.

- By the number of carriers: CA with a fixed number of carriers and CA with a non-fixed number of carriers
  – CA with a fixed number of carriers: The eNodeB selects a fixed number of intra- or inter-band carriers for CA to provide a larger bandwidth for a CA UE.
  – CA with a non-fixed number of carriers: To provide a larger bandwidth for a CA UE, the eNodeB selects the most suitable carriers from the UE-supported CA combinations for downlink CA based on the CA capabilities reported by the UE and based on factors such as the coverage overlap between cells, cell load status, bandwidths, and spectral efficiency.

  CA with a fixed number of carriers works in both uplink and downlink, and CA with a non-fixed number of carriers works only in the downlink.

- By radio access technology (RAT): single-RAT CA and multi-RAT CA
  – Single-RAT CA: The carriers involved in CA are all FDD carriers or all TDD carriers.
  – Multi-RAT CA: Both FDD and TDD carriers are involved in CA.

3.3 CA Application Scenarios

CA can be used between intra- or inter-eNodeB cells, involving both uplink CA and downlink CA. CA works in CA-group-based or adaptive configuration mode. For details about the two modes, see **4.4 Configuration Modes**.

3.3.1 Intra-eNodeB CA Scenarios

Intra-eNodeB CA scenarios refer to aggregation of cells that are served by the same eNodeB.

Annex J in 3GPP TS 36.300 V10.3.0 defines five intra-eNodeB CA scenarios. **Table 3-1** lists the five scenarios and takes two CCs as an example to describe when Huawei CA works in the scenarios.

Table 3-1 Intra-eNodeB CA scenarios

Scenario	CA-Group-based Configuration Mode	Adaptive Configuration Mode
Intra-eNodeB co-coverage carriers, shown in **Figure 3-2**	Supported	Supported
Intra-eNodeB different-coverage carriers, shown in **Figure 3-3**	Supported	Supported
Intra-eNodeB carriers (one for macro coverage; another for edge coverage), shown in **Figure 3-4** [a]	Supported	Supported

Scenario	CA-Group-based Configuration Mode	Adaptive Configuration Mode
Intra-eNodeB carriers (one provided by the site; another provided by remote radio heads [RRHs]), shown in **Figure 3-5**	FDD: supported when the ratio of macro cells to RRHs is 1:1 TDD: supported when the ratio of macro cells to RRHs is 1:*N* (*N* ≥ 1)	Supported when the ratio of macro cells to RRHs is 1:*N* (*N* ≥ 1)
Intra-eNodeB carriers (one provided only by the site; another provided by the site and a repeater), shown in **Figure 3-6**	Supported	Supported
a: In the scenario of intra-eNodeB carriers (one for macro coverage; another for edge coverage), adaptive configuration mode is recommended, but CA-group-based configuration mode is not.		

The following figures show the intra-eNodeB CA scenarios. In the figures, **F1** and **F2** denote two carrier frequencies.

Figure 3-2 Intra-eNodeB co-coverage carriers

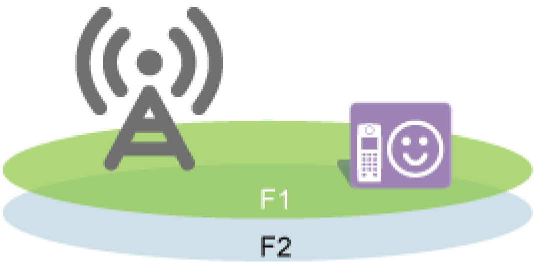


Figure 3-3 Intra-eNodeB different-coverage carriers

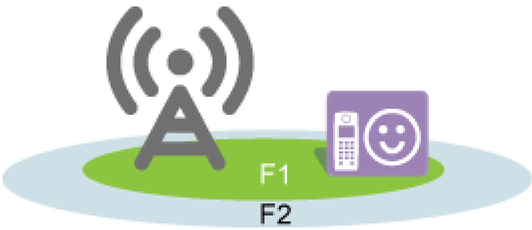


Figure 3-4 Intra-eNodeB carriers (one for macro coverage; another for edge coverage)

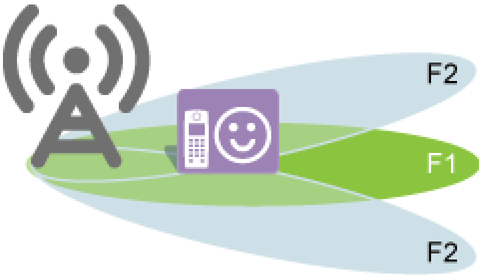


Figure 3-5 Intra-eNodeB carriers (one provided by the site; another provided by RRHs)

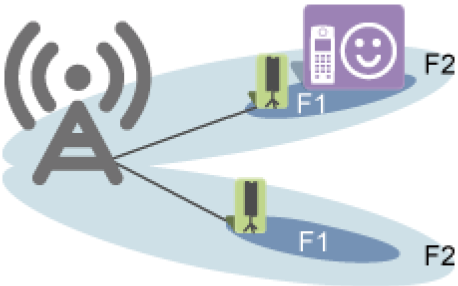
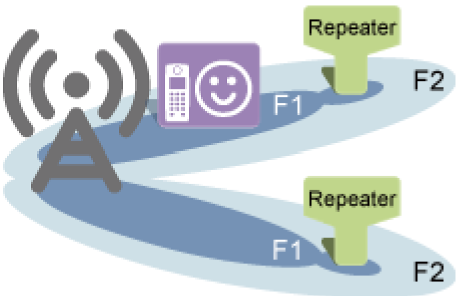


Figure 3-6 Intra-eNodeB carriers (one provided only by the site; another provided by the site and a repeater)



3.3.2 Inter-eNodeB CA Scenarios

Inter-eNodeB CA scenarios refer to aggregation of cells that are served by different eNodeBs.

Huawei eNodeBs support CA in the following inter-eNodeB scenarios:

- Relaxed backhaul
- eNodeB coordination

For downlink or uplink CA to take effect in inter-eNodeB scenarios, parameters related to one of these scenarios must be set.

3.4 Overview of CA Functions

eNodeBs offer various CA functions to CA UEs according to UE-reported CA capabilities (UE-supported band combinations in the *BandCombination* IE) and the carrier management principles described in **4 General Principles**. These functions allow CA UEs to have larger bandwidths for data transmission. **Table 3-2** provides an overview of these functions.

Table 3-2 Overview of CA functions

Function Name	Uplink	Downlink	Number of Carriers	RAT	Reference
Downlink 2CC aggregation	-	√	2	FDD TDD	**5 Downlink 2CC Aggregation**
Downlink 3CC aggregation	-	√	3	FDD TDD	**6 Downlink 3CC Aggregation**
Downlink 4CC aggregation	-	√	4	FDD TDD	**7 Downlink 4CC Aggregation**
Downlink 5CC aggregation	-	√	5	FDD TDD	**8 Downlink 5CC Aggregation**
Downlink massive CA (FDD)	-	√	6 to 8	FDD	**9 Downlink Massive CA (FDD)**
Uplink 2CC aggregation	√	-	2	FDD TDD	**10 Uplink 2CC Aggregation**
Flexible CA	-	√	2 to 8	FDD TDD	**11 Flexible CA**
Intelligent selection of serving cell combinations	√	√	Downlink: 2 to 5 Uplink: 2	FDD TDD	**12 Intelligent Selection of Serving Cell Combinations**
Downlink FDD+TDD CA	-	√	2 to 8	FDD TDD	**13 Downlink FDD+TDD CA**
Uplink FDD +TDD CA	√	-	2	FDD TDD	**14 Uplink FDD+TDD CA**
Beamforming in SCells (TDD)	√	√	Downlink: 2 to 5 Uplink: 2	TDD	**15 Beamforming in SCells (TDD)**

Function Name	Uplink	Downlink	Number of Carriers	RAT	Reference
Inter-eNodeB CA based on relaxed backhaul	-	√	Downlink: 2 to 8 Uplink: 2	FDD TDD	**16 Inter-eNodeB CA Based on Relaxed Backhaul**
Inter-eNodeB CA based on eNodeB coordination	-	√	Downlink: 2 to 8 Uplink: 2	FDD TDD	**17 Inter-eNodeB CA Based on eNodeB Coordination**
SCC buffer optimization	-	√	2 to 8	FDD	**18 SCC Buffer Optimization**

NOTE

- For multi-RAT CA to take effect on an FDD+TDD network, downlink FDD+TDD CA or uplink FDD+TDD CA must be enabled.
- For details about how CA collaborates with other key technologies, see **19 Collaboration Between CA and Other Key Technologies**.

4 General Principles

4.1 Related Concepts

PCell

A primary cell (PCell) is a cell on which a CA UE camps. In this cell, the CA UE works in the same way as it does in any other cell.

Unless otherwise stated, all switches described in this document need to be turned on only in the PCell.

SCell

A secondary cell (SCell) is a cell that an eNodeB configures for a CA UE through an RRC connection reconfiguration message from the PCell. It provides the CA UE with additional radio resources. In an SCell, there can be downlink transmission only or both downlink and uplink transmission.

CC

Component carriers (CCs) are the carriers that are aggregated for a CA UE.

PCC

The primary component carrier (PCC) is the carrier of the PCell.

SCC

A secondary component carrier (SCC) is the carrier of an SCell.

PCC Anchoring

The eNodeB selects the highest-priority cell from candidates as the PCell for a CA UE.

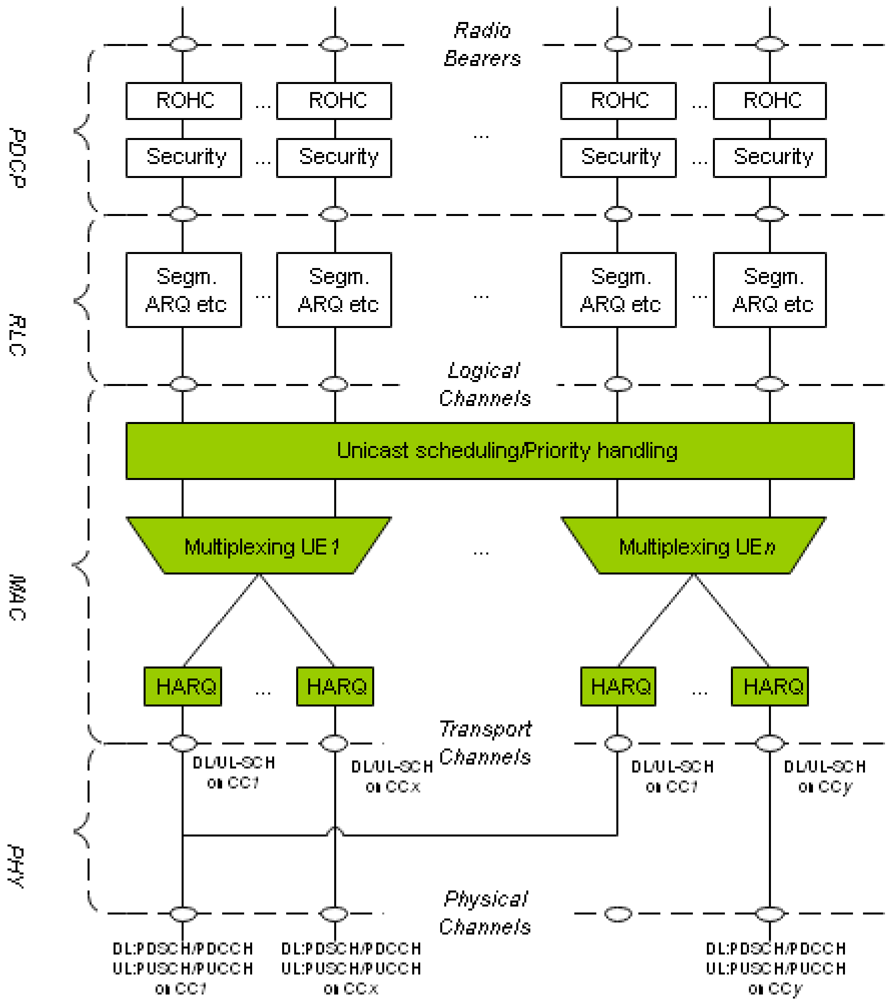
4.2 Protocol Stack Architecture

The protocol stack with CA enabled has the following characteristics:

- Each radio bearer has only one Packet Data Convergence Protocol (PDCP) entity and one Radio Link Control (RLC) entity. The number of CCs at the physical layer is not visible from the RLC layer.
- User-plane data scheduling at the Media Access Control (MAC) layer is performed separately for individual CCs.
- Each CC has an independent set of Uu interface transport channels, as well as separate hybrid automatic repeat request (HARQ) entities and retransmission processes.

Figure 4-1 shows the protocol stack.

Figure 4-1 Protocol stack with CA enabled



ROHC: robust header compression

The following constraints apply to LTE-Advanced CA:

- In the uplink or downlink, a CA UE can have no more than 32 CCs aggregated, with each CC having a maximum bandwidth of 20 MHz.

  Currently, Huawei CA supports downlink aggregation of a maximum of eight CCs and uplink aggregation of a maximum of two CCs.
- CA UEs support asymmetric CA. A UE may use different numbers of CCs in the uplink and downlink. However, the number of CCs in the downlink must always be at least equal to the number of CCs in the uplink. In addition, the uplink CCs must be included in the set of downlink CCs.
- Each CC uses the same frame structure as 3GPP Release 8 carriers, ensuring backward compatibility.
- UEs that comply with 3GPP releases earlier than Release 15 are allowed to transmit and receive data on Release 15 CCs.
- For TDD, the cells to be aggregated must use the same uplink-downlink configuration: either configuration 1 or configuration 2.

4.3 CA-related Events

CA-related events include events A1 to A6. These events are used for PCC anchoring and SCell management.

- Event A1

  Event A1 indicates that the signal quality of the PCell exceeds a specific threshold.

  Event A1 is triggered if the condition (*Ms* – *Hys* > *Thresh*) is met throughout *TimeToTrig*. For details about event A1, see *Mobility Management in Connected Mode*.
- Event A2

  Event A2 indicates that the signal quality of the PCell drops below a specific threshold.

  Event A2 is triggered if the condition (*Ms* + *Hys* < *Thresh*) is met throughout *TimeToTrig*. For details about event A2, see *Mobility Management in Connected Mode*.
- Event A3

  Event A3 indicates that the signal quality of the PCell's neighboring cell exceeds that of the PCell by a certain amount.

  Event A3 is triggered if the condition (*Mn* + *Ofn* + *Ocn* – *Hys* > *Ms* + *Ofs* + *Ocs* + *Off*) is met throughout *TimeToTrig*. For details about event A3, see *Mobility Management in Connected Mode*.
- Event A4

  Event A4 indicates that the signal quality of an inter-frequency neighboring cell exceeds a specific threshold.

  Event A4 is triggered if the condition (*Mn* + *Ofn* + *Ocn* – *Hys* > *Thresh*) is met throughout *TimeToTrig*. For details about event A4, see *Mobility Management in Connected Mode*.
- Event A5

  Event A5 indicates that the signal quality of the PCell drops below threshold 1 and the signal quality of a neighboring cell exceeds threshold 2.

Event A5 is triggered if the conditions (*Ms* + *Hys* < *Thresh1*) and (*Mn* + *Ofn* + *Ocn* – *Hys* > *Thresh2*) are met throughout *TimeToTrig*. For details about event A5, see *Mobility Management in Connected Mode*.

Inter-frequency measurement for PCC anchoring is triggered by event A5. For details about PCC anchoring, see **4.6.1 PCC Anchoring**.

Inter-frequency measurement for CA UE handovers except PCC anchoring may involve a change from event A4 to event A5 in the relevant measurement configurations. That is because, A5 measurement involves neighboring cells of the PCell and those of SCells so that an SCell can be a neighboring cell of the PCell, and therefore it is possible that the UE will be handed over from the PCell to the SCell. However, A4 measurement does not involve serving cells, so that no SCell can be a neighboring cell of the PCell; therefore, the UE cannot be handed over from its PCell to an SCell. The change from event A4 to event A5 affects threshold 1 but not threshold 2 for event A5. Whether to enable changes from event A4 to event A5 is controlled by the **CaA5HoEventSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter and the **CaA5HoEventEnhSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter. It is recommended that both options be selected.

– If the **CaA5HoEventSwitch** option is selected, the eNodeB changes event A4 to event A5 in the inter-frequency measurement configurations related to any type of handover when both the network and the UE support CA.

  - Option **CaA5HoEventEnhSwitch** deselected
    - If the data configuration indicates that event A4 is used for inter-frequency measurements, reference signal received power (RSRP) threshold 1 for event A5 to which event A4 is changed always takes the value of –43 dBm and RSRQ threshold 1 for this event A5 always takes the value of –3 dB.
    - If the data configuration indicates that event A5 is used for inter-frequency measurements, RSRP threshold 1 for event A5 related to coverage-based inter-frequency handovers always takes the value of –43 dBm and RSRQ threshold 1 for this event A5 always takes the value of –3 dB. In addition, RSRP threshold 1 and RSRQ threshold 1 for event A5 related to other handover types take the values of the **InterFreqHoGroup**.***InterFreqHoA5Thd1Rsrp*** and **InterFreqHoGroup**.***InterFreqHoA5Thd1Rsrq*** parameters, respectively.

  - Option **CaA5HoEventEnhSwitch** selected
    - If the data configuration indicates that event A4 is used for inter-frequency measurements, RSRP threshold 1 for event A5 to which event A4 is changed always takes the value of –43 dBm and RSRQ threshold 1 for this event A5 always takes the value of –3 dB.
    - If the data configuration indicates that event A5 is used for inter-frequency measurements, RSRP threshold 1 and RSRQ threshold 1 for event A5 take the values of the **InterFreqHoGroup**.***InterFreqHoA5Thd1Rsrp*** and **InterFreqHoGroup**.***InterFreqHoA5Thd1Rsrq*** parameters, respectively.

– If the **CaA5HoEventSwitch** option is deselected:

- Option **CaA5HoEventEnhSwitch** deselected

  Handover algorithms take actions on inter-frequency measurement triggering as described in **Table 4-1**.

Table 4-1 Inter-frequency measurement triggering by different handover algorithms

<table>
<tr><th>Handover Algorithm</th><th>Change from A4 to A5</th><th>Description</th></tr>
<tr><td>Coverage-based inter-frequency handover</td><td rowspan="7">• If SCells have been configured for the UE, event A4 is changed to event A5.
• Otherwise, the change is not performed.</td><td>After the change, threshold 1 for event A5 takes the value of the InterFreqHoGroup.InterFreqHoA5Thd1Rsrp parameter.</td></tr>
<tr><td>Uplink-quality-based inter-frequency handover</td><td rowspan="6">After the change, threshold 1 for event A5 takes the value of:
• –43 dBm if the data configuration indicates that event A4 is used for inter-frequency measurements.
• InterFreqHoGroup.InterFreqHoA5Thd1Rsrp if the data configuration indicates that event A5 is used for inter-frequency measurements.</td></tr>
<tr><td>Distance-based inter-frequency handover</td></tr>
<tr><td>Inter-frequency handover for UE transfer in the event of carrier disabling</td></tr>
<tr><td>Inter-frequency handover for unlimited-service (US) UE assurance</td></tr>
<tr><td>Outgoing inter-frequency handover of non-US UEs</td></tr>
<tr><td>Load-based inter-frequency handover</td></tr>
<tr><td>Frequency-priority-based handover</td><td rowspan="3">Event A4 is not changed to event A5.</td><td rowspan="3">The measurement configurations related to the handover event determined by relevant algorithms are directly delivered.</td></tr>
<tr><td>Service-based inter-frequency handover</td></tr>
<tr><td>Inter-frequency handover back to the home public land mobile network (HPLMN) based on the subscriber profile ID (SPID)</td></tr>
</table>

Handover Algorithm	Change from A4 to A5	Description
Inter-frequency handover based on evolved multimedia broadcast/ multicast service (eMBMS) interest indications		
Speed-based inter-frequency handover		
Redirection to specified frequencies for high-mobility UEs (TDD only)		
Service-request-based handover (TDD only)		

- Option **CaA5HoEventEnhSwitch** selected

  During inter-frequency measurement triggering, event A4 is changed to event A5 for all types of handover only if SCells have been configured for the UE. In this event A5 measurement configuration, threshold 1 for event A5 always takes the value of –43 dBm.

- Event A6

  Event A6 indicates that the signal quality of an SCell's intra-frequency neighboring cell exceeds that of the SCell by a certain amount. After receiving A6 reports, the eNodeB changes the SCell to a new one while keeping the PCell unchanged.

  Event A6 is triggered if the condition ($Mn + Ocn - Hys > Ms + Ocs + Off$) is met throughout *TimeToTrig*.

  Table 4-2 describes these variables.

Table 4-2 Variables involved in the entering condition for event A6

Variable	Description
Mn	RSRP measurement result of a neighboring cell
Ocn	Cell-specific offset, specified by the **EutranInterFreqNCell**.***CellIndividualOffset*** parameter on the PCell side, for the neighboring cell
Hys	Hysteresis for event A6, which is always 1 dB
Ms	RSRP measurement result of the serving cell
Ocs	Cell-specific offset, specified by the **EutranInterFreqNCell**.***CellIndividualOffset*** parameter on the PCell side, for the serving cell

Variable	Description
Off	Offset for event A6, which is specified by the **CaMgtCfg**.***CarrAggrA6Offset*** parameter

If *TimeToTrig* for event A6 (specified by the **CaMgtCfg**.***CaA6TimeToTrigger*** parameter) is set to a value greater than the timer value (3s) for gap-assisted measurements, measurement results cannot be reported.

For details about event A6, see section 5.5.4.6a "Event A6 (Neighbour becomes offset better than SCell)" in 3GPP TS 36.331 V10.12.0.

The **CaMgtCfg**.***CaSccFreqMeasOptSw*** parameter is available to specify whether to set up measurement gaps for a UE during measurement configuration for an inter-frequency handover when all of the measurement objects are the SCC frequencies of the UE.

- If this switch is on, measurement gaps are not set up for the UE in this round of inter-frequency measurement.

  If there are UE compatibility issues on the network, the **UESPECIFICPARA** MO can be used to configure UE blacklist and whitelist control to avoid affecting KPIs. An example of these issues is that, if measurement gaps are not set up for the SCC frequencies of a UE, the UE does not send measurement reports related to these frequencies.
- If this switch is off, measurement gaps are set up for the UE in this round of inter-frequency measurement.

4.4 Configuration Modes

CA works in CA-group-based and adaptive configuration modes. The adaptive mode is recommended. A configuration mode change may result in changes in the combinations of cells that can be aggregated, the number of license units to be deducted, and inter-cell routes. These changes cause SCell configuration or removal, which in turn results in changes in the values of the following counters:

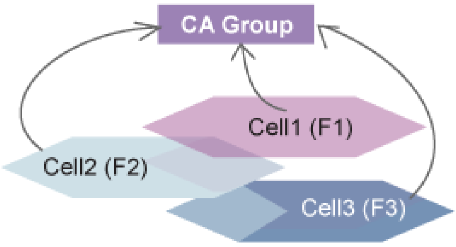
- **L.Thrp.bits.DL.CAUser**
- **L.Traffic.User.PCell.DL.Avg**
- **L.Traffic.User.SCell.DL.Avg**
- **L.Thrp.Time.DL.CAUser**

CA-Group-based Configuration Mode

CA-group-based configuration mode is enabled by deselecting the **FreqCfgSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter.

This mode, as shown in **Figure 4-2**, works when multiple cells are configured in a CA group, defined by the **CaGroup** MO on the eNodeB. Only the cells in the group can be aggregated. Blind SCell configuration is also supported in this mode. It requires the **SccBlindCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter to be selected and the **CaGroupSCellCfg**.***SCellBlindCfgFlag*** parameter to be set to **TRUE**. For details about blind SCell configuration, see **4.6.3.1 SCell Configuration**.

Figure 4-2 CA-group-based configuration mode



If CA-group-based configuration mode is used, adhere to the following configuration rules:

- Rules on the number of cells in a CA group
  - No more than nine cells can be configured into a CA group whose **CaGroup**.***CaGroupTypeInd*** is set to **FDD**. It is recommended that the nine inter-frequency cells cover the same area. For any one of the nine cells, there are eight candidate SCells.
  - No more than nine cells can be configured into a CA group whose **CaGroup**.***CaGroupTypeInd*** is set to **TDD**.
  - No more than nine cells can be configured into a CA group whose **CaGroup**.***CaGroupTypeInd*** is set to **FDDTDD**. It is recommended that the nine inter-frequency cells cover the same area. For any one of the nine cells, there are eight candidate SCells.
- Rules on SCell configuration
  - If the PCell of a CA UE is a higher-frequency cell that covers a relatively small area, it is recommended that a lower-frequency cell covering a relatively large area be defined as a blind-configurable candidate SCell. This prevents the CA UE in the higher-frequency cell from performing an unnecessary inter-frequency measurement of the lower-frequency cell, which will always meet the triggering condition for event A4.
  - If the PCell of a CA UE is a lower-frequency cell that covers a relatively large area, it is recommended that blind configuration be disabled (A4 measurements should be used) for a higher-frequency cell that covers a relatively small area. If blind configuration is enabled, the CA UE served by the lower-frequency cell will encounter low scheduling efficiency in the higher-frequency cell. In this case, the data rate of the CA UE will not reach twice that of a non-CA UE in differentiated scheduling mode.
  - If the **CaGroupSCellCfg**.***SpidGrpId*** parameter is set to a value other than **65535** for a candidate SCell, this cell cannot be configured as an SCell for UEs in this SPID group.
- Rules on intra-frequency cells in a CA group
  - An identical value of the **CaGroupCell**.***PreferredPCellPriority*** parameter is recommended for all these cells. If different PCell priorities are configured for the intra-frequency cells, the eNodeB takes the highest one as the priority for all these cells.
  - An identical value of the **CaGroupSCellCfg**.***SCellPriority*** parameter is recommended for all these cells. If different SCell priorities are configured

for the intra-frequency cells, the eNodeB takes the highest one as the priority for all these cells.

– An identical value of the **CaGroupSCellCfg**.***SpidGrpId*** parameter is recommended for all these cells.

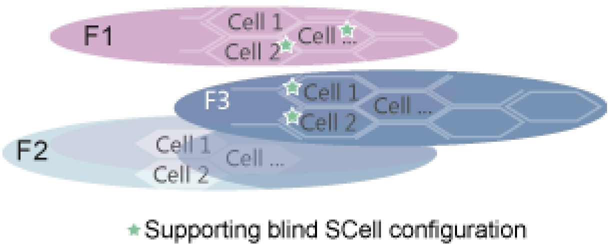
- A maximum of 1152 **CaGroupSCellCfg** MOs, which represent candidate SCells, can be configured for a single eNodeB. Ensure that the candidate SCells are included in the same CA group as the associated PCells. Otherwise, the candidate SCell configurations will not take effect. Delete the invalid MOs, as these MOs represent wasted MO capacity.

Adaptive Configuration Mode

Adaptive configuration mode is enabled by selecting both the **FreqCfgSwitch** and **AdpCaSwitch** options of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter.

This mode, as shown in **Figure 4-3**, works when the candidate PCC and SCC frequencies are defined using the **PccFreqCfg** and **SccFreqCfg** MOs on the eNodeB. Only cells on these frequencies can be aggregated. Blind SCell configuration is also supported in this mode. It requires the **CaGroupSCellCfg**.***SCellBlindCfgFlag*** parameter to be set to **TRUE**. For details about blind SCell configuration, see **4.6.3.1 SCell Configuration**.

Figure 4-3 Adaptive configuration mode



If adaptive configuration mode is used, adhere to the following configuration rules:

- Rules on the numbers of PCC frequencies and SCC frequencies
  – If the **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is deselected, no more than 16 PCC frequencies can be configured on a single eNodeB.
    - If both the **SMART_CARRIER_SELECTION_SW** and **LOW_FREQ_SCS_OPT_SW** options of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter are deselected, a maximum of 16 SCC frequencies can be specified for each PCC frequency, but no more than 8 of these SCC frequencies can be specified for a single operator.
    - If either the **SMART_CARRIER_SELECTION_SW** or **LOW_FREQ_SCS_OPT_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter is selected, a maximum of 16 SCC frequencies can be specified for each PCC frequency, and all of these SCC frequencies can be specified for a single operator. For details, see *Multi-carrier Unified Scheduling*.

  - If the **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected, no more than 9 PCC frequencies can be configured on a single eNodeB. For each PCC frequency, a total of 16 SCC frequencies can be specified, but no more than 8 of these SCC frequencies can be specified for a single operator.
  - Ensure that the total number of configured PCC and SCC frequencies does not exceed 17. Any additional frequencies will not be selected for CA.
- Rules on SCell configuration
  - On each SCC frequency, only one cell can be specified as a blind-configurable candidate SCell. If there are multiple blind-configurable candidate SCells on an SCC frequency, only one of them can take effect. It is uncertain which one will take effect. The cell that takes effect shifts after eNodeB restart or cell deactivation. This shift will affect the CA UE distribution.
  - If the **SccFreqCfg**.***SpidGrpId*** parameter is set to a value other than **65535** for a candidate SCC, this carrier cannot be configured as an SCC for UEs in this SPID group.
  - If the **SccFreqCfg**.***UlSpidGrpId*** parameter is set to a value other than **65535** for a candidate SCC, this carrier can be configured as an uplink SCC only for UEs in this SPID group.
  - If the PCell of a CA UE is a higher-frequency cell that covers a relatively small area, it is recommended that a lower-frequency cell covering a relatively large area be defined as a blind-configurable candidate SCell. This prevents the CA UE in the higher-frequency cell from performing an unnecessary inter-frequency measurement of the lower-frequency cell, which will always meet the triggering condition for event A4.
  - If the PCell of a CA UE is a lower-frequency cell that covers a relatively large area, it is recommended that blind configuration be disabled (A4 measurements should be used) for a higher-frequency cell that covers a relatively small area. If blind configuration is enabled, the CA UE served by the lower-frequency cell will encounter low scheduling efficiency in the higher-frequency cell. In this case, the data rate of the CA UE will not reach twice that of a non-CA UE in differentiated scheduling mode.
- No more than 1152 **CaGroupSCellCfg** MOs can be configured for a single eNodeB. The actual number of MOs that can take effect may be smaller than this upper limit due to insufficient base station resources (such as CA route capacity and backplane bandwidth).
- Route setup principles for CA
  - Cell relationships (that is, static routes) can be set up between a cell and a number of other cells using **CaGroupSCellCfg** MOs. The maximum number of other cells varies, as described in **Table 4-3**.

Table 4-3 Cell relationships that can be configured for each cell

Dl2CCAckResShareSw of CellAlgoSwitch.*PucchAlgoSwitch*	CaRouteNumberExtensionSwitch of CaMgtCfg.*CellCaAlgoSwitch*	Cell Relationships
Selected	Selected	A cell can be associated with up to 48 other cells.[a]
Deselected	Deselected	A cell can be associated with up to 8 other cells.
Selected	Deselected	A cell can be associated with up to 24 other cells.
Deselected	Selected	This scenario does not exist because the **Dl2CCAckResShareSw** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter must be selected before the **CaRouteNumberExtensionSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter can be selected.
a: This specification is valid only when UMPT boards are used as main control boards and UBBP boards are used as BBPs. In this scenario, the CPU usage of the main control boards and BBPs increases, which may cause service drops.		

Static routes and dynamic routes share the route capacity. When the number of **CaGroupSCellCfg** MOs (indicating the number of static routes) has reached the maximum capacity (48, 8, or 24), no more routes will be set up, including static routes created using the MOs and dynamic routes created based on A4 or A6 measurements. Therefore, no more SCells can be configured or changed to accompany a PCell. The cells within the capacity are not necessarily active SCells. They may be SCells configured but not activated, cells to which routes fail to be set up, or invalid cells.

Fixed routing does not introduce delay. To prevent excessive configurations, which waste inter-board routing bandwidth, it is recommended that one static route be configured for each cell for CA in adaptive configuration mode. Currently, there is no check on the destination cell status of a route when the route is set up.

Dynamic routing, in either intra- or inter-eNodeB cases, causes certain delay in SCell configuration.

– Aging of dynamic routes

  If a dynamic route has been set up from a cell (acting as a PCell) to another cell (acting as an SCell) for CA and CA is not performed between the cells for any CA UE throughout a period longer than the value of the **CaMgtCfg**.***SCellAgingTime*** parameter, the eNodeB considers the route to be outdated. It immediately releases this route so that the route no longer accounts for a unit of the dynamic route capacity.

– Penalty on dynamic route setup

  The eNodeB can impose a penalty on the dynamic setup of some low-value routes to relieve the route shortage for CA, reduce the number of messages transmitted in the eNodeB, and increase the contention-based preamble response rate.

  Contention-based preamble response rate = ((**L.RA.GrpA.Resp** + **L.RA.GrpB.Resp**)/(**L.RA.GrpA.Att** + **L.RA.GrpB.Att**)) x 100%

  This function takes effect when both of the following conditions are met:

  - An LTE cell acts as the PCell, and the **CaMgtCfg**.***CaRouteConfigPenaltyOfs*** parameter of the cell is set to a value other than **0**.
  - An LTE cell or licensed assisted access (LAA) cell acts as the SCell. In the case of an LAA cell, the **LaaRoutePunishmentSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter must be selected. For details about LAA cells, see *Licensed Assisted Access (TDD)*.

  With this function, the eNodeB checks whether the duration of a route from a cell acting as a PCell to a cell acting as an SCell for CA is less than the sum of the **CaMgtCfg**.***SCellAgingTime*** and **CaMgtCfg**.***CaRouteConfigPenaltyOfs*** parameter values.

  - If the duration is less than the sum, the eNodeB considers the route to be of low value and will not dynamically set up this route again within a period that is equal to **CaMgtCfg**.***CaRouteConfigPenaltyWeight*** multiplied by **CaMgtCfg**.***SCellAgingTime***.
  - If the duration is greater than or equal to the sum, the eNodeB will not prohibit dynamic setup of this route for CA.

  When the penalty function takes effect on a network, CA takes effect for a smaller number of UEs, as indicated by smaller values of the **L.CA.DLSCell.Add.Att**, **L.CA.DLSCell.Add.Succ**, and **L.Traffic.User.PCell.DL.Avg** counters for cells that act as PCells in the penalties. As a result, cell throughput decreases.

– Dynamic route setup in the case of an insufficient backplane bandwidth

  Routes set up for CA consume the backplane bandwidth of each eNodeB, which is shared by all features. If there is not enough backplane bandwidth, the routes will fail to be set up. If three consecutive attempts to set up a route to a single cell fail because of an insufficient backplane bandwidth, the eNodeB prohibits the subsequent 30 attempts on the route to this cell. This cell will not be configured as an SCell if route setup fails. As a result, the value of the **L.CA.DLSCell.Add.Att** counter decreases, but the value of the **L.CA.DLSCell.Add.Succ** counter is not affected.

NOTE

In the event of insufficient CA route capacity or backplane bandwidth, the eNodeB will request CA routes again during a version upgrade or reset. As a result, the UE distribution in SCells changes, leading to variations in the related performance indicators, such as the user-perceived rate, spectral efficiency, number of SCell additions, and number of CA UEs. To expand the CA route capacity and reduce the preceding impact on performance indicators due to the UE distribution changes in SCells, you are advised to select both the **Dl2CCAckResShareSw** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter and the **CaRouteNumberExtensionSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter.

4.5 Band Combinations

Currently, eNodeBs support CA in the following types of band combinations:

- 3GPP-defined band combinations

  For details, see **4.5.1 3GPP-defined Band Combinations**.

- Private band combinations

  For details, see **4.5.2 Private Band Combinations**.

None of the band combinations are direction-specific. Both uplink and downlink CA work with the band combinations.

During initial access, an incoming RRC connection reestablishment, or an incoming handover, a CA UE reports its CA band combination capability to the eNodeB. The eNodeB then selects multiple intra- or inter-band carriers for CA based on carrier management principles. Currently, network-requested CA band combination capability signaling is supported, allowing a UE to report a maximum of 384 (128 + 256) CA band combinations. For details, see **4.5.3 Network-requested CA Band Combination Capability Signaling**.

4.5.1 3GPP-defined Band Combinations

eNodeBs of the current version support CA in 3GPP-defined band combinations, with the following exceptions:

- FDD-only CA is possible for cells with 1.4 MHz or 3 MHz bandwidth, but 1.4 MHz cells cannot act as PCells. In 1.4 MHz cells, there are only six resource blocks (RBs) available. If these cells act as PCells for CA UEs, they cannot accommodate the significant increase in the PUCCH load caused by CA.
- TDD-only CA is not possible for cells with 1.4 MHz, 3 MHz, or 5 MHz bandwidth.
- FDD+TDD CA is not possible for FDD cells with 1.4 MHz or 3 MHz bandwidth or TDD cells with 1.4 MHz, 3 MHz, or 5 MHz bandwidth.

The supported 3GPP-defined band combinations vary as follows:

If the main control board of the eNodeB is a UMPT, the eNodeB supports CA in the band combinations defined in section 5.6A.1 "Channel bandwidths per operating band for CA" of 3GPP TS 36.101 V18.3.0.

4.5.2 Private Band Combinations

Band combinations owned by operators and those supported by UEs may not be defined by 3GPP specifications. Huawei eNodeBs support operator-defined band combinations and whitelist control over such band combinations based on UE identification so that CA works in these private band combinations for UEs.

Operator-defined Band Combinations

When operators own certain band combinations that have not been defined by 3GPP specifications, they can define band combinations to meet their requirements for band combination customization.

Currently, TDD-only CA works in the following operator-defined band combinations:

- Band 40 (10 MHz) + Band 40 (20 MHz) for non-contiguous 2CC aggregation
- Band 41 (10 MHz/20 MHz) + Band 41 (20 MHz) + Band 42 (20 MHz) for 3CC aggregation, where the two carriers in Band 41 are non-contiguous

In these combinations:

- In each pair of parentheses, a slash (/) denotes "or".
- If CA works between a combination of contiguous carriers, it also works between the contiguous carriers that constitute any subset of this combination.

  For example, if CA works between four contiguous carriers in Band 40 (20 MHz + 20 MHz + 20 MHz + 20 MHz), it also works in subsets Band 40 (20 MHz + 20 MHz) and Band 40 (20 MHz + 20 MHz + 20 MHz).
- Before setting band combinations for 3CC or 4CC aggregation, ensure that the two or three carriers in the subsets of the combinations support CA. If any subset is not defined as a CA band combination by 3GPP, add it as an operator-defined one.

Operators can configure the **PrivateCaBandComb** and **PrivateBand** MOs to define band combinations for 2CC to 8CC aggregation.

A **PrivateCaBandComb** MO defines an operator-defined CA band combination, and a **PrivateBand** MO defines a frequency band in the combination. The parameters in the **PrivateCaBandComb** and **PrivateBand** MOs are described in **Table 4-4** and **Table 4-5**, respectively.

Only one **PrivateCaBandComb** MO can be configured for 8CC aggregation. If multiple MOs are configured, only the combination with the smallest **PrivateCaBandComb**.***PrivateCaCombId*** parameter value takes effect.

Table 4-4 Parameters in the **PrivateCaBandComb** MO

Parameter ID	Parameter Description
PrivateCaBandComb.***PrivateCaCombId***	Identifies an operator-defined CA band combination. This ID is unique within an eNodeB.
PrivateCaBandComb.***MaxAggregatedBw***	Specifies the maximum aggregated bandwidth that the band combination supports.
PrivateCaBandComb.***BwCombSetId***	Identifies a bandwidth combination set for the band combination. A band combination may support multiple bandwidth combination sets.
PrivateCaBandComb.***NsaFlag***	Specifies whether the combination is an NSA band combination.

Table 4-5 Parameters in the **PrivateBand** MO

Parameter ID	Parameter Description
PrivateBand.***PrivateCaCombId***	Identifies an operator-defined CA band combination within an eNodeB.
PrivateBand.***CombBandIndex***	Identifies a frequency band within the operator-defined CA band combination. This index is unique within the band combination. The values of this parameter in the **PrivateBand** MOs for an operator-defined band combination must be consecutive. Otherwise, the band combination would be invalid. The value of this parameter must be **1** in the first MO.
PrivateBand.***CombBandId***	Specifies the indicator of the band.
PrivateBand.***CombBandBw***	Specifies the set of bandwidths allowed in the band.
PrivateBand.***BandType***	Specifies whether the band is used for LTE or NR.

The **PrivateCaBandComb** MO needs to be configured prior to the **PrivateBand** MO. The two MOs are associated through the **PrivateBand**.***PrivateCaCombId*** and **PrivateCaBandComb**.***PrivateCaCombId*** parameters.

Take the operator-defined LTE CA band combination 7a (10 MHz) + 7a (10 MHz) + 28a (10 MHz) as an example. Assume that the

PrivateCaBandComb.***PrivateCaCombId*** parameter is set to **1** for the band combination. Each of the three bands in this band combination is associated with the band combination by setting the **PrivateBand**.***PrivateCaCombId*** parameter to **1** for each band.

The following MM-based configuration example presents the parameter settings for the three bands.

```
//Setting PrivateCaCombId, MaxAggregatedBw, and BwCombSetId
ADD PRIVATECABANDCOMB: PrivateCaCombId=1, MaxAggregatedBw=30, BwCombSetId=0;
//Setting private bands
ADD PRIVATEBAND: PrivateCaCombId=1, CombBandIndex=1, CombBandId=7,
CombBandBw=Bandwidth_10M-1, BandType=LTE;
ADD PRIVATEBAND: PrivateCaCombId=1, CombBandIndex=2, CombBandId=7,
CombBandBw=Bandwidth_10M-1, BandType=LTE;
ADD PRIVATEBAND: PrivateCaCombId=1, CombBandIndex=3, CombBandId=28,
CombBandBw=Bandwidth_10M-1, BandType=LTE;
```

The following configuration constraints apply to operator-defined band combinations:

- If a 3GPP-defined band combination supported by eNodeBs conflicts with an operator-defined band combination, the operator-defined one takes precedence. To avoid UE faults caused by conflicts between 3GPP-defined and operator-defined bandwidth combination sets, operators are advised to set **PrivateCaBandComb**.**BwCombSetId** to be different from any 3GPP-defined set ID (for details, see section 5.6A.1 "Channel bandwidths per operating band for CA" in 3GPP TS 36.101 V17.3.0).
- It is recommended that the **PrivateCaBandComb**.***BwCombSetId*** parameter of an operator-defined band combination be set to the same value as the combination set ID of the UEs supporting this band combination. Otherwise, the band combination does not take effect.
- If a subset of an operator-defined band combination is not defined in 3GPP specifications or if the operator has special requirements for band combinations, data reconfiguration will be required.
- Ensure that there is no interference between the operator-defined bands for FDD+TDD CA. For details about how to meet the requirement, contact Huawei engineers.

Whitelist Control over Operator-defined Band Combinations Based on UE Identification

When certain UEs support band combinations that have not been defined by 3GPP specifications, whitelist control over operator-defined band combinations based on UE identification can be used so that CA works in these private band combinations for the UEs.

For this whitelist control function to take effect, operators must configure the **PrivateCaBandComb** and **PrivateBand** MOs to define the UE-supported band combinations on eNodeBs. The operators then must configure the **UeCompat** MO for UE compatibility handling. Ensure that the **PrivateCaBandComb**.**PrivateCaCombId** and **UeCompat**.***WhiteLstCaCombSwitch*** parameter settings are consistent.

For example, if band combinations with **PrivateCaBandComb**.**PrivateCaCombId** of **2** and **3** are defined and the **PRIVATECACOMBID_2** and **PRIVATECACOMBID_3** options of the **UeCompat**.**WhiteLstCaCombSwitch** parameter are selected, the two band combinations take effect only for whitelisted UEs.

This whitelist control mechanism takes effect only for band combinations whose value of **PrivateCaBandComb**.**PrivateCaCombId** is within the range of 0–31.

4.5.3 Network-requested CA Band Combination Capability Signaling

3GPP Release 10 stipulates that a UE can report a maximum of 128 CA band combinations. As the supply of commercial CA-capable UEs increases and UE maturity improves, the number of band combinations supported by a single UE has been growing. 3GPP Release 11 has incorporated network-requested CA band combination capability signaling, allowing a single UE to report a maximum of 384 (128 + 256) CA band combinations. This function is controlled by the **SupportNetReqCaComboSwitch** option of the **GlobalProcSwitch**.***ProtocolSupportSwitch*** parameter. For details about network-requested CA band combination capability signaling, see section 5.6.3.3 "Reception of the UECapabilityEnquiry by the UE" in 3GPP TS 36.331 V13.1.0.

With network-requested CA band combination capability signaling, UEs that support more than 128 CA band combinations can enter the CA state. The procedure for reporting CA band combinations depends on the settings of the **CaBandCombQueryOptSwitch** option of the **GlobalProcSwitch**.***ProtocolMsgOptSwitch*** parameter and the **GlobalProcSwitch**.***UeEutraCapbleSizeThld*** parameter, as presented in **Table 4-6**.

Table 4-6 Relationships between reporting procedures and parameter settings

CaBandCombQueryOptSwitch of GlobalProcSwitch.*ProtocolMsgOptSwitch*	GlobalProcSwitch.*UeEutraCapbleSizeThld*	Procedure
Selected	Non-zero value	For details, see **Parameter Setting Combination A**.
Selected	**0**	For details, see **Parameter Setting Combination B**.
Deselected	Non-zero value	For details, see **Parameter Setting Combination C**.
Deselected	**0**	For details, see **Parameter Setting Combination D**.

Operators are advised to enable network-requested CA band combination capability signaling across the network when CA is in use. Otherwise, certain CA band combinations may fail to be configured.

After network-requested CA band combination capability signaling is enabled, reporting of compressed CA band combination fields can also be enabled (controlled by the **UeBandCombReducedR13Sw** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter) so that the eNodeB is compatible with the reporting. After this reporting function is enabled, the additional procedure for querying CA band combination capabilities has a negative impact on performance indicators such as the number of signaling messages, access delay, and E-RAB setup success rate.

NOTE

If the **CaBandCombQueryOptSwitch** option of the **GlobalProcSwitch**.***ProtocolMsgOptSwitch*** parameter and the **SupportNetReqCaComboSwitch** option of the **GlobalProcSwitch**.***ProtocolSupportSwitch*** parameter are both selected, the number of UEs in the downlink 4CC and 5CC aggregation states rises and the number of UEs in the uplink 2CC aggregation state decreases.

Parameter Setting Combination A

A UE that supports more than 128 CA band combinations can enter the CA state in certain scenarios and the procedure for the UE to report CA band combinations is scenario-specific, as described in **Table 4-7**.

Table 4-7 Scenario-specific procedures for a UE to report CA band combinations

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
1	During the initial access of a UE to a cell, the EPC does not send the E-UTRAN capabilities of the UE to the eNodeB.	1. The eNodeB includes the requestedFrequencyBands IE (which contains a maximum of 16 bands) in the UECapabilityEnquiry message and sends this message to the UE. 2. If the UE supports network-requested CA band combination capability signaling, it includes the requestedBands, supportedBandCombination, and supportedBandCombinationAdd IEs in the UECapabilityInformation message and sends this message to the eNodeB. The supportedBandCombination and supportedBandCombinationAdd IEs convey up to 128 and 256 CA band combinations for the requested bands, respectively. These two IEs include a maximum of 384 combinations in total.	• The number of signaling messages involved does not increase. • The CA band combinations stored on the eNodeB for this UE vary depending on the number of bands conveyed by the requestedFrequencyBands IE. Therefore, the memory usage of the eNodeB changes. • The eNodeB includes the requestedFrequencyBands IE in the UECapabilityEnquiry message, without knowing if the UE supports network-requested CA band combination capability signaling. If the UE does not support it, exceptions such as service drops, CA failures, or performance deterioration may occur.

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
2	During the initial access of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the EPC to the eNodeB includes the requestedBands IE, but none of the CA bands conveyed by this IE are supported by the eNodeB.	1. The eNodeB includes the requestedFrequencyBands IE (which contains a maximum of 16 bands) in the UECapabilityEnquiry message and sends this message to the UE. 2. The UE includes the requestedBands, supportedBandCombination, and supportedBandCombinationAdd IEs in the UECapabilityInformation message and sends this message to the eNodeB. The supportedBandCombination and supportedBandCombinationAdd IEs convey up to 128 and 256 CA band combinations for the requested bands, respectively. These two IEs include a maximum of 384 combinations in total.	• As the eNodeB needs to store more CA band combinations for each UE, the memory usage of the eNodeB increases slightly. • The additional procedure for querying CA band combination capabilities has a negative impact on performance indicators such as the number of signaling messages, access delay, and E-RAB setup success rate. For example: – **L.Signal.Num.Uu** increases. – **L.E-RAB.InitEst.TimeAvg** increases. – **L.E-RAB.InitEst.TimeMax** increases. – **L.E-RAB.Est.TimeAvg** increases. – **L.E-RAB.Est.TimeMax** increases. – **E-RAB Setup Success Rate** decreases.

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
3	During the initial access of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the EPC to the eNodeB does not include the requestedBands IE. In addition, the size of the UE-EUTRA-Capability IE exceeds the value of the **GlobalProcSwitch**.***UeEutraCapbleSizeThld*** parameter, or the number of CA band combinations included in the E-UTRAN capability information reaches 128.	Same as scenario 2 in **Parameter Setting Combination A**	Same as scenario 2 in **Parameter Setting Combination A**

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
4	During an RRC connection reestablishment or handover of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the source eNodeB includes the requestedBands IE, but none of the CA bands conveyed by this IE are supported by the target eNodeB.	Same as scenario 2 in **Parameter Setting Combination A**	Same as scenario 2 in **Parameter Setting Combination A**

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
5	During an RRC connection reestablishment or handover of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the source eNodeB does not include the requestedBands IE. In addition, the size of the UE-EUTRA-Capability IE exceeds the value of the **GlobalProcSwitch.*UeEutraCapbleSizeThld*** parameter, or the number of CA band combinations included in the E-UTRAN capability information reaches 128.	Same as scenario 2 in **Parameter Setting Combination A**	Same as scenario 2 in **Parameter Setting Combination A**

Parameter Setting Combination B

A UE that supports more than 128 CA band combinations can enter the CA state in certain scenarios and the procedure for the UE to report CA band combinations is scenario-specific, as described in **Table 4-8**.

Table 4-8 Scenario-specific procedures for a UE to report CA band combinations

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
1	During the initial access of a UE to a cell, the EPC does not send the E-UTRAN capabilities of the UE to the eNodeB.	1. The eNodeB includes the requestedFrequencyBands IE (which contains a maximum of 16 bands) in the UECapabilityEnquiry message and sends this message to the UE. 2. If the UE supports network-requested CA band combination capability signaling, it includes the requestedBands, supportedBandCombination, and supportedBandCombinationAdd IEs in the UECapabilityInformation message and sends this message to the eNodeB. The supportedBandCombination and supportedBandCombinationAdd IEs convey up to 128 and 256 CA band combinations for the	● The number of signaling messages involved does not increase. ● The CA band combinations stored on the eNodeB for this UE vary depending on the number of bands conveyed by the requestedFrequencyBands IE. Therefore, the memory usage of the eNodeB changes. ● The eNodeB includes the requestedFrequencyBands IE in the UECapabilityEnquiry message, without knowing if the UE supports network-requested CA band combination capability signaling. If the UE does not support it, exceptions such as service drops, CA failures, or performance deterioration may occur.

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
		requested bands, respectively. These two IEs include a maximum of 384 combinations in total.	

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
2	During the initial access of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the EPC to the eNodeB includes the requestedBands IE, but none of the CA bands conveyed by this IE are supported by the eNodeB.	1. The eNodeB includes the requestedFrequencyBands IE (which contains a maximum of 16 bands) in the UECapabilityEnquiry message and sends this message to the UE. 2. The UE includes the requestedBands, supportedBandCombination, and supportedBandCombinationAdd IEs in the UECapabilityInformation message and sends this message to the eNodeB. The supportedBandCombination and supportedBandCombinationAdd IEs convey up to 128 and 256 CA band combinations for the requested bands, respectively. These two IEs include a maximum of 384 combinations in total.	• As the eNodeB needs to store more CA band combinations for each UE, the memory usage of the eNodeB increases slightly. • The additional procedure for querying CA band combination capabilities has a negative impact on performance indicators such as the number of signaling messages, access delay, and E-RAB setup success rate. For example: – **L.Signal.Num.Uu** increases. – **L.E-RAB.InitEst.TimeAvg** increases. – **L.E-RAB.InitEst.TimeMax** increases. – **L.E-RAB.Est.TimeAvg** increases. – **L.E-RAB.Est.TimeMax** increases.

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
3	During the initial access of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the EPC to the eNodeB does not include the requestedBands IE. In addition, the number of CA band combinations included in the E-UTRAN capability information reaches 128.	Same as scenario 2 in **Parameter Setting Combination B**	– **E-RAB Setup Success Rate** decreases.
4	During an RRC connection reestablishment or handover of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the source eNodeB includes the requestedBands IE, but none of the CA bands conveyed by this IE are supported by the target eNodeB.	Same as scenario 2 in **Parameter Setting Combination B**	

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
5	During an RRC connection reestablishment or handover of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the source eNodeB does not include the requestedBands IE. In addition, the number of CA band combinations included in the E-UTRAN capability information reaches 128.	Same as scenario 2 in **Parameter Setting Combination B**	

Parameter Setting Combination C

A UE that supports more than 128 CA band combinations can enter the CA state in certain scenarios and the procedure for the UE to report CA band combinations is scenario-specific, as described in **Table 4-9**.

Table 4-9 Scenario-specific procedures for a UE to report CA band combinations

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
1	During the initial access of a UE to a cell, the EPC does not send the E-UTRAN capabilities of the UE to the eNodeB.	1. The eNodeB sends a UECapabilityEnquiry message to the UE to query its CA band combination capability. 2. The UE reports the supported CA band combinations (a maximum of 128 combinations) to the eNodeB through the 3GPP Release 10 IE supportedBandCombination. If the UE complies with 3GPP Release 11, it also sends the freqBandRetrieval-r11 IE to report its capability of network-requested CA band combination capability signaling. 3. If the UE complies with 3GPP Release 11, and if the size of the UE-reported UE-EUTRA-Capability IE exceeds the value of the	● As the eNodeB needs to store more CA band combinations for each UE, the memory usage of the eNodeB increases slightly. ● The additional procedure for querying CA band combination capabilities has a negative impact on performance indicators such as the number of signaling messages, access delay, and E-RAB setup success rate. For example: – **L.Signal.Num.Uu** increases. – **L.E-RAB.InitEst.TimeAvg** increases. – **L.E-RAB.InitEst.TimeMax** increases. – **L.E-RAB.Est.TimeAvg** increases. – **L.E-RAB.Est.Ti**

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
		GlobalProcSwitch.*UeEutraCapbIeSizeThld* parameter or the number of CA band combinations reported by the UE reaches 128, the eNodeB includes the requestedFrequencyBands IE (which contains a maximum of 16 bands) in a second UECapabilityEnquiry message and sends this message to the UE. Otherwise, the procedure ends. 4. The UE includes the requestedBands, supportedBandCombination, and supportedBandCombinationAdd IEs in the UECapabilityInformation message and sends this message to the eNodeB. The supportedBandCombination and supportedBandCombinationAdd IEs convey up to 128 and 256	**meMax** increases. – **E-RAB Setup Success Rate** decreases.

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
		CA band combinations for the requested bands, respectively. These two IEs include a maximum of 384 combinations in total.	

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
2	During the initial access of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the EPC to the eNodeB includes the requestedBands IE, but none of the CA bands conveyed by this IE are supported by the eNodeB.	1. The eNodeB includes the requestedFrequencyBands IE (which contains a maximum of 16 bands) in the UECapabilityEnquiry message and sends this message to the UE. 2. The UE includes the requestedBands, supportedBandCombination, and supportedBandCombinationAdd IEs in the UECapabilityInformation message and sends this message to the eNodeB. The supportedBandCombination and supportedBandCombinationAdd IEs convey up to 128 and 256 CA band combinations for the requested bands, respectively. These two IEs include a maximum of 384 combinations in total.	

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
3	During the initial access of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the EPC to the eNodeB does not include the requestedBands IE. In addition, the size of the UE-EUTRA-Capability IE exceeds the value of the **GlobalProcSwitch.*UeEutraCapbleSizeThld*** parameter, or the number of CA band combinations included in the E-UTRAN capability information reaches 128.	Same as scenario 2 in **Parameter Setting Combination C**	
4	During an RRC connection reestablishment or handover of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the source eNodeB includes the requestedBands IE, but none of the CA bands conveyed by this IE are supported by the target eNodeB.	Same as scenario 2 in **Parameter Setting Combination C**	

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
5	During an RRC connection reestablishment or handover of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the source eNodeB does not include the requestedBands IE. In addition, the size of the UE-EUTRA-Capability IE exceeds the value of the **GlobalProcSwitch.*UeEutraCapbleSizeThld*** parameter, or the number of CA band combinations included in the E-UTRAN capability information reaches 128.	Same as scenario 2 in **Parameter Setting Combination C**	

Parameter Setting Combination D

A UE that supports more than 128 CA band combinations can enter the CA state in certain scenarios and the procedure for the UE to report CA band combinations is scenario-specific, as described in **Table 4-10**.

Table 4-10 Scenario-specific procedures for a UE to report CA band combinations

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
1	During the initial access of a UE to a cell, the EPC does not send the E-UTRAN capabilities of the UE to the eNodeB.	1. The eNodeB sends a UECapabilityEnquiry message to the UE to query its CA band combination capability. 2. The UE reports the supported CA band combinations (a maximum of 128 combinations) to the eNodeB through the 3GPP Release 10 IE supportedBandCombination. If the UE complies with 3GPP Release 11, it also sends the freqBandRetrieval-r11 IE to report its capability of network-requested CA band combination capability signaling. 3. If the UE complies with 3GPP Release 11 and the number of CA band combinations reported by the UE reaches 128, the	• As the eNodeB needs to store more CA band combinations for each UE, the memory usage of the eNodeB increases slightly. • The additional procedure for querying CA band combination capabilities has a negative impact on performance indicators such as the number of signaling messages, access delay, and E-RAB setup success rate. For example: – **L.Signal.Num.Uu** increases. – **L.E-RAB.InitEst.TimeAvg** increases. – **L.E-RAB.InitEst.TimeMax** increases. – **L.E-RAB.Est.TimeAvg** increases. – **L.E-RAB.Est.Ti**

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
		eNodeB includes the requestedFrequencyBands IE (which contains a maximum of 16 bands) in a second UECapabilityEnquiry message and sends this message to the UE. Otherwise, the procedure ends. 4. The UE includes the requestedBands, supportedBandCombination, and supportedBandCombinationAdd IEs in the UECapabilityInformation message and sends this message to the eNodeB. The supportedBandCombination and supportedBandCombinationAdd IEs convey up to 128 and 256 CA band combinations for the requested bands, respectively. These two IEs include a maximum of 384	**meMax** increases. – **E-RAB Setup Success Rate** decreases.

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
		combinations in total.	
2	During the initial access of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the EPC to the eNodeB includes the requestedBands IE, but none of the CA bands conveyed by this IE are supported by the eNodeB.	Same as scenario 2 in **Parameter Setting Combination B**	
3	During the initial access of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the EPC to the eNodeB does not include the requestedBands IE. In addition, the number of CA band combinations included in the E-UTRAN capability information reaches 128.	Same as scenario 3 in **Parameter Setting Combination B**	

Scenario No.	Scenario Description	Reporting Procedure	Network Impacts
4	During an RRC connection reestablishment or handover of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the source eNodeB includes the requestedBands IE, but none of the CA bands conveyed by this IE are supported by the target eNodeB.	Same as scenario 4 in **Parameter Setting Combination B**	
5	During an RRC connection reestablishment or handover of a 3GPP Release 11 UE to a cell, the UE's E-UTRAN capability information sent from the source eNodeB does not include the requestedBands IE. In addition, the number of CA band combinations included in the E-UTRAN capability information reaches 128.	Same as scenario 5 in **Parameter Setting Combination B**	

Specified Band Configuration

When network-requested CA band combination capability signaling takes effect, 3GPP Release 11 UEs, as controlled, can report only the combinations of specified bands. The data source of the specified bands is now under parameter control. It varies depending on the setting of **INDEP_UE_BAND_ENQUIRY_SW** of the **GlobalProcSwitch**.***ProtocolMsgOptExtSwitch*** parameter and whether the

OperatorBand.***BandType*** parameter is set to **LTE** for any band. **Table 4-11** provides the details.

Table 4-11 Relationships between parameters and the data source of specified bands

No.	INDEP_UE_BAND_ENQUIRY_SW of GlobalProcSwitch.*ProtocolMsgOptExtSwitch*	OperatorBand.*BandType* Set to LTE for Any Band?	Data Source of Specified Bands	Description
1	Selected	Yes	**Operator Band** MOs	In either CA-group-based or adaptive configuration mode, the data source of the requestedFrequencyBands IE is as follows: ● The source is the bands identified by the **OperatorBand**.***BandId*** parameter in the **OperatorBand** MOs with the **OperatorBand**.***BandType*** parameter set to **LTE**. ● If the **OperatorBand**.***BandAttribute*** parameter is set for these bands, the eNodeB filters out the bands with **OperatorBand**.***BandAttribute*** set to **EN-DC** before including bands in the requestedFrequencyBands IE to be sent to the UE. ● If the **OperatorBand**.***CnOperatorList*** parameter is set for these bands, the eNodeB filters out the bands that do not belong to the operator of a UE from the bands identified by the **OperatorBand**.***BandId*** parameter, and then it includes the remaining bands in the requestedFrequencyBands IE to be sent to the UE. ● If all the bands identified by the **OperatorBand**.***BandId*** parameter are filtered out, the requestedFrequencyBands IE is empty. In this situation, the UE does not report any band combination, and CA does not take effect for the UE.

No.	INDEP_UE_BAND_ENQUIRY_SW of GlobalProcSwitch.*ProtocolMsgOptExtSwitch*	OperatorBand.*BandType* Set to LTE for Any Band?	Data Source of Specified Bands	Description
				By using **OperatorBand** MOs, eNodeB consistency in the source of specified bands can be maintained when different frequencies are configured on the eNodeBs.
2	Selected or deselected	No	Configured bands	The data source of the requestedFrequencyBands IE varies as follows: ● CA-group-based configuration mode The source is the complete set of bands that the cells in the CA group belong to. ● Adaptive configuration mode – The source is the bands that the frequencies identified by the **PccFreqCfg**.***PccDlEarfcn*** and **SccFreqCfg**.***SccDlEarfcn*** parameters belong to and the primary and secondary operating bands of multi-band cells. – If the **SccFreqCfg**.***CnOperatorList*** parameter is set for candidate SCCs, the eNodeB filters the bands that the frequencies identified by the **SccFreqCfg**.***SccDlEarfcn*** parameter belong to and the primary and secondary operating bands of multi-band cells for a UE based on the operator of the UE.

No.	INDEP_UE_BAND_ENQUIRY_SW of **GlobalProcSwitch**.***ProtocolMsgOptExtSwitch***	**OperatorBand**.***BandType*** Set to LTE for Any Band?	Data Source of Specified Bands	Description
3	Deselected	Yes	Union of **Operator Band** MOs and configured bands	The data source of the requestedFrequencyBands IE varies as follows: ● CA-group-based configuration mode – The source is the complete set of bands that the cells in the CA group belong to. – The source is the bands identified by the **OperatorBand**.***BandId*** parameter in the **OperatorBand** MOs with the **OperatorBand**.***BandType*** parameter set to **LTE**. If the **OperatorBand**.***BandAttribute*** parameter is set for these bands, the eNodeB filters out the bands with **OperatorBand**.***BandAttribute*** set to **EN-DC** before including bands in the requestedFrequencyBands IE to be sent to the UE. If the **OperatorBand**.***CnOperatorList*** parameter is set for these bands, the eNodeB filters out the bands that do not belong to the operator of a UE from the bands identified by the **OperatorBand**.***BandId*** parameter, and then it includes the remaining bands in the requestedFrequencyBands IE to be sent to the UE. ● Adaptive configuration mode – The source is the bands that the frequencies identified by the **PccFreqCfg**.***PccDlEarfcn*** and **SccFreqCfg**.***SccDlEarfcn***

No.	INDEP_UE_BAND_ENQUIRY_SW of GlobalProcSwitch.*ProtocolMsgOptExtSwitch*	OperatorBand.*BandType* Set to LTE for Any Band?	Data Source of Specified Bands	Description
				parameters belong to and the primary and secondary operating bands of multi-band cells. If the **SccFreqCfg**.***CnOperatorList*** parameter is set for candidate SCCs, the eNodeB filters the bands that the frequencies identified by the **SccFreqCfg**.***SccDlEarfcn*** parameter belong to and the primary and secondary operating bands of multi-band cells for a UE based on the operator of the UE. – The source is the bands identified by the **OperatorBand**.***BandId*** parameter. If the **OperatorBand**.***BandAttribute*** parameter is set for these bands, the eNodeB filters out the bands with **OperatorBand**.***BandAttribute*** set to **EN-DC** before including bands in the requestedFrequencyBands IE to be sent to the UE. If the **OperatorBand**.***CnOperatorList*** parameter is set for these bands, the eNodeB filters out the bands that do not belong to the operator of a UE from the bands identified by the **OperatorBand**.***BandId*** parameter, and then it includes the remaining bands in the requestedFrequencyBands IE to be sent to the UE.

When the number of specified bands delivered by the eNodeB to a UE increases, the UE reports more CA band combinations. As a result, the eNodeB memory consumption increases slightly, and there will be a negative impact on performance indicators such as the access delay and E-RAB setup success rate. For example:

- **L.E-RAB.InitEst.TimeAvg** increases.
- **L.E-RAB.InitEst.TimeMax** increases.
- **L.E-RAB.Est.TimeAvg** increases.
- **L.E-RAB.Est.TimeMax** increases.
- **E-RAB Setup Success Rate** decreases.

If the number of bands specified using the **OperatorBand**.***BandId*** parameter is less than the number of specified bands before this parameter is configured, the number of CA band combinations reported by the UE decreases. As a result, the number of CA UEs on the network decreases even to 0.

Note the following on capability query:

- If the **OperatorBand**.***BandType*** parameter is set, EN-DC combination capability query for NSA DC UEs will be affected. For details, see *EPC-based NSA Basics*.
- As stipulated by protocols, the requestedFrequencyBands IE can contain at most 16 bands. When more than 16 bands are specified, the UE capabilities about the bands beyond the scope of 16 bands cannot be queried.

The **QUERY_SPEC_BAND_UE_CAP_SW** option of the **UeCompat**.***WhitelistCtrlExtSwitch1*** parameter can be used to determine whether **OperatorBand** MOs take effect only for whitelisted UEs:

- If the **QUERY_SPEC_BAND_UE_CAP_SW** option of the **UeCompat**.***WhitelistCtrlExtSwitch1*** parameter is selected, specified band configuration using the **OperatorBand**.***BandId*** parameter takes effect only for whitelisted UEs. For details about whitelist control, see *Terminal Awareness Differentiation*.
- If the **QUERY_SPEC_BAND_UE_CAP_SW** option of the **UeCompat**.***WhitelistCtrlExtSwitch1*** parameter is deselected, specified band configuration using the **OperatorBand**.***BandId*** parameter takes effect for all UEs.
- Particularly, if the **QUERY_SPEC_BAND_UE_CAP_SW** option of the **UeCompat**.***WhitelistCtrlExtSwitch1*** parameter and the **CaBandCombQueryOptSwitch** option of the **GlobalProcSwitch**.***ProtocolMsgOptSwitch*** parameter are selected, the eNodeB cannot identify whether a UE is a whitelisted UE when the UE performs initial access and the core network does not deliver E-UTRAN capability information. Therefore, the eNodeB considers the UE to be a non-whitelisted UE and does not specify bands for it using the **OperatorBand**.***BandId*** parameter.

4.6 Carrier Management for RRC_CONNECTED UEs

Carrier management for RRC_CONNECTED UEs includes PCC anchoring and SCell management. SCell management includes SCell configuration, change, activation, deactivation, and removal.

- PCC anchoring for RRC_CONNECTED UEs occurs when a CA UE accesses the network at initial access, an incoming RRC connection reestablishment, or an incoming necessary handover. For details, see **4.6.1 PCC Anchoring** and **4.6.2 PCC Anchoring Enhancement**.

  PCC anchoring enhancement refers to load-based PCC anchoring. With this function, intra- or inter-eNodeB cells with routes set up between them exchange their load status, and the eNodeB selects a candidate cell not in the high load state as the PCell for a UE.
- SCell configuration occurs when a CA UE accesses the network at initial access, an incoming RRC connection reestablishment, or an incoming necessary handover. For details, see **4.6.3.1 SCell Configuration** and **4.6.3.2 SCell Configuration Enhancement**.

  SCell configuration enhancement refers to load-based SCell configuration. With this function, intra- or inter-eNodeB cells with routes set up between them exchange their load status, and the eNodeB selects candidate cells not in the high load state as SCells for a UE.
- If a CA UE whose SCell has been configured receives better signal quality from an intra-frequency neighboring cell of the SCell than from the SCell, the serving eNodeB of the PCell can change the SCell while keeping the PCell unchanged. For details, see **4.6.3.3 SCell Change**.
- SCells configured for a CA UE are activated when certain conditions are met. The PCell and SCells are not actually aggregated until the SCells are activated. For details, see **4.6.3.4 SCell Activation**.
- If SCells meet deactivation conditions, the eNodeB delivers MAC control elements (CEs) to deactivate the SCells. For details, see **4.6.3.5 SCell Deactivation**.
- If SCell signal quality deteriorates, the eNodeB may remove the SCells. For details, see **4.6.3.6 SCell Removal**.

4.6.1 PCC Anchoring

PCC anchoring aims to prioritize certain frequencies as PCCs. In this procedure, the eNodeB selects the highest-priority candidate PCell as the PCell for a CA UE. This is implemented with the following:

1. The PCell priority or PCC priority is set.
   - The PCell priority is specified by the **CaGroupCell.*PreferredPCellPriority*** parameter for CA-group-based CA.
   - The PCC priority is specified by the **PccFreqCfg.*PreferredPccPriority*** parameter for adaptive CA.
2. The procedure for PCC anchoring for RRC_CONNECTED UEs is performed.

   PCC anchoring for RRC_CONNECTED UEs is controlled by the **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter.

NOTE

- This section describes only the PCC anchoring function controlled by the **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter. It is recommended that this option be selected.
- If both the **PccAnchorSwitch** and **EnhancedPccAnchorSwitch** options of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter are selected, only the **EnhancedPccAnchorSwitch** option takes effect. The eNodeB evaluates triggering of PCC anchoring as described in **4.6.1.1 Triggering Conditions**.
- If the **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is deselected and the **PccAnchorSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected, PCC anchoring can be performed during initial access of a UE that meets the attribute conditions described in **4.6.1.1 Triggering Conditions**.

PCC anchoring for RRC_CONNECTED UEs is explained from the following aspects:

- Triggering conditions. For details, see **4.6.1.1 Triggering Conditions**.
- Procedures. For details, see:
  - **4.6.1.2 CA-Group-based PCC Anchoring Procedure**
  - **4.6.1.3 Adaptive PCC Anchoring Procedure**

NOTE

- When the **ENodeBAlgoSwitch.*FastEnhancePccAnchorSwitch*** parameter is set to **ON**, the eNodeB directly delivers the A5-related inter-frequency measurement configuration for PCC anchoring to the CA UE without performing SCell configuration or instructing the UE to perform A1 measurements, if a PCell candidate is also a blind-configurable SCell candidate (**CaGroupSCellCfg.*SCellBlindCfgFlag*** is set to **TRUE** for the cell) or all PCell candidates are working on lower frequencies than the current serving cell. That is because, in this situation, there is a high probability of successful inter-frequency measurements for PCC anchoring.
- Frequencies in bands 29, 32, 46, 67, 69, 75, and 76 cannot act as PCCs. For their cell parameter settings, see **Parameters Related to Cells in Special Bands** in **5.4.1.1.3 Common Parameters**.
- PCC anchoring is not initiated for high-mobility UEs.
- PCC anchoring for RRC_CONNECTED UEs is not recommended for LTE cells with **Cell**.***HighSpeedFlag*** set to **HIGH_SPEED**, **ULTRA_HIGH_SPEED**, or **EXTRA_HIGH_SPEED**.
- If a UE supports NSA DC and CA works in adaptive configuration mode, the PCC priorities of E-UTRA frequencies configured for NSA DC are higher than those configured for CA. The eNodeB selects a PCC frequency for the UE preferentially based on the PCC priorities configured for NSA DC. This frequency, however, may not be the optimal one for CA, which may cause CA performance to deteriorate.
- If PCC anchoring for RRC_CONNECTED UEs and management of neighboring E-UTRAN frequency measurement flags are both enabled and such a flag (that is, the **FREQ_MEAS_FLAG** option of the **EutranInterNFreq**.***AggregationAttribute*** parameter) is deselected for a neighboring E-UTRAN frequency, the eNodeB does not select this frequency as the target frequency during PCC anchoring for RRC_CONNECTED UEs.

  Management of neighboring E-UTRAN frequency measurement flags is controlled by the **LTE_NFREQ_MEAS_MGMT_SW** option of the **CellAlgoExtSwitch**.***AnrOptSwitch*** parameter. For details about this function, see *ANR Management*.
- If the **HO_TRG_FREQ_FORBID_MEAS_FLAG** option of the **EutranInterNFreq**.***AggregationAttribute*** parameter is selected for a neighboring intra-RAT frequency, this neighboring frequency will not be selected as the target frequency during PCC anchoring for RRC_CONNECTED UEs. For details, see the section on frequency selection for measurement in *Mobility Management in Connected Mode*.

4.6.1.1 Triggering Conditions

Assume that PCC anchoring for RRC_CONNECTED UEs has been enabled and PCell priorities or PCC priorities have been specified. The eNodeB delivers the A1 measurement configuration to the CA UE when both of the following requirements are met: (1) the target cell for initial access, an RRC connection reestablishment, or a handover of the UE does not have the highest PCell priority or the carrier of the cell does not have the highest PCC priority; (2) the UE meets the attribute conditions for triggering PCC anchoring. The A1 measurement configuration is related to this target cell.

The UE attribute conditions are as follows:

- The UE is not running an evolved multimedia broadcast/multicast service (eMBMS). For details about eMBMS, see *eMBMS*.
- The UE is not engaged in an emergency call. For details about emergency calls, see *Emergency Call*.
- The UE is not in the high speed mobility state. For details about high speed mobility, see *High Speed Mobility*.
- (Applicable when PCC anchoring prohibition for high-mobility UEs, controlled by the **PCC_ANCHOR_HO_FORBID_SW** option of the **CellOpHoCfg**.***HighMobiUeHoForbidSw*** parameter, is enabled) The UE is a non-high-mobility UE (that is, its moving speed is less than or equal to 30 km/h). For the network impacts of PCC anchoring prohibition for high-mobility UEs, see **5.2.2 Impacts**.

  NOTE

  In non-line-of-sight (NLOS) propagation, UEs may be mistakenly identified as high-mobility UEs, and therefore PCC anchoring prohibition for high-mobility UEs may take effect unexpectedly. If that is the case, UEs cannot be handed over from frequencies with lower priorities for PCC anchoring to those with higher priorities.
- The UE is not an enhanced Machine Type Communication (eMTC) UE. For details about eMTC, see *eMTC*.
- The bearers of the UE meet at least one of the following conditions:
  - No bearer with a QoS class identifier (QCI) of 1 has been established for the UE. That is, no VoLTE service is running on the UE. For details about VoLTE, see *VoLTE*.
  - A bearer with a QCI of 1 has been established for the UE. In addition, certain conditions related to the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter, the **QCI_VOLTE_CA_COEXIST_SW** option of the **CellQciPara**.***QciAlgoSwitch*** parameter, and the **VolteSupportCaInterFreqMeasSw** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter are met. For details about these conditions, see **19.7 VoLTE**.
  - No bearer with a QCI of 65 or 66 has been established for the UE. That is, no MCPTT voice service is running on the UE. For details about MCPTT voice services, see *PTT*.
  - A bearer with a QCI of 65 or 66 has been established for the UE, and the **McpttVoiceCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected.

The RSRP threshold for event A1 is specified by the **CaMgtCfg.*EnhancedPccAnchorA1ThdRsrp*** parameter. SCell configuration is allowed before the UE reports event A1. When receiving an A1 measurement report from a UE whose SCells have been configured, the eNodeB acts as follows:

- If the **HoWithSccCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is deselected, the eNodeB removes all the configured SCells and then initiates the PCC anchoring procedure.
- If the **HoWithSccCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected, the eNodeB initiates the PCC anchoring procedure without removing the configured SCells. For details about SCell configuration during handovers, see **SCell Configuration During Handovers** in **19.4.1 Mobility Management**.

After receiving an A1 measurement report from the CA UE, the eNodeB starts the PCC anchoring procedure if the UE meets the attribute conditions for triggering PCC anchoring. For details about the procedure, see **4.6.1.2 CA-Group-based PCC Anchoring Procedure** and **4.6.1.3 Adaptive PCC Anchoring Procedure**.

In the current serving cell of a CA UE, PCC anchoring is triggered by event A1 and event A1 is reported only once.

The PCell of a CA UE cannot be deactivated or removed. It changes only during handovers or RRC connection reestablishments.

The scenarios of PCC anchoring for CA UEs vary depending on the setting of the **CaTrafficTriggerSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter.

- If this option is selected, PCC anchoring can be triggered for CA UEs during both necessary and unnecessary incoming handovers. As a result, PCC anchoring is repeatedly triggered for CA UEs, or even ping-pong PCC anchoring occurs.
- If this option is deselected, PCC anchoring can be triggered for CA UEs only during necessary incoming handovers. This prevents repeated PCC anchoring and ping-pong PCC anchoring.

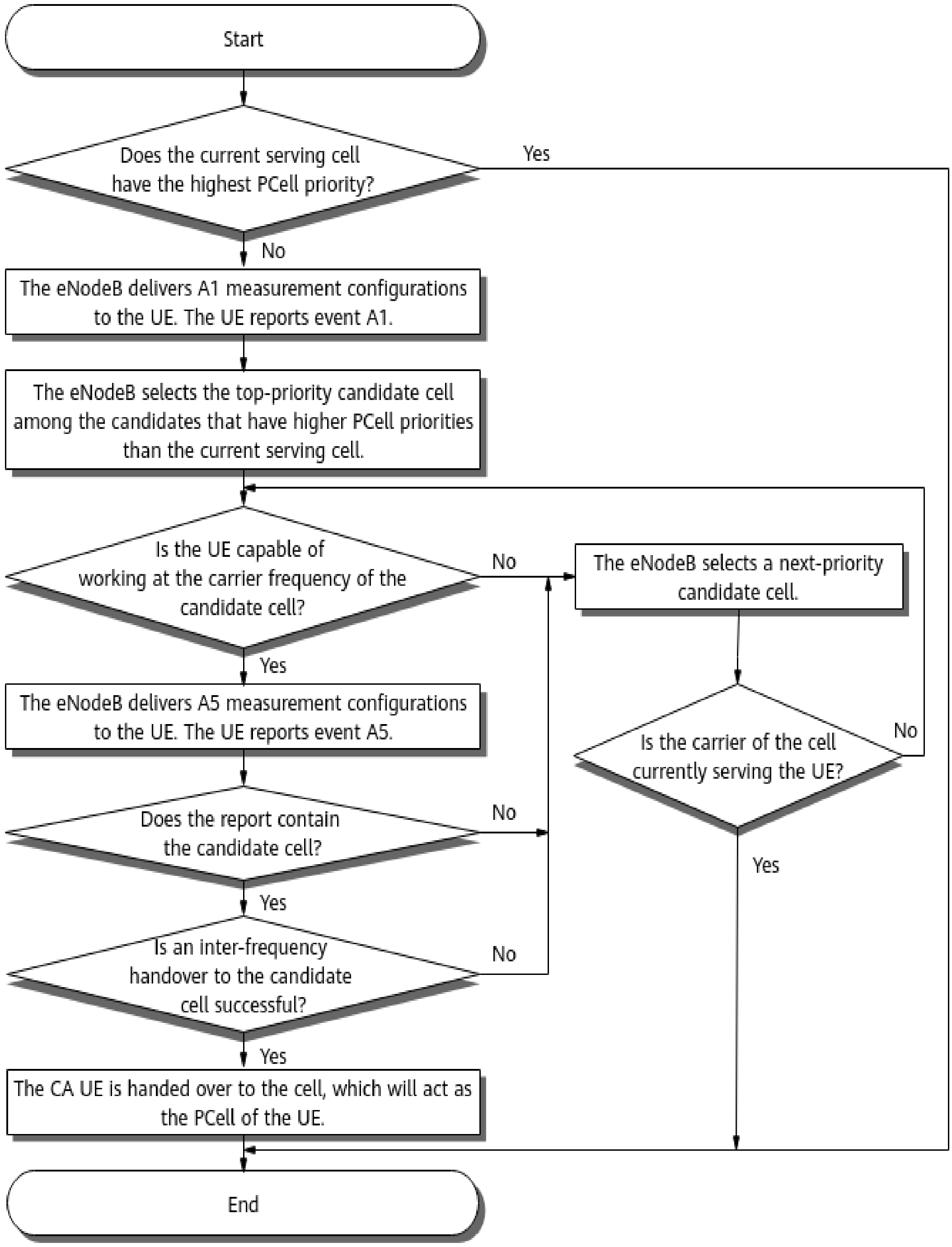
NOTE

For details about necessary handovers and unnecessary handovers, see *Mobility Management in Connected Mode*.

4.6.1.2 CA-Group-based PCC Anchoring Procedure

Figure 4-4 shows the CA-group-based PCC anchoring procedure.

Figure 4-4 CA-group-based PCC anchoring procedure



1. The eNodeB checks whether the current serving cell (for example, cell 1) has the highest PCell priority, as specified by the **CaGroupCell**.***PreferredPCellPriority*** parameter, of all the cells in the CA group.
   – If cell 1 has the highest PCell priority, the eNodeB retains cell 1 as the PCell for the UE and does not perform further PCC anchoring.
   – If cell 1 does not have the highest PCell priority, the eNodeB treats the cells that have higher PCell priorities than cell 1 in the CA group as

candidate PCells. The eNodeB then arranges the candidates in descending order of PCell priority and goes to the next step.

2. The eNodeB checks whether the UE is capable of working at the carrier frequency of the top-priority candidate PCell (for example, cell 2).
   - The UE is capable of working at the carrier frequency of cell 2.

     The eNodeB delivers the A5 measurement configuration to instruct the UE to measure cell 2, and starts a measurement timer (fixed to 3s). The A5 measurement configuration varies depending on the triggering quantity specified by the **IntraRatHoComm.*InterFreqHoA4TrigQuan*** parameter. If multiple candidate PCells have the same PCell priority, the eNodeB instructs the UE to measure all these cells. The eNodeB then processes only the first A5 measurement report from the UE and ignores subsequent reports.

     The triggering quantity parameter values and threshold 2 parameters for event A5 correspond as follows:

     - **RSRP**: The A5 measurement configuration includes only an RSRP configuration. The RSRP threshold 2 for CA is specified by the **CaGroupCell.*PCellA4RsrpThd*** parameter.
     - **RSRQ**: The A5 measurement configuration includes only a reference signal received quality (RSRQ) configuration. The RSRQ threshold 2 for CA is specified by the **CaGroupCell.*PCellA4RsrqThd*** parameter.
     - **BOTH**: The A5 measurement configuration includes both RSRP and RSRQ configurations. The UE sends an A5 measurement report when either the measured RSRP or RSRQ exceeds the specified threshold for CA. RSRP threshold 2 is specified by the **CaGroupCell.*PCellA4RsrpThd*** parameter, and RSRQ threshold 2 is specified by the **CaGroupCell.*PCellA4RsrqThd*** parameter.

     RSRP threshold 1 for event A5 is always –43 dBm, and RSRQ threshold 1 for event A5 is always –3 dB.
   - The UE is not capable of working at the carrier frequency of cell 2.

     The eNodeB moves on to the next-priority candidate PCell.
3. After receiving an A5 measurement report containing cell 2, the eNodeB hands the UE over to cell 2.

   If CA configurations are absent or incorrect in the target eNodeB or if a data link fails to be set up between cell 2 and candidate SCells, CA does not work after the handover to cell 2.

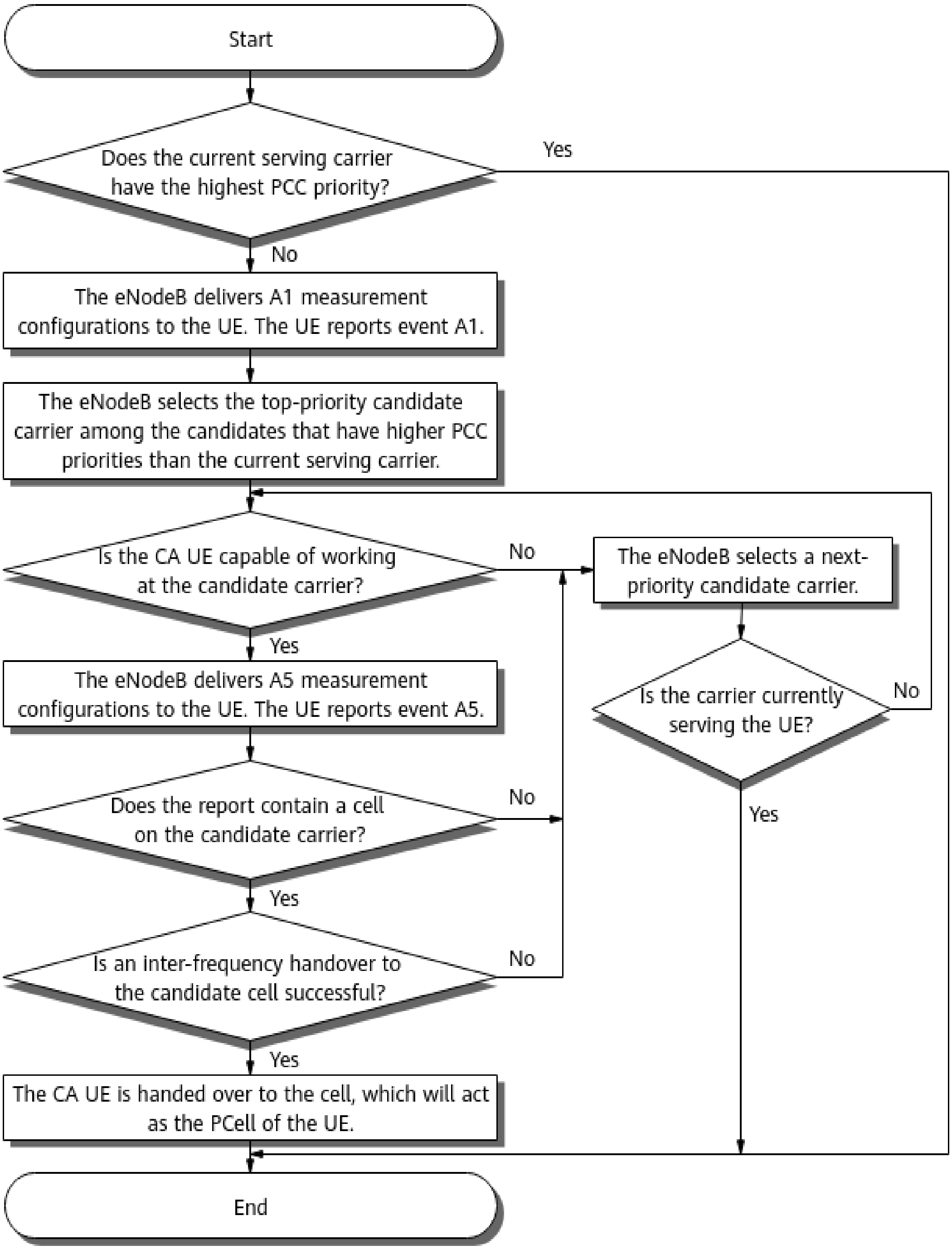
   If the eNodeB does not receive such an A5 report or if the handover to cell 2 fails, the eNodeB deletes the A5 measurement configuration and evaluates the next-priority candidate PCell in the same way.
4. If the eNodeB does not receive an A5 measurement report containing cell 2 before the measurement timer expires, the eNodeB deletes the A5 measurement configuration and evaluates the next-priority candidate PCell in the same way.

If no candidate PCell can be selected as the PCell, the eNodeB terminates the PCC anchoring procedure and cell 1 eventually serves the UE as its PCell.

4.6.1.3 Adaptive PCC Anchoring Procedure

Figure 4-5 shows the adaptive PCC anchoring procedure. Unlike CA-group-based PCC anchoring, adaptive PCC anchoring involves candidate PCCs for which only frequency-level information is configured. Therefore, the eNodeB can determine the PCell only after receiving A5 measurement reports from the CA UE.

Figure 4-5 Adaptive PCC anchoring procedure



1. The eNodeB checks whether the current serving carrier (for example, carrier 1) has the highest PCC priority, as specified by the

PccFreqCfg.***PreferredPccPriority*** parameter, of all the carriers identified by **PccFreqCfg** MOs.

- If carrier 1 has the highest PCC priority, the eNodeB retains the cell on carrier 1 as the PCell for the UE and does not perform further PCC anchoring.
- If carrier 1 does not have the highest PCC priority, the eNodeB treats higher-priority carriers as candidate PCCs for the UE, arranges them in descending order of PCC priority, and goes to the next step.

2. The eNodeB checks whether the UE is capable of working at the top-priority candidate PCC (for example, carrier 2).
   - The UE is capable of working at carrier 2.

     The eNodeB delivers the A5 measurement configuration to instruct the UE to measure carrier 2, and starts a measurement timer (fixed to 3s). The A5 measurement configuration varies depending on the triggering quantity specified by the **IntraRatHoComm.*InterFreqHoA4TrigQuan*** parameter. If multiple candidate PCCs have the same PCC priority, the eNodeB instructs the UE to measure all these carrier frequencies. After receiving A5 reports, the eNodeB selects only the first cell that meets PCC anchoring conditions from the reported cells.

     NOTE

     If no **SccFreqCfg** MO has been configured for a **PccFreqCfg** MO, the eNodeB does not deliver the relevant A5 measurement configurations.

     The triggering quantity parameter values and threshold 2 parameters for event A5 correspond as follows:

     - **RSRP**: The A5 measurement configuration includes only an RSRP configuration. The RSRP threshold 2 for CA is specified by the **PccFreqCfg.*PccA4RsrpThd*** parameter.

       It is recommended that the **PccFreqCfg.*PccA4RsrpThd*** parameter be set to a value greater than the RSRP threshold for coverage-based inter-frequency handovers (specified by the **InterFreqHoGroup**.***InterFreqHoA4ThdRsrp*** parameter), to prevent a decrease in the handover success rate.

     - **RSRQ**: The A5 measurement configuration includes only an RSRQ configuration. The RSRQ threshold 2 for CA is specified by the **PccFreqCfg.*PccA4RsrqThd*** parameter.

     - **BOTH**: The A5 measurement configuration includes both RSRP and RSRQ configurations. The UE sends an A5 measurement report when either the measured RSRP or RSRQ exceeds the specified threshold for CA. RSRP threshold 2 is specified by the **PccFreqCfg.*PccA4RsrpThd*** parameter, and RSRQ threshold 2 is specified by the **PccFreqCfg.*PccA4RsrqThd*** parameter.

     RSRP threshold 1 for event A5 is always –43 dBm, and RSRQ threshold 1 for event A5 is always –3 dB.

   - The UE is not capable of working at carrier 2.

     The eNodeB moves on to the next-priority candidate PCC.

3. After the eNodeB receives an A5 measurement report containing a cell on carrier 2, the eNodeB hands the UE over to the cell.

If CA configurations are absent or incorrect in the target eNodeB or if a data link fails to be set up between the cell and candidate SCells, CA does not work after the handover to that cell.

If the eNodeB does not receive such an A5 report or if the handover to the cell fails, the eNodeB deletes the A5 measurement configuration and evaluates the next-priority candidate PCC in the same way.

4. If the eNodeB does not receive an A5 measurement report containing a cell on carrier 2 before the measurement timer expires, the eNodeB deletes the A5 measurement configuration and evaluates the next-priority candidate PCC in the same way.

If no cells on the candidate PCCs meet the conditions for triggering event A5 or no inter-frequency handover is successful, the eNodeB terminates the PCC anchoring procedure and the cell on carrier 1 eventually serves the UE as its PCell.

4.6.2 PCC Anchoring Enhancement

This section describes an enhancement to PCC anchoring for RRC_CONNECTED UEs: load-based PCC anchoring. With this function, intra- or inter-eNodeB cells with routes set up between them exchange their load status, and the eNodeB selects a candidate cell not in the high load state as the PCell.

A candidate cell whose load status is unknown is always considered to be in the high PCell load state. A candidate cell whose load status is known is considered to be in the high PCell load state only if it meets one of the following conditions:

- The number of CA UEs that treat the cell as their PCell (referred to as PCell-served UEs in this document for clarity) exceeds the high PCell load threshold.
- The CPU usage of the BBP serving the cell is greater than 70%.

In these conditions:

- High PCell load threshold = Maximum allowed number of PCell-served UEs in the cell (specified by the **CaMgtCfg.*CellMaxPccNumber*** parameter) x Percentage specified by the **CaMgtCfg.*PccUserNumberOffloadThd*** parameter
- CPU usage of a BBP = **L.Traffic.Board.UPlane.CPULoad.AVG**

This function takes effect when both of the following options are selected:

- **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter
- **PccSmartCfgSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter

For inter-eNodeB PCC anchoring, these options must be selected for both the source and target eNodeBs, and eX2 interfaces must be set up for load information exchange between these eNodeBs. Otherwise, the source eNodeB cannot obtain the load status of candidate PCells served by the target eNodeB.

This function introduces special handling to the basic PCC anchoring procedure, as described in **4.6.2.1 Enhanced CA-Group-based PCC Anchoring Procedure** and **4.6.2.2 Enhanced Adaptive PCC Anchoring Procedure**.

If the PCell load status of a candidate PCell changes, it takes several seconds for the new load status to be exchanged between BBPs or eNodeBs for inter-BBP or inter-eNodeB CA. In case the status exchange is completed later than expected when multiple UEs access the candidate PCell within a short time, the eNodeB will make an incorrect decision on the load status.

This function results in more time being spent on inter-frequency measurements during PCell selection. Therefore, performance statistics indicate a decrease in the number of UEs that are using CA.

4.6.2.1 Enhanced CA-Group-based PCC Anchoring Procedure

This function introduces the following special handling to the CA-group-based PCC anchoring procedure described in **4.6.1.2 CA-Group-based PCC Anchoring Procedure**: The eNodeB can acquire the load status of candidate PCells in advance and therefore instructs the UE to measure only the low-load candidate PCells. If all candidate PCells are in the high load state, the eNodeB performs CA-group-based PCC anchoring for the UE without considering candidate PCell loads.

4.6.2.2 Enhanced Adaptive PCC Anchoring Procedure

This function introduces special handling to the adaptive PCC anchoring procedure described in **4.6.1.3 Adaptive PCC Anchoring Procedure**. The special handling varies as follows:

- The carrier that the CA UE accesses has the highest PCC priority, and the current serving cell of the UE is in the high load state.

  The eNodeB checks whether there are candidate PCCs that have the same PCC priority (specified by the **PccFreqCfg.*PreferredPccPriority*** parameter) as the current carrier.

  – If there are such candidate PCCs, the eNodeB delivers the A5 measurement configuration, instructing the UE to measure them (at most three candidates in a single configuration). After receiving a measurement report from the UE, the eNodeB checks whether there are low-load cells on the candidate PCCs.

    ■ If there is a low-load cell among cells on the candidate PCCs, the eNodeB selects the cell and instructs the UE to perform an inter-frequency handover to the cell. If there are multiple low-load cells, the eNodeB randomly selects one and instructs the UE to perform an inter-frequency handover to it.

    ■ If there are no such cells, no further adaptive PCC anchoring will occur.

  – If there are no such candidate PCCs, no further adaptive PCC anchoring will occur.

- The carrier that the CA UE accesses has the highest PCC priority, and the current serving cell of the UE is in the low load state.

  The eNodeB does not perform the subsequent adaptive PCC anchoring procedure.

- A candidate PCC has a higher PCC priority than the carrier that the CA UE accesses, or multiple candidate PCCs have an identical PCC priority that is higher than the priority of the current serving carrier.

  The eNodeB delivers the A5 measurement configuration to the UE, instructing the UE to measure the top-priority candidate PCCs (at most three candidates in a single configuration). After receiving a measurement report from the UE, the eNodeB selects the highest-RSRP cell on each frequency as a candidate PCell and checks whether there are low-load cells among the candidate PCells.

  – If there is a low-load cell, the eNodeB selects the cell and instructs the UE to perform an inter-frequency handover to the cell. If there are multiple low-load cells, the eNodeB randomly selects one and instructs the UE to perform an inter-frequency handover to it.
  – If there are no low-load cells, the eNodeB delivers another A5 measurement configuration to the UE, instructing the UE to measure the next-priority candidate PCCs. The eNodeB repeats the preceding actions until it has tried all candidate PCCs with priorities at least as high as the current serving carrier. If no cell is finally selected as the target PCell, the UE still camps on the initial PCell. The PCC anchoring procedure ends.

4.6.3 SCell Management

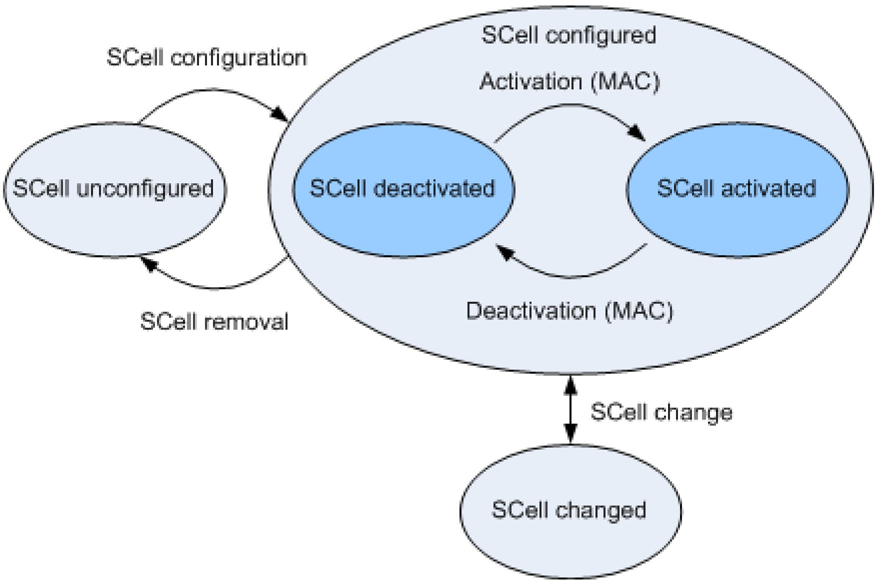
SCell management includes SCell configuration, change, activation, deactivation, and removal.

- SCell configuration is triggered when a CA UE initiates RRC connection setup during initial access, an incoming RRC connection reestablishment, or an incoming handover. For SCell configuration procedures, see **4.6.3.1 SCell Configuration** and **4.6.3.2 SCell Configuration Enhancement**.

  SCell configuration enhancement refers to load-based SCell configuration. With this function, intra- or inter-eNodeB cells with routes set up between them exchange their load status, and the eNodeB selects candidate cells not in the high load state as SCells for a UE.
- After an SCell is configured, it may be changed, activated, deactivated, or removed. For details about these procedures, see **4.6.3.3 SCell Change**, **4.6.3.4 SCell Activation**, **4.6.3.5 SCell Deactivation**, and **4.6.3.6 SCell Removal**.

Figure 4-6 shows the transitions between the SCell states.

Figure 4-6 SCell state transitions



4.6.3.1 SCell Configuration

To prioritize certain cells or frequencies as SCells or SCCs, an operator can set high SCell priorities or SCC priorities so that, by SCell configuration, the eNodeB will select the highest-priority cell as an SCell for a CA UE. The SCell priorities are specified by the **CaGroupSCellCfg.*SCellPriority*** parameter and are required in CA-group-based configuration mode. The SCC priorities are specified by the **SccFreqCfg.*SccPriority*** parameter and are required in adaptive configuration mode.

For the triggering conditions of this function, see **4.6.3.1.1 Triggering Conditions**. For the procedure of this function, see **4.6.3.1.2 CA-Group-based SCell Configuration Procedure** and **4.6.3.1.3 Adaptive SCell Configuration Procedure**.

4.6.3.1.1 Triggering Conditions

An eNodeB starts SCell configuration for a CA UE if all the following conditions are met:

- The RRC connection of the UE has been set up in any of the following scenarios:
  - Initial access
  - Incoming RRC connection reestablishment
  - Incoming handover
- SRB2 and the default DRB have been set up.
- The number of SCells for the UE has not reached the maximum allowed value.
- The UE is not running an emergency call.
- The UE is not engaged in a CS fallback procedure.

The triggering of the SCell configuration procedure is also dependent on the following factors:

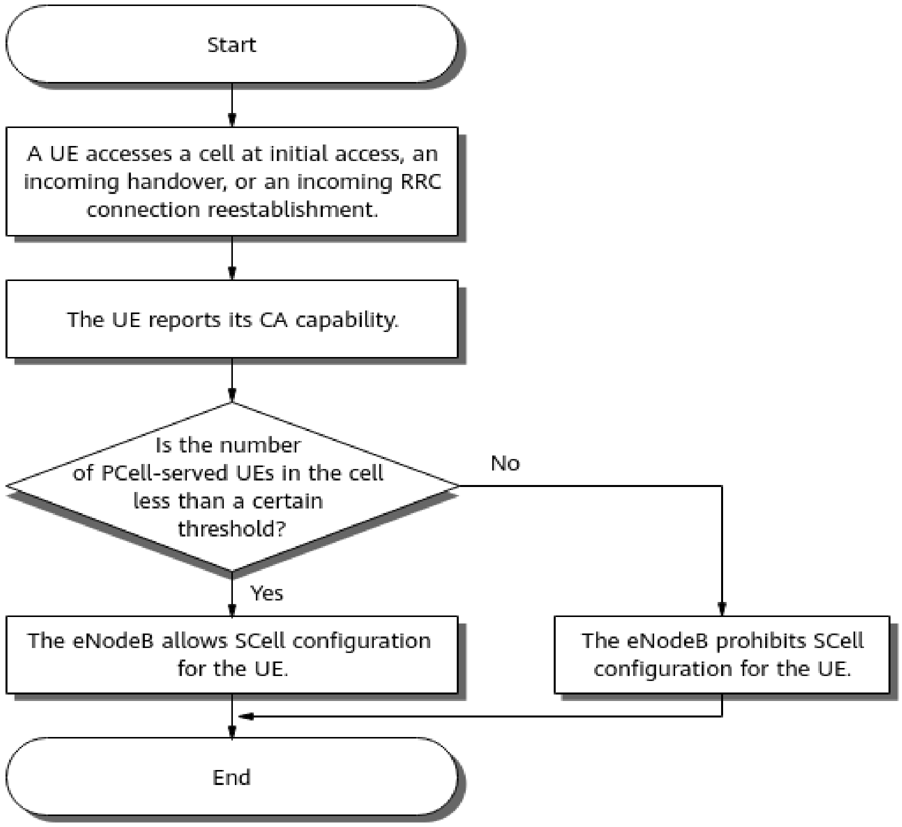
- Setting of the **CaTrafficTriggerSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter
  - Selected

    The SCell configuration procedure is triggered only when the traffic volume of the CA UE meets the SCell activation conditions described in **4.6.3.4 SCell Activation**. That means blind SCell configuration does not take effect in this case.
  - Deselected

    The SCell configuration procedure is triggered when the traffic volume of the CA UE meets the SCell activation conditions described in **4.6.3.4 SCell Activation** or an RRC connection is set up.
- Maximum permissible number of PCell-served UEs in the cell

  Figure 4-7 illustrates the control procedure.

Figure 4-7 Control of the maximum permissible number of PCell-served UEs over SCell configuration



After a CA UE accesses a cell (during initial access, an incoming RRC connection reestablishment, or an incoming handover) and reports its CA capability to the eNodeB, the eNodeB checks whether the number of PCell-served UEs in the cell has reached the threshold specified by the **CaMgtCfg.*CellMaxPccNumber*** parameter.

– If the number has not reached the threshold, the eNodeB allows SCell configuration for this UE.
– If the number has reached the threshold, the eNodeB prohibits SCell configuration for this UE.

  If a CA UE reverts to the single carrier state after the number of PCell-served UEs reaches the threshold, the eNodeB does not immediately allow SCell configuration for other UEs. Instead, the eNodeB allows SCell configuration for other UEs only when the number of PCell-served UEs falls below 90% of the threshold.

NOTE

- When comparing the number of UEs with the value of **CaMgtCfg.*CellMaxPccNumber***, the eNodeB does not consider the number of UEs that treat the cell as their SCell.
- When the value of **CaMgtCfg.*CellMaxPccNumber*** is decreased to less than the number of PCell-served UEs that have been configured with SCells, the eNodeB does not take the initiative to remove the SCells for the UEs.
- When the eNodeB configures an SCell for a CA UE, the cell radio network temporary identifier (C-RNTI) of the UE may have been used by another UE in this cell. If that is the case, a C-RNTI collision occurs. The eNodeB then proactively initiates an intra-cell handover for one of these UEs to reconfigure the C-RNTI for the UE. The probabilities of C-RNTI collisions before and after a version upgrade may differ and yield the following impacts:
  - If the probability of C-RNTI collisions before the upgrade is higher than that after the upgrade, the **L.HHO.IntraCell.ExecAttOut** counter will produce a smaller value for the cell that the UE accesses, after the upgrade.
  - If the probability of C-RNTI collisions before the upgrade is lower than that after the upgrade, the **L.HHO.IntraCell.ExecAttOut** counter will produce a larger value for the cell that the UE accesses, after the upgrade.
- If a neighboring E-UTRAN frequency measurement flag is cleared (that is, the **FREQ_MEAS_FLAG** option of the **EutranInterNFreq**.***AggregationAttribute*** parameter is deselected) for a neighboring E-UTRAN frequency, the eNodeB does not select this frequency as a candidate frequency.

  Management of neighboring E-UTRAN frequency measurement flags is controlled by the **LTE_NFREQ_MEAS_MGMT_SW** option of the **CellAlgoExtSwitch**.***AnrOptSwitch*** parameter. For details about this function, see *ANR Management*.

A UE enters the SCell configuration procedure after the procedure is triggered. For details about the SCell configuration procedure, see **4.6.3.1.2 CA-Group-based SCell Configuration Procedure** and **4.6.3.1.3 Adaptive SCell Configuration Procedure**.

The CPU usage of the main control board (indicated by the **VS.BBUBoard.CPULoad.Mean** counter) and that of the BBP (indicated by the **L.Traffic.Board.UPlane.CPULoad.AVG** counter), together with the number of UEs, have the impacts listed in **SCell Configuration Control based on the CPU Usage of the Main Control Board or BBP** and **UE-Number-based SCell Configuration Control** on the SCell configuration procedure.

Network configuration modifications (for example, MML-based addition, removal, or modification of CA groups or PCC/SCC frequencies, and cell activation) do not take effect immediately for CA UEs that entered the RRC_CONNECTED state before the modifications are made. The modifications take effect for these CA UEs only when another SCell configuration procedure is triggered for them.

Assume that the CA-group-based configuration mode is used and a cell in the CA group is deactivated. After this cell is re-activated, it can be configured as an SCell for a CA UE only when the UE re-accesses the network.

SCell Configuration Control based on the CPU Usage of the Main Control Board or BBP

CA UEs consume more CPU resources of the main control board and BBP than non-CA UEs. When the CPU usage of the main control board or BBP is high, continuing to admit CA UEs may result in the CPU being unable to respond to all UEs' requests in a timely manner. To address this, SCell configuration can be controlled based on the CPU usage of the main control board and BBP. Specifically, when such CPU usage exceeds a specified threshold, the eNodeB restricts SCell configuration for CA UEs to prevent the CPU usage from increasing rapidly. **Table 4-12** and **Table 4-13** describe the impacts of the CPU usage of the main control board and BBP on the SCell configuration procedure.

Table 4-12 Impacts of the CPU usage of the main control board on the SCell configuration procedure

CPU Usage of the Main Control Board (Unit: %)	Impact on the SCell Configuration Procedure
[0, (CA flow control threshold – 10))	No impact
[(CA flow control threshold – 10), CA flow control threshold)	There are two possible scenarios: • The CPU usage of the main control board has never reached the CA flow control threshold. There is no impact. • The CPU usage of the main control board has ever reached the CA flow control threshold. The SCell configuration probability is 95%.
[CA flow control threshold, (CA flow control threshold + 1))	The SCell configuration probability is 80%.
[(CA flow control threshold + 1), (CA flow control threshold + 2))	The SCell configuration probability is 60%.
[(CA flow control threshold + 2), (CA flow control threshold + 3))	The SCell configuration probability is 40%.
[(CA flow control threshold + 3), (CA flow control threshold + 4))	The SCell configuration probability is 20%.

CPU Usage of the Main Control Board (Unit: %)	Impact on the SCell Configuration Procedure
[(CA flow control threshold + 4), (CA flow control threshold + 5))	The SCell configuration probability is 10%.
[(CA flow control threshold + 5), 100]	SCell configuration is not performed.

NOTE

- The CA flow control threshold is specified by the **eNodeBFlowCtrlPara.*CaFlowCtrlThld*** parameter.
- The SCell configuration probability is a probability based on which SCell configuration is performed.

Table 4-13 Impacts of the CPU usage of the BBP on the SCell configuration procedure

CPU Usage of the BBP (Unit: %)	Impact on the SCell Configuration Procedure
[0, 70)	No impact
[70, 90)	SCell configuration is not allowed for CA UEs that access the network through initial access, incoming handovers, or incoming RRC connection reestablishments.
[90, 100]	SCell configuration is not allowed for any CA UE.

UE-Number-based SCell Configuration Control

CA UEs consume more of the BBP's capacity than non-CA UEs. When the BBP is already serving a large number of UEs, continuing to admit CA UEs may exhaust the UE number capacity of the BBP. To address this, UE-number-based SCell configuration control is introduced. This function restricts the number of CCs that can be configured for a UE by considering the number of UEs already on the BBP and that in the cell, as well as configuring the thresholds for prohibiting CA with varying number of CCs aggregated. This helps to prevent the rapid exhaustion of the BBP's UE number capacity.

1. Calculating the UE proportion of a cell based on the number of existing UEs on the BBP and that in the cell

UE proportion of a cell = Max (Cell's capacity proportion, BBP's capacity proportion)

– Cell's capacity proportion: indicates the proportion of the number of UEs (including SCC UEs) that have accessed a cell to the maximum number of UEs that can access each cell according to the capacity of the BBP serving the cell.

– BBP's capacity proportion: indicates the proportion of the number of UEs that have accessed the BBP serving a cell to the maximum number of UEs that the BBP can accommodate.

2. Restricting the number of CCs that can be configured for UEs based on the UE proportion of the cell and the configured thresholds for prohibiting CA with varying number of CCs aggregated

   If the UE proportion of the cell exceeds a configured threshold for prohibiting CA with a specific number of CCs aggregated, the corresponding number of CCs cannot be configured for UEs in the cell. If the UE proportion of the cell is less than or equal to a configured threshold for prohibiting CA with a specific number of CCs aggregated minus 5% (if this result is less than 1%, it is treated as 1%), the restriction mechanism will be exited and the corresponding number of CCs can be configured.

Prohibition Threshold	Parameter	Impact on the SCell Configuration Procedure
Forbid 2CC UE Ratio Threshold	**CaMgtCfg**.***Forbid2ccUeRatioThld***	• If the UE proportion of the cell exceeds the value of **CaMgtCfg**.***Forbid2ccUeRatioThld***, UEs in the cell are prohibited from being configured with 2CC or higher CA. In addition, the cell cannot be configured as an SCell for UEs in other cells with UE-number-based SCell configuration control enabled. • If the UE proportion of the cell exceeds the value of **CaMgtCfg**.***Forbid3ccUeRatioThld***, UEs in the cell are prohibited from being configured with 3CC or higher CA. • If the UE proportion of the cell exceeds the value of **CaMgtCfg**.***Forbid4ccUeRatioThld***, UEs in the cell are prohibited from being configured with 4CC or higher CA. • If the UE proportion of the cell exceeds the value of **CaMgtCfg**.***Forbid5ccUeRatioThld***, UEs in the cell are prohibited from being configured with 5CC or higher CA.
Forbid 3CC UE Ratio Threshold	**CaMgtCfg**.***Forbid3ccUeRatioThld***	
Forbid 4CC UE Ratio Threshold	**CaMgtCfg**.***Forbid4ccUeRatioThld***	
Forbid 5CC UE Ratio Threshold	**CaMgtCfg**.***Forbid5ccUeRatioThld***	

Prohibition Threshold	Parameter	Impact on the SCell Configuration Procedure
• The corresponding parameter takes effect and affects the SCell configuration procedure only when it is set to a value other than **100%**. • UE-number-based SCell configuration control takes effect only when at least one of the preceding four parameters is set to a value other than **100%**.		

NOTE

- The minimum threshold for exiting UE-number-based SCell configuration control is 1%. If the configured prohibition threshold is also 1%, any UE number fluctuation may lead to frequent entry and exit of such control.
- If a prohibition threshold of a cell changes, online UEs are not affected immediately. When SCell configuration is triggered for an online UE again, the number of CCs that can be configured for the UE is recalculated based on the number of UEs in the cell and the prohibition threshold. However, if the number of SCells already configured for the UE exceeds the limit defined by the prohibition threshold, the eNodeB does not proactively remove the SCells.

Network configuration modifications (for example, MML-based addition, removal, or modification of CA groups or PCC/SCC frequencies, and cell activation) do not take effect immediately for CA UEs that entered the RRC_CONNECTED state before the modifications are made. The modifications take effect for these CA UEs only when another SCell configuration procedure is triggered for them.

Assume that the CA-group-based configuration mode is used and a cell in the CA group is deactivated. After this cell is re-activated, it can be configured as an SCell for a CA UE only when the UE re-accesses the network.

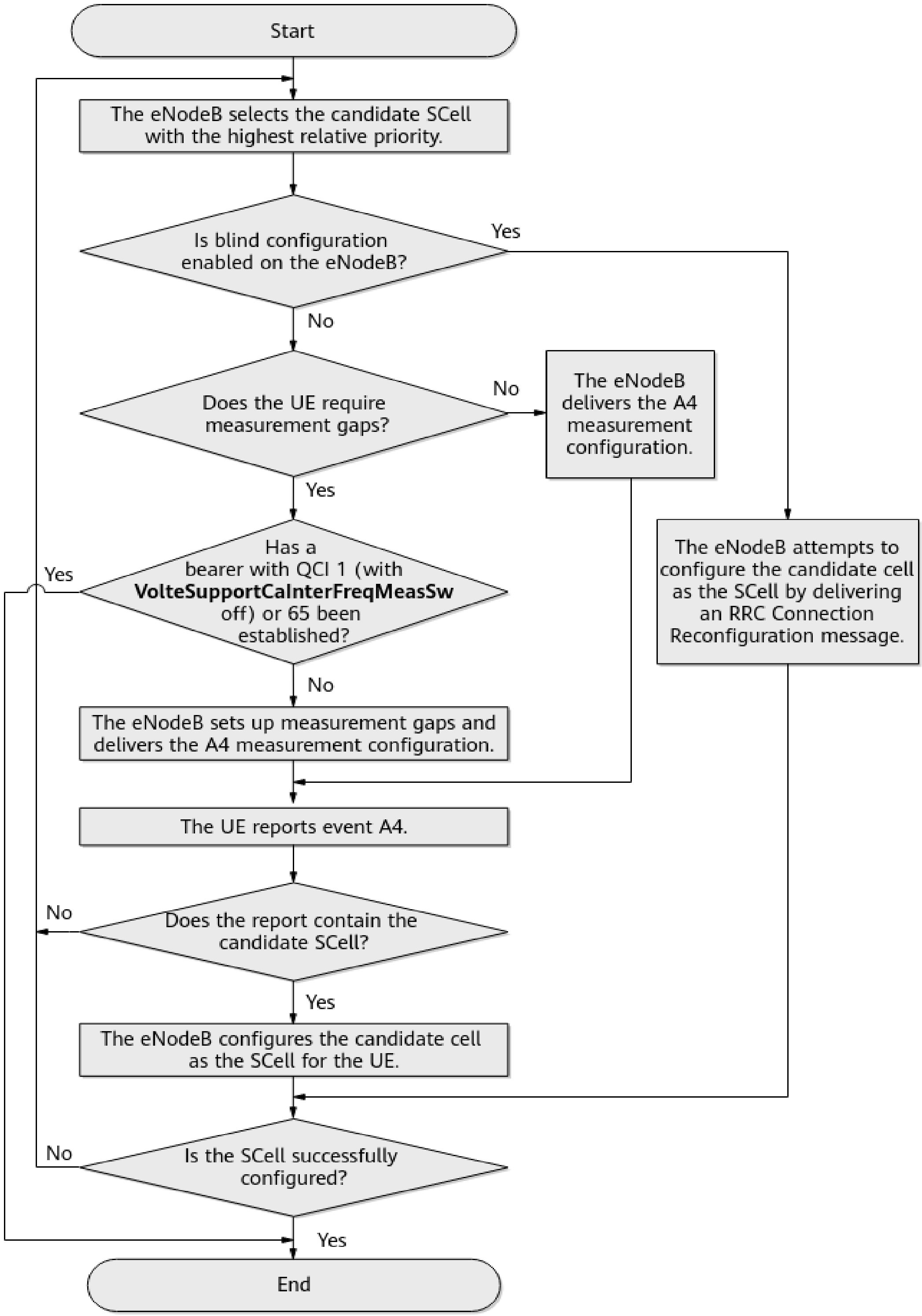
4.6.3.1.2 CA-Group-based SCell Configuration Procedure

CA-group-based SCell configuration can be used for uplink 2CC aggregation and for downlink 2CC, 3CC, 4CC, or 5CC aggregation. In the downlink, if 2CC, 3CC, 4CC, and 5CC aggregation are all supported, SCell configuration is performed to preferentially have the maximum supported number of CCs aggregated. For the configuration requirements for these downlink CA capabilities, see **5 Downlink 2CC Aggregation**, **6 Downlink 3CC Aggregation**, **7 Downlink 4CC Aggregation**, and **8 Downlink 5CC Aggregation**.

Uplink 2CC aggregation and downlink 2CC aggregation use the same SCell configuration procedure. **Figure 4-8** uses uplink/downlink 2CC aggregation as an example to describe an SCell configuration procedure. For downlink *n*CC aggregation, there are (*n*–1) candidate SCells in a selection.

For aggregation of any number of CCs, if the number of successfully configured SCells is less than the required number after all candidates have been tried, the SCell configuration procedure ends. The eNodeB then periodically evaluates whether the downlink or uplink traffic volume of the UE meets the SCell activation conditions, which are described in **4.6.3.4 SCell Activation**. If the SCell activation conditions are met, the eNodeB performs an SCell configuration procedure again, attempting to configure SCells. This periodic configuration process continues until the number of successfully configured SCells reaches the required number. The period is specified by the **CaMgtCfg**.***SccCfgInterval*** parameter.

Figure 4-8 CA-group-based SCell configuration procedure for uplink/downlink 2CC aggregation



Operator-list-based UE filtering is supported in CA-group-based SCell configuration. The operator list is specified by the **CaGroup**.***CnOperatorList*** parameter.

- If the operator list (**CaGroup**.***CnOperatorList***) is set for the CA group to which the PCell of a CA UE belongs, the eNodeB checks whether the operator of the CA UE is on the list specified by this parameter.
  - If the operator of the CA UE is not on the list specified by the **CaGroup**.***CnOperatorList*** parameter,

    SCell configuration is not allowed for the CA UE.
  - If the operator of the CA UE is on the list specified by the **CaGroup**.***CnOperatorList*** parameter,

    The eNodeB treats any cell that meets all the following conditions as a candidate SCell for the PCell of the CA UE:
    - The cell is in the same CA group as the PCell.
    - The cell has not been configured as an SCell for the UE.
    - The cell belongs to the public land mobile network (PLMN) serving the UE, or to an equivalent PLMN.
- If the operator list is not set for the CA group to which the PCell of a CA UE belongs, the eNodeB treats any cell that meets all the following conditions as a candidate SCell for the PCell of the CA UE:
  - The cell is in the same CA group as the PCell.
  - The cell has not been configured as an SCell for the UE.
  - The cell belongs to the PLMN serving the UE, or to an equivalent PLMN.

The eNodeB arranges all candidate SCells in descending order of SCell priority (specified by the **CaGroupSCellCfg.*SCellPriority*** parameter) and attempts to select a candidate SCell as an SCell for the UE. Cells with the priority value of **0** cannot be configured as SCells.

The SCell configuration procedure varies depending on the parameter settings related to blind SCell configuration.

- The **SccBlindCfgSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter is selected, and the **CaGroupSCellCfg.*SCellBlindCfgFlag*** parameter is set to **TRUE** for a top-priority candidate SCell.

  The eNodeB does not send the A4 measurement configuration related to the operating frequency of this candidate SCell. Instead, the eNodeB directly delivers an RRC Connection Reconfiguration message to configure this candidate cell as an SCell for the UE. If multiple candidate SCells have the same top SCell priority and have their blind configuration flag set to **TRUE**, the eNodeB randomly selects one cell from these candidate SCells and attempts to configure this cell as an SCell for the UE in a blind manner.
  - If the SCell is configured successfully, the SCell configuration procedure ends.
  - If the SCell fails to be configured, the eNodeB evaluates next-priority candidate SCells in the same way.

  It is recommended that blind SCell configuration be enabled to have SCells quickly configured.

NOTE

- For UEs that support NSA DC, if any NR cell is defined as a blind-configurable candidate SCell on the eNodeB, the eNodeB preferentially configures the NR cell as an SCell in a blind manner. After the NR SCell is configured, the eNodeB then performs blind E-UTRA SCell configuration.
- The following cells cannot be configured as SCells in a blind manner:
  - Cells that are shut down by energy saving features
  - Deactivated cells

- The **SccBlindCfgSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter is deselected. Alternatively, the **SccBlindCfgSwitch** option is selected, but the **CaGroupSCellCfg.*SCellBlindCfgFlag*** parameter is set to **FALSE** for all top-priority candidate SCells.

  The eNodeB does not support blind SCell configuration. In this situation, the eNodeB delivers the A4 measurement configuration related to the operating frequencies of top-priority candidate SCells. After receiving an A4 measurement report from the UE, the eNodeB determines whether to configure the reported candidate cell as an SCell for the UE.

  Entering condition for event A4: ($Mn + Ofn + Ocn - Hys > Thresh$) is true throughout a period specified by *TimeToTrig*.

  In the formula:

  – *Mn* is the measurement result of a neighboring cell.

  – *Ofn* and *Ocn* are detailed in *Mobility Management in Connected Mode*.

  – *TimeToTrig* is the time-to-trigger for CA event A4. It is specified by the **CAMGTCFG**.***CaA4TimeToTrigger*** parameter.

  – *Hys* is fixed at 1 dB.

  – *Thresh* is the threshold for event A4. This threshold is equal to the sum of the **CaMgtCfg.*CarrAggrA4ThdRsrp*** and **CaGroupSCellCfg.*SCellA4Offset*** parameter values configured on the PCell side.

  The configuration procedure is as follows:

  a. The eNodeB delivers the A4 measurement configuration, instructing the UE to measure all top-priority candidate SCells. In the measurement configuration, the eNodeB may set up measurement gaps for the UE, depending on the inter-frequency measurement capabilities reported by the UE.

     The **AutoGapSwitch** option of the **ENodeBAlgoSwitch.*HoModeSwitch*** parameter affects the decision process of the inter-frequency measurement gap setup.

     ■ If this option is selected, the eNodeB determines whether to set up the gaps based on the reported UE capabilities.

     ■ If this option is deselected, the eNodeB sets up the gaps, without considering UE capabilities.

     It is recommended that this option be deselected to ensure that only cells with satisfactory signal quality can be configured as SCells.

     ■ The UE requires measurement gaps.

- If a bearer with a QCI of 1 has been established for the UE, then:

  The procedure goes as described in **19.7 VoLTE**.

- If a bearer with a QCI of 65 or 66 has been established for the UE and:

  If the **McpttVoiceCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is deselected, the SCell configuration procedure ends.

  If the **McpttVoiceCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected, the eNodeB sets up measurement gaps in the measurement configuration.

- The UE does not require measurement gaps.

  The eNodeB does not set up measurement gaps in the measurement configuration.

After delivering the A4 measurement configuration related to candidate SCells to the UE, the eNodeB may receive an A2 measurement report, indicating unsatisfactory signal quality of the PCell, for an inter-frequency or inter-RAT handover. In such a case, the eNodeB will not configure any candidate cell as an SCell for the UE, even if the eNodeB later receives an A4 measurement report that contains the candidate cell.

b. After receiving an A4 measurement report from the UE, the eNodeB checks the cells contained in the report.

   If the report contains a candidate SCell, the eNodeB sends an RRC Connection Reconfiguration message to configure the cell as an SCell for the UE. If the SCell is configured successfully, the procedure ends.

   If the eNodeB does not receive a report that contains a candidate SCell or the SCell fails to be configured, the eNodeB continues to evaluate next-priority candidate SCells.

4.6.3.1.3 Adaptive SCell Configuration Procedure

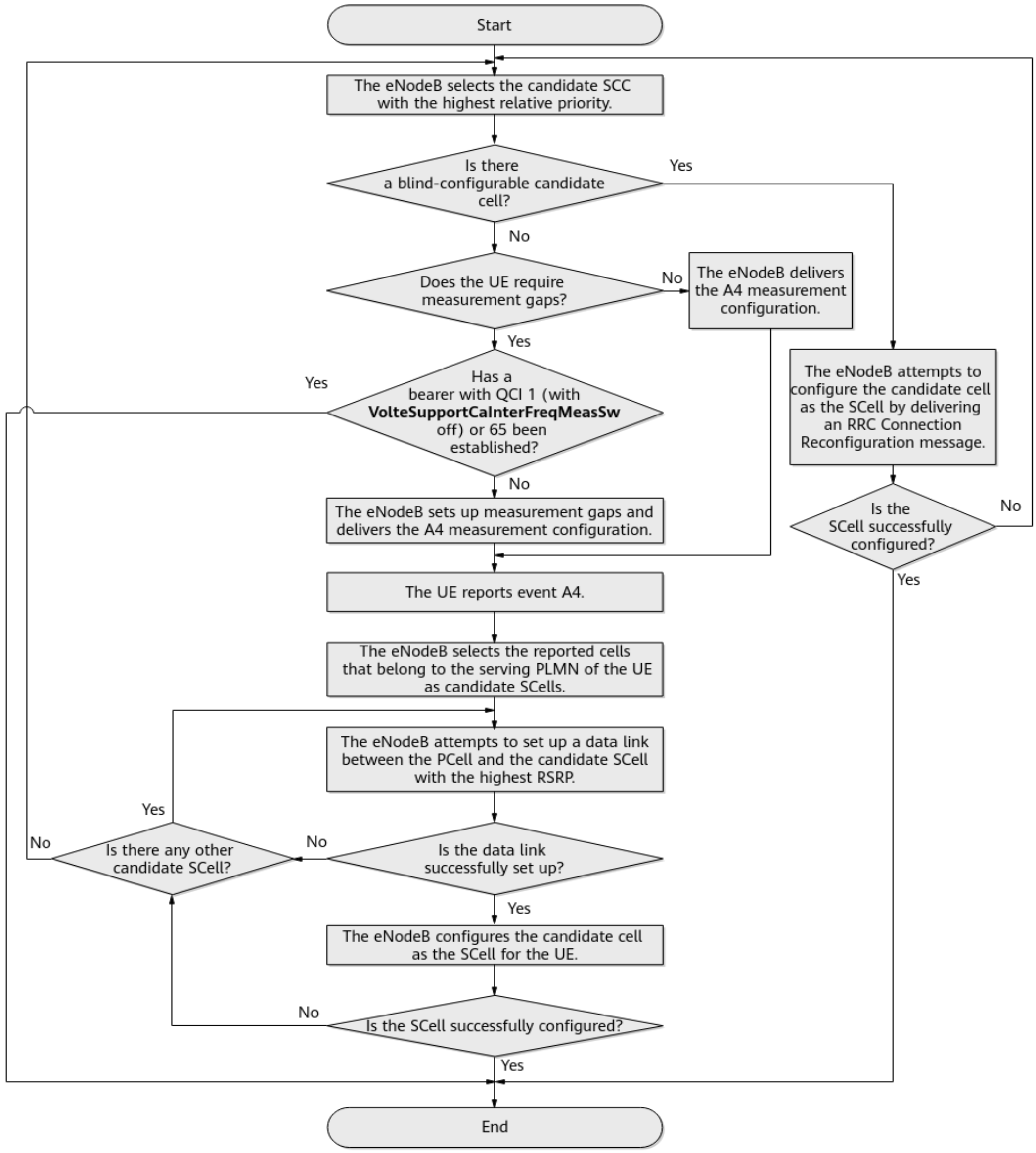
Adaptive SCell configuration can be used for uplink 2CC aggregation and for downlink 2CC, 3CC, 4CC, 5CC, or 6CC–8CC aggregation. In the downlink, if 2CC, 3CC, 4CC, and 5CC aggregation are all supported, SCell configuration is performed to preferentially have the maximum supported number of CCs aggregated. For the configuration requirements for these downlink CA capabilities, see **5 Downlink 2CC Aggregation**, **6 Downlink 3CC Aggregation**, **7 Downlink 4CC Aggregation**, and **8 Downlink 5CC Aggregation**.

If both downlink CA and uplink 2CC aggregation are supported, an uplink SCell is configured during the downlink SCell configuration procedure. The prerequisite for configuring a cell as an uplink SCell is that the cell is configured as a downlink SCell.

The SCell configuration procedure for uplink 2CC aggregation is the same as that for downlink 2CC aggregation. **Figure 4-9** uses uplink/downlink 2CC aggregation as an example to describe an SCell configuration procedure. For downlink *n*CC aggregation, there are (*n*–1) candidate SCells in a selection.

For aggregation of any number of CCs, if the number of successfully configured SCells is less than the required number after all candidates have been tried, the SCell configuration procedure ends. The eNodeB then periodically evaluates whether the downlink or uplink traffic volume of the UE meets the SCell activation conditions, which are described in 4.6.3.4 SCell Activation. If the SCell activation conditions are met, the eNodeB performs an SCell configuration procedure again, attempting to configure SCells. This periodic configuration process continues until the number of successfully configured SCells reaches the required number. The period is specified by the **CaMgtCfg**.***SccCfgInterval*** parameter.

Figure 4-9 Adaptive SCell configuration procedure for uplink/downlink 2CC aggregation



From the carriers defined in the **SccFreqCfg** MOs for the PCC of a CA UE, the eNodeB selects carriers depending on whether operator information has been specified:

- Operator information has been specified (by the **SccFreqCfg**.***CnOperatorList*** parameter) for any of these carriers.

  The eNodeB selects the carriers used by the operator that is serving the UE.

Among the selected carriers, the eNodeB treats those that have not been configured as SCCs for the UE and are supported by the UE for CA as candidate SCCs. The eNodeB then arranges all candidate SCCs in descending order of SCC priority (specified by the **SccFreqCfg**.***SccPriority*** parameter) and attempts to select a cell on a candidate SCC as an SCell for the UE, as described in the follow-up procedure.

NOTE

If cells on a frequency with a certain priority differ in their downlink bandwidths, the eNodeB checks the CA combination with only one of the bandwidths when checking whether a UE supports the CA combinations related to this frequency. In this case, the bandwidths supported by the UE may not include the bandwidth checked by the eNodeB. If this is the case, the eNodeB does not configure the UE-supported CA combination for the UE. To mitigate this issue, an identical setting of bandwidth is recommended for cells on the same frequency on a CA-enabled network.

- No operator information has been specified for any of these carriers.

  The eNodeB treats those that have not been configured as SCCs for the UE and are supported by the UE for CA as candidate SCCs. The eNodeB then arranges all candidate SCCs in descending order of SCC priority and attempts to select a cell on a candidate SCC as an SCell for the UE, as described in the follow-up procedure.

1. The eNodeB checks whether a blind-configurable candidate cell (with **CaGroupSCellCfg**.***SCellBlindCfgFlag*** set to **TRUE**) on the top-priority candidate SCC has been specified for the PCell of the UE.
   - Such a cell has been specified.

     The eNodeB delivers an RRC Connection Reconfiguration message to configure this cell as an SCell for the UE, without delivering measurement configurations.

     NOTE

     - If there are multiple blind-configurable candidate SCells on different candidate SCCs, the eNodeB randomly selects one to configure as an SCell.
     - For UEs that support NSA DC, if any NR cell is defined as a blind-configurable candidate SCell on the eNodeB, the eNodeB preferentially configures the NR cell as an SCell in a blind manner. After the NR SCell is configured, the eNodeB then performs blind E-UTRA SCell configuration.
     - The following cells cannot be configured as SCells in a blind manner:
       - Cells that are shut down by energy saving features
       - Deactivated cells
   - No such cell has been specified or the blind configuration fails.

     The eNodeB delivers the A4 measurement configuration to the UE. The entering condition for event A4 is as follows: (*Mn* + *Ofn* + *Ocn* – *Hys* > *Thresh*) is true throughout a period specified by *TimeToTrig*. If multiple candidate SCCs have the same priority, the eNodeB will perform the subsequent steps on all candidates.

     In the formula:

     - *Mn* is the measurement result of a neighboring cell.
     - *Ofn* and *Ocn* are detailed in *Mobility Management in Connected Mode*.

- *TimeToTrig* is the time-to-trigger for CA event A4. It is specified by the **CAMGTCFG**.***CaA4TimeToTrigger*** parameter.
- *Hys* is fixed at 1 dB.
- *Thresh* is the threshold for event A4.

  If the SCell configuration is performed for the UE during initial access, an incoming handover, or an incoming RRC connection reestablishment, the threshold for event A4 is equal to the sum of the **CaMgtCfg**.***CarrAggrA4ThdRsrp***, **SccFreqCfg**.***SccA4Offset***, and **SccFreqCfg**.***SccA2RsrpThldExtendedOfs*** parameter values configured on the PCell side.

  If the SCell configuration is periodically triggered for the UE based on the traffic volume, the threshold for event A4 is equal to the sum of **CaMgtCfg**.***CarrAggrA4ThdRsrp*** and **SccFreqCfg**.***SccA4Offset*** parameter values configured on the PCell side.

2. The eNodeB delivers the A4 measurement configuration, instructing the UE to measure the candidate SCC. In addition, the eNodeB may set up measurement gaps for the UE, depending on the inter-frequency measurement capabilities reported by the UE.
   - The UE requires measurement gaps.
     - If a bearer with a QCI of 1 has been established for the UE, then:

       The procedure goes as described in **19.7 VoLTE**.
     - If a bearer with a QCI of 65 or 66 has been established for the UE and:

       If the **McpttVoiceCaSwitch** option of the **CaMgtCfg**.**CellCaAlgoSwitch** parameter is deselected, the SCell configuration procedure ends.

       If the **McpttVoiceCaSwitch** option of the **CaMgtCfg**.**CellCaAlgoSwitch** parameter is selected, the eNodeB sets up measurement gaps in the measurement configuration.

NOTE

- If no neighboring cell has been configured on the candidate SCC, the eNodeB sets up measurement gaps in the measurement configuration only if all the following options are selected: **CaBasedEventAnrSwitch** and **IntraRatEventAnrSwitch** options of the **ENodeBAlgoSwitch**.***AnrSwitch*** parameter, **CaAnrGapOptSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter, and **INTRA_RAT_ANR_SW** option of the **CellAlgoSwitch**.***AnrFunctionSwitch*** parameter.
- For adaptive configuration of inter-eNodeB SCells in a relaxed backhaul scenario, during initial configuration at the local eNodeB, the control-plane link and user-plane path are set up and checked for the eX2 or X2 interface between the eNodeBs and information is exchanged between the PCell and the candidate SCell. This takes a longer time than intra-eNodeB CA. Furthermore, the SCell cannot be configured using a single round of gap-assisted measurements. Therefore, the eNodeB instructs this UE to perform two rounds of gap-assisted measurements.

  The first round is used for eX2 or X2 setup and check as well as information exchange between the PCell and the candidate SCell. The second round is used for SCell configuration. The interval between the two rounds is determined by the **CaMgtCfg**.**SccCfgInterval** parameter.

  After the eX2 or X2 interface is set up, the eNodeB sets a default delay, which is less than 4 ms. It takes 8s to finish the measurement of the actual delay. If the actual delay is longer than the default delay, there is no traffic of the UE in the SCell during delay measurement. The SCell traffic of the UE will return to normal after the delay measurement.

  After the eX2 or X2 interface is set up, the eNodeB instructs subsequent CA UEs to perform only one round of gap-assisted measurements during SCell configuration.

– The UE does not require measurement gaps.

  The eNodeB does not set up measurement gaps in the measurement configuration.

The **AutoGapSwitch** option of the **ENodeBAlgoSwitch**.***HoModeSwitch*** parameter affects the decision process of the inter-frequency measurement gap setup.

– If this option is selected, the eNodeB determines whether to set up the gaps based on the reported UE capabilities.

– If this option is deselected, the eNodeB sets up the gaps, without considering UE capabilities.

It is recommended that this option be deselected to ensure that only cells with satisfactory signal quality can be configured as SCells.

After delivering the A4 measurement configuration related to the candidate SCC to the UE, the eNodeB may receive an A2 measurement report, indicating unsatisfactory signal quality of the PCell, for an inter-frequency or inter-RAT handover. In such a case, the eNodeB will not configure any cell on the candidate SCC as an SCell for the UE, even if the eNodeB later receives an A4 measurement report containing that cell.

3. After receiving an A4 measurement report that contains cells on the candidate SCC, the eNodeB selects the reported cells that belong to the serving PLMN of the UE or an equivalent PLMN as candidate SCells. The eNodeB then arranges the candidate SCells in descending order of RSRP and acts as follows:

- If the PCell can set up a data link to the top-priority candidate SCell, the eNodeB sends an RRC Connection Reconfiguration message to configure the candidate cell as an SCell for the UE. If the SCell is configured successfully, the procedure ends. If the SCell fails to be configured, the eNodeB tries the next-priority candidate SCell.
- If the PCell cannot set up a data link to the top-priority candidate SCell, the eNodeB tries the next-priority candidate SCell.
- If none of the candidate SCells can be configured as an SCell for the UE, the subsequent procedure depends on the **MultiCarrierFlexCaSwitch** option setting of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter.
  - If this option is selected, the eNodeB evaluates the next-priority candidate SCC.
  - If this option is deselected, the SCell configuration procedure ends.

4.6.3.2 SCell Configuration Enhancement

This section describes an enhancement to SCell configuration: load-based SCell configuration. With this function, intra- or inter-eNodeB cells with routes set up between them exchange their load status, and the eNodeB selects candidate cells not in the high load state as SCells for a UE.

This function is recommended when there are more operating frequencies on a network than the number of frequencies to be aggregated and at least two of those frequencies have the same priority.

This function is activated if the **SccSmartCfgSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter is selected.

For inter-eNodeB CA, this option must be selected only for the serving eNodeB of the PCell.

The evaluation of load status indicators for candidate SCells varies depending on the setting of **CellMLB.*MlbTriggerMode***. It is recommended that the following parameters be set consistently for all of the cells on the network.

- Set to **PRB_ONLY**
  - High-load entering condition: physical resource block (PRB) usage ≥ **CellMLB.*InterFreqMlbThd*** + **CellMLB.*InterFreqOffloadOffset*** throughout 5s
  - Low load condition: before the cell enters the high load state or after it exits the high load state

    High-load leaving condition: PRB usage < **CellMLB.*InterFreqMlbThd*** + **CellMLB.*InterFreqOffloadOffset*** – **CellMLB.*LoadOffset*** throughout 5s
- Set to **UE_NUMBER_ONLY**
  - High-load entering condition: Number of UEs in the cell ≥ **CellMLB.*InterFreqMlbUeNumThd*** + **CellMLB.*InterFrqUeNumOffloadOffset*** throughout 5s

    For the definition of the number of UEs in a cell, see *Intra-RAT Mobility Load Balancing*.
  - Low load condition: before the cell enters the high load state or after it exits the high load state

High-load leaving condition: Number of UEs in the cell < **CellMLB.*InterFreqMlbUeNumThd*** + **CellMLB.*InterFrqUeNumOffloadOffset*** – **CellMLB.*MlbUeNumOffset*** throughout 5s

- Set to **PRB_OR_UE_NUMBER**
  - A cell enters the high load state if it meets either of the following conditions:
    - (PRB usage ≥ **CellMLB.*InterFreqMlbThd*** + **CellMLB.*InterFreqOffloadOffset***) throughout 5s
    - (Number of UEs in the cell ≥ **CellMLB.*InterFreqMlbUeNumThd*** + **CellMLB.*InterFrqUeNumOffloadOffset***) throughout 5s
  - The cell is in the low load state if both of the following conditions are met:
    - The cell meets the low load condition used when this parameter is set to **PRB_ONLY**.
    - The cell meets the low load condition used when this parameter is set to **UE_NUMBER_ONLY**.

This function introduces special handling to the procedures described in **4.6.3.1.2 CA-Group-based SCell Configuration Procedure** and **4.6.3.1.3 Adaptive SCell Configuration Procedure**. The special handling varies as follows:

- If blind-configurable candidate SCells have been set for the PCell of a CA UE, blind SCell configuration preferentially takes place for the UE.
  - If the number of blind-configurable candidate SCells under a given priority is greater than the number of SCells to be configured, the eNodeB checks the load status indicator of each candidate and delivers an RRC Connection Reconfiguration message to preferentially configure low-load cells as SCells. If the candidates are in the same load state (either high or low), the eNodeB randomly selects cells from them as SCells.
  - If the number of blind-configurable candidate SCells under a given priority is less than or equal to the number of SCells to be configured, the eNodeB starts SCell configuration without comparing the load states of the candidates.
- If blind-configurable candidate SCells have not been set for the PCell of a CA UE, there are multiple candidate SCells with the same priority, and:
  - The number of these candidates is greater than the number of SCells to be configured, then:

    The eNodeB collects the measurement reports about all these candidates. (If the eNodeB does not receive measurement reports about any cell on a frequency within 3s, it considers the UE to be outside of the coverage area of the frequency and instructs the UE to stop the measurements.) The eNodeB then checks the load status indicator of each candidate. Based on the load status indicators, the eNodeB delivers an RRC Connection Reconfiguration message to preferentially configure low-load cells as SCells for the UE. This measurement-based SCell configuration procedure has a longer delay than the blind SCell configuration procedure. If the candidates are in the same load state (either high or low), the eNodeB selects the cells with the greatest RSRP values as SCells.

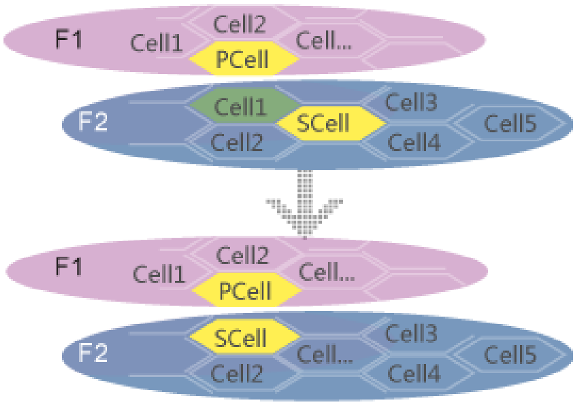
– The number of these candidates is less than or equal to the number of SCells to be configured, then:

  The eNodeB does not check the load status indicators of candidates. Instead, it starts configuring SCells upon receiving measurement reports.

4.6.3.3 SCell Change

If a CA UE whose SCells have been configured receives better signal quality from an intra-frequency neighboring cell of an SCell than from that SCell, the serving eNodeB of the PCell can change the SCell without changing the PCell. **Figure 4-10** illustrates this function.

Figure 4-10 SCell change



This function is controlled by the **SccModA6Switch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter. It is recommended that this function be always activated.

NOTE

When the **SccModA6Switch** option is selected, it is recommended that the **SCellModCaMeasRmvSwitch** option of the **GlobalProcSwitch**.***ProtocolCompatibilitySw*** parameter also be selected. With this setting, the eNodeB will delete the existing A6 measurement configurations before the SCell is changed. This ensures that these measurement configurations can be automatically deleted by the UE and prevents service drops due to repeated measurement configurations.

CA-Group-based SCell Change

If the SCell change function has been activated and cells on the same frequency are included in a CA group, the eNodeB sends an RRC Connection Reconfiguration message to a CA UE after configuring one of the cells as an SCell for that UE. This message contains an A6 measurement configuration related to this SCell. The offset for event A6 is specified by the **CaMgtCfg.*CarrAggrA6Offset*** parameter configured on the PCell side.

After receiving an A6 measurement report from the CA UE, the eNodeB evaluates the reported cells in the CA group and selects the one with the highest RSRP value. The eNodeB then sends an RRC Connection Reconfiguration message to change the SCell to the selected cell. If the SCell change fails, the eNodeB selects the cell with the next highest RSRP value among the reported cells and makes another SCell change attempt.

Adaptive SCell Change

If the SCell change function has been activated, the eNodeB sends an RRC Connection Reconfiguration message to a CA UE after configuring SCells for the UE. The message contains an A6 measurement configuration related to each SCC of the UE. The offset for event A6 is specified by the **CaMgtCfg.*CarrAggrA6Offset*** parameter configured on the PCell side.

After receiving an A6 measurement report related to an SCell (for example, cell 1) from the UE, the eNodeB arranges the reported candidate SCells in descending order of RSRP and proceeds as follows:

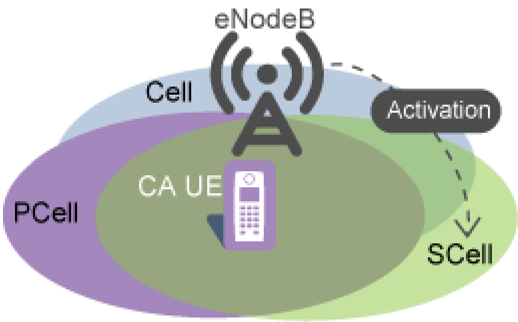
- If the PCell can set up a data link to the top-priority candidate SCell (for example, cell 2), the eNodeB sends an RRC Connection Reconfiguration message to change the SCell from cell 1 to cell 2 for the UE.
- If the PCell cannot set up a data link to cell 2, the eNodeB tries the next-priority candidate SCell.

If the PCell cannot set up a data link to any candidate SCell, the eNodeB does not change the SCell for the UE.

4.6.3.4 SCell Activation

In accordance with section 6.1.3.8 "Activation/Deactivation MAC Control Element" in 3GPP TS 36.321 V11.1.0, the eNodeB delivers a MAC CE to activate a configured SCell as long as either uplink or downlink traffic volume meets the activation conditions. The CCs are not actually aggregated for the UE until the SCell is activated. **Figure 4-11** illustrates SCell activation.

Figure 4-11 SCell activation



If the **CaInstantlyJudgeSwitch** option of the **CaMgtCfg.*CellCaAlgoSwitch*** parameter is selected, the eNodeB uses instantaneous, millisecond-level values to evaluate traffic volume conditions for SCell activation. If this option is deselected, the eNodeB uses filtered second-level values to evaluate the traffic volume conditions for SCell activation. **Table 4-14** describes how the eNodeB evaluates SCell activation.

Table 4-14 Evaluation of SCell activation

CA UE Status	Activation Condition	Activation Evaluation
Downlink SCells have been configured for the UE.	Downlink traffic volume conditions for activation: ● Data volume buffered at the RLC layer > max(Uu data rate at the RLC layer x Buffer delay threshold, Buffer length threshold) ● Delay of the first RLC protocol data unit (PDU) > Buffer delay threshold	When the downlink traffic volume meets the downlink traffic volume conditions for activation, the eNodeB performs SCell activation for downlink CA.
An uplink SCell has been configured for the UE.	Downlink traffic volume conditions for activation: ● Data volume buffered at the RLC layer > max(Uu data rate at the RLC layer x Buffer delay threshold, Buffer length threshold) ● Delay of the first RLC PDU > Buffer delay threshold Uplink traffic volume conditions for activation: ● The reported uplink buffer status of the CA UE is greater than the buffer length threshold throughout the time-to-trigger for activation for uplink CA. ● Transmission time interval (TTI) bundling does not take effect for the CA UE in the uplink.	● When the downlink traffic volume meets the downlink traffic volume conditions for activation, the eNodeB performs SCell activation for downlink CA. ● When the uplink traffic volume meets the uplink traffic volume conditions for activation, the eNodeB performs SCell activation for uplink CA. The SCells activated for uplink CA can only be those that act as SCells in both the uplink and downlink.

In the formulas:

- The buffer delay threshold is specified by:

  – **CaMgtCfg.*ActiveBufferDelayThd*** if the UE is a non-NSA UE.
  – **NsaDcMgmtConfig**.***NsaDcLteScellActBfrDelThld*** if the UE is an NSA UE.
- The buffer length threshold is specified by:
  – **CaMgtCfg.*ActiveBufferLenThd*** if the UE is a non-NSA UE.
  – **NsaDcMgmtConfig**.***NsaDcLteScellActBfrLenThld*** if the UE is an NSA UE.
- The time-to-trigger for activation for uplink CA is specified by the **CaMgtCfg.*UlCaActiveTimeToTrigger*** parameter.

If the **CaMgtCfg**.***ActiveBufferLenThd*** parameter is set to **0**:

- If an SCell has been configured for a CA UE, the eNodeB activates the SCell as long as the eNodeB prepares to send data to the CA UE. However, if the traffic volume of the UE is low, the eNodeB deactivates the SCell (because the deactivation condition is met) immediately after it activates the SCell for the UE. The number of UEs with SCells activated on the affected BBP increases, and the BBP is likely to enter flow control due to heavy load. This affects LTE UE access or other LTE services.
- If no SCell has been configured for a CA UE, the eNodeB repeatedly attempts to configure an SCell without considering **CaMgtCfg**.***SccCfgInterval*** when the eNodeB prepares to send data to the UE. This consumes a large amount of radio resources.

To ensure synchronization between the UE and the eNodeB, SCells are activated on the UE and eNodeB sides simultaneously. If the eNodeB sends a MAC CE for SCell activation to the UE in subframe n, then the SCell is activated on the UE and eNodeB sides in subframe ($n + x$). The value of x is stipulated by physical layer protocols.

- For FDD, x is equal to 8.
- For TDD, x is greater than or equal to 8.

When cells are heavily loaded, SCell activation can be prohibited to ensure user experience. This prohibition function is enabled when the **CaMgtCfg**.***ScellNoActivationUeNumThld*** parameter is set to a non-zero value. This parameter specifies the threshold for determining heavy cell load.

- If the number of UEs in a cell exceeds the value of this parameter, the eNodeB determines that the cell is heavily loaded. In this case, new SCell activation quotas will not be provided for CA UEs. When the existing SCell activation quota is reached (that is, the number of UEs that have this cell activated as their SCell has reached the quota), the activation quota becomes 0 and this cell will no longer be activated as an SCell for any CA UE.
- If the number of UEs in a cell is a little smaller than the value of this parameter, simultaneous SCell activation for a number of CA UEs will cause the number of UEs in the cell to exceed the value of this parameter even if no UEs are admitted to the cell through initial access, incoming RRC connection reestablishments, or incoming handovers.

The eNodeB measures the number of UEs in each cell every second.

The following UEs are counted into the number of UEs in a cell:

- Non-CA UEs

- CA UEs that treat the cell as their PCell
- CA UEs for which the cell has been activated as an SCell

For FDD, it is recommended that this function be enabled when the cell bandwidth is 1.4 MHz, 3 MHz, or 5 MHz.

For TDD, it is recommended that this function be enabled when the cell bandwidth is 5 MHz.

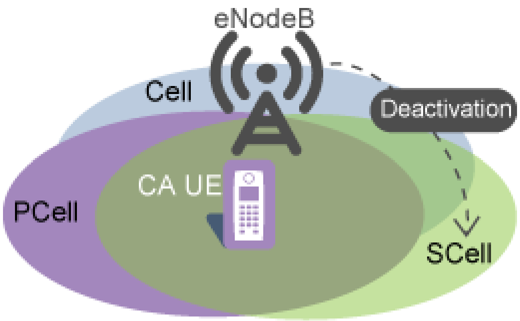
This function is not compatible with LBBPc boards.

4.6.3.5 SCell Deactivation

SCell deactivation for CA UEs, as illustrated in **Figure 4-12**, can be triggered by:

- Traffic volume
- Channel quality
- Residual block errors
- Number of RLC retransmissions

Figure 4-12 SCell deactivation



Traffic Volume–triggered SCell Deactivation

This function takes effect when the **CaMgtCfg.CarrierMgtSwitch** parameter is set to **ON** on the PCell side. With this function enabled, SCell deactivation is evaluated as described in **Table 4-15**.

Table 4-15 Evaluation of SCell deactivation

CA UE Status	Deactivation Condition	Deactivation Evaluation
Downlink SCells have been activated for the UE.	Downlink traffic volume conditions for deactivation: ● Uu data rate at the RLC layer ≤ Throughput threshold for deactivation ● Data volume buffered at the RLC layer ≤ Buffer length threshold for deactivation	When each E-RAB of the CA UE meets the downlink traffic volume conditions for deactivation, the eNodeB delivers a MAC CE to deactivate the SCells of the UE.
An uplink SCell has been activated for the UE.	Downlink traffic volume conditions for deactivation: ● Uu data rate at the RLC layer ≤ Throughput threshold for deactivation ● Data volume buffered at the RLC layer ≤ Buffer length threshold for deactivation Uplink traffic volume conditions for deactivation: ● Data size indicated in the buffer status report (BSR) from the CA UE ≤ Buffer length threshold for deactivation ● Uplink Uu data rate of the CA UE at the RLC layer ≤ Throughput threshold for deactivation	The eNodeB delivers a MAC CE to deactivate the SCell only when both the uplink and downlink traffic volumes meet their respective conditions. When only the downlink traffic volume meets the deactivation conditions, the eNodeB sends a MAC CE to deactivate all downlink-only SCells for the UE.
Note: It is recommended that the buffer length threshold for deactivation be less than the buffer length threshold for activation.		

In the formulas:

- The throughput threshold for deactivation is specified by the **CaMgtCfg**.**DeactiveThroughputThd** parameter.

- The buffer length threshold for deactivation is specified by the **CaMgtCfg**.***DeactiveBufferLenThd*** parameter.
- The buffer length threshold for activation is specified by the **CaMgtCfg**.***ActiveBufferLenThd*** parameter.

If the **CaTrafficTriggerSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected for the serving eNodeB of the PCell, the eNodeB sends an RRC Connection Reconfiguration message to remove SCells immediately after deactivating these SCells based on the traffic volume.

If the **NsaDcMgmtConfig**.***ScgAdditionBufferLenThld*** parameter is set to **0**, the eNodeB does not remove SCells for NSA CA UEs based on traffic volume.

Channel Quality–triggered SCell Deactivation

This function is recommended when blind SCell configuration is enabled or A2 measurement is disabled. In such scenarios, the channel quality indicator (CQI) for SCells will be low if CA UEs are located in weak coverage areas or are experiencing strong interference. The low CQI will cause a decrease in **Cell Downlink Average Throughput** and an increase in the number of RRC connection reestablishments, the uplink and downlink initial block error rate (IBLER) values, the packet loss rate, and the service drop rate.

This function takes effect when the **CaMgtCfg**.**CarrierMgtSwitch** parameter is set to **ON** on the PCell side.

If the **CaSccSuspendSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected, channel quality–triggered SCell deactivation does not take effect. Specifically, when the conditions for such SCell deactivation are met with this option selected, SCells enter the suspension state to stop scheduling but will not be deactivated.

Channel quality–triggered SCell deactivation does not take effect when intra-base-station CA SCC scheduling stop based on spectral efficiency (controlled by the **SCC_SPCT_EFF_STOP_SCH_SW** option of the **CellDlSchExtParam**.***DlSchExtSwitch*** parameter) is in effect. For details about intra-base-station CA SCC scheduling stop based on spectral efficiency, see *Downlink Scheduling*.

Therefore, before enabling this function, deselect the **CaSccSuspendSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter and the **SCC_SPCT_EFF_STOP_SCH_SW** option of the **CellDlSchExtParam**.***DlSchExtSwitch*** parameter.

With this function enabled:

- In a non-relaxed backhaul scenario:

  The eNodeB sends a MAC CE to deactivate an SCell for a CA UE when the spectral efficiency corresponding to the SCell's CQI reported by the UE falls below the Release 8 spectral efficiency corresponding to the **CaMgtCfg**.***SccDeactCqiThd*** parameter value specified for the SCell in single-codeword transmission.

  If **CaMgtCfg**.***SccDeactCqiThd*** is set to **0** for the SCell, the serving eNodeB of the PCell does not deactivate the SCell based on channel quality.
- In a relaxed backhaul scenario:

  The eNodeB sends a MAC CE to deactivate an SCell for a CA UE when the spectral efficiency corresponding to the SCell's CQI reported by the UE falls below the Release 8 spectral efficiency corresponding to the **CaMgtCfg**.***RelaxedBHSccDeactCqiThd*** parameter value specified for the SCell in single-codeword transmission.

If **CaMgtCfg.*RelaxedBHSccDeactCqiThd*** is set to **0** for the SCell, the serving eNodeB of the PCell does not deactivate the SCell based on channel quality.

After an SCell is deactivated based on channel quality, the CA UE no longer reports the SCell's CQI to the eNodeB. The eNodeB may subsequently activate the SCell based on traffic volume or voice service requirements. The **CaMgtCfg**.***SccReactivationTime*** parameter can be used to prevent excessively frequent SCell activation and deactivation. This parameter specifies the minimum interval between SCell deactivation and traffic-volume- or voice-service-requirements-based reactivation. This parameter is PCell-specific. It takes effect in the PCell, which makes activation decisions.

After deactivating an SCell based on channel quality, the serving eNodeB of the PCell does not immediately remove this SCell, regardless of whether the **CaTrafficTriggerSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected.

Residual Block Error–triggered SCell Deactivation

- In the downlink:

  The processing of residual block error–triggered SCell deactivation depends on the setting of the **CaMgtCfg**.***SccQuietTime*** parameter.

  – **CaMgtCfg**.***SccQuietTime*** set to **0**

    When the eNodeB detects downlink residual block errors in an SCell for a CA UE for 10 consecutive times, the eNodeB immediately delivers a MAC CE to deactivate the SCell.

  – **CaMgtCfg**.***SccQuietTime*** set to a value other than **0**

    When the eNodeB detects downlink residual block errors in an SCell for a CA UE for 10 consecutive times, the eNodeB does not deliver a MAC CE to deactivate the SCell within the period specified by the **CaMgtCfg**.***SccQuietTime*** parameter.

    The eNodeB stops scheduling the UE in the SCell and periodically sends a 1-byte packet to detect the SCell.

    ▪ ACK returned

      The eNodeB resumes the scheduling of the UE in the SCell.

    ▪ NACK returned

      The eNodeB adjusts the CQI and then continues the periodic detection. The CQI adjustment varies with the setting of the **CaMgtCfg**.***SccDetectCqiDecreaseStep*** parameter.

      ○ **CaMgtCfg**.***SccDetectCqiDecreaseStep*** set to **0**

        The eNodeB decreases the CQI by the basic adjustment value.

      ○ **CaMgtCfg**.***SccDetectCqiDecreaseStep*** set to a value other than **0**

        The eNodeB decreases the CQI by the sum of the basic adjustment value and the **CaMgtCfg**.***SccDetectCqiDecreaseStep*** parameter value.

If the number of detection failures reaches the maximum, the eNodeB delivers a MAC CE to deactivate the SCell.

- In the uplink:

  The processing of residual block error–triggered SCell deactivation depends on the setting of the **CaMgtCfg**.***SccQuietTime*** parameter and the **SccDeactByUlDtxSwitch** option setting of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter.

  – **CaMgtCfg**.***SccQuietTime*** set to **0**

    ▪ If the **SccDeactByUlDtxSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is deselected, the eNodeB immediately delivers a MAC CE to deactivate an SCell for a CA UE when it detects uplink residual block errors in the SCell for 40 consecutive times.

    ▪ If the **SccDeactByUlDtxSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is selected, the eNodeB delivers a MAC CE to deactivate an SCell for a CA UE when it detects uplink residual block errors in the SCell for 40 consecutive times or detects that the eNodeB-maintained SCell activation status is different from the actual status.

  – **CaMgtCfg**.***SccQuietTime*** set to a value other than **0**

    This function does not take effect. That is, the eNodeB does not initiate the SCell deactivation procedure based on uplink residual block errors in SCells.

    NOTE

    - To detect the actual SCell activation status, operators must select the **SccDeactByUlDtxSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter for each eNodeB involved in CA (including the serving eNodeB of the PCell and the serving eNodeB of each SCell). If operators select this option only for the serving eNodeB of the PCell or an SCell, the probability of successfully detecting activation status inconsistency decreases.
    - Operators are advised to select the **SccDeactByUlDtxSwitch** option in scenarios other than high uplink load. If this option is selected in high uplink load scenarios, the data rates in SCells will drop.

RLC Retransmission Count–triggered SCell Deactivation

During downlink data transmission to a CA UE in an SCell with a high downlink RBLER, there might be more RLC retransmissions due to an increased number of MAC retransmissions. If that is the case, the service drop rate increases. To address this issue, RLC retransmission count–triggered SCell deactivation is introduced.

- In a relaxed backhaul scenario:

  The procedure for evaluating SCell deactivation varies depending on the value of the **CaMgtCfg**.***ScellDeactRlcRetransNumThd*** parameter.

  – If this parameter is set to **0** for the PCell of a CA UE:

    The eNodeB checks whether the total number of RLC retransmissions for the CA UE in its PCell and SCells exceeds 20.

    ▪ If the number exceeds 20, the eNodeB deactivates all the SCells of the CA UE.

  - If the number does not exceed 20, the eNodeB does not deactivate the SCells.

- If this parameter is set to a non-zero value for the PCell of a CA UE:

  The SCell deactivation procedure is dependent on the triggering threshold, which is equal to min(*A*, *B*), where *A* is the value of the **CaMgtCfg**.***ScellDeactRlcRetransNumThd*** parameter and *B* is the rounded-up integer value of (0.6 x Actual maximum number of RLC ARQ retransmissions for the CA UE).

  - If the triggering threshold is less than 20:

    The eNodeB checks whether the total number of RLC retransmissions for the CA UE in its PCell and SCells exceeds the triggering threshold.

    - If it exceeds the triggering threshold, the eNodeB notifies each of the SCells of this circumstance. Among these SCells, each SCell with **CaMgtCfg**.***ScellDeactRlcRetransNumThd*** set to a non-zero value checks whether the number of MAC retransmissions reaches the maximum for two consecutive times within 1s after it receives the notification. If it does, the eNodeB deactivates the SCell. If it does not, the eNodeB does not deactivate the SCell. If the eNodeB has not deactivated all the SCells of the CA UE, the eNodeB deactivates all the remaining SCells of the UE when the number of retransmissions for the UE exceeds 20.
    - If it does not exceed the triggering threshold, the eNodeB does not deactivate the SCells.

  - If the triggering threshold is greater than 20:

    The eNodeB checks whether the total number of RLC retransmissions for the CA UE in its PCell and SCells exceeds 20.

    - If the number exceeds 20, the eNodeB deactivates all the SCells of the CA UE.
    - If the number does not exceed 20, the eNodeB does not deactivate the SCells.

  - If the triggering threshold is equal to 20:

    The following procedures are both entered:

    - Procedure 1: The eNodeB deactivates all the SCells of the CA UE.
    - Procedure 2: The eNodeB sends a notification to each SCell of the CA UE. Among these SCells, each SCell with **CaMgtCfg**.***ScellDeactRlcRetransNumThd*** set to a non-zero value checks whether the number of MAC retransmissions reaches the maximum for two consecutive times within 1s after it receives the notification. If it does, the eNodeB deactivates the SCell. If it does not, the eNodeB does not deactivate the SCell.

    If procedure 1 is triggered first, the eNodeB directly deactivates all the SCells of the CA UE. If procedure 2 is triggered first, the eNodeB first deactivates the SCells that meet the conditions, and then deactivates all the remaining SCells of the CA UE according to procedure 1.

- In a non-relaxed backhaul scenario:

  RLC retransmission count–triggered SCell deactivation takes effect only if the **CaMgtCfg**.***ScellDeactRlcRetransNumThd*** parameter is set to non-zero values for both the PCell and SCells of a CA UE.

  The eNodeB evaluates whether to deactivate SCells for the CA UE based on whether the total number of RLC retransmissions for the CA UE in its PCell and SCells exceeds the triggering threshold, which is equal to min(*A*, *B*), where *A* is the value of the **CaMgtCfg**.***ScellDeactRlcRetransNumThd*** parameter and *B* is the rounded-up integer value of (0.6 x Actual maximum number of RLC ARQ retransmissions for the CA UE).

  – If it exceeds the triggering threshold, the eNodeB notifies each of the SCells of this circumstance. Among these SCells, each SCell with **CaMgtCfg**.***ScellDeactRlcRetransNumThd*** set to a non-zero value checks whether the number of MAC retransmissions reaches the maximum for two consecutive times within 1s after it receives the notification.

    ▪ If it does in an SCell, the eNodeB deactivates the SCell.

    ▪ If it does not in an SCell, the eNodeB does not deactivate the SCell.

  – If it does not exceed the triggering threshold, the eNodeB does not deactivate the SCells.

4.6.3.6 SCell Removal

To have SCells removed when their signal quality degrades, enable the SCell removal function.

This function is controlled by the **SccA2RmvSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter.

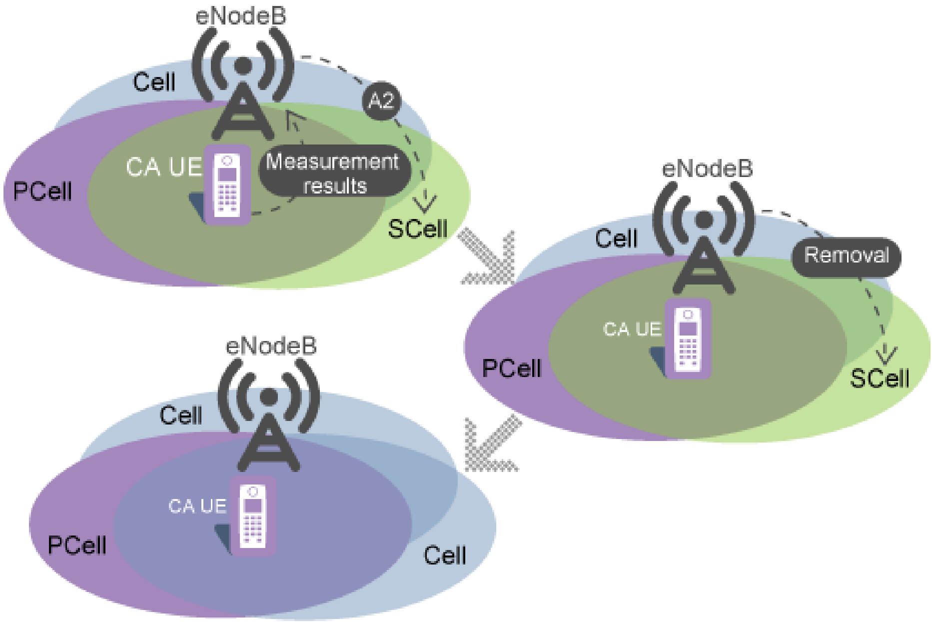
When an SCell meets the triggering conditions for event A2, the eNodeB removes the SCell. If the eNodeB removes *n* SCells for a CA UE in the uplink or downlink *m*CC aggregation state, the UE will be served by (*m*–*n*) carriers.

NOTE

- When a bearer for an emergency call or with QCI 1, 65, 66, or an enhanced extended QCI is set up for a CA UE whose SCells have been configured, the eNodeB does not remove the SCells.
- When the **SccA2RmvSwitch** option is selected, it is recommended that the **SCellModCaMeasRmvSwitch** option of the **GlobalProcSwitch**.***ProtocolCompatibilitySw*** parameter also be selected. With this setting, the eNodeB will delete the existing A2 measurement configurations before the SCell is removed. This ensures that these measurement configurations can be automatically deleted by the UE and prevents service drops due to repeated measurement configurations.

SCell removal is performed by the eNodeBs serving the PCells of CA UEs. **Figure 4-13** illustrates this function.

Figure 4-13 SCell removal



SCell removal works in CA-group-based mode and in adaptive mode, as described in **CA-Group-based SCell Removal** and **Adaptive SCell Removal**.

In either mode, SCells can be removed for CA UEs to relieve UE overheating, as described in **SCell Removal for CA UE Overheating Protection**.

CA-Group-based SCell Removal

With this function enabled in CA-group-based configuration mode, the eNodeB delivers A2 measurement configurations to a UE after configuring an SCell for the UE based on A4 measurements. After receiving an A2 measurement report that contains the SCell, the eNodeB sends an RRC Connection Reconfiguration message to remove the SCell.

Event A2 is triggered if its entering condition (*Ms* + *Hys* < *Thresh*) is true throughout *TimeToTrig*.

If an SCell has been configured for a CA UE in a blind manner, the eNodeB checks the **SCellBlindA2Switch** setting of the **GlobalProcSwitch**.*ProtocolCompatibilitySw* parameter before making a measurement configuration delivery decision. If the **SCellBlindA2Switch** option is selected, the eNodeB sends A2 measurement configurations, in which the threshold for event A2 takes the 3GPP-defined minimum value –140 dBm, to the UE. After receiving an A2 measurement report that contains the SCell, the eNodeB sends an RRC Connection Reconfiguration message to remove the SCell. If the **SCellBlindA2Switch** option is deselected, the eNodeB does not send A2 measurement configurations to the UE, which means the SCell will not be removed.

NOTE

- The variables in the entering condition for event A2 are described as follows:
  - *Ms* is the measurement result of the serving cell.
  - *Hys* is fixed at 1 dB.
  - *Thresh* is the threshold for event A2. The threshold is equal to the sum of the **CaMgtCfg.*CarrAggrA2ThdRsrp*** and **CaGroupSCellCfg.*SCellA2Offset*** parameters configured on the PCell side.
  - *TimeToTrig* is the time-to-trigger for CA event A2. It is specified by the **CAMGTCFG**.***CaA2TimeToTrigger*** parameter.
- In CA-group-based configuration mode, the following setting constraint applies: The threshold for event A4 must be higher than the threshold for event A2. In addition, only the values in the range of [–143, –43] take effect for both thresholds.
- After SCells are removed for a UE in CA-group-based configuration mode, the eNodeB can configure SCells for the UE again either in a blind manner or based on A4 measurements, when the traffic volume meets the SCell configuration conditions. Blind SCell configuration (if enabled by selecting the **SccBlindCfgSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter and setting the **CaGroupSCellCfg.*SCellBlindCfgFlag*** parameter to **TRUE**) takes priority. The eNodeB configures SCells based on A4 measurements if blind SCell configuration is disabled.

Adaptive SCell Removal

With this function enabled in adaptive configuration mode, the eNodeB delivers A2 measurement configurations to a UE after configuring an SCell for the UE based on A4 measurements or in a blind manner. After receiving an A2 measurement report that contains the SCell, the eNodeB sends an RRC Connection Reconfiguration message to remove the SCell. Events A4 and A2 are triggered as follows:

- For details about the entering conditions for event A4, see **4.6.3.1.3 Adaptive SCell Configuration Procedure**.
- Event A2 is triggered if its entering condition (*Ms* + *Hys* < *Thresh*) is true throughout *TimeToTrig*.

The threshold for CA event A4 must be higher than the threshold for CA event A2. If this requirement is not fulfilled and the **SccA2RmvSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter is selected, the eNodeB will remove an SCell for a CA UE based on CA event A2 immediately after having configured it. Then, if the traffic volume of the UE meets the SCell activation conditions throughout a certain period of time, the eNodeB configures the same SCell again. The SCell is configured and removed repeatedly.

If the **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter is selected and the **CaMgtCfg**.***FastScellSelAftScellRmvSw*** parameter is set to **ON**, the eNodeB performs an SCell configuration procedure for a CA UE immediately after all SCells are removed for the UE based on coverage conditions, increasing the CA UE throughput. This procedure is the same as the adaptive SCell configuration procedure that occurs when the **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter is selected.

NOTE

- The variables in the entering condition for event A2 are described as follows:
  - *Ms* is the measurement result of the serving cell.
  - *Hys* is fixed at 1 dB.
  - *Thresh* is the threshold for CA event A2.

    The threshold for CA event A2 is equal to the sum of the **CaMgtCfg.*CarrAggrA2ThdRsrp***, **SccFreqCfg.*SccA2Offset***, and **SccFreqCfg**.***SccA2RsrpThldExtendedOfs*** parameters configured on the PCell side.

    If the sum is less than –140 dBm, this threshold takes the value of –140 dBm. If the sum is greater than –43 dBm, this threshold takes the value of –43 dBm. This threshold ranges from –140 dBm to –43 dBm.
  - *TimeToTrig* is the time-to-trigger for CA event A2. It is specified by the **CAMGTCFG**.***CaA2TimeToTrigger*** parameter.
- After SCells are removed for a UE in adaptive configuration mode, the eNodeB can configure SCells for the UE again only based on A4 measurements, when the traffic volume meets the SCell configuration conditions.

SCell Removal for CA UE Overheating Protection

The downlink data rates of CA UEs increase with the number of carriers, leading to a rise in CPU load and a higher risk in heat dissipation. After a UE has been continuously running at a high speed, it is likely to experience a service drop because of overheating. This affects user experience.

To resolve this issue, 3GPP Release 14 proposed a CA UE overheating protection mechanism. Upon overheating, a CA UE reports overheating assistance information (conveyed by the OverheatingAssistance IE) to the eNodeB, so as to inform the eNodeB of the overheating and request assistance from the eNodeB. For details about the OverheatingAssistance IE, see section 5.6.10.3 "Actions related to transmission of UEAssistanceInformation message" in 3GPP TS 36.331 V15.4.0.

This function is controlled by the **OverheatingProtectionSwitch** option of the **GlobalProcSwitch**.***ProtocolSupportSwitch*** parameter.

With this function enabled, the eNodeB includes the OverheatingAssistanceConfig IE in the RRCConnectionReconfiguration message used to configure SCells for a CA UE for the first time during initial access, an incoming RRC connection reestablishment, or an incoming necessary handover of the UE. This IE is used to activate reporting of overheating assistance information from the UE.

- When the UE becomes overheated, it reports the OverheatingAssistance IE to the eNodeB. This IE contains the reducedMaxCCs IE, indicating the temporary maximum number of carriers supported by the UE. After receiving the IE, the eNodeB removes all the SCells of the UE and saves the reducedMaxCCs IE reported by the UE. The number of SCells to be configured for the UE will not exceed the value of reducedMaxCCs.
- When the UE recovers from overheating, it reports the OverheatingAssistance IE to the eNodeB again. However, this IE no longer contains the reducedMaxCCs IE. The eNodeB removes the reducedMaxCCs IE from its storage space and continues to configure SCells for the UE.

With this function, CA UEs that support overheating protection can exit the CA state or have fewer SCells configured when they experience overheating. As a result, the throughput of the CA UEs and the number of their SCells decrease.

This function requires that CA UEs be capable of reporting overheating assistance information. This capability is indicated by the overheatingInd-r14 IE in the UE-EUTRA-Capability IE of the UECapabilityInformation message sent from the UEs.

4.7 Carrier Management for RRC_IDLE UEs

If the **IdleModePccAnchorSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected, the eNodeB performs the procedure of PCC anchoring for RRC_IDLE UEs when the RRC connections of CA UEs are released normally. (This does not occur at abnormal RRC connection releases, for example, due to S1 faults, an eNodeB overload, or an MME overload.)

The procedure varies with priority settings:

- SPID- or operator-specific cell-reselection priorities are configured.

  The eNodeB includes the configured priorities in the idleModeMobilityControlInfo IE of the RRC Connection Release message.

- Neither SPID- nor operator-specific cell-reselection priorities are configured.
  - If both PCell/PCC priorities and common cell-reselection priorities are specified, the eNodeB performs PCC anchoring as follows:
    i. The eNodeB generates a PCC anchor frequency list for an RRC connection release. This list is based on the band combinations supported by the CA UE and PCell/PCC priorities. In the list, these band combinations are arranged in descending order of PCell/PCC priority.

       If duplex-mode-based PCC anchoring is enabled, duplex mode priorities take precedence over PCell/PCC priorities when the eNodeB arranges the band combinations in the list.

    ii. The eNodeB includes the list in the idleModeMobilityControlInfo IE of an RRC Connection Release message and, in that message, sets the length of timer T320 to a fixed value of 180 minutes.

    iii. The eNodeB sends the message to the UE, instructing the UE to camp in the specified frequency band.

    In case certain frequencies with common cell-reselection priorities are not assigned PCell/PCC priorities by operators, these frequencies are appended to those with PCell/PCC priorities and arranged in descending order of common cell-reselection priority.

    For the parameters that specify common cell-reselection priorities, see *Idle Mode Management*.
  - If only PCell/PCC priorities or common cell-reselection priorities are specified, the eNodeB performs the following PCC anchoring procedure with the PCC anchor frequency list including only the candidate frequencies assigned the priorities and supported by the UE.
    i. The eNodeB generates a PCC anchor frequency list for an RRC connection release. This list is based on the band combinations supported by the CA UE and PCell/PCC priorities. In the list, these

band combinations are arranged in descending order of PCell/PCC priority.

If duplex-mode-based PCC anchoring is enabled, duplex mode priorities take precedence over PCell/PCC priorities when the eNodeB arranges the band combinations in the list.

ii. The eNodeB includes the list in the idleModeMobilityControlInfo IE of an RRC Connection Release message and, in that message, sets the length of timer T320 to a fixed value of 180 minutes.

iii. The eNodeB sends the message to the UE, instructing the UE to camp in the specified frequency band.

– If neither PCell/PCC priorities nor common cell-reselection priorities are specified, or if the UE does not support any candidate frequencies, the eNodeB performs a relevant idle mode management procedure. For details, see *Idle Mode Management*.

NOTE

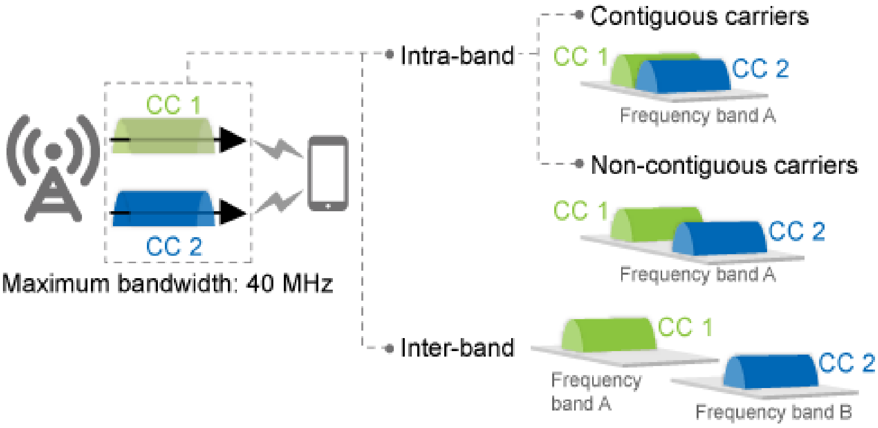
If the **HO_TRG_FREQ_FORBID_MEAS_FLAG** option of the **EutranInterNFreq**.***AggregationAttribute*** parameter is selected for a neighboring intra-RAT frequency, this neighboring frequency will not be selected as the target frequency during PCC anchoring for RRC_IDLE UEs. For details, see the section on frequency selection for measurement in *Mobility Management in Connected Mode*.

5 Downlink 2CC Aggregation

5.1 Principles

This function aggregates two intra- or inter-band carriers, as shown in **Figure 5-1**, to provide higher bandwidth.

Figure 5-1 Downlink 2CC aggregation



This function works between intra-eNodeB cells, between inter-eNodeB cells in eNodeB coordination scenarios, and between inter-eNodeB cells in relaxed backhaul scenarios.

The switch control over this function varies as described in **Table 5-1**.

Table 5-1 Switch setting requirements of downlink 2CC aggregation

Configuration Mode	Max. Downlink Bandwidth of Two CCs	CaDl2CCExtSwitch of CaMgtCfg.*CellCaAlgoSwitch*
CA-group-based (FDD)	Up to 20 MHz	No requirements

Configuration Mode	Max. Downlink Bandwidth of Two CCs	CaDl2CCExtSwitch of CaMgtCfg.*CellCaAlgoSwitch*
CA-group-based (FDD)	20 MHz < Bandwidth ≤ 40 MHz	No requirements
CA-group-based (TDD)	Up to 30 MHz	No requirements
CA-group-based (TDD)	30 MHz < Bandwidth ≤ 40 MHz	No requirements
Adaptive (FDD)	Up to 20 MHz	No requirements
Adaptive (FDD)	20 MHz < Bandwidth ≤ 40 MHz	Select this option.
Adaptive (TDD)	Up to 30 MHz	No requirements
Adaptive (TDD)	30 MHz < Bandwidth ≤ 40 MHz	Select this option.

5.2 Network Analysis

5.2.1 Benefits

This function enables CA UEs to reach higher downlink peak data rates.

Table 5-2 lists the theoretical peak data rates that a CA UE can reach using downlink 2CC aggregation.

For FDD, these values assume a TBS suitable for the 20 MHz cell bandwidth (equivalent to 100 RBs in the frequency domain).

For TDD, these values assume a TBS suitable in the following scenario:

- Bandwidth: 20 MHz (equivalent to 100 RBs in the frequency domain)
- Uplink-downlink subframe configuration: 2
- Special subframe configuration: 7

Table 5-2 Theoretical peak data rates for downlink 2CC aggregation (unit: Mbit/s)

RAT	2x2 MIMO + 64QAM	2x2 MIMO + 256QAM	4x4 MIMO + 64QAM	4x4 MIMO + 256QAM
FDD	299.6	391.6	599.7	783.3
TDD	224	286	434	572

The peak data rate that CA can achieve for a CA UE is subject to:

- Peak data rate capability of the board where the PCell for the CA UE is located

  For example, if the PCell and SCells of a CA UE are served by an LBBPd1 board that supports a downlink peak data rate of 450 Mbit/s, the peak data rate that CA can achieve for the CA UE will not exceed 450 Mbit/s.

- Capability of the CA UE

  If the UE capability is limited, the actual peak data rates will be lower than the theoretical values. The UE capability is indicated by ue-categoryDL. For details about this IE, see section 4.1A "ue-CategoryDL and ue-CategoryUL" in 3GPP TS 36.306 V15.2.0.

5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.3.3 Function Impacts**.

Downlink 2CC aggregation has the following impacts on the network:

- System capacity

  A CA UE with SCells configured has only one RRC connection to the network. It consumes one sales unit of the license for the number of RRC_CONNECTED UEs. However, the CA UE consumes a hardware resource unit in each of its serving cells. In an extreme case with *n*CC aggregation for all UEs on the network, the maximum number of UEs that can access the network decreases to 1/*n* (*n* is an integer.) When all hardware resources are used, the eNodeB preferentially releases CA UEs in their SCells to maximize the number of UEs that can access the network.

- Resource usage
  - Physical resource block (PRB) usage
    - Overall PRB usage of the entire network

      On commercial networks, most services accessed by CA UEs are burst services. There is little probability that PRBs in all of the CCs of a UE will be exhausted simultaneously. When CA is enabled, cell loads can be rapidly balanced through carrier management and scheduling, utilizing idle resources and increasing the overall PRB usage of the network.

    - PRB usage of SCells

      If the **Dl2CCAckResShareSw** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter is selected, more UEs on the network enter the CA state but the user-perceived data rates of CA UEs decrease. In addition, due to possible errors in the estimated scheduling priorities of CA UEs, the PRBs in SCells for the UEs may not be fully utilized when the SCells are each serving a small number of non-CA UEs and the non-CA UE traffic is light.

  - SRI period

    It is recommended that the **SriPeriodCfgOptSW** option of the **CellPucchAlgo**.***SriAlgoSwitch*** parameter be selected. If it is deselected and the SRI period is prolonged, the RAB setup success rate and the service drop rate may deteriorate.

- PUCCH, physical uplink shared channel (PUSCH), and physical downlink control channel (PDCCH) overheads

  Each CA UE sends the acknowledgement/negative acknowledgement (ACK/NACK) and channel state information (CSI) related to its SCells in its PCell. When the PUSCH is not scheduled, the UE sends the information over the PUCCH. When the PUSCH is scheduled, the UE sends the information over the PUSCH. Therefore, PUCCH overhead or PUSCH associated signaling overhead increases. For UEs in the CA state, there might be more NACKs/DTXs among HARQ feedback, and therefore PDCCH overhead might increase.

- CPU usage of the main control board and BBPs

  The CPU usage of the main control board and BBPs changes as the number of SCCs increases.

- Network performance
  - Overall throughput in the entire network

    CA does not directly affect network capacity. However, when resources on a network have not been exhausted, CA increases the resource usage and overall network throughput.

  - Number of CQI reports

    In accordance with section 10.1.1 "PUCCH format information" of 3GPP TS 36.213 V10.10.0, CA UEs cannot send CSI reports together with ACK/NACK over the PUCCH with format 1b. In the case of simultaneous transmission, the UEs discard CSI reports as stipulated by 3GPP specifications. CSI reports include periodic CQI reports. Therefore, when periodic CQI reports are to be transmitted together with ACK/NACK within the same subframe, the CQI reports are discarded. The number of CQI reports from CA UEs decreases, as indicated by the values of the **L.ChMeas.CQI.DL.0** to **L.ChMeas.CQI.DL.15** counters, after CA is enabled.

    When the number of CQI reports from CA UEs decreases, the total number of CQI reports in the entire network changes, depending on the radio conditions of the CA UEs and the ratio of CA UEs to all UEs. For example, if CA UEs are located in cell centers and account for a large proportion of all UEs, the total number of CQI reports may also decrease. If CA UEs are not located in cell centers or they account for a small proportion, the total number of CQI reports may increase or remain unchanged.

  - Downlink cell performance

    The downlink IBLER of a cell will fluctuate if the **2CCDlCaEnhanceSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected.

  - Key performance indicator (KPI) fluctuation

    The PCell and SCells of each CA UE may not have the same channel quality. Therefore, CQIs fluctuate after SCells are configured for CA UEs.

    If the **CaMgtCfg**.***ScellNoActivationUeNumThld*** parameter is set to a non-zero value, SCell activation prohibition upon heavy cell load is enabled. This parameter specifies the threshold for determining heavy cell load. When the number of UEs in a cell exceeds the value of this parameter, the eNodeB determines that the cell is heavily loaded. In this case, new SCell activation quotas will not be provided for CA UEs. When the existing SCell activation quota is reached (that is, the number of UEs that have this cell activated as their SCell has reached the quota), the activation quota becomes 0 and this cell will no longer be activated as an SCell for any CA UE. This prohibition function helps increase the average downlink perceived data rates of non-CA UEs in the cell, especially those at the cell edge, but it decreases the average downlink perceived data rates of CA UEs that treat the cell as their SCell. In addition, the overall average downlink data rate of the network decreases.

- Data rates of CA UEs
  - When resources on a network have not been exhausted, CA in the network increases the data rates of CA UEs.
  - When resources on a network have been exhausted, the data rates of CA UEs are dependent on scheduling policies (described in **19.5.1 Scheduling**) and UE locations.

    In the full load situation, activating an SCell may result in a decrease in the overall throughput of this cell if the UE is located at the edge of this cell.
  - If the **CaMgtCfg**.***CaMimoPriorityStrategySw*** parameter is set to **MIMO_PRIOR**, the user-perceived data rates of CA UEs may decrease because of the low probability of using rank 3 or 4 on commercial networks.
- VoLTE UEs

  After CA takes effect, the mode of ACK feedback for downlink transmissions changes. As a result, more bits are transmitted over the PUSCH. The demodulation performance of uplink data on the PUSCH deteriorates to a certain extent. This impact is relatively noticeable for VoLTE services, which use small packets. To minimize the impact on uplink data demodulation, eNodeBs reduce MCS indexes so that IBLER values converge on the target value. However, the reduction in MCS indexes causes more RBs to be occupied by cell-edge VoLTE UEs.

- With downlink 2CC aggregation in effect, PCC anchoring prohibition for high-mobility UEs may have the following impacts on the network:
  - The E-RAB service drop rate, which is equal to (**L.E-RAB.AbnormRel.eNBTot** + **L.E-RAB.AbnormRel.HOOut**)/**L.E-RAB.SuccEst** x 100%, decreases.
  - The number of RRC connection reestablishment requests due to reconfiguration failures, which is indicated by the **L.RRC.ReEst.ReconfFail.Att** counter, decreases.
  - The numbers of CA UE handover attempts (**L.HHO.PrepAttOut.CAUser.PCC** and **L.HHO.InterFddTdd.PrepAttOut.CAUser.PCC**), CA UE handover executions (**L.HHO.ExecAttOut.CAUser.PCC** and **L.HHO.InterFddTdd.ExecAttOut.CAUser.PCC**), and successful CA UE handovers (**L.HHO.ExecSuccOut.CAUser.PCC** and **L.HHO.InterFddTdd.ExecSuccOut.CAUser.PCC**) decrease.

5.3 Requirements

5.3.1 Licenses (FDD)

- LAOFD-001001 LTE-A Introduction
  - Adaptive configuration mode
    i. An eNodeB identifies all its served cells on the PCC frequencies configured on the eNodeB.
    ii. Each of the cells on PCC frequencies consumes one sales unit of the license for LAOFD-001001 LTE-A Introduction.
    iii. The eNodeB identifies the SCC frequencies configured for the PCC frequencies.
    iv. The eNodeB identifies all the cells on these SCC frequencies.
    v. If a cell on an SCC frequency participates in downlink 2CC aggregation, the cell consumes one sales unit of the license for LAOFD-001001 LTE-A Introduction.

    NOTE

    When inter-eNodeB CA based on relaxed backhaul or inter-eNodeB CA based on eNodeB coordination is enabled, SCells may be served by neighboring eNodeBs. If this is the case, these cells consume the license for LAOFD-001001 LTE-A Introduction at the neighboring eNodeBs.
  - CA-group-based configuration mode
    i. An eNodeB identifies all the cells in the CA groups configured on the eNodeB.
    ii. In a CA group that consists of at least two inter-frequency FDD cells, each cell consumes one sales unit of the license for LAOFD-001001 LTE-A Introduction.
- LAOFD-001002 Carrier Aggregation for Downlink 2CC in 40MHz
  - Adaptive configuration mode

    If the **CaDl2CCExtSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for a cell that has consumed the license for LAOFD-001001 LTE-A Introduction, the cell also consumes one sales unit of the license for LAOFD-001002 Carrier Aggregation for Downlink 2CC in 40MHz.
  - CA-group-based configuration mode

    In a CA group where the total bandwidth of all cells is greater than 20 MHz and less than or equal to 40 MHz, a cell that has consumed the license for LAOFD-001001 LTE-A Introduction also consumes one sales unit of the license for LAOFD-001002 Carrier Aggregation for Downlink 2CC in 40MHz.

Table 5-3 lists the license models and sales units for these features.

Table 5-3 License models and sales units

Feature ID	Feature Name	Model	Sales Unit
LAOFD-001001	LTE-A Introduction	LT1SA020CA00	per cell

Feature ID	Feature Name	Model	Sales Unit
LAOFD-001002	Carrier Aggregation for Downlink 2CC in 40MHz	LT1SA040CA00	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Licenses (TDD)

Each TDD cell involved in downlink 2CC aggregation has the following license requirements:

- If the aggregated bandwidth does not exceed 30 MHz, each TDD cell requires one sales unit of the following feature:
  - TDLAOFD-001001 LTE-A Introduction
- If the aggregated bandwidth exceeds 30 MHz but does not exceed 40 MHz, each TDD cell requires one sales unit for each of the following features:
  - TDLAOFD-001001 LTE-A Introduction
  - TDLAOFD-001002 Carrier Aggregation for Downlink 2CC in 40MHz

Table 5-4 lists the license models and sales units for these features.

Table 5-4 License models and sales units

Feature ID	Feature Name	Model	Sales Unit
TDLAOFD-001001	LTE-A Introduction	LT1SLTEAID01	per cell
TDLAOFD-001002	Carrier Aggregation for Downlink 2CC in 40MHz	LT1SC2C40M00	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.3 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.3.1 Prerequisite Functions

None

5.3.3.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
FDD	Super combined cell	**SuperCombCellSwitch** option of the **CellAlgoSwitch**.***SfnAlgoSwitch*** parameter	*Super Combined Cell (FDD)*
FDD	Cell radius greater than 100 km	None	*Extended Cell Range*
FDD TDD	In-band relay	**InBandRelayDeNbSwitch** option of the **CellAlgoSwitch**.***RelaySwitch*** parameter, **InBandRelayReNbSwitch** option selected on the relay BTS (ReBTS)	*Relay*
TDD	Enhanced symbol power saving	**MBSFNShutDownSwitch** option of the **CellAlgoSwitch**.***DlSchSwitch*** parameter	*Energy Conservation and Emission Reduction*
FDD TDD	Intelligent multi-RF-module coordinated energy saving[a]	**MIE_BASED_EE_CARR_SEL_SW** option of the **EnodebPwrSavingAlgo**.***PwrSavingAlgoSwitch*** parameter	*Energy Conservation and Emission Reduction*
a: Intelligent energy-efficiency-oriented carrier selection and CA in CA-group-based configuration mode cannot be both enabled.			

5.3.3.3 Function Impacts

- Functions in the category "RAN functions"

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	User-number-based connected mode load equalization	**InterFreqMlbSwitch** option of the **CellAlgoSwitch**.***MlbAlgoSwitch*** parameter **CellMLB**.***MlbTriggerMode*** set to **UE_NUMBER_ONLY** **SynchronizedUE** option of the **CellMLB**.***InterFreqUeTrsfType*** parameter	*Intra-RAT Mobility Load Balancing*	If CA UE transfer is disabled (by deselecting the **CaUserLoadTransferSw** option of the **CellAlgoSwitch**.***EnhancedMlbAlgoSwitch*** parameter), the eNodeB filters out CA UEs that treat the source cell as their PCell or SCell, when the eNodeB selects UEs for UE-number-based connected mode load equalization.
FDD TDD	NSA/SA selection based on downlink traffic volume	**LTE_FDD_NSA_SA_DL_SEL_OPT_SW** or **LTE_TDD_NSA_SA_DL_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*NSA-SA Selection Based on User Experience*	The gNodeB includes the selected EN-DC combination in the *selectedbandCombinationInfoEN-DC-v1540* IE of the handover request message. The eNodeB then configures the EN-DC combination for NSA DC. This EN-DC combination remains unchanged before the SCG added by NSA/SA selection based on downlink traffic volume is released. Therefore, the eNodeB cannot add SCells based on the selection by downlink CA.

RAT	Function Name	Function Switch	Reference	Description
FDD	NSA/SA selection based on uplink coverage	**LTE_FDD_NSA_SA_UL_SEL_OPT_SW** option of the **EnodebAlgoExtSwitch**.***MultiNetworkingOptionOptSw*** parameter	*NSA-SA Selection Based on User Experience*	The gNodeB includes the selected EN-DC combination in the *selectedbandCombinationInfoEN-DC-v1540* IE of the handover request message. The eNodeB then configures the EN-DC combination for NSA DC. This EN-DC combination remains unchanged before the SCG added by NSA/SA selection based on uplink coverage is released. Therefore, the eNodeB cannot add SCells based on the selection by downlink CA.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	CSPC	FDD: **CspcAlgoPara**.***CspcAlgoSwitch*** TDD: **CspcAlgoPara**.***TddCspcAlgoSwitch***	*CSPC*	CSPC in centralized Cloud BB scenarios requires a centralized controller, which is a process deployed on a BBP in each eNodeB. The centralized controller increases the average throughput of cells on the affected frequency and the cell edge UE throughput in the network. However, it has a negative impact on the high-throughput UEs that cause co-channel interference on the cell edge UEs. If basic scheduling is used with CA, the data rate of a CA UE (a variable used to calculate the scheduling priority) is defined as the total data rate of the UE on all the aggregated carriers. It is typically higher than the data rate of a non-CA UE. As a result, the relevant CA UEs encounter a lower probability of being scheduled and a lower data rate.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Zero Guard Band Between Contiguous Intra-Band Carriers	**CONTIG_INTRA_BAND_CARR_SW** option of the **ContigIntraBandCarr**.***ContigIntraBandCarrSw*** parameter	FDD: *Seamless Intra-Band Carrier Joining (FDD)* TDD: *Seamless Intra-Band Carrier Joining (TDD)*	Contiguous intra-band CA does not take effect for a cell with Zero Guard Band Between Contiguous Intra-Band Carriers enabled even if the **ContigIntraBand-CaSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is selected. Intra-band non-contiguous CA can take effect for a cell with Zero Guard Band Between Contiguous Intra-Band Carriers enabled in spectrum overlapping scenarios, where the spacing between the center frequencies of two to-be-aggregated carriers is less than the average of their total bandwidths. For details about the spectrum requirements of cells with Zero Guard Band Between Contiguous Intra-Band Carriers disabled on intra-band non-contiguous CA, see **5.3.5 Networking**.

RAT	Function Name	Function Switch	Reference	Description
FDD	Short TTI	**SHORT_TTI_SW** option of the **CellShortTtiAlgo**.***SttiAlgoSwitch*** parameter	*Short TTI (FDD)*	CA does not take effect for UEs that support short TTI according to their reported capabilities.
TDD	Downlink 2-layer MIMO based on TM9	**TM9Switch** option of the **CellAlgoSwitch**.***EnhMIMOSwitch*** parameter	• *Beamforming (TDD)* • *Massive MIMO Basics (TDD)*	TM9 has the following impact on UEs in the 2CC aggregation state: If the channel state information-reference signal (CSI-RS) is configured for an SCC with no sounding reference signal (SRS) configured, a CA UE can measure CSI based on CSI-RS. Therefore, the UE can enter the PMI feedback-dependent TM9 (TM9wPMI) mode on the SCC, improving the average cell throughput and user experience.
TDD	Downlink 4-layer MIMO based on TM9	• **TM9Switch** option of the **CellAlgoSwitch**.***EnhMIMOSwitch*** parameter • **CellDlschAlgo**.***MaxMimoRankPara*** set to **SW_MAX_SM_RANK_4**	• *Beamforming (TDD)* • *Massive MIMO Basics (TDD)*	TM9 has the following impact on UEs in the 2CC aggregation state: If the CSI-RS is configured for an SCC with no SRS configured, a CA UE can measure CSI based on CSI-RS. Therefore, the UE can enter the TM9wPMI mode on the SCC, improving the average cell throughput and user experience.

RAT	Function Name	Function Switch	Reference	Description
TDD	Downlink 8x8 MIMO	• **TM9Switch** option of the **CellAlgoSwitch**.***EnhMIMOSwitch*** parameter • **CellDlschAlgo**.***MaxMimoRankPara*** set to **SW_MAX_SM_RANK_8** • **CellCsiRsParaCfg**.***CsiRsPortNum*** set to **CSI_RS_PORT_8**	• *Beamforming (TDD)* • *Massive MIMO Basics (TDD)*	TM9 has the following impact on UEs in the 2CC aggregation state: If the CSI-RS is configured for an SCC with no SRS configured, a CA UE can measure CSI based on CSI-RS. Therefore, the UE can enter the TM9wPMI mode on the SCC, improving the average cell throughput and user experience. Downlink 8x8 MIMO takes effect only for UEs in the 2CC aggregation state.
TDD	Inter-cell downlink D-MIMO	**InterCellDmimoJTSwitch** option of the **CellAlgoSwitch**.***DMIMOAlgoSwitch*** parameter	*D-MIMO (TDD)*	Inter-cell downlink distributed multiple-input multiple-output (D-MIMO) does not work in SCells.
TDD	Inter-frequency split	None	*Soft Split Resource Duplex (TDD)*	After inter-frequency split is enabled, CA takes effect in a smaller area, and therefore CA gains are smaller.
FDD	Superior uplink coverage	**CellAlgoExtSwitch**.***UlCoverageEnhancementSw***	*Superior Uplink Coverage (FDD)*	UEs under enhanced coverage do not support downlink CA.

RAT	Function Name	Function Switch	Reference	Description
TDD	DL CoMP cell	**IntraDlCompSwitch** option of the **CellAlgoSwitch**.***DlCompSwitch*** parameter	*DL CoMP (TDD)*	CA UEs support downlink coordinated multipoint transmission (DL CoMP) in their PCells, but not in their SCells.
FDD	DL CoMP with TM9	**Tm9JtSwitch** option of the **CellAlgoSwitch**.***DlCompSwitch*** parameter	*DL CoMP (FDD)*	Joint transmission (JT) based on TM9 can be configured for DL CoMP in both the PCells and SCells of UEs in the intra-BBU CA state. TM9 JT in SCells can take effect only after TM9 JT is configured in the corresponding PCells.
FDD	DL CoMP with TM10	**FDDHomNetDlCompSwitch** and **FDDHetNetDlCompSwitch** options of the **CellAlgoSwitch**.***DlCompSwitch*** parameter	*DL CoMP (FDD)*	DL CoMP with TM10 can be configured in both the PCells and SCells of UEs in the intra-BBU CA state.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Adaptive SFN/SDMA	**CellAlgoSwitch**.***SfnUlSchSwitch*** **CellAlgoSwitch**.***SfnDlSchSwitch***	*SFN*	Adaptive single frequency network (SFN) requires UEs to report SRSs, based on which eNodeBs select remote radio units (RRUs) for independent scheduling of the UEs. If an adaptive SFN cell is configured as an SCell for a CA UE only in the downlink, only joint scheduling can be used in this SCell. If the adaptive SFN cell is also configured as an SCell in the uplink, independent scheduling can be used in the cell.
FDD TDD	Network-assisted CRS interference cancellation	**CellAlgoSwitch**.***CrsIcSwitch***	*Network Assisted Interference Cancellation*	For CA UEs, this cell-specific reference signal interference cancellation (CRS-IC) function takes effect only in PCells.
FDD	Dynamic TDM eICIC	**CellAlgoSwitch**.***EicicSwitch*** set to **DYNAMIC**	*TDM eICIC (FDD)*	CA UEs do not support the two types of CSI measurements for eICIC in SCells and therefore do not support dynamic time-domain enhanced inter-cell interference coordination (TDM eICIC) or further enhanced inter-cell interference coordination (FeICIC) in SCells.

RAT	Function Name	Function Switch	Reference	Description
FDD	FeICIC	**ABS** option of the **CellAlgoSwitch**.***FeicicSwitch*** parameter	*TDM eICIC (FDD)*	CA UEs do not support the two types of CSI measurements for eICIC in SCells and therefore do not support dynamic time-domain enhanced inter-cell interference coordination (TDM eICIC) or further enhanced inter-cell interference coordination (FeICIC) in SCells.
FDD TDD	Terminal Awareness Differentiation	**AbnormalUeHandleSwitch** option of the **GlobalProcSwitch**.***UeCompatSwitch*** parameter	*Terminal Awareness Differentiation*	Carrier management for CA or NSA DC does not work on some CA UEs or NSA UEs due to their software or hardware defects. To prevent UE incompatibility issues from affecting network performance, enable Terminal Awareness Differentiation with the UEs blacklisted and the **CA_SWITCH_OFF** option of the **UeCompat**.***BlkLstCtrlSwitch*** parameter selected. With these settings, eNodeBs will not perform PCC anchoring, SCell configuration, or SCG addition for these UEs. For details about carrier management for NSA DC, see *EPC-based NSA Basics*.

RAT	Function Name	Function Switch	Reference	Description
FDD	Flexible bandwidth based on overlapping carriers	**DdCellGroup*.DdBandwidth***	*Flexible Bandwidth based on Overlap Carriers (FDD)*	It is not recommended that cells with RB puncturing act as PCells for CA.
FDD	Compact bandwidth	**Cell.*CustomizedBandWidthCfgInd***	*Compact Bandwidth (FDD)*	It is not recommended that cells with RB punctured from a standard bandwidth of 5 MHz act as PCells for CA.

RAT	Function Name	Function Switch	Reference	Description
TDD	SRS and PMI combination weight	**SRS_PMI_COMB_WEIGHT_SW** option of the **CellMMAlgo**.***MMAlgoOptSwitch*** parameter	*Massive MIMO Basics (TDD)*	When CA is enabled in massive MIMO cells, SRS and PMI combination weight does not take effect for UEs in their SCells among these cells. SRS and PMI weight combination does not take effect for a UE in its PCell when all of the following conditions are met: 1. The aggregated CCs are contiguous. 2. Any SCell of the UE is activated in the downlink. 3. The **NoSrsSccBfSwitch** option of the **CellAlgoSwitch**.***NoSrsSccBfAlgoSwitch*** parameter is selected for the PCell, or the **CaUl2CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for the SCell that meets condition **b**.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Automatic congestion handling	**FDD_ACH_SWITCH** option of the **ParaAutoOptCfg**.***AchEffectiveType*** parameter **TDD_ACH_SWITCH** option of the **ParaAutoOptCfg**.***AchEffectiveType*** parameter	*Automatic Congestion Handling*	The **CaMgtCfg**.***CellMaxPccNumber*** parameter of a cell is set to **0** when dynamic downlink CA control in automatic congestion handling is enabled and the triggering condition is met. In this case, if the **PccSmartCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected on the serving eNodeB and neighboring eNodeBs, CA UEs in intra- or inter-eNodeB neighboring cells are not handed over to the cell even when the number of CA UEs in the cell does not exceed a preset threshold.
TDD	Downlink user-perceived rate optimization for heavily loaded cells	**CellDlSchExtParam**.***DlHeavyLoadOptPrbThld***	*Downlink Scheduling*	When downlink user-perceived rate optimization for heavily loaded cells takes effect, the weight used for differentiated scheduling of CA UEs (specified by **CellDlschAlgo**.***CaSchWeight***) does not take effect and each CA UE on each carrier is scheduled based on downlink user-perceived rate optimization for heavily loaded cells.

RAT	Function Name	Function Switch	Reference	Description
FDD	Downlink-load-based scheduling policy adaptation	**DL_LOAD_BASED_SCH_POL_ADAPT_SW** option of the **CellDlSchExtParam**.***DlSchExtSwitch*** parameter	*Downlink Scheduling*	When downlink-load-based scheduling policy adaptation is in effect, the weight used for differentiated scheduling of CA UEs (specified by the **CellDlschAlgo**.***CaSchWeight*** parameter) becomes invalid and CA UEs are scheduled using the equivalent Max C/I algorithm on each carrier.
TDD	Downlink scheduling priority enhancement	**DL_SCH_PRI_ENHANCE_SW** option of the **CellDlSchExtParam**.***DlSchExtSwitch*** parameter	*LTE AHR (TDD)*	When both downlink scheduling priority enhancement and CA are enabled in a cell and there are lots of services on the SCCs, **User Downlink Average Throughput** of the cell decreases.
TDD	Downlink slot awareness scheduling	**CellDlSchExtParam**.***DlSlotAwareSchPrbThld***	*LTE AHR (TDD)*	In the case of CA, downlink slot awareness scheduling takes effect on PCCs but not SCCs of CA UEs.

RAT	Function Name	Function Switch	Reference	Description
FDD	LTE FDD and NR Flash Dynamic Power Sharing	**LTE_NR_DYN_POWER_SHARING_SW** option of the **CellDynPowerSharing**.***DynamicPowerSharingSwitch*** parameter	*LTE and NR Power Allocation*	The longer the duration in which LTE cells in an LTE and NR power sharing group work for CA, the lower the proportion of duration in which power sharing takes effect. Proportion of duration in which the function takes effect = Number of TTIs in which power sharing takes effect/Total number of TTIs in which scheduling is performed
FDD	Hybrid power sharing	**LTE_HYBRID_POWER_SHARING_SW** option of the **CellDynPowerSharing**.***DynamicPowerSharingSwitch*** parameter	*Smart 8T8R (FDD)*	The longer the duration in which cells in a power sharing group work for CA, the lower the proportion of time hybrid power sharing takes effect.

RAT	Function Name	Function Switch	Reference	Description
FDD	RuralLink wireless fronthaul solution	**CellTypeExt**.***WirelessFhCellFlag***	*RuralLink Solution*	RuralLink cells are incompatible with CA. ● With X2 present between base stations: A RuralLink cell broadcasts its incompatibility with CA through the X2 interface, but there is delay in the broadcast. Before the broadcast is complete, CA-capable neighboring cells are not informed of the RuralLink cell's incompatibility with CA. If a CA UE is handed over to the RuralLink cell for PCC anchoring, CA stops taking effect for the UE and the SCells of the UE are released. In this case, the CA UE becomes a non-CA UE. ● With X2 absent between base stations: CA-capable neighboring cells cannot be informed of RuralLink cells' incompatibility with CA. If a CA UE is handed over to the RuralLink cell for PCC

RAT	Function Name	Function Switch	Reference	Description
				anchoring, CA stops taking effect for the UE and the SCells of the UE are released. In this case, the CA UE becomes a non-CA UE. ● If CA fails between an inter-base-station neighboring cell that serves as the PCell and a RuralLink cell that serves as an SCell, the CA capacity consumed by the inter-base-station neighboring cell will be released after a period of time, which is specified by the **CaMgtCfg**.***SCellAgingTime*** parameter.

RAT	Function Name	Function Switch	Reference	Description
TDD	SRS resource migration	**REMOTE_INTRF_BF_ENH_SW** option of the **UlInterfSuppressCfg**.***RemoteIntrfDlEnhSwitch*** parameter	*Interference Detection and Suppression*	When the conditions for the SRS resource migration function to take effect and those for enhanced DL ACK demodulation for 2CC aggregation (controlled by the **2CCDlCaEnhanceSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter) are met at the same time, the SRS resource migration function takes precedence. Details are as follows: ● When the SRS resource migration function takes effect, enhanced DL ACK demodulation for 2CC aggregation stops taking effect. ● When the SRS resource migration function stops taking effect, enhanced DL ACK demodulation for 2CC aggregation becomes taking effect.

RAT	Function Name	Function Switch	Reference	Description
FDD	Downlink-load-based scheduling priority optimization (FDD)	**DL_LOAD_BASED_SCH_PRI_OPT_SW** option of the **CellDlSchExtParam**.***DlSchExtSwitch*** parameter	*Downlink Scheduling*	When downlink-load-based scheduling priority optimization is in effect, the weight used for differentiated scheduling of CA UEs (**CellDlschAlgo**.***CaSchWeight***) becomes invalid and CA UEs are scheduled using the equivalent Max C/I algorithm on each carrier.

- Functions related to RAN services

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Emergency call	None	*Emergency Call*	When a CA UE is running an emergency call service, the eNodeB does not start SCell configuration for the UE, which prevents RRC signaling from affecting the service quality of the ongoing emergency call. After the emergency call service is finished, the eNodeB attempts to configure SCells for the UE if the UE traffic volume and SCell configuration interval conditions are fulfilled.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Next Generation eCall over LTE	**NG_ECALL_SW** option of the **Emc**.***EmergCallOptimizationSw*** parameter	*Emergency Call*	When a CA UE is running an emergency call service, the eNodeB does not start SCell configuration for the UE, which prevents RRC signaling from affecting the service quality of the ongoing emergency call. After the emergency call service is finished, the eNodeB attempts to configure SCells for the UE if the UE traffic volume and SCell configuration interval conditions are fulfilled.
FDD TDD	VoIP semi-persistent scheduling	**SpsSchSwitch** option of the **CellAlgoSwitch**.***UlSchSwitch*** parameter **SpsSchSwitch** option of the **CellAlgoSwitch**.***DlSchSwitch*** parameter	*VoLTE*	As stipulated in 3GPP TS 36.321, semi-persistent scheduling takes effect only in the PCells for CA UEs.
FDD TDD	Downlink voice packet duplication for VoLTE UEs	**VOLTE_DL_PKT_DUPLICATE_SW** option of the **VolteAlgoConfig**.***VolteOptSwitch*** parameter	*VoLTE*	When downlink voice packet duplication for VoLTE UEs is enabled, the eNodeB can activate SCells based on voice services, in addition to traffic volume.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	CS fallback	None	*CS Fallback*	To prevent unnecessary RRC signaling exchange, the eNodeB does not start the SCell configuration procedure for CA UEs that are engaged in CS fallback procedures.
FDD TDD	Specified service carrier	**Cell**.***SpecifiedCellFlag***	*WBB*	The following restriction applies to wireless broadband (WBB) and mobile broadband (MBB) UEs: CA does not allow WBB-service-specified cells to be used as SCells for MBB UEs, or MBB-service-prioritized cells to be used as SCells for WBB UEs. If both CA and specified service carrier are enabled, CA service experience of these UEs or related KPIs may be affected.
TDD	SRS interference avoidance	**WTTxSRSIntrfAvoidanceSw** option of the **SRSCfg**.***SrsCfgPolicySwitch*** parameter	*Massive MIMO Optimization in WTTx Scenarios (TDD)*	If the WTTx SRS interference avoidance function is set to different states (enabled and disabled) for the PCell and an SCell of a CA UE, the CA UE may fail to send SRSs in this SCell.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	eMBMS	**CellMBMSCfg**.***MBMSSwitch***	*eMBMS*	When this function is enabled, CA UEs can receive Multimedia Broadcast multicast service Single Frequency Network (MBSFN) subframes in their PCells but not in their SCells. Therefore, this function works for CA UEs only in PCells.
FDD TDD	Location service	**ENodeBAlgoSwitch**.***LcsSwitch***	*LCS*	Reference signal time difference (RSTD) measurements, which are used for positioning based on Observed Time Difference Of Arrival (OTDOA), increase the ACK/NACK loss rate. A higher ACK/NACK loss rate results in scheduling performance deterioration, which may affect user experience with CA.
FDD TDD	Out-of-band relay	**CellAlgoSwitch**.***RelaySwitch***	*Relay*	In out-of-band relay scenarios, RRNs support downlink 2CC aggregation and uplink 2CC aggregation if the **Dl2CCAckResShareSw** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter is deselected, and do not support them if this option is selected.

- Functions related to CIoT

RAT	Function Name	Function Switch	Reference	Description
FDD	Dynamic multi-carrier management	**PRB_DYNAMIC_MGMT_SW** option of the **NbPrbDynamicMgmt**.***NbPrbDynMgmtAlgoSw*** parameter	*Dynamic Multi-Carrier Management (FDD)*	LTE cells do not use shared PRBs for preallocation.

- Functions related to CloudAIR

RAT	Function Name	Function Switch	Reference	Description
TDD	LTE spectrum coordination enhancement	**WbbCaMultiCarrierCoordSw** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	*LTE Spectrum Coordination*	Inter-CC load transfer triggered by cell load does not take effect for WBB CA UEs or RRNs on which LTE spectrum coordination enhancement has taken effect. Inter-CC load transfer triggered by cell load is controlled by the **DlCaLbAlgoSwitch** option of the **ENodeBAlgoSwitch**.***CaLbAlgoSwitch*** parameter.

RAT	Function Name	Function Switch	Reference	Description
TDD	LTE spectrum coordination enhancement	**WbbCaMultiCarrierCoordSw** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	*LTE Spectrum Coordination*	Inter-CC load transfer triggered by cell load differences does not take effect for WBB CA UEs or RRNs on which LTE spectrum coordination enhancement has taken effect. Inter-CC load transfer triggered by cell load differences is controlled by the **CaLoadBalancePreAllocSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter.
FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_SPECTRUM_SHR**	*LTE FDD and NR Dynamic Spectrum Sharing*	This function reduces the number of downlink RBs available for LTE. Therefore, the throughput of UEs in the downlink FDD CA state decreases.
FDD	DSS and NR Flexible Refarming	**DSS_FLEX_REFARM_SW** option of the **SpectrumCloud**.***SpectrumCloudEnhSwitch*** parameter	*LTE FDD and NR Dynamic Spectrum Sharing*	This function reduces the number of downlink RBs available for LTE. Therefore, the throughput of UEs in the downlink FDD CA state decreases.

RAT	Function Name	Function Switch	Reference	Description
FDD	GSM and LTE Spectrum Concurrency	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **GL_SPECTRUM_CONCURRENCY**	*GSM and LTE Spectrum Concurrency*	LTE cells with GSM and LTE Spectrum Concurrency enabled are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured. As a result, the LTE network throughput is affected.
FDD TDD	Centralized power sharing	**LTE_CENTRAL_POWER_SHARING_SW** option of the **CellDynPowerSharing**.***DynamicPowerSharingSwitch*** parameter	*Dynamic Power Sharing Between LTE Carriers*	The longer the duration in which cells in a centralized power sharing group work for CA, the lower the proportion of duration in which centralized power sharing takes effect. Proportion of duration in which the function takes effect = Number of TTIs in which power sharing takes effect/ Total number of TTIs in which scheduling is performed
FDD	LTE-NR multi-dimensional scheduling optimization	**LNR_MULTI_D_OPT_SCH_SW** option of the **CellMultiCarrUniSch**.***LnrMultiCarrUnifiedSchSw*** parameter	*LTE FDD and NR Inter-RAT Unified Scheduling*	When precise scheduling for CA UEs takes effect, LTE-NR multi-dimensional scheduling optimization yields fewer benefits than expected.

- Functions related to network infrastructure

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Intra-RAT ANR	Event-triggered ANR: **IntraRatEventAnrSwitch** option of the **ENodeBAlgoSwitch**.***AnrSwitch*** parameter Fast ANR: **IntraRatFastAnrSwitch** option of the **ENodeBAlgoSwitch**.***AnrSwitch*** parameter	*ANR Management*	During fast or event-triggered automatic neighbor relation (ANR), the eNodeB determines whether to select CA UEs to perform measurements based on the **ANR**.***CaUeChoseMode*** parameter setting.
FDD TDD	Inter-RAT ANR	• GERAN: **GeranFastAnrSwitch** and **GeranEventAnrSwitch** options of the **ENodeBAlgoSwitch**.***AnrSwitch*** parameter • UTRAN: **UtranFastAnrSwitch** and **UtranEventAnrSwitch** options of the **ENodeBAlgoSwitch**.***AnrSwitch*** parameter	*ANR Management*	During fast or event-triggered ANR, the eNodeB determines whether to select CA UEs to perform measurements based on the **ANR**.***CaUeChoseMode*** parameter setting.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Intelligent power-off of carriers in the same coverage	**CellShutdown.CellShutdownSwitch** set to **ON**	*Energy Conservation and Emission Reduction*	Capacity cells in the intelligent power-off of carriers in the same coverage state do not participate in carrier aggregation. In CA-group-based configuration mode, after such a capacity cell in a CA group exits this power saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again.
FDD TDD	Dynamic carrier shutdown	**CARRIER_DYN_SHUTDOWN_SW** option of the **CellShutdown**.***CarrierShutdownEnhancedSw*** parameter	*Energy Conservation and Emission Reduction*	A capacity cell in the dynamic carrier shutdown state can be configured as an SCell for UEs in the basic cell for dynamic carrier shutdown.
FDD TDD	Multi-carrier coordinated energy saving	**CellShutdown**.***CellShutdownSwitch*** set to **ON_MULTI_CARRIER_HIER_SHUTDOWN**	*Energy Conservation and Emission Reduction*	Capacity cells in the multi-carrier coordinated energy saving state do not participate in carrier aggregation. In CA-group-based configuration mode, after such a capacity cell in a CA group exits this power saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again.

RAT	Function Name	Function Switch	Reference	Description
FDD	RF module regular time sleep	**eNodeBAutoPowerOff.*AutoPowerOffSwitch*** set to **ON**	*Energy Conservation and Emission Reduction*	Cells that are shut down due to RF module regular time sleep do not participate in carrier aggregation. In CA-group-based configuration mode, after such a cell in a CA group exits this power saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again.
FDD TDD	Automatic cell shutdown	**CellAutoShutdown.*CellAutoShutdownSwitch*** set to **ON**	*Energy Conservation and Emission Reduction*	Cells in the automatic cell shutdown state do not participate in carrier aggregation. In CA-group-based configuration mode, after such a cell in a CA group exits this power saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Low power consumption mode	**CellLowPower.*LowPwrSwitch*** set to **ON**	*Green Site*	Cells that are shut down due to low power consumption mode do not participate in carrier aggregation. In CA-group-based configuration mode, after such a cell in a CA group exits this power saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again.
FDD TDD	SOC level-based low power consumption mode	• **IPWRM.*SOCLVLADJSW*** • **SOCLVL_BASED_LOW_PWR_SW** option of the **CellLowPower**.***LowPwrConsumeModeExtSw*** parameter	*Green Site*	Cells that are shut down due to SOC level-based low power consumption mode do not participate in carrier aggregation. In CA-group-based configuration mode, after such a cell in a CA group exits this power saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again.

RAT	Function Name	Function Switch	Reference	Description
FDD	Intelligent power-off of carriers in the same coverage as UMTS networks	**InterRatCellShutdown.*ForceShutdownSwitch*** set to **ON**	*Energy Conservation and Emission Reduction*	LTE cells that are shut down due to intelligent power-off of carriers in the same coverage as UMTS networks do not participate in carrier aggregation. In CA-group-based configuration mode, after such a cell in a CA group exits this power saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	pRRU deep dormancy	• **RRU**.***DORMANCYSW*** set to **ON** • **EnodebMpruEs**.***MpruDormancyDlEarfcn*** set to a value other than **-1** • **UE_DET_PRRU_DEEP_DORMANCY_SW** option of the **EnodebMpruEs**.***DormancyAlgoSwitch*** parameter selected	*Multi-RAT Coordinated Energy Saving*	• Cells in the power saving state due to pRRU deep dormancy do not participate in carrier aggregation. In CA-group-based configuration mode, after such a cell in a CA group exits this power saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again. • After a cell exits the scheduled pRRU deep dormancy state, it does not participate in CA if the number of routes is insufficient. That is because the route for CA will be released for a cell in the scheduled pRRU deep dormancy state; after this cell exits this state, it may not obtain a route if the number of routes is insufficient.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Dynamic carrier shutdown phase 2	**CellPwrSavingAlgo**.***CarrierDynShutdownPh2Sw*** set to **CAPACITY_CELL**	*Energy Conservation and Emission Reduction*	During dynamic carrier shutdown phase 2, a capacity cell can be configured as an SCell only for UEs in intra-base-station cells.
FDD	GUL joint carrier shutdown	• **InterRatCellShutdown**.***ForceShutdownSwitch*** set to **ON** • **InterRatCellShutdown**.***ShutDownType*** set to **BY_UTRAN_WITH_LOAD**	*Multi-RAT Coordinated Energy Saving*	LTE cells that are shut down due to GUL joint carrier shutdown do not participate in carrier aggregation. In CA-group-based configuration mode, after such a cell in a CA group exits this power saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again.
FDD TDD	Cell Switch-off Based on Automatic Co-coverage Identification	**CellShutdown.*CellShutdownSwitch*** set to **ON_CO_COV_AUTO_IDENT**	*Energy Conservation and Emission Reduction*	Carrier aggregation cannot be performed for capacity cells in the intelligent power-off of carriers mode.
FDD	UL coordinated channel shutdown	**CellRfShutdown**.***MultiRatJointChnShutdownSw***	*Multi-RAT Coordinated Energy Saving*	If carrier aggregation is enabled after Multi-RAT Coordinated Channel Shutdown takes effect, the cell is likely to exit the RF channel shutdown state. As a result, the energy saving time and energy saving gains decrease.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	RF module deep dormancy	**RRU**.***DORMANCYSW*** set to **ON** or **ON_DEEP_SUPER**	*Multi-RAT Coordinated Energy Saving*	● Cells in the RF module deep dormancy state do not participate in CA. In CA-group-based configuration mode, after such a cell in a CA group exits this power saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again. ● After a cell exits the RF module deep dormancy state, it does not participate in CA if the number of routes is insufficient. That is because the route for CA will be released for a cell in the RF module deep dormancy state; after this cell exits this state, it may not obtain a route if the number of routes is insufficient.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	RF module super dormancy	**RRU**.***DORMANCYSW*** set to **ON_SUPER** or **ON_DEEP_SUPER**	*Multi-RAT Coordinated Energy Saving*	● Cells in the RF module super dormancy state do not participate in CA. In CA-group-based configuration mode, after such a cell in a CA group exits this power saving state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again. ● After a cell exits the RF module super dormancy state, it does not participate in CA if the number of routes is insufficient. That is because the route for CA will be released for a cell in the RF module super dormancy state; after this cell exits this state, it may not obtain a route if the number of routes is insufficient.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	iSDU-controlled RF module power-off super dormancy	**RRU**.***DORMANCYSW*** set to **ON_SUPER** or **ON_DEEP_SUPER**	*Multi-RAT Coordinated Energy Saving*	● Cells in the RF module dormancy state do not participate in CA. In CA-group-based configuration mode, after a cell in a CA group exits the iSDU-controlled RF module power-off dormancy state, this cell cannot be configured as an SCell for an existing CA UE until the UE accesses the network again. ● After a cell exits the RF module dormancy state, it does not participate in CA if the number of routes is insufficient. The route for CA will be released for a cell in the RF module dormancy state. After a cell exits the iSDU-controlled RF module power-off dormancy state, it may not obtain a route if the number of routes is insufficient and therefore, cannot participate in CA.

RAT	Function Name	Function Switch	Reference	Description
FDD	Physical Resource Slicing	**PHY_RES_SLICING_SWITCH** option of the **CellAlgoExtSwitch**.***OperatorFunSwitch*** parameter	*Physical Resource Slicing (FDD)*	When both CA and Physical Resource Slicing are enabled, UEs in their SCells are associated with PRB groups based on the configurations of these cells. If no SPID is configured for a UE in an SCell or the UE is not associated with any PRB group in the SCell, the SPID of the UE is considered to be 0.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Ultra-low-latency scheduling for intra-eNodeB CA	**ULTRA_LOW_LATENCY_SCH_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter	*Multi-carrier Unified Scheduling*	If the PCell is not heavily loaded and both energy saving based on proactive scheduling and precise scheduling for CA UEs have taken effect, enabling ultra-low-latency scheduling for intra-eNodeB CA may slow down data transmission for CA UEs, leading to a decrease in the downlink user-perceived rate. The specific causes are as follows: ● After energy saving based on proactive scheduling takes effect, scheduling in the PCell may be delayed. ● Compared with precise scheduling for CA UEs, data split with ultra-low latency reduces the amount of data distributed to SCells.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Energy saving based on proactive scheduling	**SymbolPwrSaving**.***TrigBndlSchDlAvgPrbThld***	*Energy Conservation and Emission Reduction*	If the PCell is not heavily loaded and both energy saving based on proactive scheduling and precise scheduling for CA UEs have taken effect, enabling ultra-low-latency scheduling for intra-eNodeB CA may slow down data transmission for CA UEs, leading to a decrease in the downlink user-perceived rate. The specific causes are as follows: ● After energy saving based on proactive scheduling takes effect, scheduling in the PCell may be delayed. ● Compared with precise scheduling for CA UEs, data split with ultra-low latency reduces the amount of data distributed to SCells.

5.3.4 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

CA works between intra- or inter-BBP cells.

CA has the following requirements on the slots where boards can be installed:

- Intra-BBP CA
  - If the LTE BBP is installed in a BBU3900, it can be deployed in any slot (0 to 5) of the BBU. To facilitate future capacity expansion, it is recommended that the LTE BBP be deployed in slot 2 or 3.
  - If the BBU is not BBU3900, there are no special requirements.
- Inter-BBP CA

  The BBPs are deployed in a single BBU.
  - If the BBPs are installed within a BBU3900, the BBPs must be installed in the following slots:

    An LBBPd or LTE UBBP must be installed in slot 2 or 3 and configured by running the **ADD BRD** command. This ensures successful setup of SRIO routes between the BBPs. (This requirement does not apply to inter-eNodeB CA based on relaxed backhaul.)

    Whether the SRIO routes can be successfully set up between the BBPs is irrelevant to whether cells are deployed on the BBPs and whether the cells belong to the same operator.
  - If the BBPs are installed within a BBU3910, the UBBPg must be installed in slots 0 to 3.
  - If the BBU is neither BBU3900 nor BBU3910, there are no special requirements.

RF Modules

To meet the time alignment requirements (see **5.3.5 Networking**), intra-band CA requires one of the following radio frequency (RF) module configurations:

- One multi-carrier RF module
- Multiple active antenna units (AAUs) of the same product model
- Multiple RRUs of the same product model
- Multiple radio frequency units (RFUs) with the same hardware version number

Inter-band CA has no special requirements for RF modules.

5.3.5 Networking

Frequencies

- There must be at least two frequencies on the live network.
- Two intra-band contiguous carriers are to-be-aggregated carriers with a center frequency spacing less than or equal to the nominal channel spacing. Two intra-band non-contiguous carriers are to-be-aggregated carriers with a center frequency spacing greater than the nominal channel spacing.

$$\text{Nominal channel spacing} = \left\lfloor \frac{BW_{Channel(1)} + BW_{Channel(2)} - 0.1\left|BW_{Channel(1)} - BW_{Channel(2)}\right|}{0.6} \right\rfloor 0.3 \ [\text{MHz}]$$

$BW_{Channel(1)}$ and $BW_{Channel(2)}$ are the bandwidths of two carriers.

- Two intra-band contiguous carriers to be aggregated must have a center frequency spacing that equals the nominal channel spacing. For example, the channel spacing between two contiguous 20 MHz carriers must be 19.8 MHz, and that between two contiguous carriers with a bandwidth combination of 20 MHz and 10 MHz must be 14.4 MHz.
- Two intra-band non-contiguous carriers to be aggregated must have a center frequency spacing at least the average of their total bandwidths.

Coverage

The coverage areas provided by the cells to be aggregated must overlap. The areas where CA takes effect are dependent on the overlapping ranges. The smaller the overlapping range, the smaller the CA area.

Cells

- TDD cells for CA must use uplink-downlink configuration 1 or 2 and one of special subframe configurations 4, 5, 6, 7, and 9. A TDD cell using special subframe configuration 4 cannot work with another TDD cell using special subframe configuration 5, 6, 7, or 9 for CA.
- Any cells possible for CA must have the same cyclic prefix (CP) length: either normal or extended.
- Candidate SCells must be inter-frequency neighboring cells of PCells.

  There is one exception: After a CA UE reports its CA capabilities during initial access, the eNodeB sends an RRC Connection Reconfiguration message to configure a blind-configurable candidate SCell, if any, as an SCell to work with the PCell. This candidate SCell is not necessarily an inter-frequency neighboring cell of the PCell. However, in any subsequent SCell configuration procedure for this UE, the candidate SCells must be inter-frequency neighboring cells of the PCell.
- Intra-frequency cell requirements
  - Inter-BBP data exchange for this function is bandwidth-consuming. If this function is activated together with CoMP or SFN, the available backplane bandwidth may be insufficient. To prevent this, it is recommended that intra-frequency cells be configured on the same BBP.
  - Network planning must mitigate physical cell identifier (PCI) conflicts. When the PCI of a candidate SCell conflicts with the PCI of an intra-frequency neighboring cell, the eNodeB will configure this candidate cell as an SCell only if all of the following options have been selected:
    - **CaBasedEventAnrSwitch** and **IntraRatEventAnrSwitch** options of the **ENodeBAlgoSwitch**.***AnrSwitch*** parameter
    - **INTRA_RAT_ANR_SW** option of the **CellAlgoSwitch**.***AnrFunctionSwitch*** parameter

5.3.6 Others

- UEs

UEs must comply with 3GPP Release 10 or later and support the frequency bands of the carriers to be aggregated and their bandwidths. UEs must also support the peak data rates that CA can achieve.

NOTE

As stipulated in section 4.3.5.2 "supportedBandCombination" of 3GPP TS 36.306 V10.1.0, CA UEs must report the supportedBandCombination IE, which contains the UE-supported band combinations and bandwidth classes for eNodeBs to perform CA.

- EPC

  For this function to reach its theoretical peak data rate described in **5.2.1 Benefits**, the maximum bit rate that each UE subscribes to in the EPC cannot be lower than this theoretical value.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

5.4.1.1.1 CA-Group-based Configuration Mode

Table 5-5 describes the parameters used for function activation. For the parameters used for function optimization, see both **Table 5-6** and **5.4.1.1.3 Common Parameters**.

Table 5-5 Parameters for activating downlink 2CC aggregation (CA-group-based)

Parameter Name	Parameter ID	Setting Notes
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	Deselect the **FreqCfgSwitch** option.
CA Group Identity	**CaGroup**.***CaGroupId***	Set this parameter based on site conditions.
CA Group Type Indication	**CaGroup**.***CaGroupTypeInd***	Set this parameter based on site conditions.
CA Group Identity	**CaGroupCell**.***CaGroupId***	Set this parameter based on site conditions.
eNodeB ID	**CaGroupCell**.***eNodeBId***	Set this parameter based on site conditions.
Local Cell ID	**CaGroupCell**.***LocalCellId***	Set this parameter based on site conditions.

Table 5-6 Parameters for optimizing downlink 2CC aggregation (CA-group-based)

Parameter Name	Parameter ID	Setting Notes
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	Select the **SccBlindCfgSwitch** option only if operators intend to use blind SCell configuration.
Preferred PCell Priority	**CaGroupCell**.***PreferredPCellPriority***	This parameter specifies the PCell priority of a cell. ● The greater the value is, the higher the probability that CA UEs treat the cell as their PCell. To raise this probability, increase the value of this parameter. ● The smaller the value is, the lower the probability that CA UEs treat the cell as their PCell.
PCell A4 RSRP Threshold	**CaGroupCell**.***PCellA4RsrpThd***	This parameter specifies the RSRP threshold for CA event A4 during PCC anchoring. If the measured RSRP of an inter-frequency neighboring cell is greater than this parameter value, the CA UE reports event A4. ● The greater the value, the lower the probability of an inter-frequency handover for PCC anchoring. To narrow the coverage area where the cell is the PCC anchor, increase the value of this parameter. ● The smaller the value, the higher the probability of an inter-frequency handover for PCC anchoring. However, if the value is too small, the handover may fail.

Parameter Name	Parameter ID	Setting Notes
PCell A4 RSRQ Threshold	**CaGroupCell**.***PCellA4RsrqThd***	This parameter specifies the RSRQ threshold for CA event A4 during PCC anchoring. If the measured RSRQ of an inter-frequency neighboring cell is greater than this parameter value, the CA UE reports event A4. ● The greater the value, the lower the probability of an inter-frequency handover for PCC anchoring. To narrow the coverage area where the cell is the PCC anchor, increase the value of this parameter. ● The smaller the value, the higher the probability of an inter-frequency handover for PCC anchoring. However, if the value is too small, the handover may fail.
SCell Local Cell ID	**CaGroupSCellCfg**.***SCellLocalCellId***	This parameter specifies the local ID of a candidate SCell.
SCell eNodeB ID	**CaGroupSCellCfg**.***SCelleNodeBId***	This parameter specifies the eNodeB ID of the candidate SCell.
SCell Priority	**CaGroupSCellCfg**.***SCellPriority***	This parameter specifies the priority of the candidate SCell. ● The greater the value is, the higher the probability that the cell is configured as an SCell that accompanies the specified PCell. ● The smaller the value is, the lower the probability that the cell is configured as an SCell that accompanies the specified PCell. The value **0** indicates that the cell cannot be configured as an SCell.
SCell Blind Configuration Flag	**CaGroupSCellCfg**.***SCellBlindCfgFlag***	To enable blind SCell configuration, set this parameter to **TRUE**.

Parameter Name	Parameter ID	Setting Notes
SCell A4 Offset	**CaGroupSCellCfg**.***SCellA4Offset***	This parameter specifies the offset used to calculate the RSRP threshold for CA event A4 related to SCell measurements. This threshold is equal to the sum of the **CaMgtCfg**.***CarrAggrA4ThdRsrp***, **CaGroupSCellCfg**.***SCellA4Offset***, and **SccFreqCfg**.***SccA2RsrpThldExtendedOfs*** parameter values configured on the PCell side. When the cell specified by the **LocalCellId** parameter is acting as the PCell, event A4 is triggered if the candidate SCell specified by the **SCelleNodeBId** and **SCellLocalCellId** parameters meets the event triggering condition. ● The greater the value of the **CaGroupSCellCfg**.***SCellA4Offset*** parameter, the higher the threshold for CA event A4, and the lower the probability of configuring an SCell. ● The smaller the value of the **CaGroupSCellCfg**.***SCellA4Offset*** parameter, the lower the threshold for CA event A4, and the higher the probability of configuring an SCell. To adjust the CA event A4 threshold for different SCells, set the **CaGroupSCellCfg**.***SCellA4Offset*** parameter according to operators' policies.

Parameter Name	Parameter ID	Setting Notes
SCell A2 Offset	**CaGroupSCellCfg**.*SCellA2Offset*	This parameter specifies the offset used to calculate the RSRP threshold for CA event A2 related to SCell measurements. This threshold is equal to the sum of the **CaMgtCfg**.*CarrAggrA2ThdRsrp*, **CaGroupSCellCfg**.*SCellA2Offset*, and **SccFreqCfg**.*SccA2RsrpThldExtendedOfs* parameter values configured on the PCell side. When the cell specified by the **LocalCellId** parameter is acting as the PCell, event A2 is triggered if the SCell specified by the **SCelleNodeBId** and **SCellLocalCellId** parameters meets the event triggering condition. ● The greater the value, the higher the threshold for CA event A2, and the higher the probability of removing an SCell. ● The smaller the value, the lower the threshold for CA event A2, and the lower the probability of removing an SCell. To adjust the CA event A2 threshold for different SCells, set this parameter according to operators' policies.

Moreover, take the following necessary actions for this function to take effect:

- Prepare data as described in **16.4.1.1 Data Preparation** if downlink 2CC aggregation is to be deployed in a relaxed backhaul scenario.
- Prepare data as described in **17.4.1.1 Data Preparation** if downlink 2CC aggregation is to be deployed in an eNodeB coordination scenario.

5.4.1.1.2 Adaptive Configuration Mode

Table 5-7 describes the parameters used for function activation. For the parameters used for function optimization, see both **Table 5-8** and **5.4.1.1.3 Common Parameters**.

Table 5-7 Parameters for activating downlink 2CC aggregation (adaptive)

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	FreqCfgSwitch	Select this option.
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	AdpCaSwitch	Select this option.
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	CaDl2CCExtSwitch	Select this option for each possible pair of PCell and SCell on carriers that meet either of the following conditions: ● For FDD: 20 MHz < aggregated bandwidth of two FDD carriers ≤ 40 MHz ● For TDD: 30 MHz < aggregated bandwidth of two TDD carriers ≤ 40 MHz
CA Algorithm Extend Switch	**ENodeBAlgoSwitch**.***CaAlgoExtSwitch***	FreqBaseCaLicAlarmSwitch	Select this option.
PCC Downlink EARFCN	**PccFreqCfg**.***PccDlEarfcn***	None	Set this parameter based on site conditions.
PCC Downlink EARFCN	**SccFreqCfg**.***PccDlEarfcn***	None	Set this parameter based on site conditions.
SCC Downlink EARFCN	**SccFreqCfg**.***SccDlEarfcn***	None	Set this parameter based on site conditions.

Table 5-8 Parameters for optimizing downlink 2CC aggregation (adaptive)

Parameter Name	Parameter ID	Setting Notes
Preferred PCC Priority	**PccFreqCfg**.***PreferredPccPriority***	This parameter specifies the priority of a candidate PCC. ● The greater the value is, the higher the probability that CA UEs treat the carrier as their PCC. ● The smaller the value is, the lower the probability that CA UEs treat the carrier as their PCC.

Parameter Name	Parameter ID	Setting Notes
PCC A4 RSRP Threshold	**PccFreqCfg**.***PccA4RsrpThd***	This parameter specifies the RSRP threshold for CA event A4 during PCC anchoring. If the measured RSRP of an inter-frequency neighboring cell is greater than this parameter value, the CA UE reports event A4. ● The greater the value, the lower the probability of an inter-frequency handover for PCC anchoring. To narrow the area where the carrier is the PCC anchor, increase the value of this parameter. ● The smaller the value, the higher the probability of an inter-frequency handover for PCC anchoring. However, if the value is too small, the handover may fail. It is recommended that this parameter be set to a value greater than **InterFreqHoGroup**.***InterFreqHoA4ThdRsrp*** to prevent a decrease in the handover success rate.
PCC A4 RSRQ Threshold	**PccFreqCfg**.***PccA4RsrqThd***	This parameter specifies the RSRQ threshold for CA event A4 during PCC anchoring. If the measured RSRQ of an inter-frequency neighboring cell is greater than this parameter value, the CA UE reports event A4. ● The greater the value, the lower the probability of an inter-frequency handover for PCC anchoring. To narrow the area where the carrier is the PCC anchor, increase the value of this parameter. ● The smaller the value, the higher the probability of an inter-frequency handover for PCC anchoring. However, if the value is too small, the handover may fail.
PCC Downlink EARFCN	**SccFreqCfg**.***PccDlEarfcn***	This parameter specifies the downlink EARFCN of a PCC. Set this parameter based on the network plan.
SCC Downlink EARFCN	**SccFreqCfg**.***SccDlEarfcn***	This parameter specifies the downlink EARFCN of a candidate SCC. Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
SCC Priority	**SccFreqCfg**.***SccPriority***	This parameter specifies the priority with which the candidate SCC identified by **SccDlEarfcn** acts as an SCC for the PCC identified by **PccDlEarfcn**. ● The greater the value, the higher the priority level. To raise the probability that CA UEs treat the carrier as their SCC, increase the value of this parameter. ● The smaller the value, the lower the priority level.
SCC A2 Offset	**SccFreqCfg**.***SccA2Offset***	This parameter specifies the offset of the threshold for CA event A2 relative to the **CarrAggrA2ThdRsrp** parameter value in the **CaMgtCfg** MO. If the sum of the offset and **CarrAggrA2ThdRsrp** is greater than –43 dBm or less than –140 dBm, the threshold for CA event A2 takes the value –43 dBm or –140 dBm, respectively. ● The greater the value, the higher the threshold for CA event A2, and the higher the probability of removing an SCell. ● The smaller the value, the lower the threshold for CA event A2, and the lower the probability of removing an SCell. To adjust the CA event A2 threshold for different SCCs, set this parameter according to operators' policies.

Parameter Name	Parameter ID	Setting Notes
SCC A4 Offset	**SccFreqCfg**.***SccA4Offset***	This parameter specifies the offset used to calculate the RSRP threshold for CA event A4 related to SCC measurements. If SCell configuration is performed for a CA UE during initial access, an incoming handover, or an incoming RRC connection reestablishment, the actually effective RSRP threshold for triggering CA event A4 is equal to the sum of the **CaMgtCfg**.***CarrAggrA4ThdRsrp***, **SccFreqCfg**.***SccA4Offset***, and **SccFreqCfg**.***SccA2RsrpThldExtendedOfs*** parameters configured on the PCell side. If SCell configuration is periodically triggered for a CA UE based on the traffic volume, the actually effective RSRP threshold for triggering CA event A4 is equal to the sum of the **CaMgtCfg**.***CarrAggrA4ThdRsrp*** and **SccFreqCfg**.***SccA4Offset*** parameters configured on the PCell side. ● The greater the value of the **SccFreqCfg**.***SccA4Offset*** parameter, the higher the threshold for CA event A4, and the lower the probability of configuring an SCell. ● The smaller the value of the **SccFreqCfg**.***SccA4Offset*** parameter, the lower the threshold for CA event A4, and the higher the probability of configuring an SCell. To adjust the CA event A4 threshold for different SCCs, set the **SccFreqCfg**.***SccA4Offset*** parameter according to operators' policies.
CN Operator List	**SccFreqCfg**.***CnOperatorList***	This parameter specifies the operators that use an SCC.

Parameter Name	Parameter ID	Setting Notes
SCell Aging Time	**CaMgtCfg**.***SCellAgingTime***	This parameter specifies the aging time for SCells that have been dynamically configured for CA. If carrier aggregation does not occur between a pair of cells throughout the aging time, the eNodeB cancels the CA relationship between the cells. ● The greater the value, the lower the probability of canceling the CA relationship. To reduce the number of outdated SCells that were dynamically configured, increase the value of this parameter. ● The smaller the value, the higher the probability of canceling the CA relationship. If the value of this parameter is decreased, the SCells whose aging time has exceeded the new value before the change will be removed soon. If the CA route capacity is insufficient, set this parameter to a relatively small value to reduce the impact of CA-related counter changes caused by an insufficient CA route capacity after upgrades.
Heavy-Load UE Count Thld for SCell Config	**CaMgtCfg**.***HLUeCntThldForScellConfig***	This parameter specifies the UE number threshold for determining whether a cell is a high-load cell, the result of which is used to evaluate SCell configuration for CA UEs. ● A larger value of this parameter leads to a lower probability of determining a cell to be a high-load cell, a lower probability of prohibiting this cell from being configured as an SCell for load reasons, and therefore a larger number of CA UEs. ● A smaller value of this parameter leads to a higher probability of prohibiting a cell from being configured as an SCell for load reasons, and therefore a smaller number of CA UEs.

Parameter Name	Parameter ID	Setting Notes
High Load Cell Type Not As SCell	**CaMgtCfg**.***HighLoadCellTypeNotAsScell***	This parameter specifies which type of high-load cells cannot be configured as SCells.
SCell Blind Configuration Flag	**CaGroupSCellCfg**.***SCellBlindCfgFlag***	To enable blind SCell configuration, set this parameter to **TRUE**. This parameter can also be automatically set by the function of Auto Neighbor Group Configuration (ANGC) for CA. For details, see *Auto Neighbor Group Configuration*.
CA Algorithm Extend Switch	**ENodeBAlgoSwitch**.***CaAlgoExtSwitch***	Select the **CaMubfPairingAdaptOptSwitch** option. Before this option is selected, the **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch**.**CaAlgoSwitch** parameter must be deselected.
Reserved Switch Parameter 0	**EnodebRsvdParamExt**.***RsvdSwParam0***	Select the **RSVD_SW_PARAM0_BIT28** option if candidate SCells are high-load massive MIMO cells and a maximum of one TDD high-load massive MIMO cell is expected to be added as the SCell for CA UEs that can participate in pairing.

Moreover, take the following necessary actions for this function to take effect:

- Prepare data as described in **16.4.1.1 Data Preparation** if downlink 2CC aggregation is to be deployed in a relaxed backhaul scenario.
- Prepare data as described in **17.4.1.1 Data Preparation** if downlink 2CC aggregation is to be deployed in an eNodeB coordination scenario.

5.4.1.1.3 Common Parameters

Before activating CA in either configuration mode (CA-group-based or adaptive), operators can optimize the settings of the parameters described in this section based on the network plan to achieve better network performance.

eNodeB-Level Algorithm Switches

Set parameters in the **ENodeBAlgoSwitch** MO.

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	EnhancedPccAnchorSwitch	Select this option.
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	PccAnchorSwitch	This option can be deselected, as the **EnhancedPccAnchorSwitch** option is the recommended switch that controls PCC anchoring.
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	SccA2RmvSwitch	Select this option if operators intend to allow the eNodeB to deliver the A2 measurement configurations for SCells that have been configured based on A4 measurements.
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	CaTrafficTriggerSwitch	Select this option if operators intend to allow the eNodeB to configure and remove SCells based on the traffic volume of UEs.
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	HoWithSccCfgSwitch	Select this option if operators intend to allow SCells to be configured for CA UEs during their handovers.
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	SccModA6Switch	Select this option if operators intend to have SCells changed to better intra-frequency neighboring cells.
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	PccSmartCfgSwitch	Select this option if operators require the eNodeB to consider the load status indicators of candidate cells when configuring PCells. Before selecting this option, select the **PccAnchorSwitch** or **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter.
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	SccSmartCfgSwitch	Select this option if operators intend to allow the eNodeB to consider the load status indicators of candidate cells when configuring SCells.

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	IdleModePccAnchorSwitch	Select this option if operators intend to use PCC anchoring for RRC_IDLE UEs.
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	CaAdpPreSchSwitch	Select this option only if operators intend to use adaptive pre-scheduling to increase the single-CA-UE throughput in the following scenarios: ● Inter-eNodeB CA based on relaxed backhaul ● Inter-eNodeB CA based on distributed eNodeB coordination (FDD) ● Inter-eNodeB CA based on hybrid eNodeB coordination (FDD) ● Downlink 2CC aggregation with the **2CCDlCaEnhanceSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter selected (TDD) ● Aggregation of three or more downlink CCs (TDD) ● **Dl2CCAckResShareSw** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter being selected (TDD) The default status of this option is selected.
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	RelaxedBHCaArqSelectSwitch	Select this option if operators intend to increase the single-CA-UE throughput by enabling ARQ selection in inter-eNodeB CA based on relaxed backhaul.
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	CaRlcPreAllocSwitch	Select this option if operators intend to solve the problem of UE throughput decreases caused by RLC send window suspending in inter-eNodeB CA based on relaxed backhaul.

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	MtaAlgSwitch	Select this option if operators intend to use multiple timing advance groups (TAGs) in uplink CA scenarios.
Handover Signaling Optimized Switch	**ENodeBAlgoSwitch**.***HoSignalingOptSwitch***	ServiceReqInterFreqBlindHoSw	Select this option if operators intend to enable blind handovers (and SCell configuration during the blind handovers, if allowed by the data configurations at target cells) for service-request-based inter-frequency handovers of TDD+FDD CA UEs from their TDD PCells to their FDD SCells.
CA Algorithm Extend Switch	**ENodeBAlgoSwitch**.***CaAlgoExtSwitch***	FTMtaAlgSwitch	Select this option if operators intend to use multiple TAGs in uplink FDD+TDD CA scenarios.
CA Algorithm Extend Switch	**ENodeBAlgoSwitch**.***CaAlgoExtSwitch***	SmartCaFastSccCfgSwitch	Select this option if operators intend to accelerate the process of entering CA for UEs by trying blind SCell configuration during intelligent selection of serving cell combinations. Before this option is selected, the **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter must be selected.
CA Algorithm Extend Switch	**ENodeBAlgoSwitch**.***CaAlgoExtSwitch***	CaAnrGapOptSwitch	Select this option if operators require UEs to perform inter-frequency measurements on candidate SCCs where no neighboring cells have been configured.

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Extend Switch	**ENodeBAlgoSwitch**.***CaAlgoExtSwitch***	CaEnhancedPreAllocSwitch	Select this option if operators intend to enable the eNodeB to dynamically distribute traffic to the CCs of each CA UE based on the real-time traffic volume and scheduling capabilities of the CCs. Selecting this option has the following impacts on the network: ● The average user-perceived downlink data rate of CA UEs increases. **L.Thrp.bits.DL.CAUser**/ **L.Thrp.Time.DL.CAUser** ● The average user-perceived downlink data rate of all UEs increases. **L.Thrp.bits.DL**/ **L.Thrp.Time.DL** ● The value of the **L.Thrp.bits.DL.LastTTI.CAUser** counter may increase. It is recommended that this option be selected on a moderately- or lightly-loaded MBB network where the **Dl2CCAckResShareSw** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter has been selected. Otherwise, there might be a negative impact. This option cannot be selected if the **DlCaLbAlgoSwitch** option of the **ENodeBAlgoSwitch**.***CaLbAlgoSwitch*** parameter or the **CaLoadBalancePreAllocS-witch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter has been selected. The following constraints apply to the **CaEnhancedPreAllocSwitch** option:

Parameter Name	Parameter ID	Option	Setting Notes
			● In FDD, this option does not take effect for inter-eNodeB CA based on relaxed backhaul. ● In TDD, if this option is selected in inter-eNodeB CA based on relaxed backhaul, traffic distribution to CCs is not based on the real-time traffic volume. Instead, the amount of traffic distributed from the PCell to SCells is adjusted only in preallocation situations.
CA Load Balancing Algorithm Switch	**ENodeBAlgoSwitch**.***CaLbAlgoSwitch***	DlCaLbAlgoSwitch	Select this option if operators intend to use inter-CC load transfer triggered by cell load.

Parameter Name	Parameter ID	Option	Setting Notes
TRM switch	**ENodeBAlgoSwitch**.***TrmSwitch***	CoOverbookingSwitch	Select this option when the value of **L.CA.SccAddFail.PhyLinkFail.Dur** is not zero. With this option selected, more CA services can be admitted and flow control takes effect based on real-time traffic volume, so that the bandwidth for coordination can be dynamically shared by CA services. If the value of **L.CA.SccAddFail.PhyLinkFail.Dur** becomes zero, this option has taken effect. If the **CoOverbookingSwitch** option is selected when the **AdpCaSwitch** or **FreqCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter has been selected, the **AdpCaSwitch** or **FreqCfgSwitch** option must be deselected and then selected again so that the setting of **CoOverbookingSwitch** takes effect immediately. CA services are unavailable during the interval between the time the **AdpCaSwitch** or **FreqCfgSwitch** option is deselected and the time it is selected again. Therefore, it is recommended that these parameters be set when traffic is low, for example, at 2:00 a.m. In eNodeB coordination scenarios, CA requires that the setting of this option be the same for all eNodeBs.

Set parameters in the **EnodebCounterParaGrp** MO.

Parameter Name	Parameter ID	Option	Setting Notes
eNodeB Counter Algorithm Switch	**EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch***	CA_UE_MULTI_CC_STAT_OPT_SW	When some CA UEs on a network have not all their SCells activated, select this option if operators expect more accurate measurement of the counters related to the number of multi-CC-active CA UEs and the number of transmitted bits and transmission duration of such UEs.

Cell and System Information Mapping

Set parameters in the **CellSiMap** MO.

Parameter Name	Parameter ID	Option	Setting Notes
Cell System Information Switch	**CellSiMap**.***SiSwitch***	ForbidCellSiSwitch	Select this option if operators intend to configure a cell as a dedicated SCell. In this case, the cell cannot act as a PCell and UEs cannot camp on, access, or be handed over to the cell.

Cell-Level Algorithm Switches

Set parameters in the **CellAlgoSwitch** MO.

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	UlScellForbidSwitch	The uplink of cells in band 38 experiences downlink interference from cells in band 7. To allow cells in band 38 to be configured as dedicated SCells only for the downlink (not for the uplink), select this option. This option takes effect only when the **ForbidCellSiSwitch** option of the **CellSiMap**.***SiSwitch*** parameter is selected.

Cell-Level Algorithm Switches (TDD)

Set parameters in the **CellAlgoSwitch** MO.

Parameter Name	Parameter ID	Option	Setting Notes
Traffic MLB Switch	**CellAlgoSwitch**.***TrafficMlbSwitch***	UlHeavyTrafficMlbTFCASw	Select this option if operators intend to use MLB for UEs in the TDD+FDD CA state in uplink heavy-traffic scenarios.
High Speed Optimized Switch	**CellAlgoSwitch**.***HighSpeedSchOptSwitch***	HSCABlindRedirectSw	Select this option if operators intend to use blind inter-frequency redirection for high-speed CA UEs.

Carrier Management Parameters

Set parameters in the **CaMgtCfg** MO.

Parameter Name	Parameter ID	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	● Select the **CaInstantlyJudgeSwitch** option if operators intend to enable the eNodeB to activate SCells more quickly for small-packet services, such as web browsing and small-sized file transmission, and accelerate the UE throughput boost. If this option is selected, the eNodeB uses instantaneous millisecond-level traffic volume values to evaluate SCell activation. The proportion of UEs in the CA state may rise drastically, resulting in fluctuations in KPIs such as downlink IBLER and residual block error rate (RBLER). ● Deselect this option if operators intend to allow the eNodeB to activate SCells for CA UEs when the UEs are downloading large files. If this option is deselected, the eNodeB uses filtered second-level traffic volume values to evaluate SCell activation.
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	Select the **2CCDlCaEnhanceSwitch** option if operators require the eNodeB to configure PUCCH format 3 for CA UEs that support this format and are working in the downlink 2CC aggregation state. With this option selected, the eNodeB allocates: ● Two RBs to PUCCH format 3 for cells with a bandwidth of 10 MHz or higher. ● One RB to PUCCH format 3 for cells with a bandwidth of 5 MHz or lower. UE capabilities of PUCCH format 3 can be identified by observing whether UEs support aggregation of three or more CCs. If a UE supports aggregation of three or more CCs, the UE supports PUCCH format 3.

Parameter Name	Parameter ID	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	Select the **RcvA2CfgSccSwitch** option if operators intend to allow SCell configuration after the serving eNodeB of the PCell receives handover-related event A2 reports from a CA UE.
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	Select the **VolteSupportCaInterFreqMeasSw** option if operators require gap-assisted inter-frequency measurements before SCells can be configured for UEs with concurrent VoLTE and data services.
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	Select the **CaEnhAperiodicCqiRptSwitch** option if operators require that CA UEs with SCells activated use aperiodic CQI reporting to promptly send accurate precoding matrix indications (PMIs) and CQIs in closed-loop multiple-input multiple-output (MIMO) scenarios.
Carrier Aggregation A2 RSRP threshold	**CaMgtCfg**.***CarrAggrA2ThdRsrp***	This parameter specifies the RSRP threshold for event A2 used to evaluate SCell removal. ● The greater the value, the higher the probability of removing SCells. To apply CA to UEs in the cell center or at medium distances from the cell center, increase the value of this parameter. ● The smaller the value, the lower the probability of removing SCells.
CA A2 Time To Trigger	**CaMgtCfg**.***CaA2TimeToTrigger***	This parameter specifies the time-to-trigger for CA event A2. ● A larger value of this parameter results in a lower probability of reporting CA event A2. ● A smaller value of this parameter results in a higher probability of reporting CA event A2.

Parameter Name	Parameter ID	Setting Notes
Carrier Aggregation A4 RSRP threshold	**CaMgtCfg**.***CarrAggrA4ThdRsrp***	This parameter specifies the RSRP threshold for event A4 used to evaluate SCell configuration. ● The greater the value, the lower the probability of configuring SCells. To apply CA to UEs in the cell center or at medium distances from the cell center, increase the value of this parameter. ● The smaller the value, the higher the probability of configuring SCells. The value of this parameter must be greater than the value of **CaMgtCfg.*CarrAggrA2ThdRsrp***.
CA A4 Time To Trigger	**CaMgtCfg**.***CaA4TimeToTrigger***	This parameter specifies the time-to-trigger for CA event A4. ● A larger value of this parameter results in a lower probability of reporting CA event A4. ● A smaller value of this parameter results in a higher probability of reporting CA event A4.
Carrier Management Switch	**CaMgtCfg**.***CarrierMgtSwitch***	This parameter specifies the carrier management switch. ● If this parameter is set to **ON**, the eNodeB deactivates SCells for a CA UE when the traffic volume of the UE is low or the channel quality of the SCells is unsatisfactory. ● If this parameter is set to **OFF**, the eNodeB does not deactivate SCells unless they experience radio link failures.
CA Active Buffer Delay Threshold	**CaMgtCfg**.***ActiveBufferDelayThd***	This parameter specifies the buffer delay threshold used to evaluate SCell activation. ● The greater the value, the less likely SCells are to be activated. ● The smaller the value, the more likely SCells are to be activated. To raise the probability of SCell activation, reduce the value of this parameter.

Parameter Name	Parameter ID	Setting Notes
CA Active Buffer Length Threshold	**CaMgtCfg**.***ActiveBufferLenThd***	This parameter specifies the buffer length threshold used to evaluate SCell activation. ● The greater the value, the less likely SCells are to be activated. ● The smaller the value, the more likely SCells are to be activated. To raise the probability of SCell activation, reduce the value of this parameter. It is not recommended that this parameter be set to **0**. If it is set to **0**, the eNodeB activates a configured SCell as long as the eNodeB intends to send data to the CA UE.
CA Deactive Throughput Threshold	**CaMgtCfg**.***DeactiveThroughputThd***	This parameter specifies the data rate threshold used to evaluate SCell deactivation. ● The greater the value, the more likely SCells are to be deactivated. To raise the probability of SCell deactivation, increase the value of this parameter. ● The smaller the value, the less likely SCells are to be deactivated.
CA Deactive Buffer Length Threshold	**CaMgtCfg**.***DeactiveBufferLenThd***	This parameter specifies the buffer length threshold used to evaluate SCell deactivation. ● The greater the value, the more likely SCells are to be deactivated. To raise the probability of SCell deactivation, increase the value of this parameter. ● The smaller the value, the less likely SCells are to be deactivated.

Parameter Name	Parameter ID	Setting Notes
SCC Deactive CQI Threshold	**CaMgtCfg**.***SccDeactCqiThd***	This parameter specifies the CQI threshold used to evaluate SCell deactivation based on channel quality. If the channel quality of an SCell for a CA UE is lower than the channel quality corresponding to the CQI threshold in single-codeword transmission, the eNodeB deactivates the SCell. If this parameter is set to **0**, the eNodeB does not deactivate SCells based on channel quality. The value **5** is recommended. ● The greater the value, the more likely SCells are to be deactivated. To maintain active SCells only when channel quality is high, increase the value of this parameter. ● The smaller the value, the less likely SCells are to be deactivated.

Parameter Name	Parameter ID	Setting Notes
SCC Configuration Interval	**CaMgtCfg**.***SccCfgInterval***	This parameter specifies the minimum interval after which the eNodeB will attempt to configure an SCell again for a CA UE whose previous SCell configuration failed. The eNodeB makes the attempt only if the traffic volume of the UE always meets the SCell activation condition throughout a period. ● A larger value results in less frequent SCell configurations and fewer RRC Connection Reconfiguration messages transmitted to configure SCells. It also results in a smaller decrease in cell throughput if the eNodeB configures SCells based on A4 measurements and the UE requires inter-frequency measurement gaps. ● A smaller value results in more frequent SCell configurations and more RRC Connection Reconfiguration messages transmitted to configure SCells. It also results in a greater decrease in cell throughput if the eNodeB configures SCells based on A4 measurements and the UE requires inter-frequency measurement gaps. To accelerate SCell configuration, reduce the value of this parameter.
Carrier Aggregation A6 Offset	**CaMgtCfg**.***CarrAggrA6Offset***	This parameter specifies the offset for CA event A6, which is triggered when the signal quality of a neighboring cell is higher than that of an SCell. ● The greater the value, the lower the probability of changing the SCell based on event A6. To raise the signal quality requirement for SCell changes, increase the value of this parameter. ● The smaller the value, the higher the probability of changing the SCell based on event A6.

Parameter Name	Parameter ID	Setting Notes
CA A6 Report Amount	**CaMgtCfg**.***CaA6ReportAmount***	This parameter specifies the number of periodic measurement reports sent after CA event A6 is triggered. ● The greater the value, the higher the UE power consumption. However, if this parameter is set to a large value, the eNodeB can monitor the radio signal condition of the SCell's intra-frequency neighboring cells. When the signal quality in the SCell is poor, the eNodeB can change the SCell in a timely manner to prevent a decrease in the SCell transmission efficiency. To raise the probability of changing the SCell, increase the value of this parameter. ● The smaller the value, the lower the UE power consumption. However, if this parameter is set to a small value, the eNodeB may not know the radio signal condition of the SCell's intra-frequency neighboring cells. When the signal quality in the SCell is poor, the eNodeB may not change the SCell in a timely manner, which in turn lowers the SCell transmission efficiency.

Parameter Name	Parameter ID	Setting Notes
CA A6 Report Interval	**CaMgtCfg**.***CaA6ReportInterval***	This parameter specifies the interval between periodic measurement reports that are sent after CA event A6 is triggered. ● The greater the value is, the less frequently each CA UE sends A6 measurement reports, and the lower the UE power consumption is. However, if this parameter is set too large, the eNodeB may not learn the radio signal condition of the SCell's intra-frequency neighboring cells. When the signal quality in the SCell is poor, the eNodeB may not change the SCell in a timely manner and the SCell transmission efficiency will decrease. ● The smaller the value is, the more frequently each CA UE sends A6 measurement reports, and the more power the UE consumes. However, if this parameter is set to a small value, the eNodeB can monitor the radio signal condition of the SCell's intra-frequency neighboring cells. When the signal quality in the SCell is poor, the eNodeB can change the SCell in a timely manner to prevent a decrease in the SCell transmission efficiency. To accelerate SCell changes for CA UEs whose PCell remains unchanged, reduce the value of this parameter.

Parameter Name	Parameter ID	Setting Notes
Cell Maximum PCC Number	**CaMgtCfg**.***CellMaxPccNumber***	This parameter specifies the maximum permissible number of PCell-served UEs in the cell. If the number of PCell-served UEs in the cell has reached the value of this parameter, no more CA UEs can treat the cell as their PCell. If this parameter is set to **0**, no UEs are allowed to treat the cell as their PCell, but UEs can still treat the cell as their SCell. ● If this parameter is set to a larger value, more UEs are allowed to treat the cell as their PCell. To allow more UEs to treat the cell as their PCell, increase the value of this parameter. If the sum of the values of this parameter for all cells on a board exceeds the board capacity, the number of UEs that can treat a cell as their PCell is preferentially subject to the board capacity. ● If this parameter is set to a smaller value, fewer UEs are allowed to treat the cell as their PCell.
Relaxed Backhaul SCC Deactive CQI Threshold	**CaMgtCfg**.***RelaxedBHSccDeactCqiThd***	This parameter specifies the CQI threshold used to evaluate SCell deactivation based on channel quality in relaxed backhaul scenarios. If the channel quality of an SCell for a CA UE is lower than the channel quality corresponding to the CQI threshold in single-codeword transmission, the eNodeB deactivates the SCell. If this parameter is set to **0**, the eNodeB does not deactivate SCells based on channel quality. The value **5** is recommended. ● The greater the value, the more likely SCells are to be deactivated. To maintain active SCells only when channel quality is high, increase the value of this parameter. ● The smaller the value, the less likely SCells are to be deactivated.

Parameter Name	Parameter ID	Setting Notes
Measurement Cycle of SCell	**CaMgtCfg**.***MeasCycleSCell***	This parameter specifies the sampling period of the radio resource measurement on an SCell by a CA UE when the SCell is deactivated. ● A larger value of this parameter indicates a longer sampling period. It causes the CA UE to measure the radio resources of the SCell less frequently. However, this decreases the UE power consumption. ● A smaller value of this parameter indicates a shorter sampling period. It causes the CA UE to measure the radio resources of the SCell more frequently. However, this also increases the UE power consumption. If this parameter is set to **NOT_CFG**, the measurement configuration messages that the eNodeB sends to CA UEs do not contain the measCycleSCell IE. In this case, if no compatibility protection is applied to the UEs, the UE behavior is unpredictable.
Anchor Policy for FDD and TDD Multi CC	**CaMgtCfg**.***FTCAMultiCCAnchorPolicy***	This parameter specifies whether the eNodeB preferentially selects FDD or TDD carriers for PCC anchoring in aggregation of three or more carriers for FDD+TDD CA. ● To prioritize FDD carriers, set this parameter to **FDD**. ● To prioritize TDD carriers, set this parameter to **TDD**. ● If there is no duplex mode preference, set this parameter to **NULL**.

Parameter Name	Parameter ID	Setting Notes
Anchor Policy for FDD and TDD 2CC	**CaMgtCfg**.***FTCA2CCAnchorPolicy***	This parameter specifies whether the eNodeB preferentially selects FDD or TDD carriers for PCC anchoring in FDD+TDD 2CC aggregation. ● To prioritize FDD carriers, set this parameter to **FDD**. ● To prioritize TDD carriers, set this parameter to **TDD**. ● If there is no duplex mode preference, set this parameter to **NULL**.
A1 RSRP Threshold for Enhanced SCC Selection	**CaMgtCfg**.***EnhancedSccSelA1ThldRsrp***	This parameter specifies the RSRP threshold for event A1 used in enhanced SCell selection. If the measured RSRP value exceeds this threshold, an event A1 report will be sent. ● The greater the value, the lower the probability of triggering inter-frequency measurements of high-priority candidate SCells. ● The smaller the value, the higher the probability of triggering inter-frequency measurements of high-priority candidate SCells. It is recommended that this parameter be used in multi-band same-coverage scenarios and that this A1 threshold be lower than the A1 threshold for PCC anchoring. The value **-40** indicates that enhanced SCell selection is disabled.
Fast SCell Selection After SCell Removal	**CaMgtCfg**.***FastScellSelAftScellRmvSw***	Select this option if operators intend to accelerate SCell configuration after all SCells of a CA UE are removed.
VoLTE CA A2 RSRP Thld	**CaMgtCfg**.***VolteCaA2RsrpThld***	This parameter specifies the RSRP threshold for event A2 of PCells for VoLTE UEs.

Parameter Name	Parameter ID	Setting Notes
SCC Detection CQI Decrease Step	**CaMgtCfg**.***SccDetectCqiDecreaseStep***	This parameter specifies the CQI decrease step used during SCell deactivation evaluation. If UE-reported CQIs are falsely high, the eNodeB may inaccurately select MCSs, causing a sharp increase in the BLER in SCells. As a result, the SCells are deactivated, and the throughput of the CA UEs decreases. This parameter is used to resolve this problem. The **CaMgtCfg**.***SccDetectCqiDecreaseStep*** parameter takes effect when the **CaMgtCfg**.***SccQuietTime*** parameter is set to a value other than **0**. ● If the **CaMgtCfg**.***SccDetectCqiDecreaseStep*** parameter is set to a value other than **0**, the outer-loop CQI adjustment value for CQI decrease is further extended by the value of this parameter. – A smaller value of this parameter results in a slower decrease in CQIs during SCell deactivation evaluation and therefore a lower probability of resuming scheduling in the SCells for UEs that have reported falsely high CQIs. – A larger value of this parameter results in a faster decrease in CQIs during SCell deactivation evaluation and therefore a higher probability of resuming scheduling in the SCells for UEs that have reported falsely high CQIs. ● If the **CaMgtCfg**.***SccDetectCqiDecreaseStep*** parameter is set to **0**, the preceding function does not take effect. LBBPc is not compatible with this function.

Parameter Name	Parameter ID	Setting Notes
Forbid 2CC UE Ratio Threshold	**CaMgtCfg**.***Forbid2ccUeRatioThld***	This parameter specifies the threshold expressed as the proportion of UEs served by a BBP for prohibiting CA with two or more CCs aggregated from taking effect. ● A smaller value of this parameter results in a lower proportion of UEs with CA in effect, fewer BBP specifications being used by these UEs, and lower UE throughput on the entire network. ● A larger value of this parameter results in a higher proportion of UEs with CA in effect, more BBP specifications being used by these UEs, and higher UE throughput on the entire network. Configuration constraints are as follows: ● If neither **CaMgtCfg**.***Forbid2ccUeRatioThld*** nor **CaMgtCfg**.***Forbid3ccUeRatioThld*** is set to **100%**, **CaMgtCfg**.***Forbid2ccUeRatioThld*** must be set to a value greater than **CaMgtCfg**.***Forbid3ccUeRatioThld***. ● If neither **CaMgtCfg**.***Forbid2ccUeRatioThld*** nor **CaMgtCfg**.***Forbid4ccUeRatioThld*** is set to **100%**, **CaMgtCfg**.***Forbid2ccUeRatioThld*** must be set to a value greater than **CaMgtCfg**.***Forbid4ccUeRatioThld***. ● If neither **CaMgtCfg**.***Forbid2ccUeRatioThld*** nor **CaMgtCfg**.***Forbid5ccUeRatioThld*** is set to **100%**, **CaMgtCfg**.***Forbid2ccUeRatioThld*** must be set to a value greater than **CaMgtCfg**.***Forbid5ccUeRatioThld***.

Parameter Name	Parameter ID	Setting Notes
Forbid 3CC UE Ratio Threshold	**CaMgtCfg**.***Forbid3ccUeRatioThld***	This parameter specifies the threshold expressed as the proportion of UEs served by a cell and that of UEs served by a BBP for prohibiting CA with three or more CCs aggregated from taking effect. ● A smaller value of this parameter results in a lower proportion of UEs with CA in effect, fewer BBP specifications being used by these UEs, and lower UE throughput on the entire network. ● A larger value of this parameter results in a higher proportion of UEs with CA in effect, more BBP specifications being used by these UEs, and higher UE throughput on the entire network. Configuration constraints are as follows: ● If neither **CaMgtCfg**.***Forbid2ccUeRatioThld*** nor **CaMgtCfg**.***Forbid3ccUeRatioThld*** is set to **100%**, **CaMgtCfg**.***Forbid2ccUeRatioThld*** must be set to a value greater than **CaMgtCfg**.***Forbid3ccUeRatioThld***. ● If neither **CaMgtCfg**.***Forbid3ccUeRatioThld*** nor **CaMgtCfg**.***Forbid4ccUeRatioThld*** is set to **100%**, **CaMgtCfg**.***Forbid3ccUeRatioThld*** must be set to a value greater than **CaMgtCfg**.***Forbid4ccUeRatioThld***. ● If neither **CaMgtCfg**.***Forbid3ccUeRatioThld*** nor **CaMgtCfg**.***Forbid5ccUeRatioThld*** is set to **100%**, **CaMgtCfg**.***Forbid3ccUeRatioThld*** must be set to a value greater than **CaMgtCfg**.***Forbid5ccUeRatioThld***.

Parameter Name	Parameter ID	Setting Notes
Forbid 4CC UE Ratio Threshold	**CaMgtCfg**.***Forbid4ccUeRatioThld***	This parameter specifies the threshold expressed as the proportion of UEs served by a cell and that of UEs served by a BBP for prohibiting CA with four or more CCs aggregated from taking effect. ● A smaller value of this parameter results in a lower proportion of UEs with CA in effect, fewer BBP specifications being used by these UEs, and lower UE throughput on the entire network. ● A larger value of this parameter results in a higher proportion of UEs with CA in effect, more BBP specifications being used by these UEs, and higher UE throughput on the entire network. Configuration constraints are as follows: ● If neither **CaMgtCfg**.***Forbid2ccUeRatioThld*** nor **CaMgtCfg**.***Forbid4ccUeRatioThld*** is set to **100%**, **CaMgtCfg**.***Forbid2ccUeRatioThld*** must be set to a value greater than **CaMgtCfg**.***Forbid4ccUeRatioThld***. ● If neither **CaMgtCfg**.***Forbid3ccUeRatioThld*** nor **CaMgtCfg**.***Forbid4ccUeRatioThld*** is set to **100%**, **CaMgtCfg**.***Forbid3ccUeRatioThld*** must be set to a value greater than **CaMgtCfg**.***Forbid4ccUeRatioThld***. ● If neither **CaMgtCfg**.***Forbid4ccUeRatioThld*** nor **CaMgtCfg**.***Forbid5ccUeRatioThld*** is set to **100%**, **CaMgtCfg**.***Forbid4ccUeRatioThld*** must be set to a value greater than **CaMgtCfg**.***Forbid5ccUeRatioThld***.

Parameter Name	Parameter ID	Setting Notes
Forbid 5CC UE Ratio Threshold	**CaMgtCfg**.***Forbid5ccUeRatioThld***	This parameter specifies the threshold expressed as the proportion of UEs served by a cell and that of UEs served by a BBP for prohibiting CA with five or more CCs aggregated from taking effect. ● A smaller value of this parameter results in a lower proportion of UEs with CA in effect, fewer BBP specifications being used by these UEs, and lower UE throughput on the entire network. ● A larger value of this parameter results in a higher proportion of UEs with CA in effect, more BBP specifications being used by these UEs, and higher UE throughput on the entire network. Configuration constraints are as follows: ● If neither **CaMgtCfg**.***Forbid2ccUeRatioThld*** nor **CaMgtCfg**.***Forbid5ccUeRatioThld*** is set to **100%**, **CaMgtCfg**.***Forbid2ccUeRatioThld*** must be set to a value greater than **CaMgtCfg**.***Forbid5ccUeRatioThld***. ● If neither **CaMgtCfg**.***Forbid3ccUeRatioThld*** nor **CaMgtCfg**.***Forbid5ccUeRatioThld*** is set to **100%**, **CaMgtCfg**.***Forbid3ccUeRatioThld*** must be set to a value greater than **CaMgtCfg**.***Forbid5ccUeRatioThld***. ● If neither **CaMgtCfg**.***Forbid4ccUeRatioThld*** nor **CaMgtCfg**.***Forbid5ccUeRatioThld*** is set to **100%**, **CaMgtCfg**.***Forbid4ccUeRatioThld*** must be set to a value greater than **CaMgtCfg**.***Forbid5ccUeRatioThld***.

eNodeB-Level Load Balancing Parameters (TDD)

Set parameters in the **eNodeBMlb** MO.

Parameter Name	Parameter ID	Setting Notes
DL CA LB Max Component Carrier Number	**eNodeBMlb**.***DlCaLbMaxCCNum***	This parameter specifies the maximum number of downlink CCs aggregated for each CA UE whose load is to be transferred for load balancing. If this parameter is set to **2CC**, the eNodeB transfers the load of CA UEs in the downlink 2CC aggregation state. If this parameter is set to **3CC**, the eNodeB transfers the load of CA UEs in the downlink 2CC aggregation state and those in the downlink 3CC aggregation state. This parameter applies only to TDD.
CA UE Minimum Data Volume Ratio	**eNodeBMlb**.***CaUeMinDataVolRatio***	This parameter specifies the minimum required ratio (expressed as a percentage) of the data volume on a high-load CC to that on a low-load CC for each CA UE. This parameter applies only to TDD.

Cell Downlink Scheduling Parameters

Set parameters in the **CellDlschAlgo** MO.

Parameter Name	Parameter ID	Setting Notes
Downlink CA Schedule Strategy	**CellDlschAlgo**.***CaSchStrategy***	To treat CA UEs as VIP users, set this parameter to **DIFF_SCHEDULE**. Otherwise, set this parameter to **BASIC_SCHEDULE**. The scheduling method must be consistent between serving cells to prevent data transmission exceptions.
Ca Scc Doppler Measurement	**CellDlschAlgo**.***CaSccDopMeas***	To use frequency-selective scheduling in SCells, set this parameter to **FROMPCC**. Otherwise, set this parameter to **FROMSCC**.

Parameter Name	Parameter ID	Setting Notes
CA Scheduling Weight Factor	**CellDlschAlgo**.***CaSchWeight***	This parameter specifies the differentiated scheduling factor for CA UEs. ● The greater the value, the greater the effect of differentiated scheduling for CA and non-CA UEs. ● The smaller the value, the fairer the scheduling between CA and non-CA UEs.

RLC/PDCP Parameters

Set parameters in the **RlcPdcpParaGroup** MO.

Parameter Name	Parameter ID	Setting Notes
CA UE RLC Parameter Adaptive Threshold	**RlcPdcpParaGroup**.***CaUeRlcParaAdptiveThd***	Set this parameter to **10**.
CA UE Reordering Timer(ms)	**RlcPdcpParaGroup**.***CaUeReorderingTimer***	Set this parameter to **Treordering_m20**.
CA UE Status Prohibit Timer(ms)	**RlcPdcpParaGroup**.***CaUeStatProhTimer***	Set this parameter to **m20**.

Frame Offset

Set parameters in the **ENodeBFrameOffset** MO.

Parameter Name	Parameter ID	Setting Notes
FDD Frame Offset	**ENodeBFrameOffset**.***FddFrameOffset***	This parameter specifies the offset of the frame start time for all LTE FDD cells served by the eNodeB relative to the time of the reference clock. This parameter takes effect only when **CellFrameOffset**.***FrameOffset*** is not configured. Set this parameter to an identical value for the cells engaged in LTE FDD CA or LTE FDD+TDD CA. For details in LTE FDD and NR Uplink Spectrum Sharing scenarios, see *LTE FDD and NR SUL Dynamic Spectrum Sharing*. You are advised to set this parameter based on the network plan.
TDD Frame Offset	**ENodeBFrameOffset**.***TddFrameOffset***	This parameter specifies the offset of the frame start time for all LTE TDD cells served by the eNodeB relative to the time of the reference clock. This parameter takes effect only when **CellFrameOffset**.***FrameOffset*** is not configured. If uplink and downlink timeslots are not aligned between TDD systems, inter-system interference will occur. Operators can adjust this parameter to minimize the error in timeslot alignment between the TDD systems. You are advised to set this parameter based on the network plan. LTE TDD CA requires that an identical frame offset be set for all the cells engaged.

Set parameters in the **CellFrameOffset** MO.

Parameter Name	Parameter ID	Setting Notes
Frame Offset Mode	**CellFrameOffset**.***FrameOffsetMode***	This parameter specifies the type of the frame offset for the cell. There are three options: ● **CustomFrameOffset**: indicates a user-defined frame offset. This is the only value that applies to FDD cells. ● **TL_FrameOffset**: indicates the frame offset automatically determined based on the subframe configuration in a TD-SCDMA + LTE TDD dual-RAT network. ● **TL_FrameOffset_SA2_SSP5**: indicates the frame offset automatically determined based on uplink-downlink configuration 2 and special subframe configuration 5 in a TD-SCDMA + LTE TDD dual-RAT network. You are advised to set this parameter based on the network plan.
Frame Offset	**CellFrameOffset**.***FrameOffset***	This parameter specifies the offset of the frame start time for the cell relative to the time of the reference clock. You are advised to set this parameter based on the network plan. LTE TDD CA requires that an identical frame offset be set for all the cells engaged. Only the frame offset specified by **CellFrameOffset**.***FrameOffset*** takes effect if it is configured together with **ENodeBFrameOffset**.***FddFrameOffset*** or **ENodeBFrameOffset**.***TddFrameOffset***.

Compatibility Parameters

Set parameters in the **GlobalProcSwitch** MO.

Parameter Name	Parameter ID	Option	Setting Notes
Protocol Compatibility Switch	**GlobalProcSwitch**.***ProtocolCompatibilitySw***	CaGapMeasPriOffSwitch	Select this option if there are a large number of Samsung Galaxy S4 UEs on the network and blind SCell configuration is disabled.
Protocol Compatibility Switch	**GlobalProcSwitch**.***ProtocolCompatibilitySw***	CaHoReqWithR9ConfigSwitch	Select this option if the network equipment is provided by different vendors.
Protocol Compatibility Switch	**GlobalProcSwitch**.***ProtocolCompatibilitySw***	SCellModCaMeasRmvSwitch	Select this option if **SccModA6Switch** or **SccA2RmvSwitch** is on.
Protocol Compatibility Switch	**GlobalProcSwitch**.***ProtocolCompatibilitySw***	SCellBlindA2Switch	Select this option if the eNodeB needs to deliver A2 measurement configuration for SCells to a CA UE after configuring an SCell for the UE in CA-group-based configuration mode in a blind manner.

Set parameters in the **UeCompat** MO.

Parameter Name	Parameter ID	Option	Setting Notes
Black List Control Switch	**UeCompat**.***BlkLstCtrlSwitch***	CA_SWITCH_OFF	Select this option if some CA UEs or NSA UEs do not support carrier management for CA or NSA DC because of their software or hardware defects. With this option selected, these UEs are blacklisted so that the eNodeB will not perform PCC anchoring, SCell configuration, or SCG addition for them, preventing the UE incompatibility issue from affecting network performance. For details about carrier management for NSA DC, see *EPC-based NSA Basics*.

Parameter Name	Parameter ID	Option	Setting Notes
White List Private CA Band Combination	**UeCompat**.***WhiteLstCaCombSwitch***	None	This parameter specifies whether individual operator-defined band combinations take effect only for whitelisted UEs. ● If an option is selected, the specified band combination takes effect only for whitelisted UEs. For example, if **PRIVATECACOMBID_0** is selected, the operator-defined CA band combination with an ID of 0 takes effect only for whitelisted UEs. ● If an option is deselected, the specified band combination takes effect for all UEs.

Set parameters in the **ENodeBAlgoSwitch** MO.

Parameter Name	Parameter ID	Option	Setting Notes
Compatibility Control Switch	**ENodeBAlgoSwitch**.***CompatibilityCtrlSwitch***	ApCqiRptAbnormalCtrlSwitch	Select this option to mitigate failures of UEs to report aperiodic CQIs.
Compatibility Control Switch	**ENodeBAlgoSwitch**.***CompatibilityCtrlSwitch***	FddTddCaPcellDuplexFdd	Select this option if some CA UEs report band combinations for FDD+TDD CA but do not report PCell capability fields as stipulated in 3GPP specifications.
Compatibility Control Switch	**ENodeBAlgoSwitch**.***CompatibilityCtrlSwitch***	FddTddCaPcellDuplexTdd	Select this option if some CA UEs report band combinations for FDD+TDD CA but do not report PCell capability fields as stipulated in 3GPP specifications.

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	PdcchOverlapSrchSpcSwitch	Deselect this option if most CA UEs on the network do not comply with 3GPP TS 36.213 V10.9.0 or later specifications. This compliance violation is indicated by RRC connection reconfiguration failures at a large number of UEs. Select this option if most CA UEs comply with these specifications.

Set parameters in the **CellSrlte** MO.

Parameter Name	Parameter ID	Setting Notes
SRLTE Switch	**CellSrlte**.***SrlteSwitch***	Set this parameter to **ON** and set the **CaMgtCfg**.***SccQuietTime*** parameter properly to mitigate CA UE throughput decreases caused by SCell deactivation for Single Radio LTE (SRLTE) or Simultaneous LTE (SLTE) UEs. These UEs periodically check for signals from an inter-RAT system. During the check, the UEs do not receive signals from their SCells, causing sharp block error rate (BLER) increases in the SCells and then SCell deactivation.

Cell-specific WTTx Parameters

Set parameters in the **CellWttxParaCfg** MO.

Parameter Name	Parameter ID	Option	Setting Notes
WBB Performance Optimization Switch	**CellWttxParaCfg**.***WbbPerformanceOptSwitch***	DL_CA_SCC_CONFIG_ACT_OPT_SW	With this option selected, the eNodeB determines whether to allow the massive MIMO cell to be configured and activated as an SCell for CA UEs based on the uplink CA capabilities of the UEs. ● The massive MIMO cell cannot be configured or activated as an SCell for any CA UE that does not support uplink CA. ● The massive MIMO cell can be configured and activated as an SCell for CA UEs that support uplink CA.

Cell Operator-specific Handover Parameters

Set parameters in the **CellOpHoCfg** MO.

Parameter Name	Parameter ID	Option	Setting Notes
High Mobility UE Handover Forbidden Sw	**CellOpHoCfg**.***HighMobiUeHoForbidSw***	PCC_ANCHOR_HO_FORBID_SW	Select this option to enable PCC anchoring prohibition for high-mobility UEs. With this function enabled, PCC anchoring does not take effect for high-mobility UEs.

Parameters Related to Cells in Special Bands

- Cells in bands 29, 32, 67, 69, 75, and 76

  These cells are **DL_ONLY** cells, which transmit only downlink data. UEs can neither camp on the cells in idle mode nor access or be handed over to the cells in connected mode. These cells can serve only as SCells for CA UEs.

  Cells and sector equipment in each of these bands must be configured as follows:

  – **Cell**.***WorkMode*** must be set to **DL_ONLY**, **Cell**.***FreqBand*** must be set to **29** (take band 29 as an example), and **Cell**.***UlEarfcn*** must be left unconfigured. In addition, **SECTOREQM**.***ANTTYPE1*** (and **SECTOREQM**.***ANTTYPE2***, if two antennas are configured) must be set to **TX_MODE**.

- **EutranInterNFreq**.***AnrInd*** must be set to **NOT_ALLOWED** for frequencies in the band so that cells on these frequencies will not be managed by ANR. Therefore, for CA to take effect, operators must manually configure cells in the band as inter-frequency neighboring cells of candidate PCells that can operate with this band for CA.
- **EutranInterNFreq**.***MlbTargetInd*** must be set to **NOT_ALLOWED** for frequencies in the band so that they will not be treated as targets for MLB.

- Cells in band 46

  TDD licensed assisted access (LAA) cells are all deployed in band 46. Cells in this band can only serve as SCells. For details about LAA, see *Licensed Assisted Access (TDD)*.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before setting CA parameters, perform the following steps:

```
//Adding neighboring E-UTRA frequencies
ADD EUTRANINTERNFREQ: LocalCellId=0, DlEarfcn=37900, UlEarfcnCfgInd=NOT_CFG,
CellReselPriorityCfgInd=NOT_CFG, SpeedDependSPCfgInd=NOT_CFG, MeasBandWidth=MBW100,
PmaxCfgInd=NOT_CFG, QqualMinCfgInd=NOT_CFG;
ADD EUTRANINTERNFREQ: LocalCellId=1, DlEarfcn=38098, UlEarfcnCfgInd=NOT_CFG,
CellReselPriorityCfgInd=NOT_CFG, SpeedDependSPCfgInd=NOT_CFG, MeasBandWidth=MBW100,
PmaxCfgInd=NOT_CFG, QqualMinCfgInd=NOT_CFG;
//Optional; required only when there are inter-eNodeB inter-frequency E-UTRAN cells
ADD EUTRANEXTERNALCELL: Mcc="460", Mnc="20", eNodeBId=1234, CellId=1, DlEarfcn=37900,
UlEarfcnCfgInd=NOT_CFG, PhyCellId=101, Tac=1;
ADD EUTRANEXTERNALCELL: Mcc="460", Mnc="20", eNodeBId=1234, CellId=0, DlEarfcn=38098,
UlEarfcnCfgInd=NOT_CFG, PhyCellId=101, Tac=1;
//Optional; required only when neighboring E-UTRAN cells are working in RAN sharing mode
ADD EUTRANEXTERNALCELLPLMN: Mcc="460", Mnc="20", eNodeBId=1234, CellId=1, ShareMcc="460",
ShareMnc="22";
ADD EUTRANEXTERNALCELLPLMN: Mcc="460", Mnc="20", eNodeBId=1234, CellId=0, ShareMcc="460",
ShareMnc="22";
//Setting up inter-frequency neighbor relationships. In this step, set No remove indicator to
FORBID_RMV_ENUM. Otherwise, ANR may automatically remove neighbor relationships, in which case CA
does not work.
ADD EUTRANINTERFREQNCELL: LocalCellId=0, Mcc="460", Mnc="20", eNodeBId=1234, CellId=1,
NoRmvFlag=FORBID_RMV_ENUM;
ADD EUTRANINTERFREQNCELL: LocalCellId=1, Mcc="460", Mnc="20", eNodeBId=1234, CellId=0,
NoRmvFlag=FORBID_RMV_ENUM;
//Setting up intra-frequency neighbor relationships
ADD EUTRANINTRAFREQNCELL: LocalCellId=0, Mcc="460", Mnc="20", eNodeBId=1234, CellId=2;
ADD EUTRANINTRAFREQNCELL: LocalCellId=2, Mcc="460", Mnc="20", eNodeBId=1234, CellId=0;
```

5.4.1.2.1 CA-Group-based Configuration Mode (FDD)

Activation Command Examples

```
//Setting the options of the CA algorithm switch
MOD ENODEBALGOSWITCH:
CaAlgoSwitch=SccBlindCfgSwitch-1&FreqCfgSwitch-0&SccA2RmvSwitch-1&HoWithSccCfgSwitch-0&SccModA
6Switch-0&EnhancedPccAnchorSwitch-1;
MOD GLOBALPROCSWITCH: ProtocolCompatibilitySw=SCellBlindA2Switch-1&SCellModCaMeasRmvSwitch-1;
//Adding a CA group and setting its attribute
ADD CAGROUP: CaGroupId=0, CaGroupTypeInd=FDD;
//Adding cells to the CA group
ADD CAGROUPCELL: CaGroupId=0, LocalCellId=0, eNodeBId=1234, PreferredPCellPriority=2,
PCellA4RsrpThd=-105, PCellA4RsrqThd=-20;
ADD CAGROUPCELL: CaGroupId=0, LocalCellId=1, eNodeBId=1234, PreferredPCellPriority=1,
PCellA4RsrpThd=-105, PCellA4RsrqThd=-20;
//Configuring candidate SCells for the CA group. Ensure that each candidate SCell has been configured as
an inter-frequency neighboring cell of the corresponding PCell.
ADD CAGROUPSCELLCFG: LocalCellId=0, SCelleNodeBId=1234, SCellLocalCellId=1, SCellBlindCfgFlag=TRUE,
SCellPriority=2, SCellA4Offset=0, SCellA2Offset=0;
ADD CAGROUPSCELLCFG: LocalCellId=1, SCelleNodeBId=1234, SCellLocalCellId=0, SCellBlindCfgFlag=TRUE,
SCellPriority=3, SCellA4Offset=0, SCellA2Offset=0;
//Turning on CA_UE_MULTI_CC_STAT_OPT_SW
MOD ENODEBCOUNTERPARAGRP: EnodebCounterAlgoSwitch=CA_UE_MULTI_CC_STAT_OPT_SW-1;
//(Optional) Enabling PCC anchoring prohibition for high-mobility UEs if PCC anchoring is not expected for
high-mobility UEs (using operator 0 as an example)
MOD CELLOPHOCFG: LocalCellId=0, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-1;
MOD CELLOPHOCFG: LocalCellId=1, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-1;
//(Optional) Enabling UE-number-based SCell configuration control if required in the event of a large
number of UEs
MOD CAMGTCFG: LocalCellId=0, Forbid5ccUeRatioThld=60, Forbid4ccUeRatioThld=70,
Forbid3ccUeRatioThld=80, Forbid2ccUeRatioThld=90;
MOD CAMGTCFG: LocalCellId=1, Forbid5ccUeRatioThld=60, Forbid4ccUeRatioThld=70,
Forbid3ccUeRatioThld=80, Forbid2ccUeRatioThld=90;
//Activating the cells
ACT CELL: LocalCellId=0;
ACT CELL: LocalCellId=1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the subordinate switches of the CA algorithm switch
MOD ENODEBALGOSWITCH:
CaAlgoSwitch=SccBlindCfgSwitch-0&FreqCfgSwitch-0&SccA2RmvSwitch-0&HoWithSccCfgSwitch-0&SccModA
6Switch-0&EnhancedPccAnchorSwitch-0;
MOD GLOBALPROCSWITCH: ProtocolCompatibilitySw=SCellBlindA2Switch-0&SCellModCaMeasRmvSwitch-0;
//Turning off CA_UE_MULTI_CC_STAT_OPT_SW
MOD ENODEBCOUNTERPARAGRP: EnodebCounterAlgoSwitch=CA_UE_MULTI_CC_STAT_OPT_SW-0;
//(Optional) Disabling PCC anchoring prohibition for high-mobility UEs
MOD CELLOPHOCFG: LocalCellId=0, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-0;
MOD CELLOPHOCFG: LocalCellId=1, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-0;
//(Optional) Disabling UE-number-based SCell configuration control
MOD CAMGTCFG: LocalCellId=0, Forbid5ccUeRatioThld=100, Forbid4ccUeRatioThld=100,
Forbid3ccUeRatioThld=100, Forbid2ccUeRatioThld=100;
MOD CAMGTCFG: LocalCellId=1, Forbid5ccUeRatioThld=100, Forbid4ccUeRatioThld=100,
Forbid3ccUeRatioThld=100, Forbid2ccUeRatioThld=100;
//Removing cells from the CA group
RMV CAGROUPCELL: CaGroupId=0, LocalCellId=0, eNodeBId=1234;
RMV CAGROUPCELL: CaGroupId=0, LocalCellId=1, eNodeBId=1234;
//Removing the CA group
RMV CAGROUP: CaGroupId=0;
```

5.4.1.2.2 CA-Group-based Configuration Mode (TDD)

Activation Command Examples

```
//Setting the options of the CA algorithm switch
MOD ENODEBALGOSWITCH:
CaAlgoSwitch=SccBlindCfgSwitch-1&FreqCfgSwitch-0&SccA2RmvSwitch-1&HoWithSccCfgSwitch-0&SccModA
6Switch-0&EnhancedPccAnchorSwitch-1;
MOD GLOBALPROCSWITCH: ProtocolCompatibilitySw=SCellBlindA2Switch-1&SCellModCaMeasRmvSwitch-1;
//Adding a CA group and setting its attribute
ADD CAGROUP: CaGroupId=0, CaGroupTypeInd=TDD;
//Adding cells to the CA group
ADD CAGROUPCELL: CaGroupId=0, LocalCellId=0, eNodeBId=1234, PreferredPCellPriority=2,
PCellA4RsrpThd=-105, PCellA4RsrqThd=-20;
ADD CAGROUPCELL: CaGroupId=0, LocalCellId=1, eNodeBId=1234, PreferredPCellPriority=1,
PCellA4RsrpThd=-105, PCellA4RsrqThd=-20;
//Configuring candidate SCells for the CA group. Ensure that each candidate SCell has been configured as
an inter-frequency neighboring cell of the corresponding PCell.
ADD CAGROUPSCELLCFG: LocalCellId=0, SCelleNodeBId=1234, SCellLocalCellId=1, SCellBlindCfgFlag=TRUE,
SCellPriority=2, SCellA4Offset=0, SCellA2Offset=0;
ADD CAGROUPSCELLCFG: LocalCellId=1, SCelleNodeBId=1234, SCellLocalCellId=0, SCellBlindCfgFlag=TRUE,
SCellPriority=3, SCellA4Offset=0, SCellA2Offset=0;
//Turning on CA_UE_MULTI_CC_STAT_OPT_SW
MOD ENODEBCOUNTERPARAGRP: EnodebCounterAlgoSwitch=CA_UE_MULTI_CC_STAT_OPT_SW-1;
//(Optional) Enabling PCC anchoring prohibition for high-mobility UEs if PCC anchoring is not expected for
high-mobility UEs (using operator 0 as an example)
MOD CELLOPHOCFG: LocalCellId=0, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-1;
MOD CELLOPHOCFG: LocalCellId=1, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-1;
//(Optional) Enabling UE-number-based SCell configuration control if required in the event of a large
number of UEs
MOD CAMGTCFG: LocalCellId=0, Forbid5ccUeRatioThld=60, Forbid4ccUeRatioThld=70,
Forbid3ccUeRatioThld=80, Forbid2ccUeRatioThld=90;
MOD CAMGTCFG: LocalCellId=1, Forbid5ccUeRatioThld=60, Forbid4ccUeRatioThld=70,
Forbid3ccUeRatioThld=80, Forbid2ccUeRatioThld=90;
//Activating the cells
ACT CELL: LocalCellId=0;
ACT CELL: LocalCellId=1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the subordinate switches of the CA algorithm switch
MOD ENODEBALGOSWITCH:
CaAlgoSwitch=SccBlindCfgSwitch-0&FreqCfgSwitch-0&SccA2RmvSwitch-0&HoWithSccCfgSwitch-0&SccModA
6Switch-0&EnhancedPccAnchorSwitch-0;
MOD GLOBALPROCSWITCH: ProtocolCompatibilitySw=SCellBlindA2Switch-0&SCellModCaMeasRmvSwitch-0;
//Turning off CA_UE_MULTI_CC_STAT_OPT_SW
MOD ENODEBCOUNTERPARAGRP: EnodebCounterAlgoSwitch=CA_UE_MULTI_CC_STAT_OPT_SW-0;
//(Optional) Disabling PCC anchoring prohibition for high-mobility UEs
MOD CELLOPHOCFG: LocalCellId=0, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-0;
MOD CELLOPHOCFG: LocalCellId=1, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-0;
//(Optional) Disabling UE-number-based SCell configuration control
MOD CAMGTCFG: LocalCellId=0, Forbid5ccUeRatioThld=100, Forbid4ccUeRatioThld=100,
Forbid3ccUeRatioThld=100, Forbid2ccUeRatioThld=100;
MOD CAMGTCFG: LocalCellId=1, Forbid5ccUeRatioThld=100, Forbid4ccUeRatioThld=100,
Forbid3ccUeRatioThld=100, Forbid2ccUeRatioThld=100;
//Removing cells from the CA group
RMV CAGROUPCELL: CaGroupId=0, LocalCellId=0, eNodeBId=1234;
RMV CAGROUPCELL: CaGroupId=0, LocalCellId=1, eNodeBId=1234;
//Removing the CA group
RMV CAGROUP: CaGroupId=0;
```

5.4.1.2.3 Adaptive Configuration Mode (FDD)

Activation Command Examples

```
//Setting the options of the CA algorithm switch
MOD ENODEBALGOSWITCH:
CaAlgoSwitch=SccBlindCfgSwitch-0&FreqCfgSwitch-1&SccA2RmvSwitch-1&HoWithSccCfgSwitch-0&SccModA6Switch-1&AdpCaSwitch-1&EnhancedPccAnchorSwitch-1;
MOD GLOBALPROCSWITCH: ProtocolCompatibilitySw=SCellModCaMeasRmvSwitch-1;
//Adding candidate PCCs and setting their attributes
ADD PCCFREQCFG: PccDlEarfcn=123, PreferredPccPriority=1, PccA4RsrpThd=-105, PccA4RsrqThd=-20;
ADD PCCFREQCFG: PccDlEarfcn=456, PreferredPccPriority=2, PccA4RsrpThd=-105, PccA4RsrqThd=-20;
//Adding a candidate SCC for each candidate PCC and setting SCC attributes
ADD SCCFREQCFG: PccDlEarfcn=123, SccDlEarfcn=567, SccPriority=2, SccA2Offset=0, SccA4Offset=0,
CnOperatorList="111111";
ADD SCCFREQCFG: PccDlEarfcn=456, SccDlEarfcn=789, SccPriority=3, SccA2Offset=0, SccA4Offset=0,
CnOperatorList="111111";
//(Optional) Setting the blind SCell configuration flag
ADD CAGROUPSCELLCFG: LocalCellId=0, SCelleNodeBId=1234, SCellLocalCellId=1, SCellBlindCfgFlag=TRUE;
ADD CAGROUPSCELLCFG: LocalCellId=1, SCelleNodeBId=1234, SCellLocalCellId=0, SCellBlindCfgFlag=TRUE;
//Turning on FreqBaseCaLicAlarmSwitch
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=FreqBaseCaLicAlarmSwitch-1;
//(Optional) Turning on CaDl2CCExtSwitch for any pair of cells on two possible CCs if the aggregated
bandwidth of the two cells is between 20 MHz and 40 MHz (inclusive)
MOD CAMGTCFG: LocalCellId=0, SCellAgingTime=15, CellCaAlgoSwitch=CaDl2CCExtSwitch-1;
MOD CAMGTCFG: LocalCellId=1, SCellAgingTime=15, CellCaAlgoSwitch=CaDl2CCExtSwitch-1;
//Turning on CA_UE_MULTI_CC_STAT_OPT_SW
MOD ENODEBCOUNTERPARAGRP: EnodebCounterAlgoSwitch=CA_UE_MULTI_CC_STAT_OPT_SW-1;
//(Optional) Enabling PCC anchoring prohibition for high-mobility UEs if PCC anchoring is not expected for
high-mobility UEs (using operator 0 as an example)
MOD CELLOPHOCFG: LocalCellId=0, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-1;
MOD CELLOPHOCFG: LocalCellId=1, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-1;
//(Optional) Enabling UE-number-based SCell configuration control if required in the event of a large
number of UEs
MOD CAMGTCFG: LocalCellId=0, Forbid5ccUeRatioThld=60, Forbid4ccUeRatioThld=70,
Forbid3ccUeRatioThld=80, Forbid2ccUeRatioThld=90;
MOD CAMGTCFG: LocalCellId=1, Forbid5ccUeRatioThld=60, Forbid4ccUeRatioThld=70,
Forbid3ccUeRatioThld=80, Forbid2ccUeRatioThld=90;
//Activating the cells
ACT CELL: LocalCellId=0;
ACT CELL: LocalCellId=1;
//(Optional) Enabling adaptive handling of CA for MU beamforming. If a candidate SCell is a high-load
massive MIMO cell, it cannot be configured as an SCell for a CA UE that cannot participate in pairing.
MOD ENODEBALGOSWITCH:
CaAlgoSwitch=FreqCfgSwitch-1&AdpCaSwitch-1&CaSmartSelectionSwitch-0,CaAlgoExtSwitch=CaMubfPairingAdaptOptSwitch-1;
MOD CAMGTCFG: LocalCellId=0, HLUeCntThldForScellConfig=100,
HighLoadCellTypeNotAsScell=MASSIVE_MIMO_CELL;
//(Optional) Enabling adaptive handling of CA for MU beamforming. If a candidate SCell is a high-load
normal cell, the cell cannot be configured as an SCell for CA UEs.
MOD ENODEBALGOSWITCH: CaAlgoSwitch=FreqCfgSwitch-1&AdpCaSwitch-1&CaSmartSelectionSwitch-0,
CaAlgoExtSwitch=CaMubfPairingAdaptOptSwitch-1;
MOD CAMGTCFG: LocalCellId=0, HLUeCntThldForScellConfig=100,
HighLoadCellTypeNotAsScell=NORMAL_CELL;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the subordinate switches of the CA algorithm switch
MOD ENODEBALGOSWITCH:
CaAlgoSwitch=SccBlindCfgSwitch-0&FreqCfgSwitch-0&SccA2RmvSwitch-0&HoWithSccCfgSwitch-0&SccModA6Switch-0&AdpCaSwitch-0&EnhancedPccAnchorSwitch-0;
```

```
MOD GLOBALPROCSWITCH: ProtocolCompatibilitySw=SCellModCaMeasRmvSwitch-0;
//Turning off CA_UE_MULTI_CC_STAT_OPT_SW
MOD ENODEBCOUNTERPARAGRP: EnodebCounterAlgoSwitch=CA_UE_MULTI_CC_STAT_OPT_SW-0;
//(Optional) Disabling PCC anchoring prohibition for high-mobility UEs
MOD CELLOPHOCFG: LocalCellId=0, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-0;
MOD CELLOPHOCFG: LocalCellId=1, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-0;
//(Optional) Disabling UE-number-based SCell configuration control
MOD CAMGTCFG: LocalCellId=0, Forbid5ccUeRatioThld=100, Forbid4ccUeRatioThld=100,
Forbid3ccUeRatioThld=100, Forbid2ccUeRatioThld=100;
MOD CAMGTCFG: LocalCellId=1, Forbid5ccUeRatioThld=100, Forbid4ccUeRatioThld=100,
Forbid3ccUeRatioThld=100, Forbid2ccUeRatioThld=100;
//(Optional) Disabling adaptive handling of CA for MU beamforming
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=CaMubfPairingAdaptOptSwitch-0;
//Removing SCC configurations
RMV SCCFREQCFG: PccDlEarfcn=123, SccDlEarfcn=567;
RMV SCCFREQCFG: PccDlEarfcn=456, SccDlEarfcn=789;
//Removing PCC configurations
RMV PCCFREQCFG: PccDlEarfcn=123;
RMV PCCFREQCFG: PccDlEarfcn=456;
```

5.4.1.2.4 Adaptive Configuration Mode (TDD)

Activation Command Examples

```
//Setting the options of the CA algorithm switch
MOD ENODEBALGOSWITCH:
CaAlgoSwitch=SccBlindCfgSwitch-0&FreqCfgSwitch-1&SccA2RmvSwitch-1&HoWithSccCfgSwitch-0&SccModA
6Switch-1&AdpCaSwitch-1&EnhancedPccAnchorSwitch-1;
MOD GLOBALPROCSWITCH: ProtocolCompatibilitySw=SCellModCaMeasRmvSwitch-1;
//Adding candidate PCCs and setting their attributes
ADD PCCFREQCFG: PccDlEarfcn=37900, PreferredPccPriority=1, PccA4RsrpThd=-105, PccA4RsrqThd=-20;
ADD PCCFREQCFG: PccDlEarfcn=38098, PreferredPccPriority=2, PccA4RsrpThd=-105, PccA4RsrqThd=-20;
//Adding a candidate SCC for each candidate PCC and setting SCC attributes
ADD SCCFREQCFG: PccDlEarfcn=37900, SccDlEarfcn=38098, SccPriority=2, SccA2Offset=0, SccA4Offset=0,
CnOperatorList="111111";
ADD SCCFREQCFG: PccDlEarfcn=38098, SccDlEarfcn=37900, SccPriority=3, SccA2Offset=0, SccA4Offset=0,
CnOperatorList="111111";
//(Optional) Setting the blind SCell configuration flag
ADD CAGROUPSCELLCFG: LocalCellId=0, SCelleNodeBId=1234, SCellLocalCellId=1, SCellBlindCfgFlag=TRUE;
ADD CAGROUPSCELLCFG: LocalCellId=1, SCelleNodeBId=1234, SCellLocalCellId=0, SCellBlindCfgFlag=TRUE;
//Turning on FreqBaseCaLicAlarmSwitch
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=FreqBaseCaLicAlarmSwitch-1;
//(Optional) Turning on CaDl2CCExtSwitch for any pair of cells on two possible CCs if the aggregated
bandwidth of the two cells is between 30 MHz and 40 MHz (inclusive)
MOD CAMGTCFG: LocalCellId=0, SCellAgingTime=15, CellCaAlgoSwitch=CaDl2CCExtSwitch-1;
MOD CAMGTCFG: LocalCellId=1, SCellAgingTime=15, CellCaAlgoSwitch=CaDl2CCExtSwitch-1;
//Turning on CA_UE_MULTI_CC_STAT_OPT_SW
MOD ENODEBCOUNTERPARAGRP: EnodebCounterAlgoSwitch=CA_UE_MULTI_CC_STAT_OPT_SW-1;
//(Optional) Enabling PCC anchoring prohibition for high-mobility UEs if PCC anchoring is not expected for
high-mobility UEs (using operator 0 as an example)
MOD CELLOPHOCFG: LocalCellId=0, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-1;
MOD CELLOPHOCFG: LocalCellId=1, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-1;
//(Optional) Enabling UE-number-based SCell configuration control if required in the event of a large
number of UEs
MOD CAMGTCFG: LocalCellId=0, Forbid5ccUeRatioThld=60, Forbid4ccUeRatioThld=70,
Forbid3ccUeRatioThld=80, Forbid2ccUeRatioThld=90;
MOD CAMGTCFG: LocalCellId=1, Forbid5ccUeRatioThld=60, Forbid4ccUeRatioThld=70,
Forbid3ccUeRatioThld=80, Forbid2ccUeRatioThld=90;
//Activating the cells
ACT CELL: LocalCellId=0;
ACT CELL: LocalCellId=1;
//(Optional) Enabling adaptive handling of CA for MU beamforming. If a candidate SCell is a high-load
massive MIMO cell, it cannot be configured as an SCell for a CA UE that cannot participate in pairing and a
maximum of one TDD high-load massive MIMO SCell can be added for CA UEs that can participate in
pairing.
```

```
MOD ENODEBALGOSWITCH:
CaAlgoSwitch=FreqCfgSwitch-1&AdpCaSwitch-1&CaSmartSelectionSwitch-0,CaAlgoExtSwitch=CaMubfPairin
gAdaptOptSwitch-1;
MOD CAMGTCFG: LocalCellId=0, HLUeCntThldForScellConfig=100,
HighLoadCellTypeNotAsScell=MASSIVE_MIMO_CELL;
MOD ENODEBRSVDPARAMEXT: RsvdSwParam0=RSVD_SW_PARAM0_BIT28-1;
//(Optional) Enabling adaptive handling of CA for MU beamforming. If a candidate SCell is a high-load
normal cell, the cell cannot be configured as an SCell for CA UEs.
MOD ENODEBALGOSWITCH: CaAlgoSwitch=FreqCfgSwitch-1&AdpCaSwitch-1&CaSmartSelectionSwitch-0,
CaAlgoExtSwitch=CaMubfPairingAdaptOptSwitch-1;
MOD CAMGTCFG: LocalCellId=0, HLUeCntThldForScellConfig=100,
HighLoadCellTypeNotAsScell=NORMAL_CELL;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the subordinate switches of the CA algorithm switch
MOD ENODEBALGOSWITCH:
CaAlgoSwitch=SccBlindCfgSwitch-0&FreqCfgSwitch-0&SccA2RmvSwitch-0&HoWithSccCfgSwitch-0&SccModA
6Switch-0&AdpCaSwitch-0&EnhancedPccAnchorSwitch-0;
MOD GLOBALPROCSWITCH: ProtocolCompatibilitySw=SCellModCaMeasRmvSwitch-0;
//Turning off CA_UE_MULTI_CC_STAT_OPT_SW
MOD ENODEBCOUNTERPARAGRP: EnodebCounterAlgoSwitch=CA_UE_MULTI_CC_STAT_OPT_SW-0;
//(Optional) Disabling PCC anchoring prohibition for high-mobility UEs
MOD CELLOPHOCFG: LocalCellId=0, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-0;
MOD CELLOPHOCFG: LocalCellId=1, CnOperatorId=0,
HighMobiUeHoForbidSw=PCC_ANCHOR_HO_FORBID_SW-0;
//(Optional) Disabling UE-number-based SCell configuration control
MOD CAMGTCFG: LocalCellId=0, Forbid5ccUeRatioThld=100, Forbid4ccUeRatioThld=100,
Forbid3ccUeRatioThld=100, Forbid2ccUeRatioThld=100;
MOD CAMGTCFG: LocalCellId=1, Forbid5ccUeRatioThld=100, Forbid4ccUeRatioThld=100,
Forbid3ccUeRatioThld=100, Forbid2ccUeRatioThld=100;
//(Optional) Disabling adaptive handling of CA for MU beamforming
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=CaMubfPairingAdaptOptSwitch-0;
//Removing SCC configurations
RMV SCCFREQCFG: PccDlEarfcn=37900, SccDlEarfcn=38098;
RMV SCCFREQCFG: PccDlEarfcn=38098, SccDlEarfcn=37900;
//Removing PCC configurations
RMV PCCFREQCFG: PccDlEarfcn=37900;
RMV PCCFREQCFG: PccDlEarfcn=38098;
```

5.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Counter Observation

If any of the counters listed in **Table 5-9** produces a non-zero value on a network that is serving CA UEs, downlink 2CC aggregation has taken effect in the network.

Table 5-9 Counters used to verify activation of downlink 2CC aggregation

Counter ID	Counter Name
1526728426	**L.Traffic.User.PCell.DL.Avg**
1526728516	**L.Traffic.User.PCell.DL.Max**
1526728427	**L.Traffic.User.SCell.DL.Avg**
1526728517	**L.Traffic.User.SCell.DL.Max**
1526728424	**L.ChMeas.PRB.DL.PCell.Used.Avg**
1526728425	**L.ChMeas.PRB.DL.SCell.Used.Avg**
1526737791	**L.Traffic.User.PCell.DL.Active.Avg**
1526737792	**L.Traffic.User.PCell.DL.Active.Max**

For the **L.Traffic.User.PCell.DL.Active.Avg** and **L.Traffic.User.PCell.DL.Active.Max** counters:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, these counters are stepped as long as an SCell is activated for a UE.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, these counters are stepped only when all SCells are activated for a UE.

For example, four carriers are configured but only two of them are active for a UE. That is, not all SCells are activated for the UE. In the presence of such a UE:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Traffic.User.PCell.DL.Active.Avg** and **L.Traffic.User.PCell.DL.Active.Max** counters are stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Traffic.User.PCell.DL.Active.Avg** and **L.Traffic.User.PCell.DL.Active.Max** counters do not count this UE in.

Message Tracing

After a CA UE accesses a cell, the eNodeB configures a cell that meets CA conditions as an SCell for the UE. When traffic conditions are met, the eNodeB activates this SCell.

You can use the MAE-Access or MML commands to verify SCell configuration and activation:

- MAE-Access
  - Observe the RRC_CONN_RECFG message traced on the Uu interface.
    - If the RRC_CONN_RECFG message contains the sCellToAddModList or SCellToAddModListExt IE, the SCell has been configured.
    - If the RRC_CONN_RECFG message contains the sCellToReleaseList or SCellToReleaseListExt IE, the SCell has been removed.
  - Observe the number of RBs and total TBS for the CA UE in the PCell and SCell. If the numbers are not zero, the SCell has been activated.
- MML commands

  Run the **DSP UEONLINEINFO** command to check the SCell configuration status of the CA UE and the cause of the last SCell configuration failure.

5.4.3 Network Monitoring

Performance Counters

- Monitoring in scenarios other than multi-operator core network (MOCN)

  Monitor the counters listed in **Table 5-10** and compare the results with the network plan to evaluate network performance.

  Specifically:
  - Monitor the **L.ChMeas.PRB.DL.Used.Avg** counter value before and after CA is enabled, to evaluate the impact of CA on the overall resource usage.
  - Calculate the average data rate of CA UEs by using the following formula: (**L.Thrp.bits.DL.CAUser** – **L.Thrp.bits.DL.LastTTI.CAUser**)/ **L.Thrp.Time.DL.RmvLastTTI.CAUser**. Then, compare this value with the data rate of non-CA UEs.
  - Calculate the downlink throughput of UEs in the downlink 2CC aggregation state (with the SCell activated) in a cell by using the following formula:

    Downlink throughput of UEs in the downlink 2CC aggregation state in a cell = Total downlink PDCP-layer traffic volume of these UEs/Total downlink PDCP-layer data transmission duration for these UEs

    Specifically:
    - Total downlink PDCP-layer traffic volume of UEs in the downlink 2CC aggregation state in a cell = Total downlink PDCP-layer traffic volume of UEs in the downlink CA state in the cell – Total downlink PDCP-layer traffic volume of all UEs in the downlink CA state except those in the downlink 2CC aggregation state in the cell
    - Total downlink PDCP-layer data transmission duration for UEs in the downlink 2CC aggregation state in a cell = Total downlink PDCP-layer data transmission duration for all UEs in the downlink CA state in the cell – Total downlink PDCP-layer data transmission duration for all UEs in the downlink CA state except those in the downlink 2CC aggregation state in the cell

For the counters related to "total downlink PDCP-layer traffic volume of all UEs in the downlink CA state except those in the downlink 2CC aggregation state in a cell" and "total downlink PDCP-layer data transmission duration for all UEs in the downlink CA state except those in the downlink 2CC aggregation state in a cell", their values vary as follows:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, these counters are stepped as long as the corresponding number of CCs are active for a UE.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, these counters are stepped only when the corresponding number of CCs are active with no SCell inactive for a UE.

For example, if only downlink 2CC aggregation and downlink 3CC aggregation are enabled in a cell, calculate the throughput of UEs in the downlink 2CC aggregation state in the cell using the following formula:

(**L.Thrp.bits.DL.CAUser – L.Thrp.bits.DL.3CC.CAUser**)/(**L.Thrp.Time.DL.CAUser – L.Thrp.Time.DL.3CC.CAUser**)

In this example, four carriers are configured but only three of them are active for a UE. That is, not all SCells are activated for the UE. Then:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Thrp.bits.DL.3CC.CAUser** and **L.Thrp.Time.DL.3CC.CAUser** counters are stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Thrp.bits.DL.3CC.CAUser** and **L.Thrp.Time.DL.3CC.CAUser** counters do not count this UE in.

– Calculate the service drop rate of CA UEs by using the following formula: **L.E-RAB.AbnormRel.CAUser**/(**L.E-RAB.AbnormRel.CAUser** + **L.E-RAB.NormRel.CAUser**) x 100%.
– Calculate the handover success rate of CA UEs by using the following formula: **L.HHO.ExecSuccOut.CAUser.PCC**/**L.HHO.ExecAttOut.CAUser.PCC** x 100%.
– Calculate the success rate of handovers from LTE FDD to LTE TDD for PCell changes of CA UEs by using the following formula: **L.HHO.InterFddTdd.ExecSuccOut.CAUser.PCC**/**L.HHO.InterFddTdd.ExecAttOut.CAUser.PCC** x 100%.

Table 5-10 Counters used to monitor performance of downlink 2CC aggregation (non-MOCN)

Counter ID	Counter Name
1526726740	**L.ChMeas.PRB.DL.Used.Avg**

Counter ID	Counter Name
1526728516	**L.Traffic.User.PCell.DL.Max**
1526728426	**L.Traffic.User.PCell.DL.Avg**
1526728517	**L.Traffic.User.SCell.DL.Max**
1526728427	**L.Traffic.User.SCell.DL.Avg**
1526728424	**L.ChMeas.PRB.DL.PCell.Used.Avg**
1526728425	**L.ChMeas.PRB.DL.SCell.Used.Avg**
1526737791	**L.Traffic.User.PCell.DL.Active.Avg**
1526737792	**L.Traffic.User.PCell.DL.Active.Max**
1526728564	**L.Thrp.bits.DL.CAUser**
1526728565	**L.Thrp.Time.DL.CAUser**
1526728514	**L.E-RAB.AbnormRel.CAUser**
1526728515	**L.E-RAB.NormRel.CAUser**
1526728520	**L.HHO.ExecSuccOut.CAUser.PCC**
1526728519	**L.HHO.ExecAttOut.CAUser.PCC**
1526728518	**L.HHO.PrepAttOut.CAUser.PCC**
1526729602	**L.HHO.InterFddTdd.PrepAttOut.CAUser.PCC**
1526729603	**L.HHO.InterFddTdd.ExecAttOut.CAUser.PCC**
1526729604	**L.HHO.InterFddTdd.ExecSuccOut.CAUser.PCC**
1526729045	**L.CA.DLSCell.Add.Att**
1526729046	**L.CA.DLSCell.Add.Succ**
1526729047	**L.CA.DLSCell.Rmv.Att**
1526729048	**L.CA.DLSCell.Rmv.Succ**
1526730592	**L.CA.DLSCell.Add.Meas.Att**
1526730593	**L.CA.DLSCell.Add.Meas.Succ**
1526730594	**L.CA.DLSCell.Rmv.Meas.Att**
1526730595	**L.CA.DLSCell.Rmv.Meas.Succ**
1526730590	**L.CA.DLSCell.Add.Blind.Att**
1526730591	**L.CA.DLSCell.Add.Blind.Succ**

Counter ID	Counter Name
1526730596	**L.CA.DLSCell.Mod.Att**
1526730597	**L.CA.DLSCell.Mod.Succ**
1526728999	**L.CA.DLSCell.Act.Att**
1526729000	**L.CA.DLSCell.Act.Succ**
1526729001	**L.CA.DLSCell.Deact.Att**
1526729002	**L.CA.DLSCell.Deact.Succ**
1526732658	**L.CA.Traffic.bits.DL.PCell**
1526729259	**L.CA.Traffic.bits.DL.SCell**
1526729003	**L.CA.DL.PCell.Act.Dur**
1526729004	**L.CA.DL.SCell.Act.Dur**
1526732656	**L.Traffic.User.SCell.Active.DL.Avg**
1526732657	**L.Traffic.User.SCell.Active.DL.Max**
1526767257	**L.CA.MeasRpts.InterEnb.Strongest.NCELL.A6.Num**

In addition, monitor the function subsets listed in **Table 5-11**, which are included in "Measurement of Cell Performance (Cell)", to collect CQI, MCS, and MAC statistics about PCells and SCells.

Table 5-11 Function subsets in "Measurement of Cell Performance (Cell)"

Function Subset	Function Subset Name
ChMeas.CQI.CA.PCell	PCell CQI Measurement
ChMeas.CQI.CA.SCell	SCell CQI Measurement
ChMeas.MCS.CA.PCell	PCell MCS Measurement
ChMeas.MCS.CA.SCell	SCell MCS Measurement
Traffic.MAC.CA.Cell	CA Cell MAC Measurement

- Monitoring in MOCN scenarios

  Monitor the counters listed in **Table 5-12** and compare the results with the network plan to evaluate network performance.

Table 5-12 Counters used to monitor performance of downlink 2CC aggregation (MOCN)

Counter ID	Counter Name
1526739796	**L.RB.DL.PCell.CAUsed.PLMN**
1526739797	**L.RB.DL.SCell.CAUsed.PLMN**
1526739766	**L.Traffic.User.PCell.DL.Avg.PLMN**
1526739767	**L.Traffic.User.SCell.DL.Avg.PLMN**
1526739743	**L.Thrp.bits.DL.CAUser.PLMN**
1526739744	**L.Thrp.Time.DL.CAUser.PLMN**
1526739798	**L.Traffic.bits.CA.DL.PCell.PLMN**
1526739799	**L.Traffic.bits.CA.DL.SCell.PLMN**
1526739800	**L.Traffic.bits.DL.PLMN**
1526739769	**L.E-RAB.AbnormRel.CAUser.PLMN**
1526739771	**L.E-RAB.NormRel.CAUser.PLMN**
1526739773	**L.CA.DLSCell.Add.Att.PLMN**
1526739774	**L.CA.DLSCell.Add.Succ.PLMN**
1526739775	**L.CA.DLSCell.Rmv.Att.PLMN**
1526739776	**L.CA.DLSCell.Rmv.Succ.PLMN**

Real-Time UE-Level Performance Monitoring on the MAE-Access

Start real-time monitoring tasks on both the PCell and SCell on the MAE-Access to check the performance of CA UEs. The UE-level monitoring items provided by the MAE-Access include:

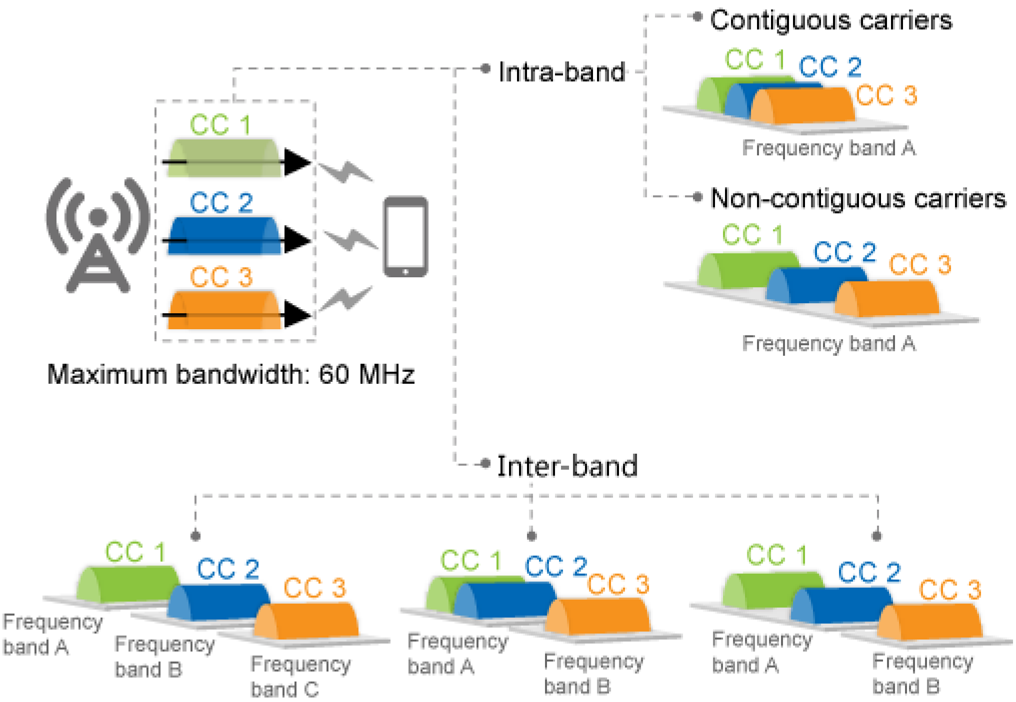
- BLER
- Quality of Channel
- Throughput
- DL Power Control
- MCS Count
- User Common Monitoring

6 Downlink 3CC Aggregation

6.1 Principles

This function aggregates three intra- or inter-band carriers, as shown in **Figure 6-1**, to provide higher bandwidth.

Figure 6-1 Downlink 3CC aggregation



This function works between intra-eNodeB cells, between inter-eNodeB cells in eNodeB coordination scenarios, and between inter-eNodeB cells in relaxed backhaul scenarios.

The switch control over this function varies as described in **Table 6-1**.

Table 6-1 Switch setting requirements of downlink 3CC aggregation

Configuration Mode	Max. Downlink Bandwidth of Three CCs	CaDl3CCSwitch of CaMgtCfg.*CellCaAlgoSwitch*	CaDl3CCExtSwitch of CaMgtCfg.*CellCaAlgoSwitch*
CA-group-based	Up to 60 MHz	Select this option.	No requirements
Adaptive (FDD)	Up to 40 MHz	Select this option.	No requirements
Adaptive (FDD)	40 MHz < Bandwidth ≤ 60 MHz	Select this option.	Select this option.
Adaptive (TDD)	Up to 60 MHz	Select this option.	No requirements

A UE in the downlink 3CC aggregation state performs simultaneous transmission of HARQ ACKs/NACKs and periodic CQI report multiplexing for the three CCs if the **CqiAdaptiveCfg.*SimulAckNackAndCqiFmt3Sw*** parameter is set to **ON**. If this parameter is set to **OFF**, the ACKs/NACKs for the three CCs cannot be transmitted simultaneously with periodic CQI reports for any of the CCs.

6.2 Network Analysis

6.2.1 Benefits

This function enables CA UEs to reach higher downlink peak data rates.

Table 6-2 lists the theoretical peak data rates that a CA UE can reach using downlink 3CC aggregation.

For FDD, these values assume a TBS suitable for the 20 MHz cell bandwidth (equivalent to 100 RBs in the frequency domain).

For TDD, these values assume a TBS suitable in the following scenario:

- Bandwidth: 20 MHz (equivalent to 100 RBs in the frequency domain)
- Uplink-downlink subframe configuration: 2
- Special subframe configuration: 7

Table 6-2 Theoretical peak data rates for downlink 3CC aggregation (unit: Mbit/s)

RAT	2x2 MIMO + 64QAM	2x2 MIMO + 256QAM	4x4 MIMO + 64QAM	4x4 MIMO + 256QAM
FDD	449.3	587.4	899.6	1175.0
TDD	336	429	651	858

The peak data rate that CA can achieve for a CA UE is subject to:

- Peak data rate capability of the board where the PCell for the CA UE is located

  For example, if the PCell and SCells of a CA UE are served by an LBBPd1 board that supports a downlink peak data rate of 450 Mbit/s, the peak data rate that CA can achieve for the CA UE will not exceed 450 Mbit/s.
- Capability of the CA UE

  If the UE capability is limited, the actual peak data rates will be lower than the theoretical values. The UE capability is indicated by ue-categoryDL. For details about this IE, see section 4.1A "ue-CategoryDL and ue-CategoryUL" in 3GPP TS 36.306 V15.2.0.

6.2.2 Impacts

The impacts between this function and other functions are described in **6.3.3.3 Function Impacts**.

This function requires additional RBs for PUCCH format-3 overhead. Therefore, the downlink IBLER of the PCell fluctuates. The impact of this function on the PUCCH overhead varies as follows:

- Scenario: The **PucchSwitch** option of the **CellAlgoSwitch.*PucchAlgoSwitch*** parameter is selected to enable adaptive allocation of PUCCH resources.

  The cell spares one additional RB for PUCCH format-3 overhead. As a result, the number of RBs available for the PUSCH decreases by at least one. (This number must be a multiple of 2, 3, or 5. For details, see section 5.3.3 "Transform precoding" in 3GPP TS 36.211 V10.1.0 (2011-03).) Total uplink throughput decreases.
- Scenario: The **PucchSwitch** option of the **CellAlgoSwitch.*PucchAlgoSwitch*** parameter is deselected to use fixed allocation of PUCCH resources.

  For FDD, the cell changes the usage of one PUCCH RB from periodic CQI reporting to PUCCH format-3 overhead. As a result, fewer RBs of the cell are used for periodic CQI reporting, and more UEs have to use aperiodic CQI reporting. Downlink UE throughput may slightly decrease.

  For TDD, the cell changes the usage of two PUCCH RBs from periodic CQI reporting to PUCCH format-3 overhead. As a result, fewer RBs of the cell are used for periodic CQI reporting, and more UEs have to use aperiodic CQI reporting. Downlink UE throughput may slightly decrease.

6.3 Requirements

6.3.1 Licenses (FDD)

- LAOFD-080207 Carrier Aggregation for Downlink 3CC in 40MHz
  - Adaptive configuration mode
    i. An eNodeB identifies all its served cells on the PCC frequencies configured on the eNodeB.
    ii. If the **CaDl3CCSwitch** option of the **CaMgtCfg.*CellCaAlgoSwitch*** parameter is selected for a cell on a PCC frequency, the cell consumes one sales unit of the license for LAOFD-080207 Carrier Aggregation for Downlink 3CC in 40MHz.

iii. If the **CaDl3CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for at least one cell on a PCC frequency, each CA cell on the SCC frequencies configured for the PCC frequency consumes one sales unit of the license for LAOFD-080207 Carrier Aggregation for Downlink 3CC in 40MHz.

NOTE

When inter-eNodeB CA based on relaxed backhaul or inter-eNodeB CA based on eNodeB coordination is enabled, SCells may be served by neighboring eNodeBs. If this is the case, these cells consume the license for LAOFD-080207 Carrier Aggregation for Downlink 3CC in 40MHz at the neighboring eNodeBs.

– CA-group-based configuration mode

i. An eNodeB identifies all the cells in the CA groups configured on the eNodeB.

ii. If a CA group consists of at least three inter-frequency FDD cells, and if the **CaDl3CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for any of the cells in the group or the **CaDl4CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for any TDD cell in the group, then each cell in the CA group consumes one sales unit of the license for LAOFD-080207 Carrier Aggregation for Downlink 3CC in 40MHz.

- LAOFD-080208 Carrier Aggregation for Downlink 3CC in 60MHz
  - Adaptive configuration mode

    If the **CaDl3CCExtSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for a cell, the cell consumes one sales unit of the license for LAOFD-080208 Carrier Aggregation for Downlink 3CC in 60MHz.
  - CA-group-based configuration mode

    An eNodeB identifies all the cells in the CA groups configured on the eNodeB. Each cell in a CA group that meets all the following conditions consumes one sales unit of the license for LAOFD-080208 Carrier Aggregation for Downlink 3CC in 60MHz:

    - The CA group consists of at least three inter-frequency FDD cells.
    - The **CaDl3CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for any of the cells in the CA group, or the **CaDl4CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for any TDD cell in the CA group.
    - The CA group contains a combination of three cells that exceeds 40 MHz total bandwidth.

Table 6-3 lists the license models and sales units for these features.

Table 6-3 License models and sales units

Feature ID	Feature Name	Model	Sales Unit
LAOFD-080207	Carrier Aggregation for Downlink 3CC in 40MHz	LT1SCAD40M00	per cell
LAOFD-080208	Carrier Aggregation for Downlink 3CC in 60MHz	LT1SCAD60M00	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.3.2 Licenses (TDD)

Each TDD cell involved in downlink 3CC aggregation consumes one sales unit of the license for TDLAOFD-081405 Carrier Aggregation for Downlink 3CC.

Table 6-4 lists the license model and sales unit for the feature.

Table 6-4 License model and sales unit

Feature ID	Feature Name	Model	Sales Unit
TDLAOFD-081405	Carrier Aggregation for Downlink 3CC	LT1SCAD3CC00	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.3.3 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.3.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
FDD TDD	Downlink 2CC aggregation	None	**5 Downlink 2CC Aggregation**

6.3.3.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
FDD	PUCCH measurement	**PucchMeasOptSwitch** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter	*SFN*

6.3.3.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
FDD	UMTS and LTE Zero Bufferzone	**UMTS_LTE_ZERO_BUFFER_ZONE_SW** option of the **ULZeroBufferZone**.***ZeroBufZoneSwitch*** parameter	*UMTS and LTE Zero Bufferzone*	There are fewer PUSCH and SRS resources in a cell in the bufferzone than in a common cell. Therefore, when the LTE bandwidth is 5 MHz or 10 MHz, using a cell in the bufferzone as a PCell for CA is not recommended. If the cell is used as a PCell, CA performance deteriorates.
FDD	UMTS and LTE Spectrum Sharing (LTE FDD)	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **UL_SPECTRUM_SHARING**	*UMTS and LTE Spectrum Sharing*	Cells with 5 MHz bandwidth are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured.

RAT	Function Name	Function Switch	Reference	Description
FDD	UMTS and LTE Spectrum Sharing Based on DC-HSDPA	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **DC_HSDPA_BASED_UL_SPECTRUM_SHR**	*UMTS and LTE Spectrum Sharing Based on DC-HSDPA*	Cells with 5 MHz bandwidth are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured.
FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_SPECTRUM_SHR**	*LTE FDD and NR Dynamic Spectrum Sharing*	When LTE FDD and NR Flash Dynamic Spectrum Sharing and downlink 3CC aggregation are both enabled, LTE UEs cannot access the network if the percentage of resources to be preferentially allocated to LTE is set to 0.
FDD	Hybrid DSS Based on Asymmetric Bandwidth	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_SPECTRUM_SHR**, and the **LNR_SPECTRUM_SHR_ASYM_SW** option of the **SpectrumCloud**.***SpectrumCloudEnhSwitch*** parameter selected	*LTE FDD and NR Hybrid Dynamic Spectrum Sharing*	When Hybrid DSS Based on Asymmetric Bandwidth and downlink 3CC aggregation are both enabled, LTE UEs cannot access the network if the percentage of resources to be preferentially allocated to LTE is set to 0.

RAT	Function Name	Function Switch	Reference	Description
TDD	SRS resource migration	**REMOTE_INTRF_BF_ENH_SW** option of the **UlInterfSuppressCfg**.***RemoteIntrfDlEnhSwitch*** parameter	*Interference Detection and Suppression*	When the conditions for the SRS resource migration function to take effect and those for downlink 3CC aggregation are met at the same time, the SRS resource migration function takes precedence. Details are as follows: • When the SRS resource migration function takes effect, downlink 3CC aggregation stops taking effect. • When the SRS resource migration function stops taking effect, downlink 3CC aggregation becomes taking effect.

6.3.4 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

For details, see **Boards** in **5.3.4 Hardware**.

RF Modules

For details, see **RF Modules** in **5.3.4 Hardware**.

6.3.5 Networking

For details, see **5.3.5 Networking**.

6.3.6 Others

- UEs

  UEs must comply with 3GPP Release 12 or later and support the frequency bands of the carriers to be aggregated and their bandwidths. UEs must also support the peak data rates that CA can achieve.
- EPC

  For this function to reach a theoretical peak data rate described in **6.2.1 Benefits**, the maximum bit rate that each UE subscribes to in the EPC cannot be lower than this theoretical value.

6.4 Operation and Maintenance

6.4.1 Data Configuration (FDD)

6.4.1.1 Data Preparation

This function works in either CA-group-based or adaptive configuration mode. Prepare basic data as described in **5.4.1.1 Data Preparation**.

In addition, for either mode, prepare data as described in **Table 6-5** for function activation.

Table 6-5 Parameters for activating downlink 3CC aggregation

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg.*CellCaAlgoSwitch***	CaDl3CCSwitch	Select this option.

Moreover, for adaptive CA, prepare data as described in **Table 6-6** for each possible set of PCell and SCells on carriers whose aggregated bandwidth is between 40 MHz and 60 MHz (inclusive).

Table 6-6 Parameters for activating downlink 3CC aggregation

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg.*CellCaAlgoSwitch***	CaDl3CCExtSwitch	Select this option.

Moreover, take the following necessary actions for this function to take effect:

- Prepare data as described in **16.4.1.1 Data Preparation** if downlink 3CC aggregation is to be deployed in a relaxed backhaul scenario.
- Prepare data as described in **17.4.1.1 Data Preparation** if downlink 3CC aggregation is to be deployed in an eNodeB coordination scenario.

6.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

6.4.1.2.1 CA-Group-based Configuration Mode

Activation Command Examples

Before activating this function, add at least three cells to the CA group and configure candidate SCells for each candidate PCell according to **5.4.1.2.1 CA-Group-based Configuration Mode (FDD)**.

The activation command examples for this function are as follows:

```
//Turning on CaDl3CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1;
```

Deactivation Command Examples

```
//Turning off CaDl3CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-0;
```

6.4.1.2.2 Adaptive Configuration Mode

Activation Command Examples

Before activating this function, configure at least two candidate SCCs for each candidate PCC according to **5.4.1.2.3 Adaptive Configuration Mode (FDD)**.

The activation command examples for this function are as follows:

```
//Turning on CaDl3CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1;
//(Optional) Turning on CaDl3CCExtSwitch for any set of cells on three possible CCs if the aggregated bandwidth of all these cells is between 40 MHz and 60 MHz (inclusive)
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCExtSwitch-1;
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=CaDl3CCExtSwitch-1;
MOD CAMGTCFG: LocalCellId=2, CellCaAlgoSwitch=CaDl3CCExtSwitch-1;
```

Deactivation Command Examples

```
//Turning off CaDl3CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-0;
```

6.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.
- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Data Configuration (TDD)

6.4.2.1 Data Preparation

This function works in either CA-group-based or adaptive configuration mode. Prepare basic data as described in **5.4.1.1 Data Preparation**.

In addition, for either mode, prepare data as described in **Table 6-7** for function activation.

Table 6-7 Parameters for activating downlink 3CC aggregation

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	CaDl3CCSwitch	Select this option.

Moreover, take the following necessary actions for this function to take effect:

- Prepare data as described in **16.4.1.1 Data Preparation** if downlink 3CC aggregation is to be deployed in a relaxed backhaul scenario.
- Prepare data as described in **17.4.1.1 Data Preparation** if downlink 3CC aggregation is to be deployed in an eNodeB coordination scenario.

6.4.2.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

6.4.2.2.1 CA-Group-based Configuration Mode

Activation Command Examples

Before activating this function, add at least three cells to the CA group and configure candidate SCells for each candidate PCell according to **5.4.1.2.2 CA-Group-based Configuration Mode (TDD)**.

The activation command examples for this function are as follows:

```
//Turning on CaDl3CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1;
```

Deactivation Command Examples

```
//Turning off CaDl3CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-0;
```

6.4.2.2.2 Adaptive Configuration Mode

Activation Command Examples

Before activating this function, configure at least two candidate SCCs for each candidate PCC according to **5.4.1.2.4 Adaptive Configuration Mode (TDD)**.

The activation command examples for this function are as follows:

```
//Turning on CaDl3CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1;
```

Deactivation Command Examples

```
//Turning off CaDl3CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-0;
```

6.4.2.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.3 Activation Verification

Counter Observation

If any of the counters listed in **Table 6-8** produces a non-zero value on a network that is serving CA UEs capable of downlink 3CC aggregation, downlink 3CC aggregation has taken effect in the network.

Table 6-8 Counters used to verify activation of downlink 3CC aggregation

Counter ID	Counter Name
1526732907	**L.Traffic.User.PCell.DL.3CC.Avg**
1526732908	**L.Traffic.User.PCell.DL.3CC.Max**
1526732915	**L.Traffic.User.PCell.DL.3CC.Active.Avg**
1526732916	**L.Traffic.User.PCell.DL.3CC.Active.Max**

For the **L.Traffic.User.PCell.DL.3CC.Active.Avg** and **L.Traffic.User.PCell.DL.3CC.Active.Max** counters:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, these counters are stepped as long as three carriers are active for a UE.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, these counters are stepped only when three carriers are active with no SCell inactive for a UE.

For example, four carriers are configured but only three of them are active for a UE. That is, not all SCells are activated for the UE. In the presence of such a UE:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Traffic.User.PCell.DL.3CC.Active.Avg** and **L.Traffic.User.PCell.DL.3CC.Active.Max** counters are stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Traffic.User.PCell.DL.3CC.Active.Avg** and **L.Traffic.User.PCell.DL.3CC.Active.Max** counters do not count this UE in.

Message Tracing

For the tools and the IEs to observe, see **Message Tracing** in **5.4.2 Activation Verification**.

6.4.4 Network Monitoring

In addition to the counters listed in **5.4.3 Network Monitoring**, monitor the counters in **Table 6-9** and compare the results with the network plan to evaluate network performance.

Calculate the throughput of UEs in the downlink 3CC aggregation state by using the following formula: **L.Thrp.bits.DL.3CC.CAUser**/**L.Thrp.Time.DL.3CC.CAUser**. Use the **L.CA.DL.PCell.3CC.Act.Dur** counter to observe the total downlink active duration of UEs in the 3CC aggregation state.

For the **L.Thrp.bits.DL.3CC.CAUser**, **L.Thrp.Time.DL.3CC.CAUser**, and **L.CA.DL.PCell.3CC.Act.Dur** counters:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, these counters are stepped as long as three carriers are active for a UE.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, these counters are stepped only when three carriers are active with no SCell inactive for a UE.

For example, four carriers are configured but only three of them are active for a UE. That is, not all SCells are activated for the UE. In the presence of such a UE:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Thrp.bits.DL.3CC.CAUser**, **L.Thrp.Time.DL.3CC.CAUser**, and **L.CA.DL.PCell.3CC.Act.Dur** counters are stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Thrp.bits.DL.3CC.CAUser**, **L.Thrp.Time.DL.3CC.CAUser**, and **L.CA.DL.PCell.3CC.Act.Dur** counters do not count this UE in.

Table 6-9 Counters used to monitor performance of downlink 3CC aggregation

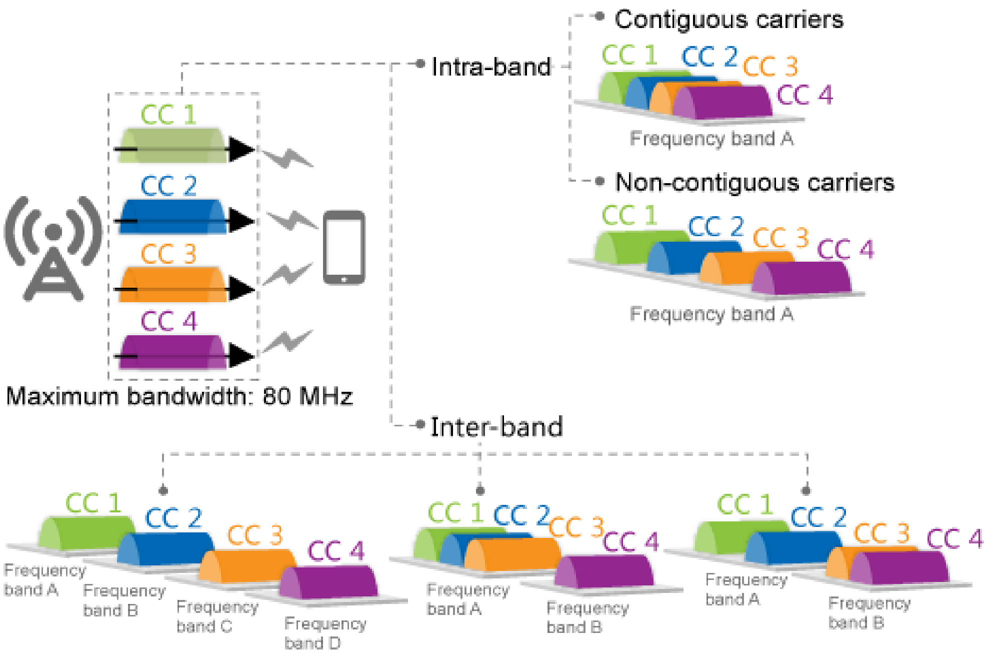
Counter ID	Counter Name
1526732907	**L.Traffic.User.PCell.DL.3CC.Avg**
1526732908	**L.Traffic.User.PCell.DL.3CC.Max**
1526732915	**L.Traffic.User.PCell.DL.3CC.Active.Avg**
1526732916	**L.Traffic.User.PCell.DL.3CC.Active.Max**
1526732917	**L.CA.DL.PCell.3CC.Act.Dur**
1526737809	**L.Thrp.Time.DL.3CC.CAUser**
1526733012	**L.Thrp.bits.DL.3CC.CAUser**

7 Downlink 4CC Aggregation

7.1 Principles

This function aggregates four intra- or inter-band carriers, as shown in **Figure 7-1**, to provide higher bandwidth.

Figure 7-1 Downlink 4CC aggregation



This function works between intra-eNodeB cells, between inter-eNodeB cells in eNodeB coordination scenarios, and between inter-eNodeB cells in relaxed backhaul scenarios.

This function is controlled by the **CaDl4CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter.

A UE in the downlink 4CC aggregation state performs simultaneous transmission of HARQ ACKs/NACKs and periodic CQI report multiplexing for the four CCs if the

CqiAdaptiveCfg.*SimulAckNackAndCqiFmt3Sw* parameter is set to **ON**. If this parameter is set to **OFF**, the ACKs/NACKs for the four CCs cannot be transmitted simultaneously with periodic CQI reports for any of the CCs.

7.2 Network Analysis

7.2.1 Benefits

This function enables CA UEs to reach higher downlink peak data rates.

Table 7-1 lists the theoretical peak data rates that a CA UE can reach using downlink 4CC aggregation.

For FDD, these values assume a TBS suitable for the 20 MHz cell bandwidth (equivalent to 100 RBs in the frequency domain).

For TDD, these values assume a TBS suitable in the following scenario:

- Bandwidth: 20 MHz (equivalent to 100 RBs in the frequency domain)
- Uplink-downlink subframe configuration: 2
- Special subframe configuration: 7

Table 7-1 Theoretical peak data rates for downlink 4CC aggregation (unit: Mbit/s)

RAT	2x2 MIMO + 64QAM	2x2 MIMO + 256QAM	4x4 MIMO + 64QAM	4x4 MIMO + 256QAM
FDD	599.1	783.3	1199.4	1566.6
TDD	448	572	868	1144

The peak data rate that CA can achieve for a CA UE is subject to:

- Peak data rate capability of the board where the PCell for the CA UE is located

  For example, if the PCell and SCells of a CA UE are served by an LBBPd1 board that supports a downlink peak data rate of 450 Mbit/s, the peak data rate that CA can achieve for the CA UE will not exceed 450 Mbit/s.
- Capability of the CA UE

  If the UE capability is limited, the actual peak data rates will be lower than the theoretical values. The UE capability is indicated by ue-categoryDL. For details about this IE, see section 4.1A "ue-CategoryDL and ue-CategoryUL" in 3GPP TS 36.306 V15.2.0.

7.2.2 Impacts

The impacts between this function and other functions are described in **7.3.3.3 Function Impacts**.

This function requires additional RBs for PUCCH format-3 overhead. Therefore, the downlink IBLER of the PCell fluctuates. The impact of this function on the PUCCH overhead varies as follows:

- Scenario: The **PucchSwitch** option of the **CellAlgoSwitch.*PucchAlgoSwitch*** parameter is selected to enable adaptive allocation of PUCCH resources.

  The cell spares one additional RB for PUCCH format-3 overhead. As a result, the number of RBs available for the PUSCH decreases by at least one. (This number must be a multiple of 2, 3, or 5. For details, see section 5.3.3 "Transform precoding" in 3GPP TS 36.211 V10.1.0 (2011-03).) Total uplink throughput decreases.

- Scenario: The **PucchSwitch** option of the **CellAlgoSwitch.*PucchAlgoSwitch*** parameter is deselected to use fixed allocation of PUCCH resources.

  For FDD, the cell changes the usage of one PUCCH RB from periodic CQI reporting to PUCCH format-3 overhead. As a result, fewer RBs of the cell are used for periodic CQI reporting, and more UEs have to use aperiodic CQI reporting. Downlink UE throughput may slightly decrease.

  For TDD, the cell changes the usage of two PUCCH RBs from periodic CQI reporting to PUCCH format-3 overhead. As a result, fewer RBs of the cell are used for periodic CQI reporting, and more UEs have to use aperiodic CQI reporting. Downlink UE throughput may slightly decrease.

7.3 Requirements

7.3.1 Licenses (FDD)

LEOFD-110303 Carrier Aggregation for Downlink 4CC and 5CC

- Adaptive configuration mode
  a. An eNodeB identifies all the cells on the PCC frequencies configured on the eNodeB.
  b. If the **CaDl4CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for a cell on a PCC frequency, the cell consumes one sales unit of the license for LEOFD-110303 Carrier Aggregation for Downlink 4CC and 5CC.
  c. If the **CaDl4CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for at least one cell on a PCC frequency, each CA cell on the SCC frequencies configured for the PCC frequency consumes one sales unit of the license for LEOFD-110303 Carrier Aggregation for Downlink 4CC and 5CC.

  NOTE

  When inter-eNodeB CA based on relaxed backhaul or inter-eNodeB CA based on eNodeB coordination is enabled, SCells may be served by neighboring eNodeBs. If this is the case, these cells consume the license for LEOFD-110303 Carrier Aggregation for Downlink 4CC and 5CC at the neighboring eNodeBs.

- CA-group-based configuration mode
  – An eNodeB identifies all the cells in the CA groups configured on the eNodeB.
  – If a CA group consists of at least four inter-frequency FDD cells and the **CaDl4CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for any of the cells in the group, each cell in the CA group

consumes one sales unit of the license for LEOFD-110303 Carrier Aggregation for Downlink 4CC and 5CC.

Table 7-2 lists the license model and sales unit for the feature.

Table 7-2 License model and sales unit

Feature ID	Feature Name	Model	Sales Unit
LEOFD-110303	Carrier Aggregation for Downlink 4CC and 5CC	LT1SCAD4A5CC	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.3.2 Licenses (TDD)

Each TDD cell involved in downlink 4CC aggregation consumes one sales unit of the license for TDLEOFD-081504 Carrier Aggregation for Downlink 4CC and 5CC.

Table 7-3 lists the license model and sales unit for the feature.

Table 7-3 License model and sales unit

Feature ID	Feature Name	Model	Sales Unit
TDLEOFD-081504	Carrier Aggregation for Downlink 4CC and 5CC	LT1SCAD4CC00	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.3.3 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.3.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
FDD TDD	Downlink 2CC aggregation	None	**5 Downlink 2CC Aggregation**
FDD TDD	Downlink 3CC aggregation	**CaDl3CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**6 Downlink 3CC Aggregation**

7.3.3.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
FDD	PUCCH measurement	**PucchMeasOptSwitch** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter	*SFN*

7.3.3.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
FDD	UMTS and LTE Zero Bufferzone	**UMTS_LTE_ZERO_BUFFER_ZONE_SW** option of the **ULZeroBufferZone**.***ZeroBufZoneSwitch*** parameter	*UMTS and LTE Zero Bufferzone*	There are fewer PUSCH and SRS resources in a cell in the bufferzone than in a common cell. Therefore, when the LTE bandwidth is 5 MHz or 10 MHz, using a cell in the bufferzone as a PCell for CA is not recommended. If the cell is used as a PCell, CA performance deteriorates.

RAT	Function Name	Function Switch	Reference	Description
FDD	UMTS and LTE Spectrum Sharing (LTE FDD)	**SpectrumCloud.*SpectrumCloudSwitch*** set to **UL_SPECTRUM_SHARING**	*UMTS and LTE Spectrum Sharing*	Cells with 5 MHz bandwidth are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured.
FDD	UMTS and LTE Spectrum Sharing Based on DC-HSDPA	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **DC_HSDPA_BASED_UL_SPECTRUM_SHR**	*UMTS and LTE Spectrum Sharing Based on DC-HSDPA*	Cells with 5 MHz bandwidth are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured.
TDD	SRS resource migration	**REMOTE_INTRF_BF_ENH_SW** option of the **UlInterfSuppressCfg**.***RemoteIntrfDlEnhSwitch*** parameter	*Interference Detection and Suppression*	When the conditions for the SRS resource migration function to take effect and those for downlink 4CC aggregation are met at the same time, the SRS resource migration function takes precedence. Details are as follows: ● When the SRS resource migration function takes effect, downlink 4CC aggregation stops taking effect. ● When the SRS resource migration function stops taking effect, downlink 4CC aggregation becomes taking effect.

7.3.4 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

For details, see **Boards** in **5.3.4 Hardware**.

RF Modules

For details, see **RF Modules** in **5.3.4 Hardware**.

7.3.5 Networking

For details, see **5.3.5 Networking**.

7.3.6 Others

- UEs

  UEs must comply with 3GPP Release 12 or later and support the frequency bands of the carriers to be aggregated and their bandwidths. UEs must also support the peak data rates that CA can achieve.

- EPC

  For this function to reach a theoretical peak data rate described in **7.2.1 Benefits**, the maximum bit rate that each UE subscribes to in the EPC cannot be lower than this theoretical value.

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

This function works in either CA-group-based or adaptive configuration mode. Prepare basic data as described in **5.4.1.1 Data Preparation**.

In addition, for either mode, prepare data as described in **Table 7-4** and **Table 7-5** for function activation and optimization.

Table 7-4 Parameters for activating downlink 4CC aggregation

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	CaDl4CCSwitch	Select this option.

Table 7-5 Parameters for optimizing downlink 4CC aggregation

Parameter Name	Parameter ID	Setting Notes
DL Beyond 3CC UE Reordering Timer	**RlcPdcpParaGroup**.***Dl4cc5ccUeReorderingTimer***	Set this parameter to **Treordering_m15**. This parameter is valid only when the **RlcPdcpParaGroup**.***RlcMode*** parameter is set to **RlcMode_AM**.
DL Beyond 3CC UE Status Prohibit Timer	**RlcPdcpParaGroup**.***Dl4cc5ccUeStatProhTimer***	Set this parameter to **m15**. This parameter is valid only when the **RlcPdcpParaGroup**.***RlcMode*** parameter is set to **RlcMode_AM**.

Moreover, take the following necessary actions for this function to take effect:

- Prepare data as described in **16.4.1.1 Data Preparation** if downlink 4CC aggregation is to be deployed in a relaxed backhaul scenario.
- Prepare data as described in **17.4.1.1 Data Preparation** if downlink 4CC aggregation is to be deployed in an eNodeB coordination scenario.

7.4.1.2 Using MML Commands

Before using MML commands, refer to **7.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Before activating this function, configure cells or frequencies according to **5.4.1.2 Using MML Commands**.

- In CA-group-based configuration mode

  Add at least four cells to the CA group and configure candidate SCells for each candidate PCell.

- In adaptive configuration mode

  Configure at least three candidate SCCs for each candidate PCC.

The activation command examples for this function are as follows:

```
//Turning on CaDl4CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl4CCSwitch-1;
//(Optional) Setting Dl4cc5ccUeReorderingTimer and Dl4cc5ccUeStatProhTimer
MOD RLCPDCPPARAGROUP: RlcPdcpParaGroupId=0, RlcMode=RlcMode_AM,
Dl4cc5ccUeReorderingTimer=Treordering_m15, Dl4cc5ccUeStatProhTimer=m15;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off CaDl4CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl4CCSwitch-0;
```

7.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

Counter Observation

If any of the counters listed in **Table 7-6** produces a non-zero value on a network that is serving CA UEs capable of downlink 4CC aggregation, downlink 4CC aggregation has taken effect in the network.

Table 7-6 Counters used to verify activation of downlink 4CC aggregation

Counter ID	Counter Name
1526737780	**L.Traffic.User.PCell.DL.4CC.Avg**
1526737781	**L.Traffic.User.PCell.DL.4CC.Max**
1526737793	**L.Traffic.User.CA.4CC.PCell.DL.Active.Avg**

Counter ID	Counter Name
1526737794	**L.Traffic.User.CA.4CC.PCell.DL.Active.Max**

For the **L.Traffic.User.CA.4CC.PCell.DL.Active.Avg** and **L.Traffic.User.CA.4CC.PCell.DL.Active.Max** counters:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, these counters are stepped as long as four carriers are active for a UE.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, these counters are stepped only when four carriers are active with no SCell inactive for a UE.

For example, five carriers are configured but only four of them are active for a UE. That is, not all SCells are activated for the UE. In the presence of such a UE:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Traffic.User.CA.4CC.PCell.DL.Active.Avg** and **L.Traffic.User.CA.4CC.PCell.DL.Active.Max** counters are stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Traffic.User.CA.4CC.PCell.DL.Active.Avg** and **L.Traffic.User.CA.4CC.PCell.DL.Active.Max** counters do not count this UE in.

Message Tracing

For the tools and the IEs to observe, see **Message Tracing** in **5.4.2 Activation Verification**.

7.4.3 Network Monitoring

In addition to the counters listed in **5.4.3 Network Monitoring**, monitor the counters in **Table 7-7** and compare the results with the network plan to evaluate network performance.

Calculate the throughput of UEs in the downlink 4CC aggregation state by using the following formula: **L.Thrp.bits.DL.4CC.CAUser**/**L.Thrp.Time.DL.4CC.CAUser**.

For the **L.Thrp.bits.DL.4CC.CAUser** and **L.Thrp.Time.DL.4CC.CAUser** counters:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, these counters are stepped as long as four carriers are active for a UE.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, these counters are stepped only when four carriers are active with no SCell inactive for a UE.

For example, five carriers are configured but only four of them are active for a UE. That is, not all SCells are activated for the UE. In the presence of such a UE:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Thrp.bits.DL.4CC.CAUser** and **L.Thrp.Time.DL.4CC.CAUser** counters are stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Thrp.bits.DL.4CC.CAUser** and **L.Thrp.Time.DL.4CC.CAUser** counters do not count this UE in.

Table 7-7 Counters used to monitor performance of downlink 4CC aggregation

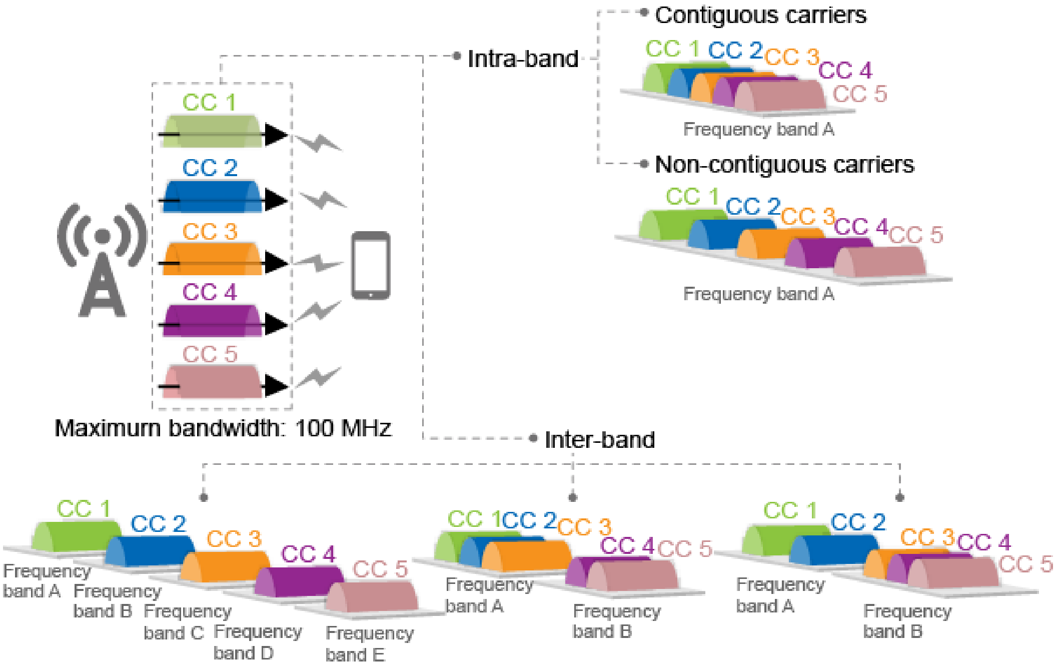
Counter ID	Counter Name
1526737780	**L.Traffic.User.PCell.DL.4CC.Avg**
1526737781	**L.Traffic.User.PCell.DL.4CC.Max**
1526737793	**L.Traffic.User.CA.4CC.PCell.DL.Active.Avg**
1526737794	**L.Traffic.User.CA.4CC.PCell.DL.Active.Max**
1526737812	**L.Thrp.Time.DL.4CC.CAUser**
1526737813	**L.Thrp.bits.DL.4CC.CAUser**

8 Downlink 5CC Aggregation

8.1 Principles

This function aggregates five intra- or inter-band carriers, as shown in **Figure 8-1**, to provide higher bandwidth.

Figure 8-1 Downlink 5CC aggregation



This function works between intra-eNodeB cells, between inter-eNodeB cells in eNodeB coordination scenarios, and between inter-eNodeB cells in relaxed backhaul scenarios.

This function is controlled by the **CaDl5CCSwitch** option of the **CaMgtCfg.*CellCaAlgoSwitch*** parameter.

A UE in the downlink 5CC aggregation state performs simultaneous transmission of HARQ ACKs/NACKs and periodic CQI report multiplexing for the five CCs if the

CqiAdaptiveCfg.*SimulAckNackAndCqiFmt3Sw* parameter is set to **ON**. If this parameter is set to **OFF**, the ACKs/NACKs for the five CCs cannot be transmitted simultaneously with periodic CQI reports for any of the CCs.

8.2 Network Analysis

8.2.1 Benefits

This function enables CA UEs to reach higher downlink peak data rates.

Table 8-1 lists the theoretical peak data rates that a CA UE can reach using downlink 5CC aggregation.

For FDD, these values assume a TBS suitable for the 20 MHz cell bandwidth (equivalent to 100 RBs in the frequency domain).

For TDD, these values assume a TBS suitable in the following scenario:

- Bandwidth: 20 MHz (equivalent to 100 RBs in the frequency domain)
- Uplink-downlink subframe configuration: 2
- Special subframe configuration: 7

Table 8-1 Theoretical peak data rates for downlink 5CC aggregation (unit: Mbit/s)

RAT	2x2 MIMO + 64QAM	2x2 MIMO + 256QAM	4x4 MIMO + 64QAM	4x4 MIMO + 256QAM
FDD	748.9	979.1	1499.3	1958.3
TDD	560	715	1085	1430

The peak data rate that CA can achieve for a CA UE is subject to:

- Peak data rate capability of the board where the PCell for the CA UE is located

  For example, if the PCell and SCells of a CA UE are served by an LBBPd1 board that supports a downlink peak data rate of 450 Mbit/s, the peak data rate that CA can achieve for the CA UE will not exceed 450 Mbit/s.
- Capability of the CA UE

  If the UE capability is limited, the actual peak data rates will be lower than the theoretical values. The UE capability is indicated by ue-categoryDL. For details about this IE, see section 4.1A "ue-CategoryDL and ue-CategoryUL" in 3GPP TS 36.306 V15.2.0.

8.2.2 Impacts

The impacts between this function and other functions are described in **8.3.3.3 Function Impacts**.

This function requires additional RBs for PUCCH format-3 overhead. Therefore, the downlink IBLER of the PCell fluctuates. The impact of this function on the PUCCH overhead varies as follows:

- Scenario: The **PucchSwitch** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter is selected to enable adaptive allocation of PUCCH resources.

  The cell spares one additional RB for PUCCH format-3 overhead. As a result, the number of RBs available for the PUSCH decreases by at least one. (This number must be a multiple of 2, 3, or 5. For details, see section 5.3.3 "Transform precoding" in 3GPP TS 36.211 V10.1.0 (2011-03).) Total uplink throughput decreases.

- Scenario: The **PucchSwitch** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter is deselected to use fixed allocation of PUCCH resources.

  For FDD, the cell changes the usage of one PUCCH RB from periodic CQI reporting to PUCCH format-3 overhead. As a result, fewer RBs of the cell are used for periodic CQI reporting, and more UEs have to use aperiodic CQI reporting. Downlink UE throughput may slightly decrease.

  For TDD, the cell changes the usage of two PUCCH RBs from periodic CQI reporting to PUCCH format-3 overhead. As a result, fewer RBs of the cell are used for periodic CQI reporting, and more UEs have to use aperiodic CQI reporting. Downlink UE throughput may slightly decrease.

8.3 Requirements

8.3.1 Licenses (FDD)

None

8.3.2 Licenses (TDD)

None

8.3.3 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.3.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
FDD TDD	Downlink 2CC aggregation	None	**5 Downlink 2CC Aggregation**
FDD TDD	Downlink 3CC aggregation	**CaDl3CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**6 Downlink 3CC Aggregation**

RAT	Function Name	Function Switch	Reference
FDD TDD	Downlink 4CC aggregation	**CaDl4CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**7 Downlink 4CC Aggregation**

8.3.3.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference
FDD	PUCCH measurement	**PucchMeasOptSwitch** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter	*SFN*

8.3.3.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
FDD	UMTS and LTE Zero Bufferzone	**UMTS_LTE_ZERO_BUFFER_ZONE_SW** option of the **ULZeroBufferZone**.***ZeroBufZoneSwitch*** parameter	*UMTS and LTE Zero Bufferzone*	There are fewer PUSCH and SRS resources in a cell in the bufferzone than in a common cell. Therefore, when the LTE bandwidth is 5 MHz or 10 MHz, using a cell in the bufferzone as a PCell for CA is not recommended. If the cell is used as a PCell, CA performance deteriorates.

RAT	Function Name	Function Switch	Reference	Description
FDD	TDM	**TDM_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Performance Enhancement*	In time division multiplexing (TDM), the ACKs/NACKs for multiple downlink subframes need to be sent in the same uplink subframe. The ACKs/NACKs for all the carriers of a UE in the downlink 5CC aggregation state cannot be sent completely. To mitigate this issue, when TDM is enabled, the following constraint applies to the maximum number of CCs that can be aggregated for downlink CA: The maximum number is 5 when the uplink-downlink subframe configuration is 0, 1, or 6. The maximum number is 4 when the configuration is 2. The maximum number is 3 when the configuration is 3 or 4. The maximum number is 2 when the configuration is 5. When the number of CCs that are actually aggregated exceeds the maximum number, SCells are removed if TDM is required. SCells can be configured later as triggered by traffic volume, so long as the number of downlink CCs does not exceed the specified maximum number.

RAT	Function Name	Function Switch	Reference	Description
FDD	UMTS and LTE Spectrum Sharing (LTE FDD)	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **UL_SPECTRUM_SHARING**	*UMTS and LTE Spectrum Sharing*	Cells with 5 MHz bandwidth are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured.
FDD	UMTS and LTE Spectrum Sharing Based on DC-HSDPA	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **DC_HSDPA_BASED_UL_SPECTRUM_SHR**	*UMTS and LTE Spectrum Sharing Based on DC-HSDPA*	Cells with 5 MHz bandwidth are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured.
TDD	SRS resource migration	**REMOTE_INTRF_BF_ENH_SW** option of the **UlInterfSuppressCfg**.***RemoteIntrfDlEnhSwitch*** parameter	*Interference Detection and Suppression*	When the conditions for the SRS resource migration function to take effect and those for downlink 5CC aggregation are met at the same time, the SRS resource migration function takes precedence. Details are as follows: ● When the SRS resource migration function takes effect, downlink 5CC aggregation stops taking effect. ● When the SRS resource migration function stops taking effect, downlink 5CC aggregation becomes taking effect.

8.3.4 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

The requirements described in **Boards** of **5.3.4 Hardware** must be fulfilled. In addition:

- Among BBPs, UBBP is recommended.
- Among main control boards, UMPT is recommended.

RF Modules

For details, see **RF Modules** in **5.3.4 Hardware**.

8.3.5 Networking

For details, see **5.3.5 Networking**.

8.3.6 Others

- UEs

  UEs must comply with 3GPP Release 12 or later and support the frequency bands of the carriers to be aggregated and their bandwidths. UEs must also support the peak data rates that CA can achieve.
- EPC

  For this function to reach a theoretical peak data rate described in **8.2.1 Benefits**, the maximum bit rate that each UE subscribes to in the EPC cannot be lower than this theoretical value.

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

This function works in either CA-group-based or adaptive configuration mode. Prepare basic data as described in **5.4.1.1 Data Preparation**.

In addition, for either mode, prepare data as described in **Table 8-2** and **Table 8-3** for function activation and optimization.

Table 8-2 Parameters for activating downlink 5CC aggregation

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	CaDl5CCSwitch	Select this option.

Table 8-3 Parameters for optimizing downlink 5CC aggregation

Parameter Name	Parameter ID	Setting Notes
DL Beyond 3CC UE Reordering Timer	**RlcPdcpParaGroup**.***Dl4cc5ccUeReorderingTimer***	Set this parameter to **Treordering_m15**. This parameter is valid only when the **RlcPdcpParaGroup**.***RlcMode*** parameter is set to **RlcMode_AM**.
DL Beyond 3CC UE Status Prohibit Timer	**RlcPdcpParaGroup**.***Dl4cc5ccUeStatProhTimer***	Set this parameter to **m15**. This parameter is valid only when the **RlcPdcpParaGroup**.***RlcMode*** parameter is set to **RlcMode_AM**.

Moreover, take the following necessary actions for this function to take effect:

- Prepare data as described in **16.4.1.1 Data Preparation** if downlink 5CC aggregation is to be deployed in a relaxed backhaul scenario.
- Prepare data as described in **17.4.1.1 Data Preparation** if downlink 5CC aggregation is to be deployed in an eNodeB coordination scenario.

8.4.1.2 Using MML Commands

Before using MML commands, refer to **8.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Before activating this function, configure cells or frequencies according to **5.4.1.2 Using MML Commands**.

- In CA-group-based configuration mode

  Add at least five cells to the CA group and configure candidate SCells for each candidate PCell.

- In adaptive configuration mode

  Configure at least four candidate SCCs for each candidate PCC.

The activation command examples for this function are as follows:

```
//Turning on CaDl5CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl5CCSwitch-1;
//(Optional) Setting Dl4cc5ccUeReorderingTimer and Dl4cc5ccUeStatProhTimer
MOD RLCPDCPPARAGROUP: RlcPdcpParaGroupId=0, RlcMode=RlcMode_AM,
Dl4cc5ccUeReorderingTimer=Treordering_m15, Dl4cc5ccUeStatProhTimer=m15;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off CaDl5CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl5CCSwitch-0;
```

8.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

Counter Observation

If any of the counters listed in **Table 8-4** produces a non-zero value on a network that is serving CA UEs capable of downlink 5CC aggregation, downlink 5CC aggregation has taken effect in the network.

Table 8-4 Counters used to verify activation of downlink 5CC aggregation

Counter ID	Counter Name
1526739805	**L.Traffic.User.PCell.DL.5CC.Avg**
1526739806	**L.Traffic.User.PCell.DL.5CC.Max**
1526740442	**L.Traffic.User.CA.5CC.PCell.DL.Active.Avg**

Counter ID	Counter Name
1526740443	**L.Traffic.User.CA.5CC.PCell.DL.Active.Max**

For the **L.Traffic.User.CA.5CC.PCell.DL.Active.Avg** and **L.Traffic.User.CA.5CC.PCell.DL.Active.Max** counters:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, these counters are stepped as long as five carriers are active for a UE.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, these counters are stepped only when five carriers are active with no SCell inactive for a UE.

For example, six carriers are configured but only five of them are active for a UE. That is, not all SCells are activated for the UE. In the presence of such a UE:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Traffic.User.CA.5CC.PCell.DL.Active.Avg** and **L.Traffic.User.CA.5CC.PCell.DL.Active.Max** counters are stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Traffic.User.CA.5CC.PCell.DL.Active.Avg** and **L.Traffic.User.CA.5CC.PCell.DL.Active.Max** counters do not count this UE in.

Message Tracing

For the tools and the IEs to observe, see **Message Tracing** in **5.4.2 Activation Verification**.

8.4.3 Network Monitoring

In addition to the counters listed in **5.4.3 Network Monitoring**, monitor the counters in **Table 8-5** and compare the results with the network plan to evaluate network performance.

Calculate the throughput of UEs in the downlink 5CC aggregation state by using the following formula: **L.Thrp.bits.DL.5CC.CAUser**/**L.Thrp.Time.DL.5CC.CAUser**.

For the **L.Thrp.bits.DL.5CC.CAUser** and **L.Thrp.Time.DL.5CC.CAUser** counters:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, these counters are stepped as long as five carriers are active for a UE.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, these counters are stepped only when five carriers are active with no SCell inactive for a UE.

For example, six carriers are configured but only five of them are active for a UE. That is, not all SCells are activated for the UE. In the presence of such a UE:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Thrp.bits.DL.5CC.CAUser** and **L.Thrp.Time.DL.5CC.CAUser** counters are stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Thrp.bits.DL.5CC.CAUser** and **L.Thrp.Time.DL.5CC.CAUser** counters do not count this UE in.

Table 8-5 Counters used to monitor performance of downlink 5CC aggregation

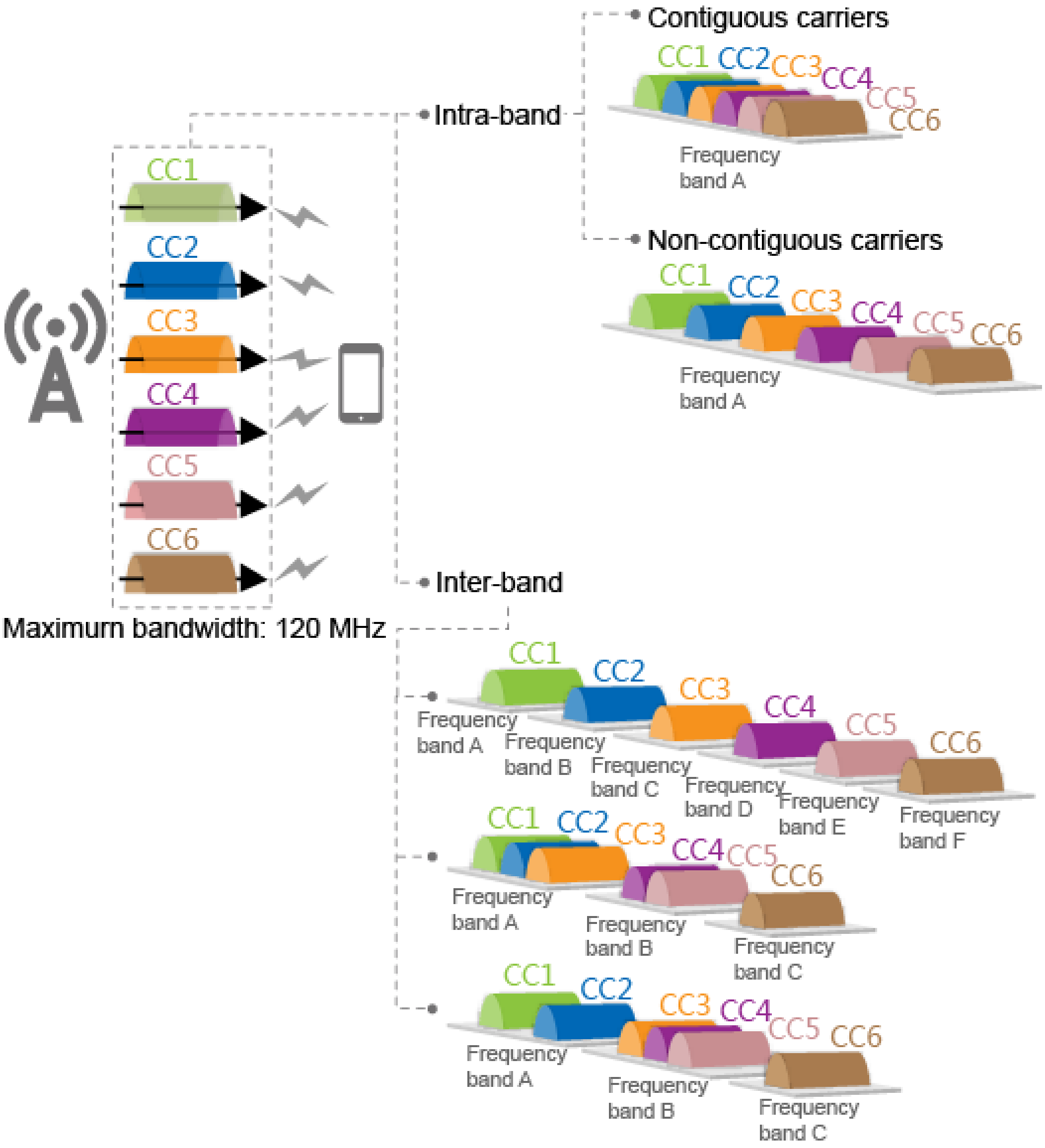
Counter ID	Counter Name
1526739805	**L.Traffic.User.PCell.DL.5CC.Avg**
1526739806	**L.Traffic.User.PCell.DL.5CC.Max**
1526740442	**L.Traffic.User.CA.5CC.PCell.DL.Active.Avg**
1526740443	**L.Traffic.User.CA.5CC.PCell.DL.Active.Max**
1526740438	**L.Thrp.Time.DL.5CC.CAUser**
1526740439	**L.Thrp.bits.DL.5CC.CAUser**

9 Downlink Massive CA (FDD)

9.1 Principles

This function aggregates six to eight intra- or inter-band carriers to provide a higher bandwidth. **Figure 9-1** uses 6CC aggregation as an example.

Figure 9-1 Downlink 6CC aggregation



This function is controlled by the **DlMassiveCaSwitch** option of the **CaMgtCfg.*CellCaAlgoSwitch*** parameter.

9.2 Network Analysis

9.2.1 Benefits

This function enables CA UEs to reach higher downlink peak data rates.

Table 9-1 lists the theoretical peak data rates that a CA UE can reach using downlink 6CC, 7CC, and 8CC aggregation.

These values assume a TBS suitable for the 20 MHz cell bandwidth (equivalent to 100 RBs in the frequency domain).

Table 9-1 Theoretical peak data rates for downlink 6CC, 7CC, and 8CC aggregation (unit: Mbit/s)

CCs	2x2 MIMO + 64QAM	2x2 MIMO + 256QAM	4x4 MIMO + 64QAM	4x4 MIMO + 256QAM
6	898	1174	1798	2350
7	1048	1370	2098	2741
8	1198	1566	2398	3132

The peak data rate that CA can achieve for a CA UE is subject to:

- Peak data rate capability of the board where the PCell for the CA UE is located

  For example, if the PCell of a CA UE is served by an LBBPd1 board that supports a downlink peak data rate of 450 Mbit/s, the peak data rate that CA can achieve for the CA UE will not exceed 450 Mbit/s in the downlink.

- Capability of the CA UE

  If the UE capability is limited, the actual peak data rates will be lower than the theoretical values. The UE capability is indicated by ue-categoryDL. For details about this IE, see section 4.1A "ue-CategoryDL and ue-CategoryUL" in 3GPP TS 36.306 V15.2.0.

9.2.2 Impacts

The impacts between this function and other functions are described in **9.3.2.3 Function Impacts**.

This function has the following impacts on the network:

- Additional RBs are required for PUCCH format-5 overhead. The impact varies as follows:
  - Scenario: The **PucchSwitch** option of the **CellAlgoSwitch.*PucchAlgoSwitch*** parameter is selected to enable adaptive allocation of PUCCH resources.

    The PCell spares two additional RBs for PUCCH format-5 overhead. As a result, the number of RBs available for the PUSCH decreases by at least two. (This number must be a multiple of 2, 3, or 5. For details, see section 5.3.3 "Transform precoding" in 3GPP TS 36.211 V10.1.0 (2011-03).) Total uplink throughput decreases.

  - Scenario: The **PucchSwitch** option of the **CellAlgoSwitch.*PucchAlgoSwitch*** parameter is deselected to use fixed allocation of PUCCH resources.

    The PCell changes the usage of two PUCCH RBs from periodic CQI reporting to PUCCH format-5 overhead. As a result, fewer RBs of the cell are used for periodic CQI reporting, and more UEs have to use aperiodic CQI reporting. Downlink UE throughput may slightly decrease.

- The total number of bits of HARQ feedback transmitted on the PUSCH increases. This reduces the spectral efficiency of the PUSCH and decreases uplink throughput.

- More messages are exchanged in inter-BBU CA scenarios, and the values of the **VS.SctpLnk.RxMeanSpeed** and **VS.SctpLnk.TxMeanSpeed** counters increase.
- If the **SMART_CARRIER_SELECTION_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter is deselected and the eNodeB supports the aggregation of six or a higher number of carriers, serving cell combinations for 6CC, 7CC, or 8CC aggregation will be preferentially configured for CA UEs that support 6CC, 7CC, or 8CC aggregation in the band combination configuration procedure for downlink massive CA. Even if downlink massive CA is enabled after intelligent selection of downlink serving cell combinations has taken effect, the eNodeB does not consider coverage overlap between cells, air interface capability, or load information during the downlink massive CA configuration procedure. As a result, it is possible that the actually effective total bandwidth of the serving cell combination for a CA UE is less than the total effective bandwidth of the serving cell combination selected when massive CA is disabled.

9.3 Requirements

9.3.1 Licenses

LEOFD-151308 Downlink Massive CA

1. An eNodeB identifies all its served cells on the PCC frequencies configured on the eNodeB.
2. If the **DlMassiveCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for a cell on a PCC frequency, the cell consumes one sales unit of the license for LEOFD-151308 Downlink Massive CA.
3. If the **DlMassiveCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for at least one cell on a PCC frequency, each CA cell on the SCC frequencies configured for the PCC frequency consumes one sales unit of the license for LEOFD-151308 Downlink Massive CA.

NOTE

When inter-eNodeB CA based on relaxed backhaul or inter-eNodeB CA based on eNodeB coordination is enabled, SCells may be served by neighboring eNodeBs. If this is the case, these cells consume the license for LEOFD-151308 Downlink Massive CA at the neighboring eNodeBs.

Table 9-2 lists the license model and sales unit for the feature.

Table 9-2 License model and sales unit

Feature ID	Feature Name	Model	Sales Unit
LEOFD-151308	Downlink Massive CA	LT1SDMCAP200	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference
Downlink 2CC aggregation	None	**5 Downlink 2CC Aggregation**
Downlink 3CC aggregation	**CaDl3CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**6 Downlink 3CC Aggregation**
Downlink 4CC aggregation	**CaDl4CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**7 Downlink 4CC Aggregation**
Downlink 5CC aggregation	**CaDl5CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**8 Downlink 5CC Aggregation**
Reporting of compressed CA band combination fields	**UeBandCombReducedR13Sw** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter	**4.5.3 Network-requested CA Band Combination Capability Signaling**
CQI reporting over PUCCH format 3	**CqiAdaptiveCfg**.***SimulAckNackAndCqiFmt3Sw***	None

9.3.2.2 Mutually Exclusive Functions

None

9.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Load-based SCell configuration	**SccSmartCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter	**4.6.3.2 SCell Configuration Enhancement**	• If smart carrier selection based on virtual grids is disabled, load-based SCell configuration does not take effect for UEs in the downlink massive CA state. • If smart carrier selection based on virtual grids is enabled, load-based SCell configuration can take effect for UEs in the downlink massive CA state.
Adaptive selection of CA or high-order MIMO	**CaMgtCfg**.***CaMimoPriorityStrategySw***	**UEs** in **19.2.2 MIMO (FDD)**	When the **CaMgtCfg**.***CaMimoPriorityStrategySw*** parameter is set to **MIMO_PRIOR** or **PEAK_RATE_PRIOR**: • If smart carrier selection based on virtual grids is disabled, UEs in the downlink massive CA state will not fall back to be served by fewer CCs. • If smart carrier selection based on virtual grids is enabled, UEs in the downlink massive CA state will fall back to be served by fewer CCs.
Uplink short-interval SPS	**CellUlschAlgo**.***IntvlOfUlSpsWithSkipping***	*Uplink Scheduling*	UEs that have entered uplink short-interval SPS do not support downlink massive CA.

Function Name	Function Switch	Reference	Description
UMTS and LTE Zero Bufferzone	**UMTS_LTE_ZERO_BUFFER_ZONE_SW** option of the **ULZeroBufferZone**.***ZeroBufZoneSwitch*** parameter	*UMTS and LTE Zero Bufferzone*	There are fewer PUSCH and SRS resources in a cell in the bufferzone than in a common cell. Therefore, when the LTE bandwidth is 5 MHz or 10 MHz, using a cell in the bufferzone as a PCell for CA is not recommended. If the cell is used as a PCell, CA performance deteriorates.
TDM	**TDM_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Performance Enhancement*	In NSA DC, after TDM takes effect in an FDD cell, the PUCCH in the cell always uses format 3. However, in downlink massive CA, the PUCCH of the cell uses format 5. Therefore, TDM and downlink massive CA cannot take effect at the same time.
LTE FDD and NR Flash Dynamic Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_SPECTRUM_SHR**	*LTE FDD and NR Dynamic Spectrum Sharing*	LTE cells with LTE FDD and NR Flash Dynamic Spectrum Sharing enabled are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured, causing the LTE network throughput to decrease.
LTE FDD and NR Uplink Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_UPLINK_SPECTRUM_SHR**	*LTE FDD and NR SUL Dynamic Spectrum Sharing*	LTE cells with LTE FDD and NR Uplink Spectrum Sharing enabled are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured. As a result, the LTE network throughput is affected.

Function Name	Function Switch	Reference	Description
UMTS and LTE Spectrum Sharing	**SpectrumCloud*.SpectrumCloudSwitch*** set to **UL_SPECTRUM_SHARING**	*UMTS and LTE Spectrum Sharing*	Downlink massive CA is not recommended for cells with UMTS and LTE Spectrum Sharing enabled.

9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

The requirements described in **Boards** of **5.3.4 Hardware** must be fulfilled. In addition:

- BBU3910A and BBU3910C cannot be used for this function.
- UBBPex2 boards cannot be used for this function.
- Only cells on UBBPd, UBBPe, UBBPg, or UBBPh boards can act as PCells.
- The main control boards of the serving eNodeBs for PCells must be UMPT boards.

RF Modules

For details, see **RF Modules** in **5.3.4 Hardware**.

9.3.4 Networking

For details, see **5.3.5 Networking**.

9.3.5 Others

- UEs

  UEs must comply with 3GPP Release 13 or later and support the frequency bands of the carriers to be aggregated and their channel bandwidths. UEs must also support the peak data rates that CA can achieve.
- EPC

  For this function to reach a theoretical peak data rate described in **9.2.1 Benefits**, the maximum bit rate that each UE subscribes to in the EPC cannot be lower than this theoretical value.

9.4 Operation and Maintenance

9.4.1 Data Configuration

9.4.1.1 Data Preparation

This function works only in adaptive configuration mode. Prepare basic data as described in **5.4.1.1.2 Adaptive Configuration Mode** and **5.4.1.1.3 Common Parameters**.

Table 9-3 and **Table 9-4** describe the parameters used for function activation and optimization, respectively.

Table 9-3 Parameters for activating downlink massive CA

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	DlMassiveCaSwitch	Select this option.

Table 9-4 Parameters for optimizing downlink massive CA

Parameter Name	Parameter ID	Setting Notes
DL Beyond 3CC UE Reordering Timer	**RlcPdcpParaGroup**.***Dl4cc5ccUeReorderingTimer***	Set this parameter to **Treordering_m15**. This parameter is valid only when the **RlcPdcpParaGroup**.***RlcMode*** parameter is set to **RlcMode_AM**.
DL Beyond 3CC UE Status Prohibit Timer	**RlcPdcpParaGroup**.***Dl4cc5ccUeStatProhTimer***	Set this parameter to **m15**. This parameter is valid only when the **RlcPdcpParaGroup**.***RlcMode*** parameter is set to **RlcMode_AM**.
Downlink RLC-SN size	**RlcPdcpParaGroup**.***DlRlcSnSize***	It is good practice to set this parameter to **RlcSnSize_size16** when the **RlcPdcpParaGroup**.***RlcMode*** parameter is set to **RlcMode_AM**.

Moreover, take the following necessary actions for this function to take effect:

- Prepare data as described in **16.4.1.1 Data Preparation** if downlink massive CA is to be deployed in a relaxed backhaul scenario.
- Prepare data as described in **17.4.1.1 Data Preparation** if downlink massive CA is to be deployed in an eNodeB coordination scenario.

9.4.1.2 Using MML Commands

Before using MML commands, refer to **9.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Before activating this function, configure at least five candidate SCCs for each candidate PCC according to **5.4.1.2.3 Adaptive Configuration Mode (FDD)**.

The activation command examples for this function are as follows:

```
//Turning on DlMassiveCaSwitch
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=DlMassiveCaSwitch-1;
//(Optional) Setting Dl4cc5ccUeReorderingTimer, Dl4cc5ccUeStatProhTimer, and DlRlcSnSize
MOD RLCPDCPPARAGROUP: RlcPdcpParaGroupId=0, RlcMode=RlcMode_AM, DlRlcSnSize=RlcSnSize_size16,
Dl4cc5ccUeReorderingTimer=Treordering_m15, Dl4cc5ccUeStatProhTimer=m15;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off DlMassiveCaSwitch
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=DlMassiveCaSwitch-0;
```

9.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.4.2 Activation Verification

Counter Observation

If any of the counters listed in **Table 9-5** produces a non-zero value on a network that is serving CA UEs capable of downlink massive CA, downlink massive CA has taken effect in the network.

Table 9-5 Counters used to verify activation of downlink massive CA

Counter ID	Counter Name
1526749452	**L.Traffic.User.PCell.DL.6CC.Avg**
1526749489	**L.Traffic.User.CA.6CC.PCell.DL.Active.Avg**
1526749453	**L.Traffic.User.PCell.DL.7CC.Avg**
1526749490	**L.Traffic.User.CA.7CC.PCell.DL.Active.Avg**
1526749454	**L.Traffic.User.PCell.DL.8CC.Avg**
1526749491	**L.Traffic.User.CA.8CC.PCell.DL.Active.Avg**

For the **L.Traffic.User.CA.6CC.PCell.DL.Active.Avg**, **L.Traffic.User.CA.7CC.PCell.DL.Active.Avg**, and **L.Traffic.User.CA.8CC.PCell.DL.Active.Avg** counters:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, these counters are stepped as long as the corresponding number of CCs are active for a UE.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, these counters are stepped only when the corresponding number of CCs are active with no SCell inactive for a UE.

For example, seven carriers are configured but only six of them are active for a UE. That is, not all SCells are activated for the UE. In the presence of such a UE:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Traffic.User.CA.6CC.PCell.DL.Active.Avg** counter is stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Traffic.User.CA.6CC.PCell.DL.Active.Avg** counter does not count this UE in.

Message Tracing

For the tools and the IEs to observe, see **Message Tracing** in **5.4.2 Activation Verification**.

9.4.3 Network Monitoring

In addition to the counters listed in **5.4.3 Network Monitoring**, monitor the counters in **Table 9-6** and compare the results with the network plan to evaluate network performance.

Calculate the throughput of UEs in the CA state as follows:

- Calculate the throughput of UEs in the downlink 6CC aggregation state by using the following formula: **L.Thrp.bits.DL.6CC.CAUser**/ **L.Thrp.Time.DL.6CC.CAUser**.
- Calculate the throughput of UEs in the downlink 7CC aggregation state by using the following formula: **L.Thrp.bits.DL.7CC.CAUser**/ **L.Thrp.Time.DL.7CC.CAUser**.
- Calculate the throughput of UEs in the downlink 8CC aggregation state by using the following formula: **L.Thrp.bits.DL.8CC.CAUser**/ **L.Thrp.Time.DL.8CC.CAUser**.

For the **L.Thrp.bits.DL.6CC.CAUser**, **L.Thrp.Time.DL.6CC.CAUser**, **L.Thrp.bits.DL.7CC.CAUser**, **L.Thrp.Time.DL.7CC.CAUser**, **L.Thrp.bits.DL.8CC.CAUser**, and **L.Thrp.Time.DL.8CC.CAUser** counters:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, these counters are stepped as long as the corresponding number of CCs are active for a UE.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, these counters are stepped only when the corresponding number of CCs are active with no SCell inactive for a UE.

For example, seven carriers are configured but only six of them are active for a UE. That is, not all SCells are activated for the UE. In the presence of such a UE:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Thrp.bits.DL.6CC.CAUser** and **L.Thrp.Time.DL.6CC.CAUser** counters are stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Thrp.bits.DL.6CC.CAUser** and **L.Thrp.Time.DL.6CC.CAUser** counters do not count this UE in.

Table 9-6 Counters used to monitor performance of downlink massive CA

Counter ID	Counter Name
1526749452	**L.Traffic.User.PCell.DL.6CC.Avg**
1526749489	**L.Traffic.User.CA.6CC.PCell.DL.Active.Avg**
1526749453	**L.Traffic.User.PCell.DL.7CC.Avg**
1526749490	**L.Traffic.User.CA.7CC.PCell.DL.Active.Avg**

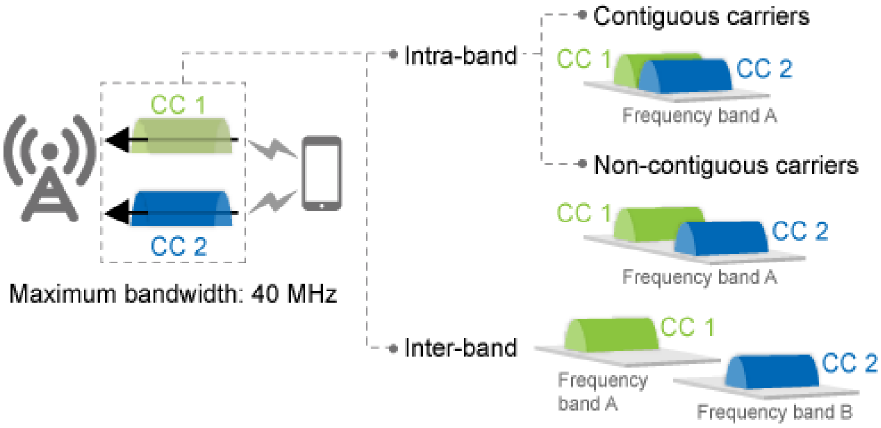
Counter ID	Counter Name
1526749454	**L.Traffic.User.PCell.DL.8CC.Avg**
1526749491	**L.Traffic.User.CA.8CC.PCell.DL.Active.Avg**
1526749443	**L.Thrp.Time.DL.6CC.CAUser**
1526749440	**L.Thrp.bits.DL.6CC.CAUser**
1526749444	**L.Thrp.Time.DL.7CC.CAUser**
1526749441	**L.Thrp.bits.DL.7CC.CAUser**
1526749445	**L.Thrp.Time.DL.8CC.CAUser**
1526749442	**L.Thrp.bits.DL.8CC.CAUser**

10 Uplink 2CC Aggregation

10.1 Principles

This function aggregates two intra- or inter-band carriers, as shown in **Figure 10-1**, to provide higher bandwidth (up to 40 MHz) for the uplink.

Figure 10-1 Uplink 2CC aggregation



This function works between intra-eNodeB cells, between inter-eNodeB cells in eNodeB coordination scenarios, and between inter-eNodeB cells in relaxed backhaul scenarios.

This function is controlled by the **CaUl2CCSwitch** option of the **CaMgtCfg.*CellCaAlgoSwitch*** parameter.

NOTE

When blacklist control for uplink CA (controlled by the **UL_CA_FORBID_SW** option of the **UeCompat**.***BlacklistControlExtSwitch2*** parameter) is enabled, uplink CA does not take effect for blacklisted UEs. When this function is disabled, no differentiated handling is performed for blacklisted UEs.

Load-based Selection of Uplink TDD Massive MIMO SCells

In TDD massive MIMO scenarios, when a TDD massive MIMO cell acts as an SCell in the uplink for a CA UE that supports uplink CA, the UE can send SRSs and participate in pairing in this cell. This increases the capacity of this cell. When the cell is in the high load state, the CA UE is more likely to be paired in the cell.

When uplink 2CC aggregation is enabled in three or more cells, load-based selection of uplink TDD massive MIMO SCells can be enabled so that the eNodeB preferentially configures a high-load massive MIMO cell as the uplink SCell for CA UEs. For details about this function, see **Load-based Selection of Uplink TDD Massive MIMO SCells** in **19.2.4 Massive MIMO (TDD)**.

Preferential Selection of Uplink TDD Massive MIMO Cells

TDD massive MIMO cells are preferentially selected for uplink CA, which helps improve the pairing capability for TDD massive MIMO. This function is controlled by the **UlCaCapbBasedPccSelectionSw** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter. If this option is selected, the eNodeB preferentially selects a TDD massive MIMO cell for uplink CA from candidates with the same priority.

Before enabling this function, ensure that the **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter has been selected. This function does not take effect when intelligent selection of serving cell combinations or smart carrier selection takes effect. For details about intelligent selection of serving cell combinations, see **12 Intelligent Selection of Serving Cell Combinations**. For details about smart carrier selection, see *Multi-carrier Unified Scheduling*.

This function increases the number of inter-frequency handovers for PCC anchoring. In addition, the performance indicators such as traffic volume and uplink performance of each frequency band fluctuate due to the changes in the carriers on which UEs camp.

10.2 Network Analysis

10.2.1 Benefits

This function enables CA UEs to reach higher uplink peak data rates.

Table 10-1 lists the theoretical peak data rates that a CA UE can reach using FDD uplink 2CC aggregation. These values assume a TBS suitable for the 20 MHz cell bandwidth (equivalent to 100 RBs in the frequency domain).

Table 10-2 lists the theoretical peak data rates that a CA UE can reach using TDD uplink 2CC aggregation. These values assume a TBS suitable in the following scenario:

- Bandwidth: 20 MHz (equivalent to 100 RBs in the frequency domain)
- Uplink-downlink subframe configuration: 2
- Special subframe configuration: 7

Table 10-1 Theoretical peak data rates for FDD uplink 2CC aggregation (unit: Mbit/s)

Non-MIMO + 16QAM	2x2 MIMO + 16QAM	Non-MIMO + 64QAM	2x2 MIMO + 64QAM	Non-MIMO + 256QAM	2x2 MIMO + 256QAM
102.0	203.7	150.8	299.6	211.1	422.1

Table 10-2 Theoretical peak data rates for TDD uplink 2CC aggregation (unit: Mbit/s)

Non-MIMO + 16QAM	Non-MIMO + 64QAM	Non-MIMO + 256QAM
20	30	46

The peak data rate that CA can achieve for a CA UE is subject to:

- Peak data rate capability of the board where the PCell for the CA UE is located

  For example, if the PCell of a CA UE is served by an LBBPd1 board that supports an uplink peak data rate of 225 Mbit/s, the peak data rate that CA can achieve for the CA UE will not exceed 225 Mbit/s in the uplink.
- Capability of the CA UE

  If the UE capability is limited, the actual peak data rates will be lower than the theoretical values. The UE capability is indicated by ue-categoryUL. For details about this IE, see section 4.1A "ue-CategoryDL and ue-CategoryUL" in 3GPP TS 36.306 V15.2.0.

10.2.2 Impacts

The impacts between this function and other functions are described in **10.3.3.3 Function Impacts**.

This function may have the following impacts on the network:

- Uplink interference increases because CA UEs consume additional uplink resources.
- If multiple timing advances (MTA) is enabled, the counters related to random access to SCells return larger values, because of non-contention-based random access to the SCells. In addition, if uplink power of CA UEs is limited, MTA measurements result in certain throughput loss.
- Uplink throughput rises when the **UlCaPuschPcOptSwitch** option of the **CellAlgoSwitch**.***UlPcAlgoSwitch*** parameter is selected, as long as uplink resources are sufficient. When this option is selected, the transmit power on each PRB decreases so that more PRBs can be allocated to UEs. This function extends the number of UEs in the uplink CA state. However, because more resources are now used by CA UEs, uplink interference rises, resulting in a higher uplink IBLER and a larger number of discontinuous transmissions (DTXs). Consequently, the **RRC Setup Success Rate** and **E-RAB Setup Success Rate** decrease and the **Service Drop Rate** increases.

- If all parameters involved in the downlink traffic volume requirements for SCell activation are set to non-zero values, there are the following impacts on the PCell: (1) The number of UEs in the downlink CA state and the total duration with downlink SCells being active for CA UEs increase; (2) The average CQI for non-CA UEs varies. In addition, as a UE in the CA state sends the CQIs for all its serving cells in the PCell, the channel quality differences between the PCell and SCells cause the average CQI in the PCell to alter when the number of CA UEs increases.
- The number of UEs in the uplink CA state increases, and the BBP CPU usage increases. In addition, the CPU usage of the main control board varies, affecting the RRC connection setup success rate and delay.
- If an FDD massive MIMO cell serves as an uplink SCell for a UE and the UE is about to send uplink SCell SRSs in DRX-defined sleep time, the eNodeB extends the On Duration timer to ensure that there are opportunities for the UE to send SRSs within the On Duration. As a result, the DRX-defined sleep time is shortened. Therefore, the following changes are possible:
  – The values of the **L.Cdrx.Enter.Num** and **L.Cdrx.Exit.Num** counters increase. These counters indicate how frequently UEs entered and exited DRX mode.
  – The value of the **L.Traffic.User.Cdrx.Avg** counter decreases. This counter indicates the average number of UEs on which DRX took effect.
  – The value of the **L.Cdrx.Active.TtiNum** counter increases. This counter indicates the length of active time.
  – The value of the **L.Cdrx.Sleep.TtiNum** counter decreases. This counter indicates the length of sleep time.
  – The value of the **L.Signal.Num.DRX.Reconfig** counter increases. This counter indicates the number of DRX reconfiguration messages.
- When a TDD cell acts as an SCell for a CA UE, the SCell offers beamforming gains in the downlink and experiences an increase in downlink cell throughput.

10.3 Requirements

10.3.1 Licenses (FDD)

LAOFD-080202 Carrier Aggregation for Uplink 2CC

- Adaptive configuration mode
  a. An eNodeB identifies all its served cells on the PCC frequencies configured on the eNodeB.
  b. If the **CaUl2CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for a cell on a PCC frequency, the cell consumes one sales unit of the license for LAOFD-080202 Carrier Aggregation for Uplink 2CC.
  c. The eNodeB identifies the SCC frequencies configured for the PCC frequencies.
  d. The eNodeB identifies all its served cells on these SCC frequencies.

e. If the **Cell**.***WorkMode*** parameter is set to a value other than **DL_ONLY** for a cell on an SCC frequency and the cell participates in uplink 2CC aggregation, the cell consumes one sales unit of the license for LAOFD-080202 Carrier Aggregation for Uplink 2CC.

NOTE

When inter-eNodeB CA based on relaxed backhaul or inter-eNodeB CA based on eNodeB coordination is enabled, SCells may be served by neighboring eNodeBs. If this is the case, these cells consume the license for LAOFD-080202 Carrier Aggregation for Uplink 2CC at the neighboring eNodeBs.

- CA-group-based configuration mode
  - An eNodeB identifies all the cells in the CA groups configured on the eNodeB.
  - If a CA group consists of at least two inter-frequency FDD cells and the **CaUl2CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for any of the cells in the group, each cell in the CA group consumes one sales unit of the license for LAOFD-080202 Carrier Aggregation for Uplink 2CC.

Table 10-3 lists the license model and sales unit for the feature.

Table 10-3 License model and sales unit

Feature ID	Feature Name	Model	Sales Unit
LAOFD-080202	Carrier Aggregation for Uplink 2CC	LT1SCAU2CC00	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

10.3.2 Licenses (TDD)

TDLAOFD-081407 Carrier Aggregation for Uplink 2CC

- Adaptive configuration mode

a. An eNodeB identifies all its served cells on the PCC frequencies configured on the eNodeB.

b. If the **CaUl2CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for a cell on a PCC frequency, the cell consumes one sales unit of the license for TDLAOFD-081407 Carrier Aggregation for Uplink 2CC.

c. The eNodeB identifies the SCC frequencies configured for the PCC frequencies.

d. The eNodeB identifies all its served cells on these SCC frequencies.

NOTE

When inter-eNodeB CA based on relaxed backhaul or inter-eNodeB CA based on eNodeB coordination is enabled, SCells may be served by neighboring eNodeBs. If this is the case, these cells consume the license for TDLAOFD-081407 Carrier Aggregation for Uplink 2CC at the neighboring eNodeBs.

- CA-group-based configuration mode
  - An eNodeB identifies all the cells in the CA groups configured on the eNodeB.
  - If a CA group consists of at least two inter-frequency TDD cells and the **CaUl2CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for any of the cells in the group, each cell in the CA group consumes one sales unit of the license for TDLAOFD-081407 Carrier Aggregation for Uplink 2CC.

Table 10-4 lists the license model and sales unit for the feature.

Table 10-4 License model and sales unit

Feature ID	Feature Name	Model	Sales Unit
TDLAOFD-081407	Carrier Aggregation for Uplink 2CC	LT1STULCA000	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

10.3.3 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.3.3.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
FDD TDD	Downlink 2CC aggregation	None	**5 Downlink 2CC Aggregation**

10.3.3.2 Mutually Exclusive Functions

None

10.3.3.3 Function Impacts

- Functions related to RAN performance

RAT	Function Name	Function Switch	Reference	Description
FDD	Uplink short-interval SPS	**CellUlschAlgo**.***IntvlOfUlSpsWithSkipping***	*Uplink Scheduling*	UEs that have entered uplink short-interval SPS do not support uplink 2CC aggregation.
FDD	UL CRA	**UL_COORD_RES_ALLOC_SWITCH** option of the **UlCsAlgoPara**.***UlCsSw*** parameter	*Uplink Coordinated Scheduling*	Uplink SCells seldom suffer from heavy loads or interference limitations. Therefore, uplink coordinated resource allocation (UL CRA) is currently not used in uplink SCells.
FDD	UL CPC	**UL_COORD_PC_SWITCH** option of the **UlCsAlgoPara**.***UlCsSw*** parameter	*Uplink Coordinated Scheduling*	Uplink SCells seldom suffer from heavy loads or interference limitations. Therefore, uplink coordinated power control (UL CPC) is currently not used in uplink SCells.
TDD	Intra-eNodeB UL CAMC	**UlCamcSw** option of the **CellAlgoSwitch**.***CamcSwitch*** parameter	*Uplink Coordinated Scheduling*	CA and uplink coordinated adaptive modulation and coding (UL CAMC) can be enabled simultaneously. However, UL CAMC takes effect for CA UEs only in their PCells.
TDD	Inter-eNodeB UL CAMC	**UlInterEnbCamcSw** option of the **CellAlgoSwitch**.***CamcSwitch*** parameter	*Uplink Coordinated Scheduling*	CA and UL CAMC can be enabled simultaneously. However, UL CAMC takes effect for CA UEs only in their PCells.
FDD	UL CIAS	**UL_INTRA_ENB_CIAS_SWITCH** option of the **UlCsAlgoPara**.***UlCoordIntrfAwareSchSw*** parameter	*Uplink Coordinated Scheduling*	If uplink 2CC aggregation and UL CIAS are both enabled, the gains offered by UL CIAS decrease when there is no SRS resources for SCC UEs.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Uplink 2x4 MU-MIMO	**UlVmimoSwitch** option of the **CellAlgoSwitch**.***UlSchSwitch*** parameter	*MIMO*	When uplink CA is disabled, the uplink 2x4 multi-user MIMO (MU-MIMO) function works for CA UEs only in their PCells. When uplink CA is enabled, this function works for CA UEs in all their serving cells, where uplink transmission from the UEs is applicable.
TDD	Uplink 2x8 MU-MIMO	**UlVmimoSwitch** option of the **CellAlgoSwitch**.***UlSchSwitch*** parameter	*MIMO*	When uplink CA is disabled, the uplink 2x8 MU-MIMO function works for CA UEs only in their PCells. When uplink CA is enabled, this function works for CA UEs in all their serving cells, where uplink transmission from the UEs is applicable.
TDD	Uplink SU-MIMO	**ULSUMIMO2LayersSwitch** option of the **CellAlgoSwitch**.***UlSuMimoAlgoSwitch*** parameter	*MIMO*	Uplink 2CC aggregation takes precedence over uplink single-user MIMO (SU-MIMO). If a UE has been selected for uplink CA, it will no longer be selected for uplink SU-MIMO. If a UE working in uplink SU-MIMO is selected for uplink CA, it exits uplink SU-MIMO.

RAT	Function Name	Function Switch	Reference	Description
TDD	Single-stream beamforming	**CellAlgoSwitch**.***BfAlgoSwitch***	*Beamforming (TDD)*	PCells allow for beamforming. The beamforming capabilities of SCells vary as follows: ● If uplink CA is enabled, uplink SCells also allow for beamforming. ● If uplink CA is disabled but beamforming in SCells is enabled, SCells without SRSs allow for single- and dual-stream beamforming but do not allow for long-SRS-period beamforming or MU beamforming. ● If neither uplink CA nor beamforming in SCells is enabled, SCells do not allow for beamforming.

RAT	Function Name	Function Switch	Reference	Description
TDD	Dual-stream beamforming	**CellAlgoSwitch**.***BfAlgoSwitch*** and **CellBf**.***MaxBfRankPara***	*Beamforming (TDD)*	PCells allow for beamforming. The beamforming capabilities of SCells vary as follows: ● If uplink CA is enabled, uplink SCells also allow for beamforming. ● If uplink CA is disabled but beamforming in SCells is enabled, SCells without SRSs allow for single- and dual-stream beamforming but do not allow for long-SRS-period beamforming or MU beamforming. ● If neither uplink CA nor beamforming in SCells is enabled, SCells do not allow for beamforming.

RAT	Function Name	Function Switch	Reference	Description
TDD	MU beamforming	**MuBfSwitch** option of the **CellAlgoSwitch**.***MuBfAlgoSwitch*** parameter	*Beamforming (TDD)*	PCells allow for beamforming. The beamforming capabilities of SCells vary as follows: ● If uplink CA is enabled, uplink SCells also allow for beamforming. ● If uplink CA is disabled but beamforming in SCells is enabled, SCells without SRSs allow for single- and dual-stream beamforming but do not allow for long-SRS-period beamforming or MU beamforming. ● If neither uplink CA nor beamforming in SCells is enabled, SCells do not allow for beamforming.
FDD TDD	UL CoMP cell	**UlJointReceptionSwitch** option of the **CellAlgoSwitch**.***UplinkCompSwitch*** parameter	*UL CoMP*	If an SCell is configured in the uplink for a CA UE, this UE will not be selected in the SCell for UL CoMP.

RAT	Function Name	Function Switch	Reference	Description
TDD	UL CoMP between massive MIMO cells	**MASSIVE_MIMO_UL_COMP_SW** option of the **CellUlCompAlgo**.***UlCompAlgoSwitch*** parameter	*Massive MIMO Enhancements (TDD)*	When uplink 2CC aggregation is enabled, UL CoMP cannot be performed for uplink CA UEs in their SCells. In scenarios where periodic SCell-only scheduling is performed for UEs, the duration in which UL CoMP takes effect is affected and performance indicators (such as the number of UL CoMP PRBs) slightly deteriorate.
FDD	Uplink interference cancellation	**UlInterSiteIcSwitch** option of the **CellAlgoSwitch**.***UplinkIcSwitch*** parameter	*Uplink Interference Cancellation (FDD)*	If an SCell is configured in the uplink for a CA UE, the eNodeB will not select the UE for uplink interference cancellation (UL IC).

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Terminal Awareness Differentiation	**AbnormalUeHandleSwitch** option of the **GlobalProcSwitch**.***UeCompatSwitch*** parameter	*Terminal Awareness Differentiation*	Certain UEs on live networks have compatibility issues with uplink CA. Whitelist or blacklist control for uplink CA can be used to prevent these UEs from delivering abnormal performance with uplink CA. ● With whitelist control for uplink CA, uplink CA takes effect only for UEs that have no compatibility issues with uplink CA. Whitelist control for uplink CA takes effect when all the following conditions are met: – The **CA_SWITCH_OFF** option of the **UeCompat**.***BlkLstCtrlSwitch*** parameter is deselected. – The **UL_CA_FORBID_SW** option of the **UeCompat**.***BlacklistControlExtSwitch2*** parameter is deselected. – The **UL_CA_SWITCH_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter is selected. ● With blacklist control for uplink CA (controlled by the **UL_CA_FORBID_SW**

RAT	Function Name	Function Switch	Reference	Description
				option of the **UeCompat**.***BlacklistControlExtSwitch2*** parameter), uplink CA does not take effect for UEs that have compatibility issues with uplink CA.
FDD	UMTS and LTE Zero Bufferzone	**UMTS_LTE_ZERO_BUFFER_ZONE_SW** option of the **ULZeroBufferZone**.***ZeroBufZoneSwitch*** parameter	*UMTS and LTE Zero Bufferzone*	There are fewer PUSCH and SRS resources in a cell in the bufferzone than in a common cell. Therefore, when the LTE bandwidth is 5 MHz or 10 MHz, using a cell in the bufferzone as a PCell for CA is not recommended. If the cell is used as a PCell, CA performance deteriorates.

RAT	Function Name	Function Switch	Reference	Description
FDD	Massive MIMO	Massive MIMO works in multiple scenarios. Its function switch varies depending on scenarios. For details, see *Massive MIMO Basics (FDD)*.	*Massive MIMO Basics (FDD)*	● Intra-eNodeB uplink FDD 2CC aggregation works in massive MIMO scenarios. However, when they work together, there will be slight decreases in the data rates achieved using uplink 2CC aggregation and the gains offered by Static Multiple Beam for FDD massive MIMO. ● When an FDD massive MIMO cell serves as the PCell, inter-eNodeB uplink FDD 2CC aggregation does not work. ● In FDD massive MIMO scenarios, cells on the UBBPex2 board cannot serve as PCells. ● When an FDD massive MIMO cell is engaged in uplink 2CC aggregation and the UE-specific SRS period in the cell is fixed (rather than changes with the number of UEs), SRS resources may fail to be allocated to UEs that access the cell. ● When an FDD massive MIMO cell acts as an uplink SCell for a CA UE and no SRS resources in the cell can be allocated to the UE, the RRC

RAT	Function Name	Function Switch	Reference	Description
				connection is repeatedly reestablished for the UE. When an FDD massive MIMO cell acts as an uplink SCell, a UE may not be configured with SRS resources in the SCell because of compatibility issues. As a result, the measurement results of this cell are inaccurate. To mitigate this issue, it is recommended that SRS measurement boundary protection in TM9 be enabled in the massive MIMO cell, with the **SectorSplitGroup**.***Tm9SrsMeasThreshold*** parameter set to -3 dB.
FDD	Intelligent beam shaping	**MM_INTELLIGENT_BEAM_SHAPING_SW** option of the **SectorSplitGroup**.***SectorSplitSwitch*** parameter	*Massive MIMO Enhancements (FDD)*	● Intra-base-station uplink 2CC aggregation is supported in massive MIMO cells. However, there will be slight decreases in the data rates of uplink 2CC aggregation and the gains of Static Multiple Beam. ● Inter-base-station uplink 2CC aggregation is not supported in massive MIMO cells that serve as PCells. ● Massive MIMO cells on the UBBPex2 cannot serve as PCells.

RAT	Function Name	Function Switch	Reference	Description
TDD	SRS coverage optimization	**WttxSrsCovOptSw** option of the **SRSCfg**.***SrsCfgPolicySwitch*** parameter	*Massive MIMO Optimization in WTTx Scenarios (TDD)*	SRS coverage optimization works in PCells but not in SCells.
TDD	MU beamforming	**MuBfSwitch** option of the **CellAlgoSwitch**.***MuBfAlgoSwitch*** parameter	*Massive MIMO Basics (TDD)*	● The number of PRBs successfully paired for MU beamforming on each layer may decrease by 5% when uplink CA is enabled and the **Dl2CCAckResShareSw** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter is selected. ● When uplink CA is enabled, up to eight layers can be used for pairing SCell-served UEs in MU beamforming. There is no restriction on the maximum number of layers for pairing SCell-served UEs and PCell-served UEs or the maximum number of layers for pairing SCell-served UEs and non-CA UEs.
TDD	SRS-expansion-based precise beamforming	**SRSCfg**.***SrsResExpansionSwitch***	*Massive MIMO Enhancements (TDD)*	In CA scenarios, if this feature is not enabled or disabled in all the PCell and SCells, the uplink CA success rate may decrease and there may be a higher probability that UEs in the SCells do not have SRS resources.

- Functions related to RAN services

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	TTI bundling	**TtiBundlingSwitch** option of the **CellAlgoSwitch**.***UlSchSwitch*** parameter	*VoLTE*	Uplink CA is incompatible with TTI bundling at the UE level. TTI bundling takes precedence over uplink CA. When an eNodeB determines to configure TTI bundling for a CA UE, the eNodeB sends an RRC Connection Reconfiguration message to remove the uplink SCell and configure TTI bundling.
FDD	Inter-eNodeB VoLTE CoMP	**UlVoiceJROverRelaxedBHSw** option of the **ENodeBAlgoSwitch**.***OverBBUsSwitch*** parameter	*VoLTE*	If an SCell is configured in the uplink for a CA UE, this UE will not be selected in the SCell for UL CoMP.
FDD TDD	Out-of-band relay	**CellAlgoSwitch**.***RelaySwitch***	*Relay*	In out-of-band relay scenarios, RRNs support downlink 2CC aggregation and uplink 2CC aggregation if the **Dl2CCAckResShareSw** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter is deselected, and do not support them if this option is selected.

- Functions related to CloudAIR (FDD)

RAT	Function Name	Function Switch	Reference	Description
FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	**SpectrumCloud.*SpectrumCloudSwitch*** set to **LTE_NR_SPECTRUM_SHR**	*LTE FDD and NR Dynamic Spectrum Sharing*	This function reduces the number of uplink RBs available for LTE. Therefore, the throughput of UEs in the uplink FDD CA state decreases.
FDD	LTE FDD and NR Uplink Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_UPLINK_SPECTRUM_SHR**	*LTE FDD and NR SUL Dynamic Spectrum Sharing*	This function reduces the number of uplink RBs available for LTE. Therefore, the throughput of UEs in the uplink FDD CA state decreases.
FDD	DSS and NR Flexible Refarming	**DSS_FLEX_REFARM_SW** option of the **SpectrumCloud**.***SpectrumCloudEnhSwitch*** parameter	*LTE FDD and NR Dynamic Spectrum Sharing*	This function reduces the number of uplink RBs available for LTE. Therefore, the throughput of UEs in the uplink FDD CA state decreases.

10.3.4 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

The requirements described in **Boards** of **5.3.4 Hardware** must be fulfilled. In addition:

For downlink massive CA plus uplink 2CC aggregation, only cells on UBBPd or later BBPs can act as the uplink SCell.

RF Modules

For details, see **RF Modules** in **5.3.4 Hardware**.

10.3.5 Networking

The requirements described in **5.3.5 Networking** must be fulfilled. In addition, if an SFN cell whose layer 1 and layer 2 are deployed in different BBUs participates in uplink CA, the versions of the eNodeBs where these BBUs are configured must be eRAN11.1 or later so that MTA can be enabled.

10.3.6 Others

- UEs

  UEs must comply with 3GPP Release 12 or later and support the frequency bands of the carriers to be aggregated and their bandwidths. UEs must also support the peak data rates that CA can achieve.
- EPC

  For this function to reach a theoretical peak data rate, the maximum bit rate that each UE subscribes to in the EPC cannot be lower than this theoretical value.
  - For the theoretical peak data rates for downlink, see the relevant "Benefits" sections.
  - For the theoretical peak data rates for uplink, see **10.2.1 Benefits**.

10.4 Operation and Maintenance

10.4.1 Data Configuration (FDD)

10.4.1.1 Data Preparation

This function works in either CA-group-based or adaptive configuration mode. Prepare basic data as described in **5.4.1.1 Data Preparation**.

In addition, for either mode, prepare data as described in **Table 10-5** and **Table 10-6** for function activation and optimization.

Table 10-5 Parameters for activating uplink 2CC aggregation

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg.*CellCaAlgoSwitch***	CaUl2CCSwitch	Select this option.

Table 10-6 Parameters for optimizing uplink 2CC aggregation

Parameter Name	Parameter ID	Option	Setting Notes
Uplink power control algorithm switch	**CellAlgoSwitch**.***UlPcAlgoSwitch***	UlCaPuschPcOptSwitch	Select this option.
CA Algorithm Extend Switch	**ENodeBAlgoSwitch**.***CaAlgoExtSwitch***	SccDeactByUlDtxSwitch	Select this option when uplink load is not heavy and UL CoMP is not enabled. UL CoMP is controlled by the **UlJointReceptionSwitch** option of the **CellAlgoSwitch**.***UplinkCompSwitch*** parameter.
Uplink schedule switch	**CellAlgoSwitch**.***UlSchSwitch***	SchedulerCtrlPowerSwitch	Select this option.
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	NackDtxIdentifySwitch	The setting of this option must be consistent between each possible PCell and each possible SCell for that PCell.

Parameter Name	Parameter ID	Option	Setting Notes
White List Control Switch	**UeCompat**.***WhiteLstCtrlSwitch***	UL_CA_SWITCH_ON	Though whitelist control for uplink CA is controlled by the **UL_CA_SWITCH_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter, it is also affected by the **CA_SWITCH_OFF** option of the **UeCompat**.***BlkLstCtrlSwitch*** parameter and the **UL_CA_FORBID_SW** option of the **UeCompat**.***BlacklistControlExtSwitch2*** parameter. ● When the **UL_CA_SWITCH_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter is selected: – If the **CA_SWITCH_OFF** option of the **UeCompat**.***BlkLstCtrlSwitch*** parameter and the **UL_CA_FORBID_SW** option of the **UeCompat**.***BlacklistControlExtSwitch2*** parameter are both deselected, whitelist control for uplink CA is enabled and uplink CA can take effect for the specified UEs. – Otherwise, whitelist control for uplink CA is disabled. ● When the **UL_CA_SWITCH_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter is deselected, whitelist control for uplink CA does not take effect.

Parameter Name	Parameter ID	Option	Setting Notes
Blacklist Control Extension Switch 2	**UeCompat**.***BlacklistControlExtSwitch2***	UL_CA_FORBID_SW	If there are UEs that have issues on compatibility with uplink CA on the network and this option is selected for these UEs, uplink CA is prohibited for these blacklisted UEs. In this case, uplink CA does not take effect for these UEs even if the **UL_CA_SWITCH_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter is selected.

Moreover, take the following necessary actions for this function to take effect:

- Prepare data as described in **16.4.1.1 Data Preparation** if uplink 2CC aggregation is to be deployed in a relaxed backhaul scenario.
- Prepare data as described in **17.4.1.1 Data Preparation** if uplink 2CC aggregation is to be deployed in an eNodeB coordination scenario.

10.4.1.2 Using MML Commands

Before using MML commands, refer to **10.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Before activating this function, configure cells or frequencies according to **5.4.1.2 Using MML Commands**.

The activation command examples for this function are as follows:

```
//Turning on CaUl2CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaUl2CCSwitch-1;
//(Optional) Turning on UlCaPuschPcOptSwitch for each possible PCell and SCell to enable PUSCH power
control optimization for uplink CA
MOD CELLALGOSWITCH: LocalCellId=0, UlPcAlgoSwitch=UlCaPuschPcOptSwitch-1;
MOD CELLALGOSWITCH: LocalCellId=1, UlPcAlgoSwitch=UlCaPuschPcOptSwitch-1;
//(Optional) Turning on SchedulerCtrlPowerSwitch for each possible PCell and SCell
MOD CELLALGOSWITCH: LocalCellId=0, UlSchSwitch=SchedulerCtrlPowerSwitch-1;
MOD CELLALGOSWITCH: LocalCellId=1, UlSchSwitch=SchedulerCtrlPowerSwitch-1;
//(Optional) Turning on NackDtxIdentifySwitch for each possible PCell and SCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=NackDtxIdentifySwitch-1;
```

```
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=NackDtxIdentifySwitch-1;
//(Optional) Turning on SccDeactByUlDtxSwitch when uplink load is not heavy and UL CoMP is not enabled (UL CoMP is controlled by the UlJointReceptionSwitch option of the CellAlgoSwitch.UplinkCompSwitch parameter.)
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=SccDeactByUlDtxSwitch-1;
//(Optional) Enabling whitelist control for uplink CA if uplink CA is required only for UEs that have no compatibility issues with uplink CA
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0, WhiteLstCtrlSwitch=UL_CA_SWITCH_ON-1;
//(Optional) Enabling uplink CA prohibition for blacklisted UEs, which have issues on compatibility with uplink CA, on the network
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,BlacklistControlExtSwitch2=UL_CA_FORBID_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off CaUl2CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaUl2CCSwitch-0;
```

10.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.

- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

10.4.2 Data Configuration (TDD)

10.4.2.1 Data Preparation

This function works in either CA-group-based or adaptive configuration mode. Prepare basic data as described in **5.4.1.1 Data Preparation**.

In addition, for either mode, prepare data as described in **Table 10-7** and **Table 10-8** for function activation and optimization.

Table 10-7 Parameters for activating uplink 2CC aggregation

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	CaUl2CCSwitch	Select this option.

Table 10-8 Parameters for optimizing uplink 2CC aggregation

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	UlCaCapbBasedPccSelectionSw	Select this option. Before selecting this option, ensure that the **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter has been selected.
Uplink power control algorithm switch	**CellAlgoSwitch**.***UlPcAlgoSwitch***	UlCaPuschPcOptSwitch	Select this option.
Uplink power control algorithm switch	**CellAlgoSwitch**.***UlPcAlgoSwitch***	CloseLoopOptPUSCHSwitch	Deselect this option.
CA Algorithm Extend Switch	**ENodeBAlgoSwitch**.***CaAlgoExtSwitch***	SccDeactByUlDtxSwitch	Select this option when uplink load is not heavy and UL CoMP is not enabled. UL CoMP is controlled by the **UlJointReceptionSwitch** option of the **CellAlgoSwitch**.***UplinkCompSwitch*** parameter.
Uplink schedule switch	**CellAlgoSwitch**.***UlSchSwitch***	PuschDtxSchOptSwitch	Select this option.
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	NackDtxIdentifySwitch	The setting of this option must be consistent between each possible PCell and each possible SCell for that PCell.

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	CaUlSmartSelectionSwitch	Select this option. This option can be selected only when the **Cell**.***TxRxMode*** parameter is set to **64T64R** or **32T32R**.

Parameter Name	Parameter ID	Option	Setting Notes
White List Control Switch	**UeCompat**.***WhiteLstCtrlSwitch***	UL_CA_SWITCH_ON	Though whitelist control for uplink CA is controlled by the **UL_CA_SWITCH_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter, it is also affected by the **CA_SWITCH_OFF** option of the **UeCompat**.***BlkLstCtrlSwitch*** parameter and the **UL_CA_FORBID_SW** option of the **UeCompat**.***BlacklistControlExtSwitch2*** parameter. ● When the **UL_CA_SWITCH_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter is selected: – If the **CA_SWITCH_OFF** option of the **UeCompat**.***BlkLstCtrlSwitch*** parameter and the **UL_CA_FORBID_SW** option of the **UeCompat**.***BlacklistControlExtSwitch2*** parameter are both deselected, whitelist control for uplink CA is enabled and uplink CA can take effect for the specified UEs. – Otherwise, whitelist control for uplink CA is disabled. ● When the **UL_CA_SWITCH_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter is deselected, whitelist control for uplink CA does not take effect.

Parameter Name	Parameter ID	Option	Setting Notes
Blacklist Control Extension Switch 2	**UeCompat**.***BlacklistControlExtSwitch2***	UL_CA_FORBID_SW	If there are UEs that have issues on compatibility with uplink CA on the network and this option is selected for these UEs, uplink CA is prohibited for these blacklisted UEs. In this case, uplink CA does not take effect for these UEs even if the **UL_CA_SWITCH_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter is selected.

10.4.2.2 Using MML Commands

Before using MML commands, refer to **10.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Before activating this function, configure cells or frequencies according to **5.4.1.2 Using MML Commands**.

The activation command examples for this function are as follows:

```
//Turning on CaUl2CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaUl2CCSwitch-1;
//(Optional) Turning on UlCaPuschPcOptSwitch and PuschDtxSchOptSwitch for each possible PCell and
SCell and turning off CloseLoopOptPUSCHSwitch if PUSCH power control optimization for uplink CA is
required
MOD CELLALGOSWITCH: LocalCellId=0,
UlPcAlgoSwitch=CloseLoopOptPUSCHSwitch-0&UlCaPuschPcOptSwitch-1,
UlSchSwitch=PuschDtxSchOptSwitch-1;
MOD CELLALGOSWITCH: LocalCellId=1,
UlPcAlgoSwitch=CloseLoopOptPUSCHSwitch-0&UlCaPuschPcOptSwitch-1,
UlSchSwitch=PuschDtxSchOptSwitch-1;
//(Optional) Turning on SccDeactByUlDtxSwitch when uplink load is not heavy and UL CoMP is not enabled
(UL CoMP is controlled by the UlJointReceptionSwitch option of the CellAlgoSwitch.UplinkCompSwitch
parameter.)
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=SccDeactByUlDtxSwitch-1;
//(Optional) Turning on NackDtxIdentifySwitch for each possible PCell and SCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=NackDtxIdentifySwitch-1;
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=NackDtxIdentifySwitch-1;
//(Optional) Turning on CaUlSmartSelectionSwitch in TDD massive MIMO scenarios
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaUlSmartSelectionSwitch-1;
```

```
//(Optional) Enabling whitelist control for uplink CA if uplink CA is required only for UEs that have no compatibility issues with uplink CA
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0, WhiteLstCtrlSwitch=UL_CA_SWITCH_ON-1;
//(Optional) Enabling uplink CA prohibition for blacklisted UEs, which have issues on compatibility with uplink CA, on the network
ADD UECOMPAT: Index=0, UeInfoType=UE_CAPABILITY, UeCapIndex=0,BlacklistControlExtSwitch2=UL_CA_FORBID_SW-1;
```

Optimization Command Examples

```
//Turning on EnhancedPccAnchorSwitch
MOD ENODEBALGOSWITCH: CaAlgoSwitch=EnhancedPccAnchorSwitch-1;
//Turning on UlCaCapbBasedPccSelectionSw
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=UlCaCapbBasedPccSelectionSw-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off CaUl2CCSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaUl2CCSwitch-0;
```

10.4.2.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.
- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

10.4.3 Activation Verification

Counter Observation

If the counters listed in **Table 10-9** produce non-zero values on a network that is serving CA UEs capable of uplink 2CC aggregation, uplink 2CC aggregation has taken effect in the network.

Table 10-9 Counters used to verify activation of uplink 2CC aggregation

Counter ID	Counter Name
1526732905	**L.Traffic.User.PCell.UL.Avg**

Counter ID	Counter Name
1526732894	**L.Traffic.User.SCell.UL.Avg**
1526733015	**L.ChMeas.PRB.UL.PCell.Used.Avg**
1526733016	**L.ChMeas.PRB.UL.SCell.Used.Avg**
1526733013	**L.Thrp.bits.UL.CAUser**
1526742185	**L.Thrp.Time.UL.CAUser**

Message Tracing

After a CA UE accesses a cell, the eNodeB configures a cell that meets CA conditions as an SCell for the UE. When traffic conditions are met, the eNodeB activates this SCell.

You can use the MAE-Access to verify SCell configuration and activation:

- Observe the RRC_CONN_RECFG message traced on the Uu interface. If the UL-Configuration-r10 IE in the RadioResourceConfigCommonSCell-r10 IE of the RRC_CONN_RECFG message contains the UL-FreqInfo-r10 IE, the SCell has been configured.
- Observe the number of RBs and total TBS for the CA UE in the PCell and SCell. If the numbers are not zero, the SCell has been activated.

10.4.4 Network Monitoring

In addition to the counters listed in **5.4.3 Network Monitoring**, monitor the counters in **Table 10-10** and compare the results with the network plan to evaluate network performance.

Table 10-10 Counters used to monitor performance of uplink 2CC aggregation

Counter ID	Counter Name
1526732905	**L.Traffic.User.PCell.UL.Avg**
1526732906	**L.Traffic.User.PCell.UL.Max**
1526732894	**L.Traffic.User.SCell.UL.Avg**
1526732895	**L.Traffic.User.SCell.UL.Max**
1526733015	**L.ChMeas.PRB.UL.PCell.Used.Avg**
1526733016	**L.ChMeas.PRB.UL.SCell.Used.Avg**
1526733017	**L.CA.Traffic.bits.UL.PCell**
1526733018	**L.CA.Traffic.bits.UL.SCell**
1526733019	**L.Traffic.User.SCell.Active.UL.Avg**

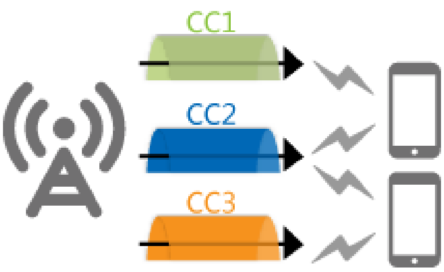
Counter ID	Counter Name
1526733020	**L.Traffic.User.SCell.Active.UL.Max**
1526767239	**L.Traffic.User.PCell.UL.Active.Avg**
1526733021	**L.CA.UL.PCell.Act.Dur**
1526742185	**L.Thrp.Time.UL.CAUser**
1526733013	**L.Thrp.bits.UL.CAUser**

11 Flexible CA

11.1 Principles

This function allows an eNodeB to select the most suitable carriers for downlink CA from a larger number of candidate carriers, as shown in **Figure 11-1**. The selections are based on the CA capabilities reported by the CA UE and on carrier management principles.

Figure 11-1 Flexible CA



This function works between intra-eNodeB cells, between inter-eNodeB cells in eNodeB coordination scenarios, and between inter-eNodeB cells in relaxed backhaul scenarios.

In CA-group-based configuration mode, this function is not under switch control. To enable this function, you only need to apply for the required licenses. For details, see **11.3.1 Licenses (FDD)** or **11.3.2 Licenses (TDD)**.

In adaptive configuration mode, this function is controlled by the **MultiCarrierFlexCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter.

- If this option is selected, the eNodeB can currently select 1–7 SCCs from a larger number of candidate carriers. It can select 7 SCCs from a maximum of 8 inter-frequency carriers. Among the candidate frequencies, the operating frequencies of the cells in the CA combination with the largest number of CCs take precedence over other frequencies.
- If this option is deselected, the eNodeB includes only a limited number of frequencies in the A4 measurement configuration delivered to the UE. This number is equal to the number of SCCs supported by the UE. When selecting

these frequencies, the eNodeB primarily considers SCC priorities (specified by the **SccFreqCfg**.***SccPriority*** parameter). It does not preferentially select the operating frequencies of the cells in the CA combination with the largest number of CCs. As a result, the eNodeB may not configure that CA combination for the UE.

The process of this function is as follows:

1. During initial access, an incoming RRC connection reestablishment, or an incoming handover, the CA UE reports its CA capabilities to the target eNodeB after setting up an RRC connection with the PCell.
2. The eNodeB selects appropriate carriers as SCCs for the UE based on the CA capabilities reported by the UE and the carrier deployment at the eNodeB.

11.2 Network Analysis

11.2.1 Benefits

This function enables an eNodeB to select the most suitable carriers for downlink CA from a larger number of candidate carriers.

For the theoretical peak data rates that a CA UE can reach using downlink 2CC, 3CC, 4CC, or 5CC aggregation, see the "Benefits" sections that describe these downlink CA functions.

For the theoretical peak data rates that a CA UE can reach using downlink FDD 6CC–8CC aggregation, see **9.2.1 Benefits**.

11.2.2 Impacts

The impacts between this function and other functions are described in **11.3.3.3 Function Impacts**.

This function has no impact on the network.

11.3 Requirements

11.3.1 Licenses (FDD)

LAOFD-070201 Flexible CA from Multiple Carriers

- Adaptive configuration mode

  If the **MultiCarrierFlexCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected for a cell, the cell consumes one sales unit of the license for LAOFD-070201 Flexible CA from Multiple Carriers.
- CA-group-based configuration mode
  - An eNodeB identifies all the cells in the CA groups configured on the eNodeB.
  - In a CA group that consists of at least three inter-frequency intra-duplex-mode cells, each cell consumes one sales unit of the license for LAOFD-070201 Flexible CA from Multiple Carriers.

Table 11-1 lists the license model and sales unit for the feature.

Table 11-1 License model and sales unit

Feature ID	Feature Name	Model	Sales Unit
LAOFD-070201	Flexible CA from Multiple Carriers	LT1SCAD2MC00	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

11.3.2 Licenses (TDD)

- In CA-group-based configuration mode

  If the cells in the CA group operate on more than two frequencies, each cell in the CA group consumes one sales unit of the license for TDLAOFD-070201 Flexible CA from Multiple Carriers.

- In adaptive configuration mode

  Each TDD cell involved in flexible CA consumes one sales unit of the license for TDLAOFD-070201 Flexible CA from Multiple Carriers.

Table 11-2 lists the license model and sales unit for the feature.

Table 11-2 License model and sales unit

Feature ID	Feature Name	Model	Sales Unit
TDLAOFD-070201	Flexible CA from Multiple Carriers	LT1SCADFMC00	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

11.3.3 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.3.3.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Downlink 2CC aggregation	None	**5 Downlink 2CC Aggregation**	Flexible CA requires this function to be activated.
FDD TDD	Downlink 3CC aggregation	**CaDl3CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**6 Downlink 3CC Aggregation**	(Optional) This function must be activated if three carriers need to be aggregated in the downlink for flexible CA.
FDD TDD	Downlink 4CC aggregation	**CaDl4CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**7 Downlink 4CC Aggregation**	(Optional) This function must be activated if four carriers need to be aggregated in the downlink for flexible CA.
FDD TDD	Downlink 5CC aggregation	**CaDl5CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**8 Downlink 5CC Aggregation**	(Optional) This function must be activated if five carriers need to be aggregated in the downlink for flexible CA.

11.3.3.2 Mutually Exclusive Functions

None

11.3.3.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_SPECTRUM_SHR**	*LTE FDD and NR Dynamic Spectrum Sharing*	The selected serving cell combination may change.

RAT	Function Name	Function Switch	Reference	Description
FDD	LTE FDD and NR Uplink Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_UPLINK_SPECTRUM_SHR**	*LTE FDD and NR SUL Dynamic Spectrum Sharing*	The selected serving cell combination may change.
FDD TDD	Smart carrier selection based on virtual grids	**SMART_CARRIER_SELECTION_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter	*Multi-carrier Unified Scheduling*	When smart carrier selection based on virtual grids takes effect, flexible CA no longer takes effect. In this case, smart carrier selection based on virtual grids takes effect for CA UEs.

11.3.4 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

For details, see **Boards** in **5.3.4 Hardware**.

RF Modules

For details, see **RF Modules** in **5.3.4 Hardware**.

11.3.5 Networking

The requirements described in **5.3.5 Networking** must be fulfilled. In addition, the number of frequencies on the live network must be greater than or equal to three.

11.3.6 Others

- UEs

  UEs must comply with 3GPP Release 12 or later and support the frequency bands of the carriers to be aggregated and their bandwidths. UEs must also support the peak data rates that CA can achieve.

- EPC

  For this function to reach a theoretical peak data rate for downlink 2CC, 3CC, 4CC, or 5CC aggregation, the maximum bit rate that each UE subscribes to in the EPC cannot be lower than this theoretical value. For the theoretical values, see the relevant "Benefits" sections.

11.4 Operation and Maintenance

11.4.1 Data Configuration

11.4.1.1 Data Preparation

This function works in either CA-group-based or adaptive configuration mode. Prepare basic data as described in **5.4.1.1 Data Preparation**.

In addition, for adaptive configuration mode, prepare data as described in **Table 11-3** for function activation.

Table 11-3 Parameters for activating flexible CA

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	MultiCarrierFlexCaSwitch	Select this option.

11.4.1.2 Using MML Commands

Before using MML commands, refer to **11.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

11.4.1.2.1 CA-Group-based Configuration Mode

Activation Command Examples

For details, see **5.4.1.2 Using MML Commands**. Add multiple cells to the CA group and configure candidate SCells for each candidate PCell.

Deactivation Command Examples

For details, see **5.4.1.2 Using MML Commands**. This function is deactivated so long as there are only two cells in the CA group.

11.4.1.2.2 Adaptive Configuration Mode

Activation Command Examples

Before activating this function, configure candidate SCCs for each candidate PCC according to **5.4.1.2 Using MML Commands**.

The activation command examples for this function are as follows:

```
//Turning on MultiCarrierFlexCaSwitch
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=MultiCarrierFlexCaSwitch-1;
```

Deactivation Command Examples

```
//Turning off MultiCarrierFlexCaSwitch
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=MultiCarrierFlexCaSwitch-0;
```

11.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.
- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.4.2 Activation Verification

For details, see **5.4.2 Activation Verification**.

11.4.3 Network Monitoring

For the counters used in scenarios of downlink 2CC to 8CC aggregation, see the corresponding "Network Monitoring" sections.

12 Intelligent Selection of Serving Cell Combinations

12.1 Principles

12.1.1 Overview

Intelligent selection of serving cell combinations can be used when CA is working in adaptive configuration mode. With this function, the eNodeB selects the most suitable serving cell combination for a CA UE based on the CA capabilities reported by the UE and based on factors such as the coverage overlap between cells, cell load status, bandwidths, and spectral efficiency. The eNodeB then informs the UE of the combination.

If CA has been deployed with PCC anchoring for RRC_CONNECTED UEs enabled before this intelligent selection function is activated, the serving cell combinations selected after the function activation might be different from those selected before the activation. If that is the case, the CA UE distribution may change. (PCC anchoring for RRC_CONNECTED UEs is controlled by the **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter.)

This function works between intra-eNodeB cells, between inter-eNodeB cells in eNodeB coordination scenarios, and between inter-eNodeB cells in relaxed backhaul scenarios.

12.1.2 Triggers

The triggering scenarios are as follows:

- During initial access, an incoming RRC connection reestablishment, or an incoming handover of a CA UE, the eNodeB initiates a procedure for intelligent selection of serving cell combinations after it receives UE capabilities reported by the UE.
- The eNodeB initiates a procedure for intelligent selection of serving cell combinations for a UE again after an E-RAB, with a QCI for which the **SMART_CA_ALLOWED** option of the **CellQciPara**.***QciAlgoSwitch*** parameter

is selected, is set up. In this situation, the eNodeB selects the most suitable serving cell combination for the UE again.

- With PCC anchoring for RRC_CONNECTED UEs enabled (by selecting the **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter), the eNodeB delivers A1 measurement configurations related to PCC anchoring for RRC_CONNECTED UEs to a CA UE during initial access, an incoming RRC connection reestablishment, or an incoming coverage-based handover of the UE. This event A1 is reported only once. After event A1 is reported, a procedure for intelligent selection of serving cell combinations is initiated.

  The RSRP threshold for event A1 involved in PCC anchoring for RRC_CONNECTED UEs is specified by the **CaMgtCfg.*EnhancedPccAnchorA1ThdRsrp*** parameter.

- With A1-based enhanced SCell selection enabled (by setting the **CaMgtCfg**.**EnhancedSccSelA1ThldRsrp** parameter to a value other than –40 dBm), the eNodeB delivers A1 measurement configurations related to A1-based enhanced SCell selection to a CA UE during initial access, an incoming RRC connection reestablishment, or an incoming coverage-based handover of the UE. This event A1 is reported only once. After event A1 is reported, a procedure for intelligent selection of serving cell combinations is initiated.

  The RSRP threshold for event A1 involved in enhanced SCell selection is specified by the **CaMgtCfg**.**EnhancedSccSelA1ThldRsrp** parameter.

- With both PCC anchoring for RRC_CONNECTED UEs and A1-based enhanced SCell selection enabled, the eNodeB first delivers A1 measurement configurations related to enhanced SCell selection to a CA UE during initial access, an incoming RRC connection reestablishment, or an incoming coverage-based handover of the UE. After event A1 is reported, a procedure for intelligent selection of serving cell combinations is initiated. When this procedure is complete, the eNodeB delivers A1 measurement configurations related to PCC anchoring for RRC_CONNECTED UEs to the UE. After event A1 is reported, another procedure for intelligent selection of serving cell combinations is initiated.

  It is recommended that the RSRP threshold for event A1 involved in enhanced SCell selection be lower than that for event A1 involved in PCC anchoring for RRC_CONNECTED UEs.

12.1.3 Intelligent Selection of Downlink Serving Cell Combinations

Intelligent selection of downlink serving cell combinations takes effect when both of the following conditions are met:

- The **MultiCarrierFlexCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected.
- The **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter is selected.

  When this option is changed from deselected to selected, it takes 3 minutes for this intelligent selection function to take effect.

The eNodeB selects a downlink serving cell combination for a CA UE as follows:

1. It selects the PCell's inter-frequency neighboring cells that overlap the PCell itself.

   The eNodeB selects an inter-frequency neighboring cell that meets one of the following conditions:

   – The **EutranInterFreqNCell.*OverlapInd*** parameter is set to **YES**.
   – The **CaGroupSCellCfg.*SCellBlindCfgFlag*** parameter is set to **TRUE**. In this case, this neighboring cell is a blind-configurable candidate SCell for the PCell.

   It is recommended that ANGC be activated to automatically set and check these two parameters. For details, see *Auto Neighbor Group Configuration*.

   If the inter-frequency neighbor relationship or the setting of either parameter is updated, it takes 3 minutes for the update to be recognized.

   If no inter-frequency neighboring cells of the PCell meet these conditions, intelligent selection does not take effect.

2. It filters out inter-frequency neighboring cells in the high-load state.

   The eNodeB considers a cell to be in the high-load state if the number of UEs in the cell is at least equal to the sum of the values of **CellMLB.*InterFreqMlbUeNumThd*** and **CellMLB.*InterFrqUeNumOffloadOffset*** for the PCell.

   For details about how to define the number of UEs in a cell, see *Intra-RAT Mobility Load Balancing*.

3. It selects the serving cell combination that has the highest total downlink air interface capability.

   If there is a noticeable load imbalance between carriers, it is recommended that intra-RAT MLB be activated to decrease the imbalance before CA is used. If MLB is not activated, a serving cell combination with a high air interface capability may have attracted a large number of UEs, which will affect CA UE experience.

   The method of calculating the air interface capability of each cell varies as follows:

   – If the **CaMgtCfg**.***MinDlAvgToBeScheduledUeNum*** parameter is set to **0**, the formula is as follows: $P \times SF \times SE - G$.
   – If the **CaMgtCfg**.***MinDlAvgToBeScheduledUeNum*** parameter is set to a non-zero value, the formula is as follows: $(P \times SF \times SE - G)/N$.

   where

   – P is the number of available downlink PRBs in the cell. It is measured by **L.ChMeas.PRB.DL.Avail**.
   – SF is specified by the **CellMLB**.***CellCapacityScaleFactor*** parameter.

     This parameter is recommended for FDD+TDD networks. For TDD cells, the impact of their uplink-downlink configurations does not need to be considered.

   – G is the bandwidth required for GBR services in the cell.
   – SE is the converted spectral efficiency in the unit of bit/RB.

     The eNodeB automatically updates the spectral efficiency of the cell with a measurement period of 1 minute.

- If the number of uplink-synchronized UEs in the cell within a measurement period is less than 10, the eNodeB does not update the spectral efficiency of the cell.
- If the number of uplink-synchronized UEs in the cell within a measurement period is greater than or equal to 10, the eNodeB calculates the spectral efficiency of the cell for the current period by using the formula of $VolumeTrans / UsedRbNum$ and converting the result into a value expressed in bit/s/Hz.

  VolumeTrans is the traffic volume within the period, and *UsedRbNum* is the total number of RBs used within the period. When measuring the traffic volume and the total number of RBs, the eNodeB does not consider the data generated and RBs used because of MCS selection with prioritized RBs. For details about MCS selection with prioritized RBs, see *Downlink Scheduling*.

– *N* is the number of active UEs in the cell. Active UEs are those whose downlink data has been buffered.

  The eNodeB samples all CA UEs every 1 ms, including those who treat the cell as their PCell and those who treat the cell as their SCell. It checks whether there is buffered downlink data for the UEs and counts those with buffered downlink data. The eNodeB then takes an average of these counts.

  - If the average is less than the value of the **CaMgtCfg**.***MinDlAvgToBeScheduledUeNum*** parameter, *N* takes the parameter value.
  - If the average is greater than or equal to the value of the **CaMgtCfg**.***MinDlAvgToBeScheduledUeNum*** parameter, *N* takes the average number.

The PCell change policy varies as follows:

– The eNodeB selects the serving cell combination with the highest total downlink air interface capability while keeping the PCell unchanged for the UE, if either of the following conditions is met:

  - Both the **EnhancedPccAnchorSwitch** and **PccAnchorSwitch** options of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter are deselected.
  - The **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected, but the CA UE has not sent any A1 measurement report to the eNodeB.

– The eNodeB selects the serving cell combination with the highest total downlink air interface capability, having the PCell either changed (as described in **4**) or unchanged for the UE, if both of the following conditions are met:

  - The **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected.
  - The eNodeB has received A1 measurement reports from the CA UE.

  In this situation, when calculating the total downlink air interface capabilities of the combinations with PCell being the same as the UE's

PCell, the eNodeB multiplies their capability values by (1 + **CaMgtCfg**.***SmartCaPccAnchoringHyst***). Note that, if the value of the **CaMgtCfg**.***SmartCaPccAnchoringHyst*** parameter is changed, the new value takes effect 3 minutes later.

4. It selects a PCell for the UE.
   - If there is only one serving cell combination with the highest total downlink air interface capability, the eNodeB selects the PCell in this combination as the PCell for the UE. PCell changes require handovers. For details about handover procedures, see **19.4.1 Mobility Management**.
   - If multiple serving cell combinations have the same total downlink air interface capability, the eNodeB selects the cell with higher duplex mode priority and the highest PCC priority as the PCell. For details about these priorities, see **4.6.1 PCC Anchoring**.
   - With the **CaMgtCfg**.***SmartCaPccAnchoringHyst*** parameter set to **0**:

     If multiple serving cell combinations have the same total downlink air interface capability, the same higher duplex mode priority, and the same highest PCC priority, the selection rule varies as follows:

     - **SmartCaPccSelSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter selected (This option takes effect about 3 minutes after it is selected.)

       The eNodeB selects the cell with the highest air interface capability as the PCell.

       If multiple cells have the highest air interface capability, the eNodeB selects a PCell as follows:

       - If the UE's PCell is included in these cells, the eNodeB keeps the PCell unchanged for the UE.
       - If the UE's PCell is not included in these cells, the eNodeB randomly selects one from these cells as the PCell.
     - **SmartCaPccSelSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter deselected
       - If the UE's PCell is included in the candidate cells, which have a higher duplex mode priority and the highest PCC priority, the eNodeB keeps the PCell unchanged for the UE.
       - If the UE's PCell is not included in the candidate cells, the eNodeB randomly selects one from the candidate cells as the PCell.
   - With the **CaMgtCfg**.***SmartCaPccAnchoringHyst*** parameter set to a non-zero value:
     - If the UE's PCell is included in the selected serving cell combination, the **SmartCaPccSelSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter does not take effect.
     - If the UE's PCell is not included in the selected serving cell combination, the eNodeB selects a PCell in the same way as it does when the **CaMgtCfg**.***SmartCaPccAnchoringHyst*** parameter is set to **0**.

12.1.4 Intelligent Selection of Uplink Serving Cell Combinations

Intelligent selection of uplink serving cell combinations takes effect when all of the following conditions are met:

- The **MultiCarrierFlexCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected.
- The **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter is selected.

  When this option is changed from deselected to selected, it takes 3 minutes for this intelligent selection function to take effect.
- The **UL_CA_COMBINATION_FIRST** option of the **CaMgtCfg**.***CaTrafficDirectionPref*** parameter is selected.

For this function to deliver its optimal performance, it is recommended that the **CaUl2CCSwitch** option of the **CaMgtCfg.*CellCaAlgoSwitch*** parameter be selected on both the PCell and SCell sides.

The eNodeB selects an uplink serving cell combination for a CA UE as follows:

1. It follows rules 1 and 2 described in **12.1.3 Intelligent Selection of Downlink Serving Cell Combinations**.
2. It preferentially selects the serving cell combination that has the largest total uplink bandwidth.
   a. The eNodeB selects the serving cell combination that has the largest total uplink bandwidth while keeping the PCell unchanged for the UE, if either of the following conditions is met:
      i. The **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter is deselected.
      ii. The **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter is selected, but the CA UE has not sent any A1 measurement report to the eNodeB.
   b. The eNodeB selects the serving cell combination that has the largest total uplink bandwidth, with the PCell allowed to be changed for the UE, if both of the following conditions are met:
      i. The **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter is selected.
      ii. The eNodeB has received A1 measurement reports from the CA UE.
3. If more than one serving cell combination has the largest total uplink bandwidth, the eNodeB selects the optimal one from them by adhering to the rules for intelligent selection of downlink serving cell combinations.

12.2 Network Analysis

12.2.1 Benefits

This function enables an eNodeB to select the most suitable carriers for CA from a larger number of candidate carriers.

For the theoretical peak data rates that a CA UE can reach using downlink 2CC, 3CC, 4CC, or 5CC aggregation or using uplink 2CC aggregation, see the "Benefits" sections that describe these CA functions.

12.2.2 Impacts

The impacts between this function and other functions are described in **12.3.3.3 Function Impacts**.

This function may have the following impacts on the network:

- Intelligent selection of serving cell combinations changes the distribution of PCells and SCells of CA UEs in frequency bands. These changes will, in turn, make alterations to performance indicators such as the traffic volume of each band and the CPU usage.
- If the **EnhancedPccAnchorSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected, intelligent selection of serving cell combinations causes an increase in the number of inter-frequency handovers for PCC anchoring. The service drop rate rises for UEs with abnormal performance in handovers.

  In this situation, if the **SmartCaPccSelSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is selected, there will be a further increase in the number of inter-frequency handovers for PCC anchoring. The service drop rate will further rise for UEs with abnormal performance in handovers.
- If intelligent selection of uplink serving cell combinations is enabled, the number of UEs in the uplink CA state increases and the CPU usage of BBPs rises. In addition, the CPU usage of the main control board varies, affecting the RRC connection setup success rate and delay.
- If PCC anchoring for NSA DC is enabled at an eNodeB, this function will select PCells for UEs that support NSA DC, after the UEs access the cells served by the eNodeB. Intelligent selection of serving cell combinations can select only SCells but cannot change PCells for these UEs.

12.3 Requirements

12.3.1 Licenses (FDD)

None

12.3.2 Licenses (TDD)

None

12.3.3 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

12.3.3.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Downlink 2CC aggregation	None	5 Downlink 2CC Aggregation	Intelligent selection of serving cell combinations requires this function to be activated.
FDD TDD	Downlink 3CC aggregation	**CaDl3CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	6 Downlink 3CC Aggregation	(Optional) This function must be activated if three carriers need to be aggregated in the downlink using intelligent selection of downlink serving cell combinations.
FDD TDD	Downlink 4CC aggregation	**CaDl4CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	7 Downlink 4CC Aggregation	(Optional) This function must be activated if four carriers need to be aggregated in the downlink using intelligent selection of downlink serving cell combinations.
FDD TDD	Downlink 5CC aggregation	**CaDl5CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	8 Downlink 5CC Aggregation	(Optional) This function must be activated if five carriers need to be aggregated in the downlink using intelligent selection of downlink serving cell combinations.
FDD TDD	Uplink 2CC aggregation	**CaUl2CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	10 Uplink 2CC Aggregation	(Optional) This function must be activated if intelligent selection of uplink serving cell combinations is required.

12.3.3.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Adaptive handling of CA for MU beamforming	**CaMubfPairingAdaptOptSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter	19.2.1 Adaptive Handling of CA for MU Beamforming	None
FDD TDD	Dedicated carrier for TM9	**Tm9DedicatedCarrier**.***Tm9CcDlEarfcn*** **Tm9DedicatedCarrier**.***WorkMode***	*Dedicated Carrier for TM9*	Intelligent selection of uplink serving cell combinations and dedicated carrier for TM9 use different selection policies for CA. The former takes priority over the latter. When intelligent selection of uplink serving cell combinations takes effect, dedicated carrier for TM9 cannot take effect.
FDD TDD	Smart carrier selection based on virtual grids	**SMART_CARRIER_SELECTION_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter	*Multi-carrier Unified Scheduling*	None
FDD TDD	Mid- and Low-Band Coordination	**LOW_FREQ_SCS_OPT_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter	*Low-Band Booster*	Mid- and Low-Band Coordination cannot be enabled if intelligent selection of serving cell combinations has been enabled.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Predictive traffic awareness-based smart carrier selection	**PRED_TRAFFIC_AWARENESS_SCS_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchEnSw*** parameter	None	If this option is selected, the **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter cannot be selected.

12.3.3.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	NSA PCC anchoring	**NSA_PCC_ANCHORING_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Performance Enhancement*	After NSA PCC anchoring is enabled, PCC anchoring for intelligent selection of serving cell combinations does not take effect. After NSA PCC anchors are selected, SCells are then selected based on intelligent selection of serving cell combinations.
FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_SPECTRUM_SHR**	*LTE FDD and NR Dynamic Spectrum Sharing*	The selected serving cell combination may change.
FDD	LTE FDD and NR Uplink Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_UPLINK_SPECTRUM_SHR**	*LTE FDD and NR SUL Dynamic Spectrum Sharing*	The selected serving cell combination may change.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	SPID-specific CA PCC anchoring policy	Multiple parameters are required for deploying SPID-specific CA PCC anchoring policies. For details, see *Flexible User Steering*.	*Flexible User Steering*	Intelligent selection of serving cell combinations (controlled by the **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter) and SPID-specific CA PCC anchoring policies cannot simultaneously take effect. If both functions are enabled, only intelligent selection of serving cell combinations takes effect.

12.3.4 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

For details, see **Boards** in **5.3.4 Hardware**.

RF Modules

For details, see **RF Modules** in **5.3.4 Hardware**.

12.3.5 Networking

The requirements described in **5.3.5 Networking** must be fulfilled. In addition:

- There must be at least three frequencies on the live network.
- For inter-eNodeB CA, X2 interfaces must be functioning.

  They are required for load status exchange, signaling exchange, and user data transmission between eNodeBs. If the X2 interfaces are not correctly configured, appropriate serving cell combinations cannot be selected for inter-eNodeB CA.

Due to X2-related protocol specifications, eNodeBs can exchange information about only eight frequencies. Certain inter-eNodeB CA combinations may fail to take effect if there are more than eight frequencies on which inter-frequency neighboring cells with **EutranInterFreqNCell**.***OverlapInd*** set to **YES** or with **CaGroupSCellCfg**.***SCellBlindCfgFlag*** set to **TRUE** are operating.

The maximum allowed number of cells on a board varies depending on the board type. If the maximum allowed numbers of cells on the boards at both ends of an X2 interface are different, certain inter-eNodeB CA combinations may fail to take effect.

12.3.6 Others

For details, see **11.3.6 Others**.

12.4 Operation and Maintenance

12.4.1 Data Configuration

12.4.1.1 Data Preparation

This function works only in adaptive configuration mode. Prepare basic data as described in **5.4.1.1.2 Adaptive Configuration Mode**.

Table 12-1 and **Table 12-2** describe the parameters used for function activation and optimization, respectively.

Table 12-1 Parameters for activating intelligent selection of serving cell combinations

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	MultiCarrierFlexCaSwitch	Select this option.
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	CaSmartSelectionSwitch	Select this option.
QCI Algorithm Switch	**CellQciPara**.***QciAlgoSwitch***	SMART_CA_ALLOWED	Select this option.
SCell Blind Configuration Flag	**CaGroupSCellCfg**.***SCellBlindCfgFlag***	None	Set this parameter to **TRUE**.

Parameter Name	Parameter ID	Option	Setting Notes
Overlap Indicator	**EutranInterFreqNCell**.***OverlapInd***	None	Set this parameter based on the network plan. If this parameter is set to **YES**, it indicates that the neighboring cell is identified as an overlapping neighboring cell. If this parameter is set to **NO**, it indicates that the default state is retained or the neighboring cell is identified as a non-overlapping neighboring cell.
Uplink schedule switch	**CellAlgoSwitch**.***UlSchSwitch***	SchedulerCtrlPowerSwitch	Select this option if intelligent selection of uplink serving cell combinations is enabled.
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	CaUl2CCSwitch	Select this option for each possible PCell and SCell if intelligent selection of uplink serving cell combinations is enabled.
CA Traffic Direction Preference	**CaMgtCfg**.***CaTrafficDirectionPref***	UL_CA_COMBINATION_FIRST	Select this option if intelligent selection of uplink serving cell combinations is enabled.

Table 12-2 Parameters for optimizing intelligent selection of serving cell combinations

Parameter Name	Parameter ID	Setting Notes
Min DL Average To-Be-Scheduled UE Number	**CaMgtCfg**.***MinDlAvgToBeScheduledUeNum***	Set this parameter to **2**.
Smart CA PCC Anchoring Hysteresis	**CaMgtCfg**.***SmartCaPccAnchoringHyst***	Set this parameter based on site conditions.
CA Algorithm Extend Switch	**ENodeBAlgoSwitch**.***CaAlgoExtSwitch***	Select the **SmartCaPccSelSwitch** option. This option takes effect when the **CaMgtCfg**.***SmartCaPccAnchoringHyst*** parameter is set to **0**.
Inter-Freq UE Number Offload Offset	**CellMLB**.***InterFrqUeNumOffloadOffset***	Set this parameter to **200**.

Parameter Name	Parameter ID	Setting Notes
MLB Match Other Feature Mode	**CellMlbHo**.***MlbMatchOtherFeatureMode***	Deselect the **HoAdmitSwitch** option.

12.4.1.2 Using MML Commands

Before using MML commands, refer to **12.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Before activating this function, configure candidate SCCs for each candidate PCC according to **5.4.1.2 Using MML Commands**. The activation command examples for this function are as follows:

```
//Turning on MultiCarrierFlexCaSwitch
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=MultiCarrierFlexCaSwitch-1;
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=MultiCarrierFlexCaSwitch-1;
//Turning on the switches controlling intelligent selection of serving cell combinations for the serving
eNodeB of each possible PCell
MOD ENODEBALGOSWITCH: CaAlgoSwitch=EnhancedPccAnchorSwitch-1&CaSmartSelectionSwitch-1,
CaAlgoExtSwitch=SmartCaPccSelSwitch-1;
//Setting InterFrqUeNumOffloadOffset
MOD CELLMLB: LocalCellId=0, InterFrqUeNumOffloadOffset=200;
MOD CELLMLB: LocalCellId=1, InterFrqUeNumOffloadOffset=200;
//Turning off HoAdmitSwitch
MOD CELLMLBHO: LocalCellId=0, MlbMatchOtherFeatureMode=HoAdmitSwitch-0;
MOD CELLMLBHO: LocalCellId=1, MlbMatchOtherFeatureMode=HoAdmitSwitch-0;
//Selecting the SMART_CA_ALLOWED option
MOD CELLQCIPARA: LocalCellId=0, Qci=1, QciAlgoSwitch=SMART_CA_ALLOWED-1;
MOD CELLQCIPARA: LocalCellId=1, Qci=1, QciAlgoSwitch=SMART_CA_ALLOWED-1;
//Setting BlindScellConfigFlag
MOD CAGROUPSCELLCFG: LocalCellId=0, SCelleNodeBId=1234, SCellLocalCellId=1, SCellBlindCfgFlag=TRUE;
MOD CAGROUPSCELLCFG: LocalCellId=1, SCelleNodeBId=1234, SCellLocalCellId=0, SCellBlindCfgFlag=TRUE;
//Setting OverlapInd
MOD EUTRANINTERFREQNCELL: LocalCellId=0, Mcc="460", Mnc="20", eNodeBId=1234, CellId=1,
OverlapInd=YES;
MOD EUTRANINTERFREQNCELL: LocalCellId=1, Mcc="460", Mnc="20", eNodeBId=1234, CellId=0,
OverlapInd=YES;
//Turning on SchedulerCtrlPowerSwitch for each possible PCell and SCell
MOD CELLALGOSWITCH: LocalCellId=0, UlSchSwitch=SchedulerCtrlPowerSwitch-1;
MOD CELLALGOSWITCH: LocalCellId=1, UlSchSwitch=SchedulerCtrlPowerSwitch-1;
//Setting the relevant parameters for each possible PCell and SCell
MOD CAMGTCFG: LocalCellId=0, MinDlAvgToBeScheduledUeNum=2, SmartCaPccAnchoringHyst=0,
CellCaAlgoSwitch=CaUl2CCSwitch-1, CaTrafficDirectionPref=UL_CA_COMBINATION_FIRST;
MOD CAMGTCFG: LocalCellId=1, MinDlAvgToBeScheduledUeNum=2, SmartCaPccAnchoringHyst=0,
CellCaAlgoSwitch=CaUl2CCSwitch-1, CaTrafficDirectionPref=UL_CA_COMBINATION_FIRST;
```


Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch controlling intelligent selection of serving cell combinations for the serving eNodeB of each possible PCell
MOD ENODEBALGOSWITCH: CaAlgoSwitch=CaSmartSelectionSwitch-0;
```

12.4.1.3 Using the MAE-Deployment

- Fast batch activation

  This function can be batch activated using the Feature Operation and Maintenance function of the MAE-Deployment. For detailed operations, see the following section in the MAE-Deployment product documentation or online help: **MAE-Deployment Operation and Maintenance** > **MAE-Deployment Guidelines** > **Enhanced Feature Management** > **Feature Operation and Maintenance**.
- Single/Batch configuration

  This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

  NOTE

  Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

12.4.2 Activation Verification

In addition to 5.4.2 Activation Verification, run the **DSP SMARTCACOMBINATION** command to check whether intelligent selection of serving cell combinations has taken effect.

12.4.3 Network Monitoring

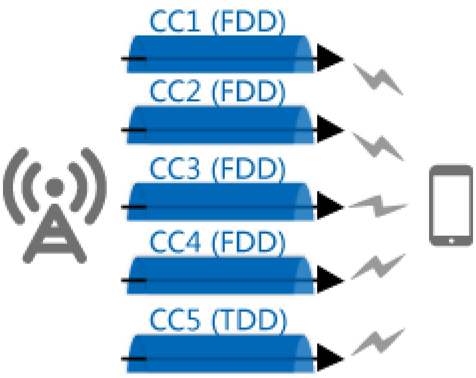
For the counters used in scenarios of downlink 2CC to 8CC aggregation, see the corresponding "Network Monitoring" sections.

13 Downlink FDD+TDD CA

13.1 Principles

Downlink FDD+TDD CA aggregates two to eight downlink FDD and TDD CCs. **Figure 13-1** uses 5CC aggregation as an example.

Figure 13-1 Downlink FDD+TDD 5CC aggregation



Downlink FDD+TDD CA is controlled by the **InterFddTddCaSwitch** option of the **CaMgtCfg.*CellCaAlgoSwitch*** parameter.

After downlink FDD+TDD CA is enabled:

- If downlink FDD+TDD massive CA is enabled, a total of six to eight FDD and TDD carriers can be aggregated in the downlink.

  Downlink FDD+TDD massive CA can be enabled by setting the **CaMgtCfg**.***FddTddCaDlMaxCcNum*** parameter to **DL_MASSIVE_CA**.

  In this situation:

  - Only FDD carriers can act as PCCs. For details about the PCC anchoring procedure, see **4.6.1.2 CA-Group-based PCC Anchoring Procedure** and **4.6.1.3 Adaptive PCC Anchoring Procedure**.
  - Both FDD and TDD carriers can act as SCCs. For details about the SCell configuration procedure, see **4.6.3.1.2 CA-Group-based SCell Configuration Procedure** and **4.6.3.1.3 Adaptive SCell Configuration Procedure**.

- If downlink FDD+TDD massive CA is disabled, a maximum of five FDD and TDD carriers can be aggregated in the downlink.

  In this situation, the PCC anchoring and SCell configuration procedures are as follows:

  – PCC anchoring procedure

    Either FDD or TDD carriers can act as PCCs. Whether an FDD or TDD cell is configured as the PCell for a UE depends on UE capabilities and the activation status of duplex-mode-priority-based PCC anchoring.

    ■ UE capabilities

      ○ The UE complies with 3GPP Release 12 or earlier and supports FDD+TDD CA.

        If the UE capability message carries the tdd-FDD-CA-PCellDuplex-r12 IE, the eNodeB configures an FDD or TDD cell as the PCell based on the reported UE capabilities.

        If the message does not carry the tdd-FDD-CA-PCellDuplex-r12 IE, the eNodeB configures an FDD or TDD cell as the PCell based on the **FddTddCaPcellDuplexFdd** or **FddTddCaPcellDuplexTdd** option setting of the **ENodeBAlgoSwitch**.***CompatibilityCtrlSwitch*** parameter.

      ○ The UE complies with 3GPP releases later than Release 12 and supports FDD+TDD CA.

        The eNodeB configures an FDD or TDD cell as the PCell based on the reported UE capabilities.

    ■ Activation status of duplex-mode-priority-based PCC anchoring

      To prioritize FDD or TDD frequencies as PCCs, duplex-mode-priority-based PCC anchoring can be enabled. The eNodeB will preferentially evaluate frequencies in the prioritized duplex mode (FDD or TDD) for PCC anchoring.

      ○ For UEs supporting aggregation of three or more FDD and TDD carriers

        The eNodeB preferentially evaluates frequencies in the duplex mode specified by the **CaMgtCfg**.***FTCAMultiCCAnchorPolicy*** parameter.

      ○ For UEs supporting FDD+TDD 2CC aggregation

        The eNodeB preferentially evaluates frequencies in the duplex mode specified by the **CaMgtCfg**.***FTCA2CCAnchorPolicy*** parameter.

      If it is not possible to select a PCC in the prioritized duplex mode, the eNodeB evaluates frequencies in the other duplex mode. For example, if FDD was prioritized by the operator but no FDD frequency could be selected as the PCC, the eNodeB would then evaluate TDD frequencies.

    For more details about the PCC anchoring procedure in this situation, see **4.6.1.2 CA-Group-based PCC Anchoring Procedure** and **4.6.1.3 Adaptive PCC Anchoring Procedure**.

  – SCell configuration procedure

Both FDD and TDD carriers can act as SCCs. For details about the SCell configuration procedure in this situation, see **4.6.3.1.2 CA-Group-based SCell Configuration Procedure** and **4.6.3.1.3 Adaptive SCell Configuration Procedure**.

Usage Scenarios

This function works in both intra- and inter-eNodeB scenarios:

- Intra-eNodeB

  In intra-eNodeB scenarios, FDD and TDD cells are served by the same eNodeB. For details, see **3.3.1 Intra-eNodeB CA Scenarios**.

- Inter-eNodeB

  Inter-eNodeB scenarios include:

  - Relaxed backhaul

    For details, see **16 Inter-eNodeB CA Based on Relaxed Backhaul**.

  - Centralized eNodeB coordination

    For details, see **17 Inter-eNodeB CA Based on eNodeB Coordination**.

Subframe Configurations

TDD cells for CA must use uplink-downlink configuration 1 or 2 and one of special subframe configurations 4, 5, 6, 7, and 9. A TDD cell using special subframe configuration 4 cannot work with another TDD cell using special subframe configuration 5, 6, 7, or 9 for CA. TDD cells served by LampSite eNodeBs can only use special subframe configurations 6 and 7.

No subframe configuration requirements are imposed on FDD cells.

Table 13-1 lists the supported FDD+TDD CC combinations, which vary depending on the PCell duplex mode and TDD subframe configuration.

Table 13-1 FDD+TDD CC combinations supported (by PCell duplex mode and TDD subframe configuration)

Duplex Mode of the PCell	Uplink-Downlink Configuration of TDD Cells	CC Combination
FDD	1 or 2	Any combination of eight CCs
TDD	1	Any combination of five CCs
TDD	2	4 TDD CCs + 1 FDD CC

For details about the principles and engineering guidelines for subframe configuration, see *Subframe Configuration (TDD)*.

13.2 Network Analysis

13.2.1 Benefits

This function deals with spectrum shortages by utilizing both FDD and TDD spectrum resources, addresses mobile broadband service competition, and improves service quality.

The theoretical peak data rates that downlink FDD+TDD CA can reach are as follows:

- Theoretical peak data rate for downlink 2CC aggregation = Theoretical peak data rate on a single FDD carrier in the downlink + Theoretical peak data rate on a single TDD carrier in the downlink
- Theoretical peak data rate for downlink 3CC aggregation = Theoretical peak data rate on two FDD carriers in the downlink + Theoretical peak data rate on a single TDD carrier in the downlink
- Theoretical peak data rate for downlink 4CC aggregation = Theoretical peak data rate on three FDD carriers in the downlink + Theoretical peak data rate on a single TDD carrier in the downlink
- Theoretical peak data rate for downlink 5CC aggregation = Theoretical peak data rate on four FDD carriers in the downlink + Theoretical peak data rate on a single TDD carrier in the downlink
- Theoretical peak data rate for downlink 6CC aggregation = Theoretical peak data rate on five FDD carriers in the downlink + Theoretical peak data rate on a single TDD carrier in the downlink
- Theoretical peak data rate for downlink 7CC aggregation = Theoretical peak data rate on six FDD carriers in the downlink + Theoretical peak data rate on a single TDD carrier in the downlink
- Theoretical peak data rate for downlink 8CC aggregation = Theoretical peak data rate on seven FDD carriers in the downlink + Theoretical peak data rate on a single TDD carrier in the downlink

13.2.2 Impacts

The impacts between this function and other functions are described in **13.3.2.3 Function Impacts**.

This function has the following impacts on the network:

- When an FDD carrier is working as the PCC and TDD carriers are working as SCCs for a CA UE:

  The downlink TDD spectrum resources are fully utilized, which increases the downlink data rate. The FDD PCC also provides better uplink coverage for the CA UE than a TDD PCC would do.
- When a TDD carrier is working as the PCC:
  - The TDD PCC can obtain beamforming gains, represented by an increased downlink data rate. However, in certain scenarios, if TTI bundling is triggered due to the limited number of bits for ACK/NACK over the PUCCH in the TDD PCell, the downlink data rate drops on each CC. TTI bundling may be triggered when:
    - The TDD PCell uses uplink-downlink configuration 2 and has participated in FDD+TDD 3CC, 4CC, or 5CC aggregation.

    - The TDD PCell uses uplink-downlink configuration 1 and has participated in FDD+TDD 4CC or 5CC aggregation.
  - Additional RBs are used for PUCCH format-3 overhead. Therefore, the downlink IBLER of the PCell fluctuates. The impact varies as follows:
    - Scenario: The **PucchSwitch** option of the **CellAlgoSwitch.*PucchAlgoSwitch*** parameter is selected to enable adaptive allocation of PUCCH resources.

      The PCell spares one additional RB for PUCCH format-3 overhead. As a result, the number of RBs available for the PUSCH decreases by at least one. (This number must be a multiple of 2, 3, or 5. For details, see section 5.3.3 "Transform precoding" in 3GPP TS 36.211 V10.1.0 (2011-03).) Total uplink throughput decreases.
    - Scenario: The **PucchSwitch** option of the **CellAlgoSwitch.*PucchAlgoSwitch*** parameter is deselected to use fixed allocation of PUCCH resources.

      The PCell changes the usage of two PUCCH RBs from periodic CQI reporting to PUCCH format-3 overhead. As a result, fewer RBs of the cell are used for periodic CQI reporting, and more UEs have to use aperiodic CQI reporting. Downlink UE throughput may slightly decrease.

Aggregation of more than five CCs has the following additional impacts on the network:

- Additional RBs are required for PUCCH format-5 overhead. The impact varies as follows:
  - Scenario: The **PucchSwitch** option of the **CellAlgoSwitch.*PucchAlgoSwitch*** parameter is selected to enable adaptive allocation of PUCCH resources.

    The PCell spares two additional RBs for PUCCH format-5 overhead. As a result, the number of RBs available for the PUSCH decreases by at least two. (This number must be a multiple of 2, 3, or 5. For details, see section 5.3.3 "Transform precoding" in 3GPP TS 36.211 V10.1.0 (2011-03).) Total uplink throughput decreases.
  - Scenario: The **PucchSwitch** option of the **CellAlgoSwitch.*PucchAlgoSwitch*** parameter is deselected to use fixed allocation of PUCCH resources.

    The PCell changes the usage of two PUCCH RBs from periodic CQI reporting to PUCCH format-5 overhead. As a result, fewer RBs of the cell are used for periodic CQI reporting, and more UEs have to use aperiodic CQI reporting. Downlink UE throughput may slightly decrease.
- The total number of bits of HARQ feedback transmitted on the PUSCH increases. This reduces the spectral efficiency of the PUSCH and decreases uplink throughput.
- More messages are exchanged in inter-BBU CA scenarios, and the values of the **VS.SctpLnk.RxMeanSpeed** and **VS.SctpLnk.TxMeanSpeed** counters increase.

13.3 Requirements

13.3.1 Licenses

Each FDD cell involved in downlink FDD+TDD CA requires one sales unit of the license for MRFD-101222 FDD+TDD Downlink Carrier Aggregation(LTE FDD).

Each TDD cell involved in downlink FDD+TDD CA requires one sales unit of the license for MRFD-101231 FDD+TDD Downlink Carrier Aggregation(LTE TDD).

If six to eight cells are involved in downlink FDD+TDD CA:

- Each of the FDD cells requires one sales unit of the license for MRFD-151309 FDD+TDD Downlink Massive CA(LTE FDD).
- Each of the TDD cells requires one sales unit of the license for MRFD-151401 FDD+TDD Downlink Massive CA(LTE TDD).

If two or more FDD cells are involved in downlink FDD+TDD CA, each of these cells also follows the licensing principles described in the corresponding "Licenses" section.

If two or more TDD cells are involved in downlink FDD+TDD CA, each of these cells also follows the licensing principles described in the corresponding "Licenses" section.

In adaptive CA configuration mode, if FDD+TDD CA is enabled for a cell on a PCC frequency, each cell on an SCC frequency in a different duplex mode than the PCC frequency (referred to as an inter-duplex-mode SCC frequency in this section for short) follows certain licensing rules. Assume that: (1) *N* denotes the value of the **CaMgtCfg**.***FddTddCaDlMaxCcNum*** parameter for a cell on the PCC frequency; (2) *M* denotes the number of inter-duplex-mode SCC frequencies among the SCC frequencies configured for the PCC frequency; (3) *G* denotes the maximum number of inter-duplex-mode SCCs that can be aggregated for downlink FDD +TDD CA. Then, *G* is equal to the smaller one between (*N* – 1) and *M*. The licensing rules for a cell on an inter-duplex-mode SCC frequency are described as follows (no repeated consumption of any feature license):

- If *G* is equal to 2 and:
  - If the cell is an FDD cell, then this cell requires one sales unit of the license for LAOFD-001001 LTE-A Introduction.
  - If the cell is a TDD cell, then this cell requires one sales unit of the license for TDLAOFD-001001 LTE-A Introduction.
- If *G* is equal to 3 and:
  - If the cell is an FDD cell, then this cell requires one sales unit of the license for LAOFD-001001 LTE-A Introduction and one sales unit of the license for LAOFD-080207 Carrier Aggregation for Downlink 3CC in 40MHz.
  - If the cell is a TDD cell, then this cell requires one sales unit of the license for TDLAOFD-001001 LTE-A Introduction and one sales unit of the license for TDLAOFD-081405 Carrier Aggregation for Downlink 3CC.
- If *G* is greater than or equal to 4 and:

  - If the cell is an FDD cell, then this cell requires one sales unit of the license for LAOFD-001001 LTE-A Introduction, one sales unit of the license for LAOFD-080207 Carrier Aggregation for Downlink 3CC in 40MHz, and one sales unit of the license for LEOFD-110303 Carrier Aggregation for Downlink 4CC and 5CC.
  - If the cell is a TDD cell, then this cell requires one sales unit of the license for TDLAOFD-001001 LTE-A Introduction, one sales unit of the license for TDLAOFD-081405 Carrier Aggregation for Downlink 3CC, and one sales unit of the license for TDLEOFD-081504 Carrier Aggregation for Downlink 4CC and 5CC.
- If *G* is between 0 (inclusive) and 2 (not inclusive), the preceding licenses are not required.

NOTE

In the case of inter-eNodeB CA based on relaxed backhaul or inter-eNodeB CA based on eNodeB coordination, the preceding licensing rules also apply to the neighboring eNodeBs that serve the cells on the SCC frequencies.

Table 13-2 lists the license models and sales units for these features.

Table 13-2 License models and sales units

Feature ID	Feature Name	Model	Sales Unit
MRFD-101222	FDD+TDD Downlink Carrier Aggregation(LTE FDD)	LT1S0FTCAF00	per cell
MRFD-101231	FDD+TDD Downlink Carrier Aggregation(LTE TDD)	LT1SFATCA000	per cell
MRFD-151309	FDD+TDD Downlink Massive CA(LTE FDD)	LT1S0FTDMCA00	per cell
MRFD-151401	FDD+TDD Downlink Massive CA(LTE TDD)	LT4SFTMCATDD	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

Table 13-3 provides certain licensing examples of downlink FDD+TDD CA.

Table 13-3 Licensing examples of downlink FDD+TDD CA

CC Combination	Scenario	FDD Licensing	TDD Licensing
1F+1T	Intra-eNodeB	Each cell requires one sales unit of the license for MRFD-101222 FDD +TDD Downlink Carrier Aggregation(LTE FDD).	Each cell requires one sales unit of the license for MRFD-101231 FDD +TDD Downlink Carrier Aggregation(LTE TDD).
1F+1T	FDD cell and TDD cell being served by different eNodeBs	Each cell requires one sales unit of the license for MRFD-101222 FDD +TDD Downlink Carrier Aggregation(LTE FDD).	Each cell requires one sales unit of the license for MRFD-101231 FDD +TDD Downlink Carrier Aggregation(LTE TDD).
2F+1T	Intra-eNodeB	Each cell requires one sales unit of the license for each of the following features: ● MRFD-101222 FDD +TDD Downlink Carrier Aggregation(LTE FDD) ● LAOFD-001001 LTE-A Introduction	Each cell requires one sales unit of the license for MRFD-101231 FDD +TDD Downlink Carrier Aggregation(LTE TDD).
2F+1T	● FDD: two intra-eNodeB cells ● FDD cells and TDD cell being served by different eNodeBs	Each cell requires one sales unit of the license for each of the following features: ● MRFD-101222 FDD +TDD Downlink Carrier Aggregation(LTE FDD) ● LAOFD-001001 LTE-A Introduction	Each cell requires one sales unit of the license for MRFD-101231 FDD +TDD Downlink Carrier Aggregation(LTE TDD).

CC Combination	Scenario	FDD Licensing	TDD Licensing
2F+1T	• FDD: two inter-eNodeB cells in a relaxed backhaul scenario • FDD cells and TDD cell being served by different eNodeBs	Each cell requires one sales unit of the license for each of the following features: • MRFD-101222 FDD +TDD Downlink Carrier Aggregation(LTE FDD) • LAOFD-001001 LTE-A Introduction In addition, the serving eNodeB of each FDD cell requires one sales unit of the license for LAOFD-080201 Inter-eNodeB CA Based on Relaxed Backhaul.	Each cell requires one sales unit of the license for MRFD-101231 FDD +TDD Downlink Carrier Aggregation(LTE TDD).
2F+2T	Intra-eNodeB	Each cell requires one sales unit of the license for each of the following features: • MRFD-101222 FDD +TDD Downlink Carrier Aggregation(LTE FDD) • LAOFD-001001 LTE-A Introduction	Each cell requires one sales unit of the license for each of the following features: • MRFD-101231 FDD +TDD Downlink Carrier Aggregation(LTE TDD) • TDLAOFD-001001 LTE-A Introduction

CC Combination	Scenario	FDD Licensing	TDD Licensing
2F+2T	• FDD: two intra-eNodeB cells • TDD: two inter-eNodeB cells in a relaxed backhaul scenario	Each cell requires one sales unit of the license for each of the following features: • MRFD-101222 FDD +TDD Downlink Carrier Aggregation(LTE FDD) • LAOFD-001001 LTE-A Introduction	Each cell requires one sales unit of the license for each of the following features: • MRFD-101231 FDD +TDD Downlink Carrier Aggregation(LTE TDD) • TDLAOFD-001001 LTE-A Introduction In addition, each of the serving eNodeBs for the TDD cells requires one sales unit of the license for TDLAOFD-081402 Inter-eNodeB CA Based on Relaxed Backhaul.
2F+2T	• FDD: two inter-eNodeB cells in an eNodeB coordination scenario • TDD: two intra-eNodeB cells	Each cell requires one sales unit of the license for each of the following features: • MRFD-101222 FDD +TDD Downlink Carrier Aggregation(LTE FDD) • LAOFD-001001 LTE-A Introduction In addition, each of the serving eNodeBs for the FDD cells requires one sales unit of the license for LAOFD-070202 Inter-eNodeB CA based on Coordinated eNodeB.	Each cell requires one sales unit of the license for each of the following features: • MRFD-101231 FDD +TDD Downlink Carrier Aggregation(LTE TDD) • TDLAOFD-001001 LTE-A Introduction

CC Combination	Scenario	FDD Licensing	TDD Licensing
2F+4T	Intra-eNodeB	Each cell requires one sales unit of the license for each of the following features: ● MRFD-101222 FDD +TDD Downlink Carrier Aggregation(LTE FDD) ● MRFD-151309 FDD +TDD Downlink Massive CA(LTE FDD) ● LAOFD-001001 LTE-A Introduction	Each cell requires one sales unit of the license for each of the following features: ● MRFD-101231 FDD +TDD Downlink Carrier Aggregation(LTE TDD) ● MRFD-151401 FDD +TDD Downlink Massive CA(LTE TDD) ● TDLAOFD-001001 LTE-A Introduction ● TDLAOFD-081405 Carrier Aggregation for Downlink 3CC ● TDLEOFD-081504 Carrier Aggregation for Downlink 4CC and 5CC
2F+4T	● FDD: two inter-eNodeB cells in an eNodeB coordination scenario ● TDD: four intra-eNodeB cells	Each cell requires one sales unit of the license for each of the following features: ● MRFD-101222 FDD +TDD Downlink Carrier Aggregation(LTE FDD) ● MRFD-151309 FDD +TDD Downlink Massive CA(LTE FDD) ● LAOFD-001001 LTE-A Introduction Moreover, each of the serving eNodeBs for the two FDD cells requires one sales unit of the license for LAOFD-070202 Inter-eNodeB CA based on Coordinated eNodeB.	Each cell requires one sales unit of the license for each of the following features: ● MRFD-101231 FDD +TDD Downlink Carrier Aggregation(LTE TDD) ● MRFD-151401 FDD +TDD Downlink Massive CA(LTE TDD) ● TDLAOFD-001001 LTE-A Introduction ● TDLAOFD-081405 Carrier Aggregation for Downlink 3CC ● TDLEOFD-081504 Carrier Aggregation for Downlink 4CC and 5CC

CC Combination	Scenario	FDD Licensing	TDD Licensing
2F+4T	• FDD: two intra-eNodeB cells • TDD: four inter-eNodeB cells in a relaxed backhaul scenario	Each cell requires one sales unit of the license for each of the following features: • MRFD-101222 FDD +TDD Downlink Carrier Aggregation(LTE FDD) • MRFD-151309 FDD +TDD Downlink Massive CA(LTE FDD) • LAOFD-001001 LTE-A Introduction	Each cell requires one sales unit of the license for each of the following features: • MRFD-101231 FDD +TDD Downlink Carrier Aggregation(LTE TDD) • MRFD-151401 FDD +TDD Downlink Massive CA(LTE TDD) • TDLAOFD-001001 LTE-A Introduction • TDLAOFD-081405 Carrier Aggregation for Downlink 3CC • TDLEOFD-081504 Carrier Aggregation for Downlink 4CC and 5CC In addition, each of the serving eNodeBs for the four TDD cells requires one sales unit of the license for TDLAOFD-081402 Inter-eNodeB CA Based on Relaxed Backhaul.

13.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

13.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Downlink 2CC aggregation	None	**5 Downlink 2CC Aggregation**	(Optional; depending on how many FDD carriers are involved in CA and how many TDD carriers are involved in CA) Downlink 2CC aggregation is required on the FDD or TDD side if at least two FDD or TDD carriers need to be aggregated in the downlink.
FDD TDD	Downlink 3CC aggregation	**CaDl3CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**6 Downlink 3CC Aggregation**	(Optional; depending on how many FDD carriers are involved in CA and how many TDD carriers are involved in CA) Downlink 3CC aggregation is required on the FDD or TDD side if at least three FDD or TDD carriers need to be aggregated in the downlink.
FDD TDD	Downlink 4CC aggregation	**CaDl4CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**7 Downlink 4CC Aggregation**	(Optional; depending on how many FDD carriers are involved in CA and how many TDD carriers are involved in CA) Downlink 4CC aggregation is required on the FDD or TDD side if at least four FDD or TDD carriers need to be aggregated in the downlink.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Downlink 5CC aggregation	**CaDl5CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**8 Downlink 5CC Aggregation**	(Optional; depending on how many FDD carriers are involved in CA and how many TDD carriers are involved in CA) Downlink 5CC aggregation is required on the FDD or TDD side if at least five FDD or TDD carriers need to be aggregated in the downlink.
FDD	Downlink massive CA	**DlMassiveCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**9 Downlink Massive CA (FDD)**	(Optional) If six or more carriers need to be aggregated in the downlink, only FDD carriers can be used as the PCCs and downlink massive CA must be enabled.
FDD TDD	Intelligent selection of serving cell combinations	**CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter	**12 Intelligent Selection of Serving Cell Combinations**	(Optional) Intelligent selection of serving cell combinations is required on the FDD or TDD side if this intelligent selection function is required for the FDD or TDD carriers involved in downlink CA.
FDD TDD	Inter-eNodeB CA based on eNodeB coordination	**FreqCfgCaOverBBUsSwitch** option of the **ENodeBAlgoSwitch**.***OverBBUsSwitch*** parameter	**17 Inter-eNodeB CA Based on eNodeB Coordination**	(Optional) Inter-eNodeB CA based on eNodeB coordination is required on the FDD or TDD side if the FDD or TDD carriers to be aggregated in the downlink are inter-eNodeB ones in an eNodeB coordination scenario.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Inter-eNodeB CA based on relaxed backhaul	FDD: **RelaxedBackhaulCaSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter TDD: **TddRelaxedBackhaulCaSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter	**16 Inter-eNodeB CA Based on Relaxed Backhaul**	(Optional) Inter-eNodeB CA based on relaxed backhaul is required on the FDD or TDD side if the FDD or TDD carriers to be aggregated in the downlink are inter-eNodeB ones in a relaxed backhaul scenario.

13.3.2.2 Mutually Exclusive Functions

None

13.3.2.3 Function Impacts

- Functions in the category "RAN functions"

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	User-number-based connected mode load equalization	**InterFreqMlbSwitch** option of the **CellAlgoSwitch**.***MlbAlgoSwitch*** parameter **CellMLB**.***MlbTriggerMode*** set to **UE_NUMBER_ONLY** **SynchronizedUE** option of the **CellMLB**.***InterFreqUeTrsfType*** parameter	*Intra-RAT Mobility Load Balancing*	If CA UE transfer is disabled (by deselecting the **CaUserLoadTrans-ferSw** option of the **CellAlgoSwitch**.***EnhancedMlbAlgoSwitch*** parameter), the eNodeB filters out CA UEs that treat the source cell as their PCell or SCell, when the eNodeB selects UEs for UE-number-based connected mode load equalization.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	CSPC	FDD: **CspcAlgoPara**.***CspcAlgoSwitch*** TDD: **CspcAlgoPara**.***TddCspcAlgoSwitch***	*CSPC*	CSPC in centralized Cloud BB scenarios requires a centralized controller, which is a process deployed on a BBP in each eNodeB. The centralized controller increases the average throughput of cells on the affected frequency and the cell edge UE throughput in the network. However, it has a negative impact on the high-throughput UEs that cause co-channel interference on the cell edge UEs. If basic scheduling is used with CA, the data rate of a CA UE (a variable used to calculate the scheduling priority) is defined as the total data rate of the UE on all the aggregated carriers. It is typically higher than the data rate of a non-CA UE. As a result, the relevant CA UEs encounter a lower probability of being scheduled and a lower data rate.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Zero Guard Band Between Contiguous Intra-Band Carriers	**CONTIG_INTRA_BAND_CARR_SW** option of the **ContigIntraBandCarr**.***ContigIntraBandCarrSw*** parameter	FDD: *Seamless Intra-Band Carrier Joining (FDD)* TDD: none	Contiguous intra-band CA does not take effect for a cell with Zero Guard Band Between Contiguous Intra-Band Carriers enabled even if the **ContigIntraBand-CaSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is selected. Intra-band non-contiguous CA can take effect for a cell with Zero Guard Band Between Contiguous Intra-Band Carriers enabled in spectrum overlapping scenarios, where the spacing between the center frequencies of two to-be-aggregated carriers is less than the average of their total bandwidths. For details about the spectrum requirements of cells with Zero Guard Band Between Contiguous Intra-Band Carriers disabled on intra-band non-contiguous CA, see **5.3.5 Networking**.

RAT	Function Name	Function Switch	Reference	Description
FDD	Short TTI	**SHORT_TTI_SW** option of the **CellShortTtiAlgo**.***SttiAlgoSwitch*** parameter	*Short TTI (FDD)*	CA does not take effect for UEs that support short TTI according to their reported capabilities.
TDD	Downlink 2-layer MIMO based on TM9	**TM9Switch** option of the **CellAlgoSwitch**.***EnhMIMOSwitch*** parameter	• *Beamforming (TDD)* • *Massive MIMO Basics (TDD)*	TM9 has the following impact on UEs in the 2CC aggregation state: If the CSI-RS is configured for an SCC with no SRS configured, a CA UE can measure CSI based on CSI-RS. Therefore, the UE can enter the TM9wPMI mode on the SCC, improving the average cell throughput and user experience.
TDD	Downlink 4-layer MIMO based on TM9	• **TM9Switch** option of the **CellAlgoSwitch**.***EnhMIMOSwitch*** parameter • **CellDlschAlgo**.***MaxMimoRankPara*** set to **SW_MAX_SM_RANK_4**	• *Beamforming (TDD)* • *Massive MIMO Basics (TDD)*	TM9 has the following impact on UEs in the 2CC aggregation state: If the CSI-RS is configured for an SCC with no SRS configured, a CA UE can measure CSI based on CSI-RS. Therefore, the UE can enter the TM9wPMI mode on the SCC, improving the average cell throughput and user experience.

RAT	Function Name	Function Switch	Reference	Description
TDD	Downlink 8x8 MIMO	• **TM9Switch** option of the **CellAlgoSwitch**.***EnhMIMOSwitch*** parameter • **CellDlschAlgo**.***MaxMimoRankPara*** set to **SW_MAX_SM_RANK_8** • **CellCsiRsParaCfg**.***CsiRsPortNum*** set to **CSI_RS_PORT_8**	• *Beamforming (TDD)* • *Massive MIMO Basics (TDD)*	TM9 has the following impact on UEs in the 2CC aggregation state: If the CSI-RS is configured for an SCC with no SRS configured, a CA UE can measure CSI based on CSI-RS. Therefore, the UE can enter the TM9wPMI mode on the SCC, improving the average cell throughput and user experience. Downlink 8x8 MIMO takes effect only for UEs in the 2CC aggregation state.
TDD	Inter-cell downlink D-MIMO	**InterCellDmimoJTSwitch** option of the **CellAlgoSwitch**.***DMIMOAlgoSwitch*** parameter	*D-MIMO (TDD)*	Inter-cell downlink D-MIMO does not work in SCells.
TDD	Inter-frequency split	None	*Soft Split Resource Duplex (TDD)*	After inter-frequency split is enabled, CA takes effect in a smaller area, and therefore CA gains are smaller.
FDD	Superior uplink coverage	**CellAlgoExtSwitch**.***UlCoverageEnhancementSw***	*Superior Uplink Coverage (FDD)*	UEs under enhanced coverage do not support downlink CA.

RAT	Function Name	Function Switch	Reference	Description
TDD	DL CoMP cell	**IntraDlCompSwitch** option of the **CellAlgoSwitch**.***DlCompSwitch*** parameter	*DL CoMP (TDD)*	CA UEs support downlink coordinated multipoint transmission (DL CoMP) in their PCells, but not in their SCells.
FDD	DL CoMP with TM9	**Tm9JtSwitch** option of the **CellAlgoSwitch**.***DlCompSwitch*** parameter	*DL CoMP (FDD)*	Joint transmission (JT) based on TM9 can be configured for DL CoMP in both the PCells and SCells of UEs in the intra-BBU CA state. TM9 JT in SCells can take effect only after TM9 JT is configured in the corresponding PCells.
FDD	DL CoMP with TM10	**FDDHomNetDlCompSwitch** and **FDDHetNetDlCompSwitch** options of the **CellAlgoSwitch**.***DlCompSwitch*** parameter	*DL CoMP (FDD)*	DL CoMP with TM10 can be configured in both the PCells and SCells of UEs in the intra-BBU CA state.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Adaptive SFN/SDMA	**CellAlgoSwitch**.***SfnUlSchSwitch*** **CellAlgoSwitch**.***SfnDlSchSwitch***	*SFN*	Adaptive SFN requires UEs to report SRSs, based on which eNodeBs select RRUs for independent scheduling of the UEs. If an adaptive SFN cell is configured as an SCell for a CA UE only in the downlink, only joint scheduling can be used in this SCell. If the adaptive SFN cell is also configured as an SCell in the uplink, independent scheduling can be used in the cell.
FDD TDD	Network-assisted CRS interference cancellation	**CellAlgoSwitch**.***CrsIcSwitch***	*Network Assisted Interference Cancellation*	For CA UEs, this cell-specific reference signal interference cancellation (CRS-IC) function takes effect only in PCells.
FDD	Dynamic TDM eICIC	**CellAlgoSwitch**.***EicicSwitch*** set to **DYNAMIC**	*TDM eICIC (FDD)*	CA UEs do not support the two types of CSI measurements for eICIC in SCells and therefore do not support dynamic time-domain enhanced inter-cell interference coordination (TDM eICIC) or further enhanced inter-cell interference coordination (FeICIC) in SCells.

RAT	Function Name	Function Switch	Reference	Description
FDD	FeICIC	**ABS** option of the **CellAlgoSwitch**.***FeicicSwitch*** parameter	*TDM eICIC (FDD)*	CA UEs do not support the two types of CSI measurements for eICIC in SCells and therefore do not support dynamic time-domain enhanced inter-cell interference coordination (TDM eICIC) or further enhanced inter-cell interference coordination (FeICIC) in SCells.
FDD TDD	Terminal Awareness Differentiation	**AbnormalUeHandleSwitch** option of the **GlobalProcSwitch**.***UeCompatSwitch*** parameter	*Terminal Awareness Differentiation*	Carrier management for CA or NSA DC does not work on some CA UEs or NSA UEs due to their software or hardware defects. To prevent UE incompatibility issues from affecting network performance, enable Terminal Awareness Differentiation with the UEs blacklisted and the **CA_SWITCH_OFF** option of the **UeCompat**.***BlkLstCtrlSwitch*** parameter selected. With these settings, eNodeBs will not perform PCC anchoring, SCell configuration, or SCG addition for these UEs. For details about carrier management for NSA DC, see *EPC-based NSA Basics*.

RAT	Function Name	Function Switch	Reference	Description
FDD	Dynamic Massive Beam	None	*Smart 8T8R (FDD)*	When passive AAU ports are used, determine whether to enable FDD+TDD CA. If the frame offset of TDD cells on the live network is not 0, you are not advised to enable FDD+TDD CA for cells served by these AAUs. If FDD+TDD CA is enabled under these conditions, channel calibration will be affected and cell performance will deteriorate.
TDD	SRS resource migration	**REMOTE_INTRF_BF_ENH_SW** option of the **UlInterfSuppressCfg**.***RemoteIntrfDlEnhSwitch*** parameter	*Interference Detection and Suppression*	When the conditions for the SRS resource migration function to take effect and those for downlink FDD+TDD CA are met at the same time, the SRS resource migration function takes precedence. Details are as follows: ● When the SRS resource migration function takes effect, downlink FDD+TDD CA stops taking effect. ● When the SRS resource migration function stops taking effect, downlink FDD+TDD CA becomes taking effect.

- Functions related to RAN services

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Emergency call	None	*Emergency Call*	When a CA UE is running an emergency call service, the eNodeB does not start SCell configuration for the UE, which prevents RRC signaling from affecting the service quality of the ongoing emergency call. After the emergency call service is finished, the eNodeB attempts to configure SCells for the UE if the UE traffic volume and SCell configuration interval conditions are fulfilled.
FDD TDD	Next Generation eCall over LTE	**NG_ECALL_SW** option of the **Emc**.***EmergCallOptimizationSw*** parameter	*Emergency Call*	When a CA UE is running an emergency call service, the eNodeB does not start SCell configuration for the UE, which prevents RRC signaling from affecting the service quality of the ongoing emergency call. After the emergency call service is finished, the eNodeB attempts to configure SCells for the UE if the UE traffic volume and SCell configuration interval conditions are fulfilled.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	VoIP semi-persistent scheduling	**SpsSchSwitch** option of the **CellAlgoSwitch**.***UlSchSwitch*** parameter **SpsSchSwitch** option of the **CellAlgoSwitch**.***DlSchSwitch*** parameter	*VoLTE*	As stipulated in section 5.10 "Semi-Persistent Scheduling" of 3GPP TS 36.321 V10.1.0, semi-persistent scheduling takes effect only in the PCells for CA UEs.
FDD TDD	CS fallback	None	*CS Fallback*	To prevent unnecessary RRC signaling exchange, the eNodeB does not start the SCell configuration procedure for CA UEs that are engaged in CS fallback procedures.
FDD TDD	Specified service carrier	**Cell**.***SpecifiedCellFlag***	*WBB*	CA does not allow WBB-service-specified cells to be used as SCells for MBB UEs, or MBB-service-prioritized cells to be used as SCells for WBB UEs. If both CA and specified service carrier are enabled, CA service experience of these UEs or related KPIs may be affected.

RAT	Function Name	Function Switch	Reference	Description
TDD	SRS interference avoidance	**WTTxSRSIntrfAvoidanceSw** option of the **SRSCfg**.***SrsCfgPolicySwitch*** parameter	*Massive MIMO Optimization in WTTx Scenarios (TDD)*	If WTTx SRS interference avoidance is set to different states (enabled and disabled) for the PCell and SCells of a CA UE, there might be no uplink SRSs for the UE in one or more of these SCells.
FDD TDD	eMBMS	**CellMBMSCfg**.***MBMSSwitch***	*eMBMS*	When this function is enabled, CA UEs can receive MBSFN subframes in their PCells but not in their SCells. Therefore, this function works for CA UEs only in PCells.
FDD TDD	Location service	**ENodeBAlgoSwitch**.***LcsSwitch***	*LCS*	Reference signal time difference (RSTD) measurements, which are used for positioning based on Observed Time Difference Of Arrival (OTDOA), increase the ACK/NACK loss rate. A higher ACK/NACK loss rate results in scheduling performance deterioration, which may affect user experience with CA.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Out-of-band relay	**CellAlgoSwitch**.***RelaySwitch***	*Relay*	In out-of-band relay scenarios, RRNs support downlink 2CC aggregation and uplink 2CC aggregation if the **Dl2CCAckResShareSw** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter is deselected, and do not support them if this option is selected.

- Functions related to CloudAIR

RAT	Function Name	Function Switch	Reference	Description
TDD	LTE spectrum coordination enhancement	**WbbCaMultiCarrierCoordSw** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	*LTE Spectrum Coordination*	Inter-CC load transfer triggered by cell load does not take effect for WBB CA UEs or RRNs on which LTE spectrum coordination enhancement has taken effect. Inter-CC load transfer triggered by cell load is controlled by the **DlCaLbAlgoSwitch** option of the **ENodeBAlgoSwitch**.***CaLbAlgoSwitch*** parameter.

RAT	Function Name	Function Switch	Reference	Description
TDD	LTE spectrum coordination enhancement	**WbbCaMultiCarrierCoordSw** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	*LTE Spectrum Coordination*	Inter-CC load transfer triggered by cell load differences does not take effect for WBB CA UEs or RRNs on which LTE spectrum coordination enhancement has taken effect. Inter-CC load transfer triggered by cell load differences is controlled by the **CaLoadBalancePreAllocSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter.
FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_SPECTRUM_SHR**	*LTE FDD and NR Dynamic Spectrum Sharing*	This function reduces the number of downlink RBs available for LTE. Therefore, the throughput of UEs in the downlink FDD CA state decreases.
FDD	GSM and LTE Spectrum Concurrency	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **GL_SPECTRUM_CONCURRENCY**	*GSM and LTE Spectrum Concurrency*	LTE cells with GSM and LTE Spectrum Concurrency enabled are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured. As a result, the LTE network throughput is affected.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	UL and DL Decoupling	**NRDUCellAlgoSwitch**.***UlDlDecouplingSwitch***	*UL and DL Decoupling* in *5G RAN Feature Documentation*	When an 8:2 slot assignment and a 3 ms frame offset are configured for C-band, and when UL and DL Decoupling is enabled together with LTE FDD and NR Uplink Spectrum Sharing, a 3 ms frame offset must be configured for LTE FDD. If LTE TDD is deployed in the same frequency band as NR TDD, FDD+TDD Downlink Carrier Aggregation cannot be enabled on the LTE side.

- Functions related to network infrastructure

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Multi-carrier coordinated energy saving	**CellShutdown**.***CellShutdownSwitch*** set to **ON_MULTI_CARRIER_HIER_SHUTDOWN**	None	Capacity cells in the multi-carrier coordinated energy saving state do not participate in carrier aggregation.
FDD TDD	Multi-RAT Coordinated Channel Shutdown	**CellRfShutdown**.***MultiRatJointChnShutdownSw***	*Multi-RAT Coordinated Energy Saving*	If carrier aggregation is enabled after Multi-RAT Coordinated Channel Shutdown takes effect, the cell is likely to exit the RF channel shutdown state. As a result, the energy saving time and energy saving gains decrease.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Intra-RAT ANR	Event-triggered ANR: **IntraRatEventAnrSwitch** option of the **ENodeBAlgoSwitch**.***AnrSwitch*** parameter Fast ANR: **IntraRatFastAnrSwitch** option of the **ENodeBAlgoSwitch**.***AnrSwitch*** parameter	*ANR Management*	During fast or event-triggered ANR, the eNodeB determines whether to select CA UEs to perform measurements based on the **ANR**.***CaUeChoseMode*** parameter setting.
FDD TDD	Inter-RAT ANR	GERAN: **GeranFastAnrSwitch** and **GeranEventAnrSwitch** options of the **ENodeBAlgoSwitch**.***AnrSwitch*** parameter UTRAN: **UtranFastAnrSwitch** and **UtranEventAnrSwitch** options of the **ENodeBAlgoSwitch**.***AnrSwitch*** parameter	*ANR Management*	During fast or event-triggered ANR, the eNodeB determines whether to select CA UEs to perform measurements based on the **ANR**.***CaUeChoseMode*** parameter setting.

Downlink FDD+TDD massive CA has the following additional impacts on functions.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Load-based SCell configuration	**SccSmartCfgSwitch** option of the **ENodeBAlgoSwitch**.*CaAlgoSwitch* parameter	**4.6.3.2 SCell Configuration Enhancement**	• If smart carrier selection based on virtual grids is disabled, load-based SCell configuration does not take effect for UEs in the downlink FDD +TDD massive CA state. • If smart carrier selection based on virtual grids is enabled, load-based SCell configuration can take effect for UEs in the downlink FDD+TDD massive CA state.
FDD	Adaptive selection of CA or high-order MIMO	**CaMgtCfg**.***CaMimoPriorityStrategySw***	**UEs** in **19.2.2 MIMO (FDD)**	When the **CaMgtCfg**.***CaMimoPriorityStrategySw*** parameter is set to **MIMO_PRIOR** or **PEAK_RATE_PRIOR**: • If smart carrier selection based on virtual grids is disabled, UEs in the downlink FDD+TDD massive CA state will not fall back to be served by fewer CCs. • If smart carrier selection based on virtual grids is enabled, UEs in the downlink FDD+TDD massive CA state will fall back to be served by fewer CCs.
FDD	Uplink short-interval SPS	**CellUlschAlgo**.***IntvlOfUlSpsWithSkipping***	*Uplink Scheduling*	UEs that have entered uplink short-interval SPS do not support downlink massive CA.

RAT	Function Name	Function Switch	Reference	Description
FDD	UMTS and LTE Zero Bufferzone	**UMTS_LTE_ZERO_BUFFER_ZONE_SW** option of the **ULZeroBufferZone**.***ZeroBufZoneSwitch*** parameter	*UMTS and LTE Zero Bufferzone*	There are fewer PUSCH and SRS resources in a cell in the bufferzone than in a common cell. Therefore, when the LTE bandwidth is 5 MHz or 10 MHz, using a cell in the bufferzone as a PCell for CA is not recommended. If the cell is used as a PCell, CA performance deteriorates.
FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	**SpectrumCloud.*****SpectrumCloudSwitch*** set to **LTE_NR_SPECTRUM_SHR**	*LTE FDD and NR Dynamic Spectrum Sharing*	LTE cells with LTE FDD and NR Flash Dynamic Spectrum Sharing enabled are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured, causing the LTE network throughput to decrease.
FDD	LTE FDD and NR Uplink Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_UPLINK_SPECTRUM_SHR**	*LTE FDD and NR SUL Dynamic Spectrum Sharing*	LTE cells with LTE FDD and NR Uplink Spectrum Sharing enabled are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured. As a result, the LTE network throughput is affected.
FDD TDD	UMTS and LTE Spectrum Sharing	**SpectrumCloud.*****SpectrumCloudSwitch*** set to **UL_SPECTRUM_SHARING**	*UMTS and LTE Spectrum Sharing*	Downlink massive CA is not recommended for cells with UMTS and LTE Spectrum Sharing enabled.

13.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)

- DBS3900 LampSite and DBS5900 LampSite

Boards

The requirements described in **Boards** of **5.3.4 Hardware** must be fulfilled. In addition:

- Do not use an FDD cell on any LBBPd4 board as the PCell for aggregation of three or more downlink FDD and TDD CCs. Otherwise, the aggregation will not work.
- When the total number of FDD and TDD carriers aggregated in the downlink exceeds five, the following constraints also apply:
  - BBU3910A and BBU3910C cannot be used for this function.
  - UBBPex2 boards cannot be used for this function.
  - Only cells on FDD UBBPd, UBBPe, UBBPg, or UBBPh boards can act as PCells.
  - The main control boards of the serving eNodeBs for PCells must be UMPT boards.

RF Modules

For details, see **RF Modules** in **5.3.4 Hardware**.

Cells

For details, see **Subframe Configurations** in **13.1 Principles**.

13.3.4 Networking

The requirements described in **5.3.5 Networking** must be fulfilled. In addition:

- eNodeBs must be time-synchronized (with the **TASM**.***CLKSYNCMODE*** parameter set to **TIME**). The time alignment error must be within:
  - 130 ns between two contiguous CCs.
  - 260 ns between two non-contiguous CCs.
- If this function is to be deployed in a relaxed backhaul scenario, the inter-eNodeB one-way delay must be less than or equal to 4 ms and the RTT must be less than or equal to 8 ms. Moreover, the requirements described in **16.3.5 Networking** must be fulfilled.
- If this function is to be deployed in an eNodeB coordination scenario, the requirements described in **17.3.5 Networking** must be fulfilled.

13.3.5 Others

- UEs

  UEs must comply with 3GPP Release 12 or later and support the frequency bands of the carriers to be aggregated and their bandwidths. UEs must also support the peak data rates that CA can achieve.

  Downlink FDD+TDD massive CA requires UEs to comply with 3GPP Release 13 or later.

- EPC

  For this function to reach a theoretical peak data rate, the maximum bit rate that each UE subscribes to in the EPC cannot be lower than this theoretical value. For the theoretical values, see **13.2.1 Benefits**.

13.4 Operation and Maintenance

13.4.1 Data Configuration

13.4.1.1 Data Preparation

This function works in either CA-group-based or adaptive configuration mode. Prepare basic data as described in **5.4.1.1 Data Preparation**.

In addition, for either mode, prepare data as described in **Table 13-4** and **Table 13-5** for function activation and optimization.

Table 13-4 Parameters for activating downlink FDD+TDD CA

Parameter Name	Parameter ID	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	Select the **InterFddTddCaSwitch** option.
FDD TDD CA DL Max CC Number	**CaMgtCfg**.***FddTddCaDlMaxCcNum***	Set this parameter based on site conditions. This parameter cannot be set to **DL_MASSIVE_CA** when the **PucchMeasOptSwitch** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter is selected.
FDD Frame Offset	**ENodeBFrameOffset**.***FddFrameOffset***	This parameter specifies the offset of the frame start time for all LTE FDD cells served by the eNodeB relative to the time of the reference clock. LTE FDD+TDD CA requires an identical frame offset setting for all the cells. Otherwise, FDD+TDD CA does not work. Run the **DSP CELLFRAMEOFFSET** command to query the frame offset actually used for each cell. If the values of **Frame Offset Effect Value(Ts)** are the same for the cells involved, the frame offsets of the cells are the same. You are advised to set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
TDD Frame Offset	**ENodeBFrameOffset**.***TddFrameOffset***	This parameter specifies the offset of the frame start time for all LTE TDD cells served by the eNodeB relative to the time of the reference clock. If uplink and downlink timeslots are not aligned between TDD systems, inter-system interference will occur. Operators can adjust this parameter to minimize the error in timeslot alignment between the TDD systems. Run the **DSP CELLFRAMEOFFSET** command to query the frame offset actually used for each cell. If the values of **Frame Offset Effect Value(Ts)** are the same for the cells involved, the frame offsets of the cells are the same. You are advised to set this parameter based on the network plan.

Table 13-5 Parameters for optimizing downlink FDD+TDD CA

Parameter Name	Parameter ID	Setting Notes
SRS-based Low Efficiency UE SINR Threshold	**CaMgtCfg**.***SrsBasedLowEffSinrThld***	• When PCC anchoring based on SRS quality is required, the value **-10** is recommended. You can adjust the value based on site conditions. To prevent ping-pong handovers, ensure that the RSRP of the co-coverage FDD cell relevant to this SINR threshold is greater than the **InterFreqHoGroup**.***InterFreqHoA2ThdRsrp*** parameter value and less than the **CaMgtCfg**.***EnhancedPccAnchorA1ThdRsrp*** parameter value for this FDD cell. If the SINR value of a CA UE is lower than this SINR threshold, the PCC can be changed from a TDD carrier to an FDD carrier that is one of the SCCs. You are advised to set this parameter for networks where TDD carriers are working in high frequency bands (for example, 3.5 GHz) and FDD carriers are working in low frequency bands (for example, 1.8 GHz). • To disable SRS-quality-based PCC anchoring, set this parameter to **255**. This parameter applies only to TDD.

Moreover, take the following necessary actions for this function to take effect:

- Prepare data as described in the relevant "Data Preparation" sections if two, three, four, or five FDD and TDD CCs need to be aggregated in the downlink.
- Prepare data as described in **17.4.1.1 Data Preparation** if downlink FDD+TDD CA is to be deployed in an eNodeB coordination scenario.
- Prepare data as described in **16.4.1.1 Data Preparation** if downlink FDD+TDD CA is to be deployed in a relaxed backhaul scenario.
- Prepare the data described in **Table 13-6** and **Table 13-7** if downlink FDD +TDD massive CA is required. Downlink FDD+TDD massive CA works only in adaptive configuration mode.

Table 13-6 Parameters for activating downlink FDD+TDD massive CA

Parameter Name	Parameter ID	Setting Notes
FDD TDD CA DL Max CC Number	**CaMgtCfg**.*FddTddCaDlMaxCcNum*	Set this parameter to **DL_MASSIVE_CA**.

Table 13-7 Parameters for optimizing downlink FDD+TDD massive CA

Parameter Name	Parameter ID	Setting Notes
DL Beyond 3CC UE Reordering Timer	**RlcPdcpParaGroup**.*Dl4cc5ccUeReorderingTimer*	Set this parameter to **Treordering_m15**. This parameter is valid only when the **RlcPdcpParaGroup**.*RlcMode* parameter is set to **RlcMode_AM**.
DL Beyond 3CC UE Status Prohibit Timer	**RlcPdcpParaGroup**.*Dl4cc5ccUeStatProhTimer*	Set this parameter to **m15**. This parameter is valid only when the **RlcPdcpParaGroup**.*RlcMode* parameter is set to **RlcMode_AM**.
Downlink RLC-SN size	**RlcPdcpParaGroup**.*DlRlcSnSize*	It is good practice to set this parameter to **RlcSnSize_size16** when the **RlcPdcpParaGroup**.*RlcMode* parameter is set to **RlcMode_AM**.

13.4.1.2 Using MML Commands

Before using MML commands, refer to **13.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Before activating this function, configure cells or frequencies according to **5.4.1.2 Using MML Commands**.

This section uses downlink FDD+TDD 2CC aggregation, downlink FDD+TDD 4CC aggregation (1 FDD carrier + 3 TDD carriers), and downlink FDD+TDD 6CC aggregation (2 FDD carriers + 4 TDD carriers) as examples to describe how to activate this function.

- Downlink FDD+TDD 2CC aggregation

```
//Turning on InterFddTddCaSwitch and setting FddTddCaDlMaxCcNum to 2CC for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=InterFddTddCaSwitch-1,
FddTddCaDlMaxCcNum=2CC;
//Setting frame offsets for the base station
MOD ENODEBFRAMEOFFSET: TddFrameOffset=0,FddFrameOffset=0;
//(Optional) Setting SrsBasedLowEffSinrThld for a TDD cell serving as the PCell
MOD CAMGTCFG: LocalCellId=0, SrsBasedLowEffSinrThld=-10;
//(Optional) Setting FTRelaxedBHCaDLMaxCcNum to DL2CC for each possible PCell and SCell, in
addition to the operations described in the section of inter-eNodeB CA based on relaxed backhaul, to
activate inter-eNodeB downlink FDD+TDD CA in a relaxed backhaul scenario
MOD CAMGTCFG: LocalCellId=0, FTRelaxedBHCaDLMaxCcNum=DL2CC;
MOD CAMGTCFG: LocalCellId=1, FTRelaxedBHCaDLMaxCcNum=DL2CC;
```

- Downlink FDD+TDD 4CC aggregation (1 FDD carrier + 3 TDD carriers)

```
//Turning on InterFddTddCaSwitch and setting FddTddCaDlMaxCcNum to 4CC for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=InterFddTddCaSwitch-1,
FddTddCaDlMaxCcNum=4CC;
//Turning on CaDl3CCSwitch for each possible SCell
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=CaDl3CCSwitch-1;
MOD CAMGTCFG: LocalCellId=2, CellCaAlgoSwitch=CaDl3CCSwitch-1;
MOD CAMGTCFG: LocalCellId=3, CellCaAlgoSwitch=CaDl3CCSwitch-1;
//Setting frame offsets for the base station
MOD ENODEBFRAMEOFFSET: TddFrameOffset=0,FddFrameOffset=0;
//(Optional) Setting SrsBasedLowEffSinrThld for a TDD cell serving as the PCell
MOD CAMGTCFG: LocalCellId=0, SrsBasedLowEffSinrThld=-10;
//(Optional) Setting FTRelaxedBHCaDLMaxCcNum to DL4CC for each possible PCell and SCell, in
addition to the operations described in the section of inter-eNodeB CA based on relaxed backhaul, to
activate inter-eNodeB downlink FDD+TDD CA in a relaxed backhaul scenario
MOD CAMGTCFG: LocalCellId=0, FTRelaxedBHCaDLMaxCcNum=DL4CC;
MOD CAMGTCFG: LocalCellId=1, FTRelaxedBHCaDLMaxCcNum=DL4CC;
MOD CAMGTCFG: LocalCellId=2, FTRelaxedBHCaDLMaxCcNum=DL4CC;
MOD CAMGTCFG: LocalCellId=3, FTRelaxedBHCaDLMaxCcNum=DL4CC;
```

- Downlink FDD+TDD 6CC aggregation (2 FDD carriers + 4 TDD carriers)

```
//Turning on InterFddTddCaSwitch and setting FddTddCaDlMaxCcNum to DL_MASSIVE_CA for each
possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=InterFddTddCaSwitch-1,
FddTddCaDlMaxCcNum=DL_MASSIVE_CA;
//Turning on CaDl5CCSwitch for each possible SCell
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=CaDl5CCSwitch-1;
MOD CAMGTCFG: LocalCellId=2, CellCaAlgoSwitch=CaDl5CCSwitch-1;
MOD CAMGTCFG: LocalCellId=3, CellCaAlgoSwitch=CaDl5CCSwitch-1;
MOD CAMGTCFG: LocalCellId=4, CellCaAlgoSwitch=CaDl5CCSwitch-1;
MOD CAMGTCFG: LocalCellId=5, CellCaAlgoSwitch=CaDl5CCSwitch-1;
//Setting frame offsets for the base station
MOD ENODEBFRAMEOFFSET: TddFrameOffset=0,FddFrameOffset=0;
//(Optional) Setting SrsBasedLowEffSinrThld for a TDD cell serving as the PCell
MOD CAMGTCFG: LocalCellId=0, SrsBasedLowEffSinrThld=-10;
//(Optional) Setting Dl4cc5ccUeReorderingTimer, Dl4cc5ccUeStatProhTimer, and DlRlcSnSize
MOD RLCPDCPPARAGROUP: RlcPdcpParaGroupId=0, RlcMode=RlcMode_AM,
DlRlcSnSize=RlcSnSize_size16, Dl4cc5ccUeReorderingTimer=Treordering_m15,
Dl4cc5ccUeStatProhTimer=m15;
//(Optional) Setting FTRelaxedBHCaDLMaxCcNum to DL_MASSIVE_CA for each possible PCell and
FDD SCell to activate inter-eNodeB downlink FDD+TDD CA in a relaxed backhaul scenario.
FTRelaxedBHCaDLMaxCcNum cannot be set to DL_MASSIVE_CA for TDD SCells.
MOD CAMGTCFG: LocalCellId=0, FTRelaxedBHCaDLMaxCcNum=DL_MASSIVE_CA;
MOD CAMGTCFG: LocalCellId=1, FTRelaxedBHCaDLMaxCcNum=DL_MASSIVE_CA;
MOD CAMGTCFG: LocalCellId=2, FTRelaxedBHCaDLMaxCcNum=DL5CC;
MOD CAMGTCFG: LocalCellId=3, FTRelaxedBHCaDLMaxCcNum=DL5CC;
MOD CAMGTCFG: LocalCellId=4, FTRelaxedBHCaDLMaxCcNum=DL5CC;
MOD CAMGTCFG: LocalCellId=5, FTRelaxedBHCaDLMaxCcNum=DL5CC;
```


Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Disabling downlink FDD+TDD 2CC aggregation
```
//Turning off InterFddTddCaSwitch for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=InterFddTddCaSwitch-0;
```
- Disabling downlink FDD+TDD 3CC, 4CC, or 5CC aggregation
```
//Setting FddTddCaDlMaxCcNum to a value less than the corresponding number of CCs. The following is an example for deactivation of downlink FDD+TDD 3CC aggregation.
MOD CAMGTCFG: LocalCellId=0, FddTddCaDlMaxCcNum=2CC;
```
- Disabling downlink FDD+TDD massive CA
```
//Setting FddTddCaDlMaxCcNum to a value other than DL_MASSIVE_CA (Take 5CC as an example.)
MOD CAMGTCFG: LocalCellId=0, FddTddCaDlMaxCcNum=5CC;
```

13.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

13.4.2 Activation Verification

In addition to the methods described in **5.4.2 Activation Verification**, observe the counters listed in this section to verify function activation.

- Downlink FDD+TDD 2CC aggregation

  If any of the counters listed in **Table 13-8** produces a non-zero value, downlink FDD+TDD 2CC aggregation has taken effect.

Table 13-8 Counters used to verify activation of downlink FDD+TDD 2CC aggregation

Counter ID	Counter Name
1526737782	**L.Traffic.User.FddTddCA.PCell.DL.Avg**
1526737783	**L.Traffic.User.FddTddCA.PCell.DL.Max**
1526737795	**L.Traffic.User.FddTddCA.PCell.DL.Active.Avg**
1526737796	**L.Traffic.User.FddTddCA.PCell.DL.Active.Max**

- Downlink FDD+TDD 3CC aggregation

  If any of the counters listed in **Table 13-9** produces a non-zero value, downlink FDD+TDD 3CC aggregation has taken effect.

Table 13-9 Counters used to verify activation of downlink FDD+TDD 3CC aggregation

Counter ID	Counter Name
1526737776	**L.Traffic.User.FddTddCA.3CC.PCell.DL.Avg**
1526737777	**L.Traffic.User.FddTddCA.3CC.PCell.DL.Max**
1526737797	**L.Traffic.User.FddTddCA.3CC.PCell.DL.Active.Avg**
1526737798	**L.Traffic.User.FddTddCA.3CC.PCell.DL.Active.Max**

- Downlink FDD+TDD 4CC aggregation

  If any of the counters listed in **Table 13-10** produces a non-zero value, downlink FDD+TDD 4CC aggregation has taken effect.

Table 13-10 Counters used to verify activation of downlink FDD+TDD 4CC aggregation

Counter ID	Counter Name
1526737778	**L.Traffic.User.FddTddCA.4CC.PCell.DL.Avg**
1526737779	**L.Traffic.User.FddTddCA.4CC.PCell.DL.Max**
1526737799	**L.Traffic.User.FddTddCA.4CC.PCell.DL.Active.Avg**
1526737800	**L.Traffic.User.FddTddCA.4CC.PCell.DL.Active.Max**

- Downlink FDD+TDD 5CC aggregation

  If any of the counters listed in **Table 13-11** produces a non-zero value, downlink FDD+TDD 5CC aggregation has taken effect.

Table 13-11 Counters used to verify activation of downlink FDD+TDD 5CC aggregation

Counter ID	Counter Name
1526741758	**L.Traffic.User.FddTddCA.5CC.PCell.DL.Avg**
1526741735	**L.Traffic.User.FddTddCA.5CC.PCell.DL.Active.Avg**

- Downlink FDD+TDD massive CA

  If any of the counters listed in **Table 13-12** produces a non-zero value on a network that is not serving any UE in the FDD- or TDD-only downlink massive CA state, downlink FDD+TDD massive CA has taken effect in the network.

Table 13-12 Counters used to verify activation of downlink FDD+TDD massive CA

Counter ID	Counter Name
1526749452	**L.Traffic.User.PCell.DL.6CC.Avg**
1526749489	**L.Traffic.User.CA.6CC.PCell.DL.Active.Avg**
1526749453	**L.Traffic.User.PCell.DL.7CC.Avg**
1526749490	**L.Traffic.User.CA.7CC.PCell.DL.Active.Avg**
1526749454	**L.Traffic.User.PCell.DL.8CC.Avg**
1526749491	**L.Traffic.User.CA.8CC.PCell.DL.Active.Avg**

For the **L.Traffic.User.FddTddCA.PCell.DL.Active.Avg**, **L.Traffic.User.FddTddCA.PCell.DL.Active.Max**, **L.Traffic.User.FddTddCA.3CC.PCell.DL.Active.Avg**, **L.Traffic.User.FddTddCA.3CC.PCell.DL.Active.Max**, **L.Traffic.User.FddTddCA.4CC.PCell.DL.Active.Avg**, **L.Traffic.User.FddTddCA.4CC.PCell.DL.Active.Max**, **L.Traffic.User.FddTddCA.5CC.PCell.DL.Active.Avg**, **L.Traffic.User.CA.6CC.PCell.DL.Active.Avg**, **L.Traffic.User.CA.7CC.PCell.DL.Active.Avg**, and **L.Traffic.User.CA.8CC.PCell.DL.Active.Avg** counters:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, these counters are stepped as long as the corresponding number of CCs are active for a UE.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, these counters are stepped only when the corresponding number of CCs are active with no SCell inactive for a UE.

For example, five FDD and TDD carriers are configured but only four of them are active for a UE in the downlink. That is, not all SCells are activated for the UE. In the presence of such a UE:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Traffic.User.FddTddCA.4CC.PCell.DL.Active.Avg** and **L.Traffic.User.FddTddCA.4CC.PCell.DL.Active.Max** counters are stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Traffic.User.FddTddCA.4CC.PCell.DL.Active.Avg** and **L.Traffic.User.FddTddCA.4CC.PCell.DL.Active.Max** counters do not count this UE in.

13.4.3 Network Monitoring

Monitoring in Non-MOCN Scenarios

In addition to the counters listed in **5.4.3 Network Monitoring**, monitor the counters in this section and compare the results with the network plan to evaluate network performance.

- Downlink FDD+TDD 2CC aggregation

  Calculate the downlink throughput of UEs in the downlink FDD+TDD 2CC aggregation state in a cell by using the following formula:

  Downlink throughput of UEs in the downlink FDD+TDD 2CC aggregation state in a cell = Total downlink PDCP-layer traffic volume of these UEs/Total downlink PDCP-layer data transmission duration for these UEs

  In the preceding formula:

  – Total downlink PDCP-layer traffic volume of UEs in the downlink FDD +TDD 2CC aggregation state in a cell = Total downlink PDCP-layer traffic volume of UEs in the downlink FDD+TDD CA state in the cell – Total downlink PDCP-layer traffic volume of all UEs in the downlink FDD+TDD CA state except those in the downlink FDD+TDD 2CC aggregation state in the cell

  – Total downlink PDCP-layer data transmission duration for UEs in the downlink FDD+TDD 2CC aggregation state in a cell = Total downlink PDCP-layer data transmission duration for all UEs in the downlink FDD +TDD CA state in the cell – Total downlink PDCP-layer data transmission duration for all UEs in the downlink FDD+TDD CA state except those in the downlink FDD+TDD 2CC aggregation state in the cell

  For the counters related to "total downlink PDCP-layer traffic volume of all UEs in the downlink FDD+TDD CA state except those in the downlink FDD +TDD 2CC aggregation state in a cell" and "total downlink PDCP-layer data transmission duration for all UEs in the downlink FDD+TDD CA state except those in the downlink FDD+TDD 2CC aggregation state in a cell", their values vary as follows:

  – If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, these counters are stepped as long as the corresponding number of CCs are active for a UE.

  – If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, these counters are stepped only when the corresponding number of CCs are active with no SCell inactive for a UE.

  For example, if only downlink FDD+TDD 2CC aggregation and downlink FDD +TDD 3CC aggregation are enabled in a cell, calculate the throughput of UEs in the downlink FDD+TDD 2CC aggregation state in the cell by using the following formula:

  (**L.Thrp.bits.DL.FddTddCAUser** – **L.Thrp.bits.DL.3CC.FddTddCAUser**)/(**L.Thrp.Time.DL.FddTddCAUser** – **L.Thrp.Time.DL.3CC.FddTddCAUser**)

  In this example, four FDD and TDD carriers are configured but only three of them are active for a UE in the downlink. That is, not all SCells are activated for the UE. Then:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Thrp.bits.DL.3CC.FddTddCAUser** and **L.Thrp.Time.DL.3CC.FddTddCAUser** counters are stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Thrp.bits.DL.3CC.FddTddCAUser** and **L.Thrp.Time.DL.3CC.FddTddCAUser** counters do not count this UE in.

Table 13-13 Counters used to monitor performance of downlink FDD+TDD 2CC aggregation (non-MOCN)

Counter ID	Counter Name
1526737782	**L.Traffic.User.FddTddCA.PCell.DL.Avg**
1526737783	**L.Traffic.User.FddTddCA.PCell.DL.Max**
1526737795	**L.Traffic.User.FddTddCA.PCell.DL.Active.Avg**
1526737796	**L.Traffic.User.FddTddCA.PCell.DL.Active.Max**
1526737807	**L.Thrp.Time.DL.FddTddCAUser**
1526737808	**L.Thrp.bits.DL.FddTddCAUser**
1526737768	**L.E-RAB.AbnormRel.FddTddCAUser**
1526737774	**L.E-RAB.NormRel.FddTddCAUser**

- Downlink FDD+TDD 3CC aggregation

  Calculate the throughput of UEs in the downlink FDD+TDD 3CC aggregation state by using the following formula: **L.Thrp.bits.DL.3CC.FddTddCAUser**/ **L.Thrp.Time.DL.3CC.FddTddCAUser**.

Table 13-14 Counters used to monitor performance of downlink FDD+TDD 3CC aggregation (non-MOCN)

Counter ID	Counter Name
1526737776	**L.Traffic.User.FddTddCA.3CC.PCell.DL.Avg**
1526737777	**L.Traffic.User.FddTddCA.3CC.PCell.DL.Max**
1526737797	**L.Traffic.User.FddTddCA.3CC.PCell.DL.Active.Avg**
1526737798	**L.Traffic.User.FddTddCA.3CC.PCell.DL.Active.Max**
1526737810	**L.Thrp.Time.DL.3CC.FddTddCAUser**

Counter ID	Counter Name
1526737811	**L.Thrp.bits.DL.3CC.FddTddCAUser**
1526737764	**L.E-RAB.AbnormRel.FddTddCAUser.3CC**
1526737770	**L.E-RAB.NormRel.FddTddCAUser.3CC**

- Downlink FDD+TDD 4CC aggregation

  Calculate the throughput of UEs in the downlink FDD+TDD 4CC aggregation state by using the following formula: **L.Thrp.bits.DL.4CC.FddTddCAUser**/ **L.Thrp.Time.DL.4CC.FddTddCAUser**.

Table 13-15 Counters used to monitor performance of downlink FDD+TDD 4CC aggregation (non-MOCN)

Counter ID	Counter Name
1526737778	**L.Traffic.User.FddTddCA.4CC.PCell.DL.Avg**
1526737779	**L.Traffic.User.FddTddCA.4CC.PCell.DL.Max**
1526737799	**L.Traffic.User.FddTddCA.4CC.PCell.DL.Active.Avg**
1526737800	**L.Traffic.User.FddTddCA.4CC.PCell.DL.Active.Max**
1526737814	**L.Thrp.Time.DL.4CC.FddTddCAUser**
1526737815	**L.Thrp.bits.DL.4CC.FddTddCAUser**
1526737765	**L.E-RAB.AbnormRel.FddTddCAUser.4CC**
1526737771	**L.E-RAB.NormRel.FddTddCAUser.4CC**

- Downlink FDD+TDD 5CC aggregation

  Calculate the throughput of UEs in the downlink FDD+TDD 5CC aggregation state by using the following formula: **L.Thrp.bits.DL.5CC.FddTddCAUser**/ **L.Thrp.Time.DL.5CC.FddTddCAUser**.

Table 13-16 Counters used to monitor performance of downlink FDD+TDD 5CC aggregation (non-MOCN)

Counter ID	Counter Name
1526741758	**L.Traffic.User.FddTddCA.5CC.PCell.DL.Avg**
1526741735	**L.Traffic.User.FddTddCA.5CC.PCell.DL.Active.Avg**
1526741754	**L.Thrp.Time.DL.5CC.FddTddCAUser**
1526741755	**L.Thrp.bits.DL.5CC.FddTddCAUser**

Counter ID	Counter Name
1526741756	**L.E-RAB.AbnormRel.FddTddCAUser.5CC**
1526741757	**L.E-RAB.NormRel.FddTddCAUser.5CC**

- Downlink FDD+TDD massive CA

  Given that the network is not serving any UE in the FDD- or TDD-only downlink massive CA state:

  – Calculate the throughput of UEs in the downlink FDD+TDD 6CC aggregation state by using the following formula: **L.Thrp.bits.DL.6CC.CAUser**/**L.Thrp.Time.DL.6CC.CAUser**.
  – Calculate the throughput of UEs in the downlink FDD+TDD 7CC aggregation state by using the following formula: **L.Thrp.bits.DL.7CC.CAUser**/**L.Thrp.Time.DL.7CC.CAUser**.
  – Calculate the throughput of UEs in the downlink FDD+TDD 8CC aggregation state by using the following formula: **L.Thrp.bits.DL.8CC.CAUser**/**L.Thrp.Time.DL.8CC.CAUser**.

Table 13-17 Counters used to monitor performance of downlink FDD+TDD massive CA

Counter ID	Counter Name
1526749452	**L.Traffic.User.PCell.DL.6CC.Avg**
1526749489	**L.Traffic.User.CA.6CC.PCell.DL.Active.Avg**
1526749453	**L.Traffic.User.PCell.DL.7CC.Avg**
1526749490	**L.Traffic.User.CA.7CC.PCell.DL.Active.Avg**
1526749454	**L.Traffic.User.PCell.DL.8CC.Avg**
1526749491	**L.Traffic.User.CA.8CC.PCell.DL.Active.Avg**
1526749443	**L.Thrp.Time.DL.6CC.CAUser**
1526749440	**L.Thrp.bits.DL.6CC.CAUser**
1526749444	**L.Thrp.Time.DL.7CC.CAUser**
1526749441	**L.Thrp.bits.DL.7CC.CAUser**
1526749445	**L.Thrp.Time.DL.8CC.CAUser**
1526749442	**L.Thrp.bits.DL.8CC.CAUser**

For the **L.Thrp.bits.DL.3CC.FddTddCAUser**, **L.Thrp.Time.DL.3CC.FddTddCAUser**, **L.Thrp.bits.DL.4CC.FddTddCAUser**, **L.Thrp.Time.DL.4CC.FddTddCAUser**, **L.Thrp.bits.DL.5CC.FddTddCAUser**, **L.Thrp.Time.DL.5CC.FddTddCAUser**, **L.Thrp.bits.DL.6CC.CAUser**, **L.Thrp.Time.DL.6CC.CAUser**, **L.Thrp.bits.DL.7CC.CAUser**, **L.Thrp.Time.DL.7CC.CAUser**, **L.Thrp.bits.DL.8CC.CAUser**, and **L.Thrp.Time.DL.8CC.CAUser** counters:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, these counters are stepped as long as the corresponding number of CCs are active for a UE.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, these counters are stepped only when the corresponding number of CCs are active with no SCell inactive for a UE.

For example, five FDD and TDD carriers are configured but only four of them are active for a UE in the downlink. That is, not all SCells are activated for the UE. In the presence of such a UE:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Thrp.bits.DL.4CC.FddTddCAUser** and **L.Thrp.Time.DL.4CC.FddTddCAUser** counters are stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Thrp.bits.DL.4CC.FddTddCAUser** and **L.Thrp.Time.DL.4CC.FddTddCAUser** counters do not count this UE in.

Monitoring in MOCN Scenarios

In addition to the counters listed in **5.4.3 Network Monitoring**, monitor the counters in **Table 13-18** and compare the results with the network plan to evaluate network performance.

Table 13-18 Counters used to monitor performance of downlink FDD+TDD CA (MOCN)

Counter ID	Counter Name
1526739768	**L.Traffic.User.FddTddCA.PCell.DL.Avg.PLMN**
1526739746	**L.Thrp.Time.DL.FddTddCAUser.PLMN**
1526739745	**L.Thrp.bits.DL.FddTddCAUser.PLMN**
1526739770	**L.E-RAB.AbnormRel.FddTddCAUser.PLMN**
1526739772	**L.E-RAB.NormRel.FddTddCAUser.PLMN**

14 Uplink FDD+TDD CA

14.1 Principles

This function aggregates two uplink FDD and TDD CCs, as shown in **Figure 14-1**.

Figure 14-1 Uplink FDD+TDD CA

This function is controlled by the **InterFddTddCaSwitch** option of the **CaMgtCfg.*CellCaAlgoSwitch*** parameter and the **CaMgtCfg**.***FddTddCaUlMaxCcNum*** parameter. The latter parameter must be set to **2CC**.

The PCC anchoring and SCell configuration procedures for uplink FDD+TDD CA are as follows:

- PCC anchoring procedure

  The PCC anchoring procedure is similar to that for downlink FDD+TDD CA. However, the treatment of UEs that comply with 3GPP Release 12 or later and support FDD+TDD CA is different. The PCC anchoring procedure for such a UE varies as follows:

  – The UE capability message carries the tdd-FDD-CA-PCellDuplex-r12 IE.

    The eNodeB configures an FDD or TDD cell as the PCell based on the reported UE capability.

  – The UE capability message does not carry the tdd-FDD-CA-PCellDuplex-r12 IE.

    The eNodeB configures an FDD or TDD cell as the PCell based on the **FddTddCaPcellDuplexFdd** or **FddTddCaPcellDuplexTdd** option setting of the **ENodeBAlgoSwitch**.***CompatibilityCtrlSwitch*** parameter.

- SCell configuration procedure

Either FDD or TDD carriers can act as SCCs. The SCell configuration procedure is the same as that for downlink FDD+TDD CA. For details, see **13.1 Principles**.

Usage Scenarios

For details, see **Usage Scenarios** in **13.1 Principles**.

Subframe Configuration

The TDD cells involved in this function must use uplink-downlink configuration 2.

If a non-zero frame offset is configured for a TDD cell, the same frame offset must be specified for the FDD cells involved in CA. FDD hardware of SRAN10.1 or later versions all support frame offset setting.

For details about the principles and engineering guidelines for subframe configuration, see *Subframe Configuration (TDD)*.

14.2 Network Analysis

14.2.1 Benefits

This function deals with spectrum shortages by utilizing both FDD and TDD spectrum resources, addresses mobile broadband service competition, and improves service quality.

Theoretical peak data rate for uplink FDD+TDD 2CC aggregation = Theoretical peak data rate on a single FDD carrier in the uplink + Theoretical peak data rate on a single TDD carrier in the uplink

14.2.2 Impacts

The impacts between this function and other functions are described in **14.3.2.3 Function Impacts**.

This function has the following impacts on the network:

- When a TDD carrier serves as an SCC for a UE that supports uplink CA, TM7/TM8/TM9 hybrid precoding can be enabled in the downlink for the SCC to obtain beamforming gains and increase the downlink cell throughput.
- When MTA is configured in FDD+TDD CA scenarios, this function increases the uplink throughput of CA UEs and improves uplink performance if there is a noticeable difference (for example, about 78 meters, which is equivalent to one TA) between the distances of a CA UE to the receive antennas of the PCell and an SCell. In addition, due to the non-contention-based random access to the SCell, the performance counters related to random access to the SCell produce larger values.
- When CA is deployed between eNodeBs in relaxed backhaul scenarios, ACKs/NACKs for the PCell and SCell may not be received promptly because of inter-eNodeB transmission delay. If that is the case, there will be a decrease of no more than 5% in the downlink data rates of CA UEs. In addition, due to inter-eNodeB transmission delay, transmission of CQIs on the PUSCH cannot be

identified promptly. To address this issue, PUSCH scheduling in the SCell yields to periodic CQI reporting. The shorter the CQI reporting period, the more frequently the yielding occurs and the greater the decrease in the uplink data rates of CA UEs. Especially when a TDD cell and an FDD cell act as the PCell and SCell respectively, the decrease is so large that PUSCH scheduling may involve only the SCell.

- When a TDD cell is working as the PCell and an FDD cell is working as an SCell for a CA UE, the uplink FDD spectral resources are fully utilized, which increases the uplink data rate. In addition, the FDD SCell provides better uplink coverage for the CA UE than a TDD cell does for a non-CA UE.

14.3 Requirements

14.3.1 Licenses

Each FDD cell involved in uplink FDD+TDD CA consumes one sales unit of the license for MRFD-111222 FDD+TDD Uplink Carrier Aggregation (LTE FDD).

Each TDD cell involved in uplink FDD+TDD CA consumes one sales unit of the license for MRFD-111232 FDD+TDD Uplink Carrier Aggregation (LTE TDD).

Table 14-1 lists the license models and sales units for these features.

Table 14-1 License models and sales units

Feature ID	Feature Name	Model	Sales Unit
MRFD-111222	FDD+TDD Uplink Carrier Aggregation (LTE FDD)	LT1SFTULCAF0	per cell
MRFD-111232	FDD+TDD Uplink Carrier Aggregation (LTE TDD)	LT1SFTULCAT0	per cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

14.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

14.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
FDD TDD	Downlink FDD +TDD CA	**InterFddTddCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**13 Downlink FDD +TDD CA**
FDD TDD	MTA	**FTMtaAlgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter	None

14.3.2.2 Mutually Exclusive Functions

None

14.3.2.3 Function Impacts

- Functions related to RAN performance

RAT	Function Name	Function Switch	Reference	Description
FDD	Uplink short-interval SPS	**CellUlschAlgo**.***IntvlOfUlSpsWithSkipping***	*Uplink Scheduling*	UEs that have entered uplink short-interval SPS do not support uplink 2CC aggregation.
FDD	UL CRA	**UL_COORD_RES_ALLOC_SWITCH** option of the **UlCsAlgoPara**.***UlCsSw*** parameter	*Uplink Coordinated Scheduling*	Uplink SCells seldom suffer from heavy loads or interference limitations. Therefore, uplink coordinated resource allocation (UL CRA) is currently not used in uplink SCells.
FDD	UL CPC	**UL_COORD_PC_SWITCH** option of the **UlCsAlgoPara**.***UlCsSw*** parameter	*Uplink Coordinated Scheduling*	Uplink SCells seldom suffer from heavy loads or interference limitations. Therefore, uplink coordinated power control (UL CPC) is currently not used in uplink SCells.

RAT	Function Name	Function Switch	Reference	Description
TDD	Intra-eNodeB UL CAMC	**UlCamcSw** option of the **CellAlgoSwitch**.***CamcSwitch*** parameter	*Uplink Coordinated Scheduling*	CA and uplink coordinated adaptive modulation and coding (UL CAMC) can be enabled simultaneously. However, UL CAMC takes effect for CA UEs only in their PCells.
TDD	Inter-eNodeB UL CAMC	**UlInterEnbCamcSw** option of the **CellAlgoSwitch**.***CamcSwitch*** parameter	*Uplink Coordinated Scheduling*	CA and uplink coordinated adaptive modulation and coding (UL CAMC) can be enabled simultaneously. However, UL CAMC takes effect for CA UEs only in their PCells.
FDD	UL CIAS	**UL_INTRA_ENB_CIAS_SWITCH** option of the **UlCsAlgoPara**.***UlCoordIntrfAwareSchSw*** parameter	*Uplink Coordinated Scheduling*	If uplink FDD+TDD CA and UL CIAS are both enabled, the gains offered by UL CIAS decrease when there are no SRS resources for SCC UEs.
FDD TDD	Uplink 2x4 MU-MIMO	**UlVmimoSwitch** option of the **CellAlgoSwitch**.***UlSchSwitch*** parameter	*MIMO*	When uplink CA is disabled, the uplink 2x4 multi-user MIMO (MU-MIMO) function works for CA UEs only in their PCells. When uplink CA is enabled, this function works for CA UEs in all their serving cells, where uplink transmission from the UEs is applicable.

RAT	Function Name	Function Switch	Reference	Description
TDD	Uplink 2x8 MU-MIMO	**UlVmimoSwitch** option of the **CellAlgoSwitch**.***UlSchSwitch*** parameter	*MIMO*	When uplink CA is disabled, the uplink 2x8 MU-MIMO function works for CA UEs only in their PCells. When uplink CA is enabled, this function works for CA UEs in all their serving cells, where uplink transmission from the UEs is applicable.
TDD	Uplink SU-MIMO	**ULSUMIMO2LayersSwitch** option of the **CellAlgoSwitch**.***UlSuMimoAlgoSwitch*** parameter	*MIMO*	Uplink 2CC aggregation takes precedence over uplink single-user MIMO (SU-MIMO). If a UE has been selected for uplink CA, it will no longer be selected for uplink SU-MIMO. If a UE working in uplink SU-MIMO is selected for uplink CA, it exits uplink SU-MIMO.

RAT	Function Name	Function Switch	Reference	Description
TDD	Single-stream beamforming	**CellAlgoSwitch**.***BfAlgoSwitch***	*Beamforming (TDD)*	PCells allow for beamforming. The beamforming capabilities of SCells vary as follows: ● If uplink CA is enabled, uplink SCells also allow for beamforming. ● If uplink CA is disabled but beamforming in SCells is enabled, SCells without SRSs allow for single- and dual-stream beamforming but do not allow for long-SRS-period beamforming or MU beamforming. ● If neither uplink CA nor beamforming in SCells is enabled, SCells do not allow for beamforming.

RAT	Function Name	Function Switch	Reference	Description
TDD	Dual-stream beamforming	**CellAlgoSwitch**.***BfAlgoSwitch*** and **CellBf**.***MaxBfRankPara***	*Beamforming (TDD)*	PCells allow for beamforming. The beamforming capabilities of SCells vary as follows: ● If uplink CA is enabled, uplink SCells also allow for beamforming. ● If uplink CA is disabled but beamforming in SCells is enabled, SCells without SRSs allow for single- and dual-stream beamforming but do not allow for long-SRS-period beamforming or MU beamforming. ● If neither uplink CA nor beamforming in SCells is enabled, SCells do not allow for beamforming.

RAT	Function Name	Function Switch	Reference	Description
TDD	MU beamforming	**MuBfSwitch** option of the **CellAlgoSwitch**.***MuBfAlgoSwitch*** parameter	*Beamforming (TDD)*	PCells allow for beamforming. The beamforming capabilities of SCells vary as follows: ● If uplink CA is enabled, uplink SCells also allow for beamforming. ● If uplink CA is disabled but beamforming in SCells is enabled, SCells without SRSs allow for single- and dual-stream beamforming but do not allow for long-SRS-period beamforming or MU beamforming. ● If neither uplink CA nor beamforming in SCells is enabled, SCells do not allow for beamforming.
FDD TDD	UL CoMP cell	**UlJointReceptionSwitch** option of the **CellAlgoSwitch**.***UplinkCompSwitch*** parameter	*UL CoMP*	If an SCell is configured in the uplink for a CA UE, this UE will not be selected in the SCell for UL CoMP.
FDD	Uplink interference cancellation	**UlInterSiteIcSwitch** option of the **CellAlgoSwitch**.***UplinkIcSwitch*** parameter	*Uplink Interference Cancellation (FDD)*	If an SCell is configured in the uplink for a CA UE, the eNodeB will not select the UE for uplink interference cancellation (UL IC).

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Terminal Awareness Differentiation	**AbnormalUeHandleSwitch** option of the **GlobalProcSwitch**.***UeCompatSwitch*** parameter	*Terminal Awareness Differentiation*	Certain UEs on live networks have compatibility issues with uplink CA. Whitelist control for uplink CA can be used to prevent these UEs from delivering abnormal performance with uplink CA. With this function, uplink CA takes effect only for UEs that have no compatibility issues with uplink CA. Whitelist control for uplink CA is enabled when both of the following conditions are met: ● The **CA_SWITCH_OFF** option of the **UeCompat**.***BlkLstCtrlSwitch*** parameter is deselected. ● The **UL_CA_SWITCH_ON** option of the **UeCompat**.***WhiteLstCtrlSwitch*** parameter is selected.

RAT	Function Name	Function Switch	Reference	Description
FDD	UMTS and LTE Zero Bufferzone	**UMTS_LTE_ZERO_BUFFER_ZONE_SW** option of the **ULZeroBufferZone.*ZeroBufZoneSwitch*** parameter	*UMTS and LTE Zero Bufferzone*	There are fewer PUSCH and SRS resources in a cell in the bufferzone than in a common cell. Therefore, when the LTE bandwidth is 5 MHz or 10 MHz, using a cell in the bufferzone as a PCell for CA is not recommended. If the cell is used as a PCell, CA performance deteriorates.
FDD	Dynamic Massive Beam	None	*Smart 8T8R (FDD)*	When passive AAU ports are used, determine whether to enable FDD+TDD CA. If the frame offset of TDD cells on the live network is not 0, you are not advised to enable FDD+TDD CA for cells served by these AAUs. If FDD+TDD CA is enabled under these conditions, channel calibration will be affected and cell performance will deteriorate.

RAT	Function Name	Function Switch	Reference	Description
TDD	MU beamforming	**MuBfSwitch** option of the **CellAlgoSwitch**.***MuBfAlgoSwitch*** parameter	*Massive MIMO Basics (TDD)*	• The number of PRBs successfully paired for MU beamforming on each layer may decrease by 5% when uplink FDD+TDD CA is enabled and the **Dl2CCAckResShareSw** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter is selected. • When uplink CA is enabled, up to eight layers can be used for pairing SCell-served UEs in MU beamforming. There is no restriction on the maximum number of layers for pairing SCell-served UEs and PCell-served UEs or the maximum number of layers for pairing SCell-served UEs and non-CA UEs.
FDD TDD	NSA networking based on EPC	**NSA_DC_CAPABILITY_SWITCH** option of the **NsaDcMgmtConfig**.***NsaDcAlgoSwitch*** parameter	*EPC-based NSA Basics*	Uplink FDD+TDD CA does not take effect for UEs running NSA DC services. If an NSA DC service is to be initiated for a UE with FDD and TDD CCs aggregated in the uplink, the uplink SCell will be removed for the UE.

- Functions related to RAN services

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	TTI bundling	**TtiBundlingSwitch** option of the **CellAlgoSwitch**.***UlSchSwitch*** parameter	*VoLTE*	Uplink CA is incompatible with TTI bundling at the UE level. TTI bundling takes precedence over uplink CA. When an eNodeB determines to configure TTI bundling for a CA UE, the eNodeB sends an RRC Connection Reconfiguration message to remove the uplink SCell and configure TTI bundling.
FDD	Inter-eNodeB VoLTE CoMP	**UlVoiceJROverRelaxedBHSw** option of the **ENodeBAlgoSwitch**.***OverBBUsSwitch*** parameter	*VoLTE*	If an SCell is configured in the uplink for a CA UE, this UE will not be selected in the SCell for UL CoMP.
FDD TDD	Out-of-band relay	**CellAlgoSwitch**.***RelaySwitch***	*Relay*	In out-of-band relay scenarios, RRNs support downlink 2CC aggregation and uplink 2CC aggregation if the **Dl2CCAckResShareSw** option of the **CellAlgoSwitch**.***PucchAlgoSwitch*** parameter is deselected, and do not support them if this option is selected.

- Functions related to CloudAIR (FDD)

RAT	Function Name	Function Switch	Reference	Description
FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_SPECTRUM_SHR**	*LTE FDD and NR Dynamic Spectrum Sharing*	This function reduces the number of uplink RBs available for LTE. Therefore, the throughput of UEs in the uplink FDD CA state decreases.
FDD	LTE FDD and NR Uplink Spectrum Sharing	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **LTE_NR_UPLINK_SPECTRUM_SHR**	*LTE FDD and NR SUL Dynamic Spectrum Sharing*	This function reduces the number of uplink RBs available for LTE. Therefore, the throughput of UEs in the uplink CA state decreases.
FDD	DSS and NR Flexible Refarming	**DSS_FLEX_REFARM_SW** option of the **SpectrumCloud**.***SpectrumCloudEnhSwitch*** parameter	*LTE FDD and NR Dynamic Spectrum Sharing*	This function reduces the number of uplink RBs available for LTE. Therefore, the throughput of UEs in the uplink FDD CA state decreases.

14.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

The requirements described in **Boards** of **13.3.3 Hardware** must be fulfilled. In addition, for downlink massive CA plus uplink 2CC aggregation, only cells on UBBPd or later BBPs can act as the uplink SCell.

RF Modules

For details, see **RF Modules** in **13.3.3 Hardware**.

Cells

For details, see **Subframe Configuration** in **14.1 Principles**.

14.3.4 Networking

For details, see **13.3.4 Networking**.

14.3.5 Others

- UEs
  - UEs must comply with 3GPP Release 12 or later and support the frequency bands of the carriers to be aggregated and their channel bandwidths. UEs must also support the peak data rates that CA can achieve.
  - UEs must support MTA.
- EPC

  For this function to reach a theoretical peak data rate, the maximum bit rate that each UE subscribes to in the EPC cannot be lower than this theoretical value. For the theoretical values, see **14.2.1 Benefits**.

14.4 Operation and Maintenance

14.4.1 Data Configuration

14.4.1.1 Data Preparation

This function works in either CA-group-based or adaptive configuration mode. Prepare basic data as described in **5.4.1.1 Data Preparation**.

In addition, for either mode, prepare data as described in **Table 14-2** and **Table 14-3** for function activation and optimization.

Table 14-2 Parameters for activating uplink FDD+TDD CA

Parameter Name	Parameter ID	Option	Setting Notes
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	InterFddTddCaSwitch	Select this option.
FDD TDD CA UL Max CC Number	**CaMgtCfg**.***FddTddCaUlMaxCcNum***	None	Set this parameter to **2CC**. This parameter must be set to **2CC** when the **CaMgtCfg**.***FTRelaxedBHCaUl-MaxCcNum*** parameter is set to **2CC**.

Parameter Name	Parameter ID	Option	Setting Notes
FDD Frame Offset	**ENodeBFrameOffset**.***FddFrameOffset***	None	This parameter specifies the offset of the frame start time for all LTE FDD cells served by the eNodeB relative to the time of the reference clock. FDD+TDD CA requires time synchronization between the cells involved. An identical frame offset must be set for all the cells. Otherwise, FDD+TDD CA does not work. Run the **DSP CELLFRAMEOFFSET** command to query the frame offset actually used for each cell. If the values of **Frame Offset Effect Value(Ts)** are the same for the cells involved, the frame offsets of the cells are the same. You are advised to set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
TDD Frame Offset	**ENodeBFrameOffset**.*TddFrameOffset*	None	This parameter specifies the offset of the frame start time for all LTE TDD cells served by the eNodeB relative to the time of the reference clock. If uplink and downlink timeslots are not aligned between TDD systems, inter-system interference will occur. Operators can adjust this parameter to minimize the error in timeslot alignment between the TDD systems. Run the **DSP CELLFRAMEOFFSET** command to query the frame offset actually used for each cell. If the values of **Frame Offset Effect Value(Ts)** are the same for the cells involved, the frame offsets of the cells are the same. You are advised to set this parameter based on the network plan.

Table 14-3 Parameters for optimizing uplink FDD+TDD CA

Parameter Name	Parameter ID	Option	Setting Notes
Uplink schedule switch	**CellAlgoSwitch**.*UlSchSwitch*	SchedulerCtrlPowerSwitch	Select this option for each possible PCell and SCell.
Uplink power control algorithm switch	**CellAlgoSwitch**.*UlPcAlgoSwitch*	CloseLoopOptPUSCHSwitch	Deselect this option because it applies only to TDD.
Cell Level CA Algorithm Switch	**CaMgtCfg**.*CellCaAlgoSwitch*	NackDtxIdentifySwitch	The setting of this option must be consistent between each possible PCell and each possible SCell for that PCell.

Moreover,

- Prepare data as described in **16.4.1.1 Data Preparation** if uplink FDD+TDD CA is to be deployed in a relaxed backhaul scenario.
- Prepare data as described in **17.4.1.1 Data Preparation** if uplink FDD+TDD CA is to be deployed in an eNodeB coordination scenario.

14.4.1.2 Using MML Commands

Before using MML commands, refer to **14.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Before activating this function, configure cells or frequencies according to **5.4.1.2 Using MML Commands**.

The activation command examples for this function are as follows:

```
//Turning on InterFddTddCaSwitch and setting FddTddCaUlMaxCcNum to 2CC for each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=InterFddTddCaSwitch-1, FddTddCaUlMaxCcNum=2CC;
//Setting frame offsets for the base station
MOD ENODEBFRAMEOFFSET: TddFrameOffset=0,FddFrameOffset=0;
//(Optional) Turning on SchedulerCtrlPowerSwitch for each possible PCell and SCell
MOD CELLALGOSWITCH: LocalCellId=0, UlSchSwitch=SchedulerCtrlPowerSwitch-1;
MOD CELLALGOSWITCH: LocalCellId=1, UlSchSwitch=SchedulerCtrlPowerSwitch-1;
//(Optional) Turning off CloseLoopOptPUSCHSwitch on the TDD side (using a TDD cell with LocalCellId
being 1 as an example)
MOD CELLALGOSWITCH: LocalCellId=1, UlPcAlgoSwitch=CloseLoopOptPUSCHSwitch-0;
//(Optional) Setting FTRelaxedBHCaULMaxCcNum to 2CC for each possible PCell and SCell to activate inter-
eNodeB uplink FDD+TDD CA in a relaxed backhaul scenario
MOD CAMGTCFG: LocalCellId=0, FTRelaxedBHCaUlMaxCcNum=2CC;
MOD CAMGTCFG: LocalCellId=1, FTRelaxedBHCaUlMaxCcNum=2CC;
//(Optional) Turning on NackDtxIdentifySwitch for each possible PCell and SCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=NackDtxIdentifySwitch-1;
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=NackDtxIdentifySwitch-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Setting FddTddCaUlMaxCcNum to 0CC
MOD CAMGTCFG: LocalCellId=0, FddTddCaUlMaxCcNum=0CC;
MOD CAMGTCFG: LocalCellId=1, FddTddCaUlMaxCcNum=0CC;
```

14.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

14.4.2 Activation Verification

Counter Observation

If the counters listed in **Table 14-4** produce non-zero values, uplink FDD+TDD CA has taken effect.

Table 14-4 Counters used to verify activation of uplink FDD+TDD CA

Counter ID	Counter Name
1526741844	**L.Traffic.User.FddTddCA.PCell.UL.Avg**
1526742034	**L.Traffic.User.FddTddCA.SCell.UL.Avg**
1526741932	**L.Traffic.User.FddTddCA.SCell.Active.UL.Avg**
1526743685	**L.Traffic.User.PCell.UL.RelaxedBackhaul-CA.Avg**
1526743658	**L.Traffic.User.SCell.UL.RelaxedBackhaul-CA.Avg**
1526743661	**L.Traffic.User.RelaxedBackhaul-CA.SCell.Active.UL.Avg**

Message Tracing

For details, see **Message Tracing** in **10.4.3 Activation Verification**.

14.4.3 Network Monitoring

In addition to the counters listed in **10.4.4 Network Monitoring** and **13.4.3 Network Monitoring**, monitor the counters in **Table 14-5** and compare the results with the network plan to evaluate network performance.

Table 14-5 Counters used to monitor performance of uplink FDD+TDD CA

Counter ID	Counter Name
1526741844	**L.Traffic.User.FddTddCA.PCell.UL.Avg**
1526742034	**L.Traffic.User.FddTddCA.SCell.UL.Avg**
1526741932	**L.Traffic.User.FddTddCA.SCell.Active.UL.Avg**
1526741933	**L.CA.UL.FddTddCA.PCell.Act.Dur**
1526741931	**L.CA.UL.FddTddCA.SCell.Act.Dur**

Counter ID	Counter Name
1526741919	**L.Thrp.bits.UL.FddTddCA.CAUser**
1526743685	**L.Traffic.User.PCell.UL.RelaxedBackhaul-CA.Avg**
1526743658	**L.Traffic.User.SCell.UL.RelaxedBackhaul-CA.Avg**
1526743661	**L.Traffic.User.RelaxedBackhaul-CA.SCell.Active.UL.Avg**

15 Beamforming in SCells (TDD)

15.1 Principles

If a CA UE does not transmit SRSs in a TDD SCell, the beamforming in SCells function allows the UE to enter transmission mode (TM) 9 without PMI, TM7, or TM8 in this SCell by calculating a weight value based on the SRSs transmitted in another TDD serving cell of the UE. In this way, the UE enjoys beamforming benefits, and the average cell traffic volume rises.

This function is controlled by the **NoSrsSccBfSwitch** option of the **CellAlgoSwitch**.***NoSrsSccBfAlgoSwitch*** parameter.

This function allows the UE to enter TM9 with PMI mode in the TDD SCell without SRSs by using the beam-related information of the TDD serving cell with SRSs, when the two cells are TDD massive MIMO cells meeting certain conditions. In this way, the UE enjoys benefits of TM9 hybrid precoding, and the average cell traffic volume rises. The following are the conditions:

- The **TM9HybridPrecodingSwitch** option of the **CellAlgoSwitch**.***EnhMIMOSwitch*** parameter is selected for both cells.
- The **CellBf**.***PdschSplitBeamConfigIndex*** parameter is set to the same value for both cells.
- The **TM9_SPLIT_SDMA_OPT_SWITCH** option of the **CellMMAlgo**.***MMAlgoOptSwitch*** parameter is selected for both cells.

If these conditions are met and the **NoSrsSccMuBfSwitch** option of the **CellAlgoSwitch**.***NoSrsSccBfAlgoSwitch*** parameter is additionally selected for the TDD SCell without SRSs, 8-port TM9 UEs in the SCell without SRSs can participate in TM9 split SDMA, increasing the average cell traffic volume. (TM9 split SDMA for SCells without SRSs is a trial function.)

For details about TM9 hybrid precoding and TM9 split SDMA, see *Massive MIMO Basics (TDD)*.

NOTE

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

15.2 Network Analysis

15.2.1 Benefits

This function allows UEs to enjoy beamforming or TM9 hybrid precoding benefits in the downlink in their TDD SCells without SRSs. The downlink cell throughput and downlink CA UE throughput increase. This function is recommended in stationary scenarios.

This function is recommended when both of the following conditions are met:

- Either FDD+TDD downlink CA or TDD-only downlink CA has been enabled. In addition, UEs capable of uplink CA must support TM7, TM8, or TM9 hybrid precoding in the downlink of SCCs.
- At least two TDD carriers are involved in CA.

15.2.2 Impacts

The impacts between this function and other functions are described in **15.3.2.3 Function Impacts**.

This function has no impact on the network.

15.3 Requirements

15.3.1 Licenses

None

15.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

15.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
TDD	Carrier aggregation	CA works in multiple scenarios. Its function switch varies depending on scenarios. For details, see the relevant sections.	*Carrier Aggregation*
TDD	TM9 hybrid precoding	**TM9HybridPrecodingSwitch** option of the **CellAlgoSwitch**.***EnhMIMOSwitch*** parameter	*Massive MIMO Basics (TDD)*
TDD	Single-stream beamforming	**BfSwitch** option of the **CellAlgoSwitch**.***BfAlgoSwitch*** parameter	*Beamforming (TDD)*
TDD	Dual-stream beamforming	**CellBf**.***MaxBfRankPara*** set to **DUAL_LAYER_BF**	*Beamforming (TDD)*
TDD	SCC transmission mode adaptation	**CellBfMimoParaCfg**.***SccBfMimoAdaptiveSwitch***	**19.2.3 MIMO (TDD)**
TDD	Adaptive Transmission Mode	**MIMO_BF_ADAPTIVE** option of the **CellBfMimoParaCfg**.***BfMimoAdaptiveSwitch*** parameter	*TMA (TDD)*

15.3.2.2 Mutually Exclusive Functions

None

15.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
TDD	TM9 hybrid precoding	**TM9HybridPrecodingSwitch** option of the **CellAlgoSwitch**.***EnhMIMOSwitch*** parameter	*Massive MIMO Basics (TDD)*	Massive MIMO PCells allow for TM9 hybrid precoding. The precoding capabilities of massive MIMO SCells vary as follows: ● If uplink CA is enabled, uplink SCells also allow for TM9 hybrid precoding. ● If uplink CA is disabled but beamforming in SCells is enabled, SCells without SRSs allow for TM9 hybrid precoding. ● If neither uplink CA nor beamforming in SCells is enabled, SCells do not allow for TM9 hybrid precoding.

RAT	Function Name	Function Switch	Reference	Description
TDD	3D beamforming	**BfSwitch** option of the **CellAlgoSwitch**.***BfAlgoSwitch*** parameter	*Massive MIMO Basics (TDD)*	PCells allow for beamforming. The beamforming capabilities of SCells vary as follows: ● If uplink CA is enabled, uplink SCells also allow for beamforming. ● If uplink CA is disabled but beamforming in SCells is enabled, SCells without SRSs allow for single- and dual-stream beamforming but do not allow for long-SRS-period beamforming or MU beamforming. ● If neither uplink CA nor beamforming in SCells is enabled, SCells do not allow for beamforming.

RAT	Function Name	Function Switch	Reference	Description
TDD	MU beamforming	**MuBfSwitch** option of the **CellAlgoSwitch**.***MuBfAlgoSwitch*** parameter	*Massive MIMO Basics (TDD)*	PCells allow for beamforming. The beamforming capabilities of SCells vary as follows: ● If uplink CA is enabled, uplink SCells also allow for beamforming. ● If uplink CA is disabled but beamforming in SCells is enabled, SCells without SRSs allow for single- and dual-stream beamforming but do not allow for long-SRS-period beamforming or MU beamforming. ● If neither uplink CA nor beamforming in SCells is enabled, SCells do not allow for beamforming.

RAT	Function Name	Function Switch	Reference	Description
TDD	Single-stream beamforming	**BfSwitch** option of the **CellAlgoSwitch**.***BfAlgoSwitch*** parameter	*Beamforming (TDD)*	PCells allow for beamforming. The beamforming capabilities of SCells vary as follows: ● If uplink CA is enabled, uplink SCells also allow for beamforming. ● If uplink CA is disabled but beamforming in SCells is enabled, SCells without SRSs allow for single- and dual-stream beamforming but do not allow for long-SRS-period beamforming or MU beamforming. ● If neither uplink CA nor beamforming in SCells is enabled, SCells do not allow for beamforming.

RAT	Function Name	Function Switch	Reference	Description
TDD	Dual-stream beamforming	**CellBf**.***MaxBfRankPara*** set to **DUAL_LAYER_BF**	*Beamforming (TDD)*	PCells allow for beamforming. The beamforming capabilities of SCells vary as follows: ● If uplink CA is enabled, uplink SCells also allow for beamforming. ● If uplink CA is disabled but beamforming in SCells is enabled, SCells without SRSs allow for single- and dual-stream beamforming but do not allow for long-SRS-period beamforming or MU beamforming. ● If neither uplink CA nor beamforming in SCells is enabled, SCells do not allow for beamforming.

RAT	Function Name	Function Switch	Reference	Description
TDD	MU beamforming	**MuBfSwitch** option of the **CellAlgoSwitch**.***MuBfAlgoSwitch*** parameter	*Beamforming (TDD)*	PCells allow for beamforming. The beamforming capabilities of SCells vary as follows: ● If uplink CA is enabled, uplink SCells also allow for beamforming. ● If uplink CA is disabled but beamforming in SCells is enabled, SCells without SRSs allow for single- and dual-stream beamforming but do not allow for long-SRS-period beamforming or MU beamforming. ● If neither uplink CA nor beamforming in SCells is enabled, SCells do not allow for beamforming.

15.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

No requirements

RF Modules

Only 8T8R and 64T64R RF modules are compatible with this function.

Cells

At least two inter-frequency TDD cells must meet all of the following conditions:

- The quotient of the larger over smaller cell operating frequencies is less than 1.06.
- The cells are served by the same RRU or AAU.
- They are normal cells and use the same number of antennas: 8T8R or 64T64R.

15.3.4 Others

No requirements

15.4 Operation and Maintenance

15.4.1 Data Configuration

15.4.1.1 Data Preparation

Table 15-1 describes the parameters used for function activation.

Table 15-1 Parameters for activating "beamforming in SCells"

Parameter Name	Parameter ID	Setting Notes
SRS-Free SCC BF Algorithm Switch	**CellAlgoSwitch**.***NoSrsSccBfAlgoSwitch***	• Select the **NoSrsSccBfSwitch** option. • Select the **NoSrsSccMuBfSwitch** option if TM9 split SDMA for SCells without SRSs is required.
Enhanced MIMO Switch	**CellAlgoSwitch**.***EnhMIMOSwitch***	Select the **TM9HybridPrecodingSwitch** option if TM9 hybrid precoding is required.
PDSCH Split Beam Config Index	**CellBf**.***PdschSplitBeamConfigIndex***	Set this parameter to **CONFIG_INDEX4** if TM9 hybrid precoding is required.
Massive MIMO Algorithm Optimization Switch	**CellMMAlgo**.***MMAlgoOptSwitch***	Select the **TM9_SPLIT_SDMA_OPT_SWITCH** option if TM9 hybrid precoding is required.

15.4.1.2 Using MML Commands

Before using MML commands, refer to **15.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on NoSrsSccBfSwitch on the TDD serving cell with SRSs and the TDD SCell without SRSs
MOD CELLALGOSWITCH: LocalCellId=0, NoSrsSccBfAlgoSwitch=NoSrsSccBfSwitch-1;
MOD CELLALGOSWITCH: LocalCellId=1, NoSrsSccBfAlgoSwitch=NoSrsSccBfSwitch-1;
//(Optional) Enabling TM9 hybrid precoding on the TDD serving cell with SRSs and the TDD SCell without
SRSs to benefit from this function
MOD CELLALGOSWITCH: LocalCellId=0, EnhMIMOSwitch=TM9HybridPrecodingSwitch-1;
MOD CELLALGOSWITCH: LocalCellId=1, EnhMIMOSwitch=TM9HybridPrecodingSwitch-1;
MOD CELLBF: LocalCellId=0, PdschSplitBeamConfigIndex=CONFIG_INDEX4;
MOD CELLBF: LocalCellId=1, PdschSplitBeamConfigIndex=CONFIG_INDEX4;
MOD CELLMMALGO: LocalCellId=0, MMAlgoOptSwitch=TM9_SPLIT_SDMA_OPT_SWITCH-1;
MOD CELLMMALGO: LocalCellId=1, MMAlgoOptSwitch=TM9_SPLIT_SDMA_OPT_SWITCH-1;
//(Optional) Turning on NoSrsSccMuBfSwitch for the TDD SCell without SRSs if TM9 split SDMA for SCells
without SRSs is required
MOD CELLALGOSWITCH: LocalCellId=1, NoSrsSccBfAlgoSwitch=NoSrsSccMuBfSwitch-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off NoSrsSccBfSwitch on the TDD serving cell with SRSs and the TDD SCell without SRSs
MOD CELLALGOSWITCH: LocalCellId=0, NoSrsSccBfAlgoSwitch=NoSrsSccBfSwitch-0;
MOD CELLALGOSWITCH: LocalCellId=1, NoSrsSccBfAlgoSwitch=NoSrsSccBfSwitch-0;
//(Optional) Disabling TM9 hybrid precoding on the TDD serving cell with SRSs and the TDD SCell without
SRSs
MOD CELLALGOSWITCH: LocalCellId=0, EnhMIMOSwitch=TM9HybridPrecodingSwitch-0;
MOD CELLALGOSWITCH: LocalCellId=1, EnhMIMOSwitch=TM9HybridPrecodingSwitch-0;
MOD CELLBF: LocalCellId=0, PdschSplitBeamConfigIndex=CONFIG_DEFAULT;
MOD CELLBF: LocalCellId=1, PdschSplitBeamConfigIndex=CONFIG_DEFAULT;
MOD CELLMMALGO: LocalCellId=0, MMAlgoOptSwitch=TM9_SPLIT_SDMA_OPT_SWITCH-0;
MOD CELLMMALGO: LocalCellId=1, MMAlgoOptSwitch=TM9_SPLIT_SDMA_OPT_SWITCH-0;
//(Optional) Turning off NoSrsSccMuBfSwitch for the TDD SCell without SRSs
MOD CELLALGOSWITCH: LocalCellId=1, NoSrsSccBfAlgoSwitch=NoSrsSccMuBfSwitch-0;
```

15.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

15.4.2 Activation Verification

If the counters listed in **Table 15-2** produce larger values, beamforming in SCells has taken effect.

Table 15-2 Counters used to verify activation of beamforming in SCells

Counter ID	Counter Name
1526737845	**L.ChMeas.PRB.TM7**
1526737846	**L.ChMeas.PRB.TM8**
1526747667	**L.ChMeas.PRB.TM9**

15.4.3 Network Monitoring

Monitor the counters listed in **Table 15-3** and compare the results with the network plan to evaluate network performance.

Table 15-3 Counters used to monitor performance of beamforming in SCells

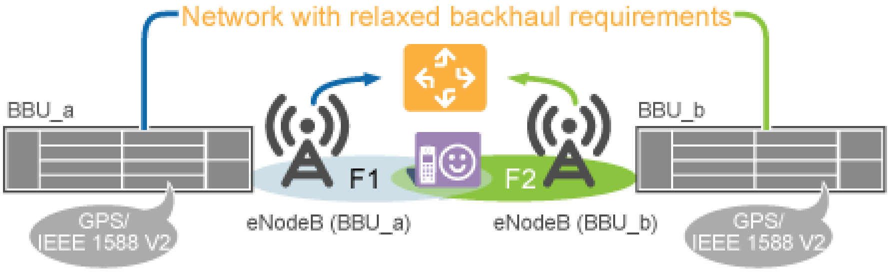
Counter ID	Counter Name
1526728564	**L.Thrp.bits.DL.CAUser**
1526728565	**L.Thrp.Time.DL.CAUser**
1526729259	**L.CA.Traffic.bits.DL.SCell**
1526729004	**L.CA.DL.SCell.Act.Dur**

16 Inter-eNodeB CA Based on Relaxed Backhaul

16.1 Principles

This function allows for downlink 2CC to 8CC aggregation and uplink 2CC aggregation between inter-eNodeB cells in relaxed backhaul scenarios (that is, non-ideal backhaul scenarios), as illustrated in **Figure 16-1**.

Figure 16-1 Inter-eNodeB CA based on relaxed backhaul



The switch control over this function varies, as described in **Table 16-1**.

Table 16-1 Switch control over inter-eNodeB CA based on relaxed backhaul

RAT	Option	Description
FDD	**RelaxedBackhaulCaSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter	This option must be selected for all the eNodeBs involved.
TDD	**TddRelaxedBackhaulCaSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter	This option must be selected for all the eNodeBs involved.

RAT	Option	Description
FDD TDD	**FreqBaseIntereNBSccSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter	This option must be selected for the serving eNodeBs of SCells.

3GPP Release 10 specifications stipulate that downlink-data demodulation results (ACK or NACK) must be reported over the PCC of each CA UE. In a relaxed backhaul scenario, the serving eNodeB of each SCell for a CA UE estimates the CQI, NACK, and scheduling priority to determine the scheduling occasion and resources for the UE in advance, without affecting the scheduling priorities and fairness among UEs. This prevents inter-eNodeB CA failures caused by transmission delay.

The MAC-layer scheduling priority of downlink RLC retransmission packets for UEs in the relaxed-backhaul-based CA state can be lifted. This lifting function is controlled by the **RelaxedBhDlRlcRetransOptSw** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter. When this option is selected, the RLC retransmission scheduling delay for UEs in the relaxed-backhaul-based CA state decreases, but the scheduling delay for other CA UEs may increase.

In relaxed backhaul scenarios, the downlink target IBLERs of the SCells can be specified by the **CaMgtCfg**.***RelaxedBhSccDlTargetIbler*** parameter. A larger value of this parameter results in a higher selected MCS index and a higher probability of retransmissions. If the spectral efficiency gains brought by the MCS index increase are more significant than the performance loss caused by the retransmission rate increase, the downlink traffic volume increases. Otherwise, the downlink traffic volume decreases. A smaller value of this parameter results in a lower selected MCS index and a lower probability of retransmissions. If the spectral efficiency loss caused by the MCS index decrease is less significant than the gains brought by the retransmission rate decrease, the downlink traffic volume increases. Otherwise, the downlink traffic volume decreases.

Inter-eNodeB cell filtering is supported. This function is enabled when the **SccMeasRptLocalCellPriorSw** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is selected. With this function, the serving eNodeB of the PCell for a CA UE does not configure any inter-eNodeB cell that is working on the same frequency as an activated cell on this local eNodeB as an SCell for the UE. This relieves transmission congestion caused by inter-eNodeB CA based on relaxed backhaul. However, the number of UEs in the relaxed-backhaul-based inter-eNodeB CA state decreases.

Usage Scenarios

The relaxed backhaul scenarios are classified into the following types:

- Inter-eNodeB one-way delay ≤ 4 ms and RTT ≤ 8 ms

  In this scenario, both downlink CA and uplink CA work between inter-eNodeB cells.

- 4 ms < inter-eNodeB one-way delay ≤ 8 ms and 8 ms < RTT ≤ 16 ms

  In this scenario, only downlink CA works.

To enable inter-eNodeB CA in this scenario, the **RelaxedBHCaEnhanceSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter must be selected.

- 8 ms < inter-eNodeB one-way delay ≤ 16 ms and 16 ms < RTT ≤ 32 ms

  In this scenario, only downlink FDD CA works.

  To enable inter-eNodeB CA in this scenario, the **RelaxedBHCaEnh2Switch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter must be selected.

The eNodeB combination can be one of the following:

- Macro+macro, macro+LampSite, or LampSite+LampSite

  The cells exchange signaling messages and service data for CA through the eX2 or X2 interfaces between the main control boards (UMPT) of these eNodeBs.

  The **GlobalProcSwitch**.***ItfTypeForNonIdealModeServ*** parameter determines which interface to use for inter-eNodeB CA based on relaxed backhaul.

For details about eX2 or X2 interfaces between eNodeBs, see *eX2 Self-Management* or *S1 and X2 Self-Management*.

Procedures

This function employs the same PCC anchoring and SCell management procedures as for intra-eNodeB CA. For details, see **4.6 Carrier Management for RRC_CONNECTED UEs** and **4.7 Carrier Management for RRC_IDLE UEs**.

This function implements SCell configuration and SCell removal with inter-eNodeB transmission quality requirements considered, in addition to the principles described in **4.6.3 SCell Management**.

The process of this function is as follows:

1. During initial access, an incoming RRC connection reestablishment, or an incoming handover, the CA UE reports its CA capabilities to the target eNodeB after setting up an RRC connection with the PCell.
2. The eNodeB selects appropriate inter-eNodeB cells as SCells for the UE.
3. After activating the SCells, the serving eNodeB of the PCell diverts downlink traffic of the UE from the PCell to the SCells and converges the uplink traffic from the SCell with the uplink traffic in the PCell.

   If inter-eNodeB transmission quality deteriorates (for example, the transmission delay does not meet requirements or transmission is interrupted), the eNodeB removes the SCells to exit inter-eNodeB CA. After transmission quality recovers, SCells can be configured again at the subsequent initial access, incoming RRC connection reestablishment, or incoming handover.

16.2 Network Analysis

16.2.1 Benefits

This function enables inter-eNodeB carriers to be aggregated to increase user-perceived data rates for CA UEs.

16.2.2 Impacts

The impacts between this function and other functions are described in **16.3.3.3 Function Impacts**.

In relaxed backhaul scenarios, when CA and other inter-eNodeB features (such as UL CoMP) are enabled together, these features share transmission bandwidth. If transmission bandwidth is insufficient, delays increase and the benefits of these features are affected.

This function may have the following impacts on the network:

- CA UEs cannot promptly acquire the real-time CQI changes about inter-eNodeB SCells. This causes slight deterioration in frequency-selective scheduling performance and an increase in IBLER.
- Due to inter-eNodeB delay, HARQ feedback is postponed, which affects the RBLER of CA UEs. In HARQ retransmission statistics, the values of the RBLER-related counters rise. CQI reports for SCells of CA UEs are also delayed, which affects the IBLER of these UEs. The IBLER-related counters produce larger values. If a UE in the inter-eNodeB CA state is located a medium or long distance from the center of its PCell or SCell, the data rate of the UE fluctuates.
- Due to the differences in the RLC data arrival time for the aggregated carriers, CA UEs have to combine and organize the received data. This process places an additional burden on the UEs' CPUs. If the CPU capacity is insufficient, the data rates of the UEs fluctuate.
- To minimize the impact of inter-eNodeB delay, RLC retransmissions occur only in PCells. If the Uu bandwidth of a PCell is used up by guaranteed bit rate (GBR) services, RLC retransmissions for the CA UE are often blocked and the data rate of the UE fluctuates.
- Due to possible errors in the estimated scheduling priorities of CA UEs, the PRBs in SCells for the UEs may not be fully utilized when the SCells are each serving a small number of non-CA UEs and the non-CA UE traffic is light.
- Routes are added between eNodeBs. As a result, SCell bandwidth and lower-layer link bandwidth specifications will be exceeded, causing **L.CA.SccAddFail.PhyLinkFail.Dur** to increase.
- Due to data exchanges between the PCC and inter-base-station SCCs, the values of IP-layer counters (such as **VS.IP.TxPackets**, **VS.IP.TxBytes**, **VS.IP.RxPackets**, and **VS.IP.RxBytes**) will increase, leading to changes of the IP packet size.
- In relaxed-backhaul-based inter-eNodeB CA scenarios, if MBSFN subframes and MTA are configured, the PCell MBSFN subframe numbers described in the following table are recommended for different SCell PRACH subframe number settings. If the MBSFN subframes are not set as recommended, uplink synchronization fails in SCells. This failure causes SCells to be deactivated and therefore affects the perceived data rates of CA UEs. If multiple SCell PRACH subframes are set within 10 ms, the intersection of the corresponding MBSFN subframe number settings is recommended.

Table 16-2 Recommended PCell MBSFN subframe numbers for different FDD SCell PRACH subframe number settings

FDD SCell PRACH Subframe Number	Recommended PCell MBSFN Subframe Numbers
0	3, 6, 7, 8
1	1, 2, 6, 7, 8
2	1, 2, 6, 7, 8
3	1, 2, 3, 6, 7, 8
4	1, 2, 3, 6, 7, 8
5	1, 2, 3, 8
6	1, 2, 3, 6, 7
7	1, 2, 3, 6, 7
8	1, 2, 3, 6, 7, 8
9	1, 2, 3, 6, 7, 8

Table 16-3 Recommended PCell MBSFN subframe numbers for different TDD SCell PRACH subframe number settings

Uplink-Downlink Configuration of the TDD SCell	TDD SCell PRACH Subframe Number	Recommended PCell MBSFN Subframe Numbers
1	2	9
1	3	4, 9
1	7	4
1	8	4, 9
2	2	3, 8, 9
	7	3, 4, 8

- In relaxed-backhaul-based inter-eNodeB CA scenarios where the serving eNodeB of the SCC is of eRAN13.1 or an earlier version, if the UE supports NSA DC and the NSA DC function has been enabled, frequent NR secondary cell group (SCG) changes trigger RRC connection reestablishment of the UE with its E-UTRA PCell. For details about the NSA DC function, see *EPC-based NSA Basics*.
- A larger number of SCCs aggregated using relaxed-backhaul-based inter-eNodeB CA for a UE causes more data to be forwarded on the RLC layer at the PCC and a higher CPU usage of the BBP (indicated by the **L.Traffic.Board.UPlane.CPULoad.AVG** counter) where the PCC is configured. If the CPU usage is too high, the peak data rate of the UE decreases.

- In inter-eNodeB CA based on relaxed backhaul scenarios, the CPU usage of the main control board has the following impacts on the SCell configuration procedure in addition to the impacts listed in **4.6.3.1.1 Triggering Conditions**:

  When configuring an inter-eNodeB SCell, the eNodeB that serves the PCell checks the CPU usage of the main control board of the eNodeB that serves this inter-eNodeB SCell.

  - The CPU usage of the main control board of the eNodeB that serves the inter-eNodeB SCell is lower than the inter-eNodeB SCell flow control threshold.

    There is no impact on the inter-eNodeB SCell configuration procedure.

  - The CPU usage of the main control board of the eNodeB that serves the inter-eNodeB SCell reaches the inter-eNodeB SCell flow control threshold.

    Within one minute, none of the cells served by the eNodeB that serves the inter-eNodeB SCell will be configured as SCells. One minute later, these cells can be configured as SCells gradually.

    NOTE

    The inter-eNodeB SCell flow control threshold is specified by the **eNodeBFlowCtrlPara.*InterEnbScellFlowCtrlThld*** parameter.

- In inter-eNodeB CA based on relaxed backhaul scenarios, if the **EX2_FLOW_CONTROL_SW** option of the **eNodeBFlowCtrlPara**.***CongFlowCtrlSwitch*** parameter is selected, SCTP congestion of the eX2 interface has the following impacts on the SCell configuration procedure.

  - If the SCTP congestion level of the eX2 interface is high, the inter-eNodeB SCell configuration probability will be adjusted per second, as indicated in **Table 16-4**.

Table 16-4 Inter-eNodeB SCell configuration probabilities at different time points

Time Point	SCell Configuration Probability
At the beginning	80%
In the 1st second	60%
In the 2nd second	40%
In the 3rd second	20%
In the 4th second	10%
In the 5th second and later	Inter-eNodeB SCells will not be configured.

- When the SCTP congestion level of the eX2 interface has changed from high to low, the SCell configuration probability is 95%.
- When the SCTP congestion status of the eX2 interface has changed from no congestion to low-level congestion, the inter-eNodeB SCell configuration procedure is not affected.
- The eX2 interface flow control status is updated every minute. Flow control will be canceled for the eX2 interface if inter-eNodeB SCell configuration is not performed or the inter-eNodeB SCell configuration probability is 95%. After the canceling, the inter-eNodeB SCell configuration procedure is not affected.

- Inter-eNodeB SCell configuration and removal in relaxed backhaul scenarios are affected by the CPU usage of the main control board (indicated by the **VS.BBUBoard.CPULoad.Max** counter). The impacts are as follows:
  - When the CPU usage of the main control board in the local eNodeB serving the PCell or in the peer eNodeB serving an SCell exceeds 60%, the local eNodeB imposes the restriction that SCell configuration can be performed only in a blind manner. When both the CPU usage of the main control board in the local eNodeB serving the PCell and that in the peer eNodeB serving an SCell become less than or equal to 50%, the local eNodeB removes the restriction on SCell configuration.
  - When the CPU usage of the main control board in the local eNodeB serving the PCell exceeds 70%, the local eNodeB removes all the SCells that have been configured in a non-blind manner.
- When inter-eNodeB CA based on relaxed backhaul is in effect and the PCell and SCell of a CA UE are served by different eNodeBs, the UE also consumes the RRC_CONNECTED UE specification quota of the main control board in the SCell's eNodeB. As a result, the actual available number of RRC_CONNECTED UEs supported by the main control board in the SCell's eNodeB cannot reach its maximum specification.

16.3 Requirements

16.3.1 Licenses (FDD)

If the **RelaxedBackhaulCaSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected for an eNodeB, the eNodeB consumes the license for LAOFD-080201 Inter-eNodeB CA Based on Relaxed Backhaul.

If this eNodeB is serving a UE with inter-eNodeB SCells aggregated with the PCell and the **FreqBaseIntereNBSccSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is selected for the neighboring eNodeB serving the SCells, this neighboring eNodeB also consumes the license for LAOFD-080201 Inter-eNodeB CA Based on Relaxed Backhaul.

NOTE

If the **FreqBaseIntereNBSccSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is selected for a local eNodeB in a relaxed backhaul scenario:

- The local eNodeB can acquire the CA-related switch settings of each neighboring eNodeB and, based on the settings, determine license consumption by local cells (cells served by the local eNodeB). For example:

  If 4CC aggregation has been enabled at a neighboring eNodeB, local cells can act as SCells. The local eNodeB then determines that each local cell on the frequencies configured as candidate SCCs on the neighboring eNodeB should consume one sales unit of the license for Carrier Aggregation for Downlink 4CC and 5CC.

  If 4CC aggregation has been enabled neither at the local eNodeB nor at neighboring eNodeBs, local cells will not act as SCells in 4CC aggregation. The local eNodeB determines that the license for Carrier Aggregation for Downlink 4CC and 5CC should not be consumed by any local cell.
- If the version of a connected neighboring eNodeB is earlier than eRAN12.1, the local eNodeB assumes that all CA features have been enabled at the neighboring eNodeB, for compatibility consideration. The local eNodeB attempts to make the consumption of all CA feature licenses take effect for each local cell. In this situation, the feature query result at the local eNodeB indicates that the configuration status of all CA features is on.

Table 16-5 lists the license model and sales unit for the feature.

Table 16-5 License model and sales unit

Feature ID	Feature Name	Model	Sales Unit
LAOFD-080201	Inter-eNodeB CA Based on Relaxed Backhaul	LT1S0IPRAN01	per eNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

16.3.2 Licenses (TDD)

- In CA-group-based configuration mode

  If a CA group includes cells that are served by different eNodeBs in a relaxed backhaul scenario, each of the eNodeBs serving these cells requires one sales unit of the license for Inter-eNodeB CA Based on Relaxed Backhaul.
- In adaptive configuration mode

  If any cell served by the local eNodeB is to be accompanied by any inter-eNodeB SCell in a relaxed backhaul scenario, each of the eNodeBs requires one sales unit of the license for Inter-eNodeB CA Based on Relaxed Backhaul.

NOTE

If the **FreqBaseIntereNBSccSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is selected for a local eNodeB in a relaxed backhaul scenario:

- The local eNodeB can acquire the CA-related switch settings of each neighboring eNodeB and, based on the settings, determine license consumption by local cells (cells served by the local eNodeB). For example:

  If 4CC aggregation has been enabled at a neighboring eNodeB, local cells can act as SCells. The local eNodeB then determines that each local cell on the frequencies configured as candidate SCCs on the neighboring eNodeB should consume one sales unit of the license for Carrier Aggregation for Downlink 4CC and 5CC.

  If 4CC aggregation has been enabled neither at the local eNodeB nor at neighboring eNodeBs, local cells will not act as SCells in 4CC aggregation. The local eNodeB determines that the license for Carrier Aggregation for Downlink 4CC and 5CC should not be consumed by any local cell.

- If the version of a connected neighboring eNodeB is earlier than eRAN12.1, the local eNodeB assumes that all CA features have been enabled at the neighboring eNodeB, for compatibility consideration. The local eNodeB attempts to make the consumption of all CA feature licenses take effect for each local cell. In this situation, the feature query result at the local eNodeB indicates that the configuration status of all CA features is on.

Table 16-6 lists the license model and sales unit for the feature.

Table 16-6 License model and sales unit

Feature ID	Feature Name	Model	Sales Unit
TDLAOFD-081402	Inter-eNodeB CA Based on Relaxed Backhaul	LT1STIBCAD00	per eNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

16.3.3 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

16.3.3.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Downlink 2CC aggregation	None	**5 Downlink 2CC Aggregation**	Inter-eNodeB CA based on relaxed backhaul requires this function to be activated.
FDD TDD	Downlink 3CC aggregation	**CaDl3CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**6 Downlink 3CC Aggregation**	(Optional) This function must be activated if three carriers need to be aggregated in the downlink for inter-eNodeB CA based on relaxed backhaul.
FDD TDD	Downlink 4CC aggregation	**CaDl4CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**7 Downlink 4CC Aggregation**	(Optional) This function must be activated if four carriers need to be aggregated in the downlink for inter-eNodeB CA based on relaxed backhaul.
FDD TDD	Downlink 5CC aggregation	**CaDl5CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**8 Downlink 5CC Aggregation**	(Optional) This function must be activated if five carriers need to be aggregated in the downlink for inter-eNodeB CA based on relaxed backhaul.
FDD	Downlink massive CA	**DlMassiveCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**9 Downlink Massive CA (FDD)**	(Optional) If six or more carriers need to be aggregated in the downlink, only FDD carriers can be used as the PCCs and downlink massive CA must be enabled.
FDD TDD	Downlink FDD+TDD CA	**InterFddTddCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	**13 Downlink FDD+TDD CA**	(Optional) This function must be activated if downlink inter-eNodeB FDD +TDD CA based on relaxed backhaul is required.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Intelligent selection of serving cell combinations	**CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter	12 Intelligent Selection of Serving Cell Combinations	(Optional) This function must be activated if it is to be used for inter-eNodeB CA based on relaxed backhaul.
FDD TDD	Uplink 2CC aggregation	**CaUl2CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	10 Uplink 2CC Aggregation	(Optional) This function must be activated if two carriers need to be aggregated in the uplink for inter-eNodeB CA based on relaxed backhaul.
FDD TDD	Uplink FDD +TDD CA	**InterFddTddCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	14 Uplink FDD+TDD CA	(Optional) This function must be activated if uplink inter-eNodeB FDD +TDD CA based on relaxed backhaul is required.

16.3.3.2 Mutually Exclusive Functions

None

16.3.3.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Downlink frequency selective scheduling	**FreqSelSwitch** option of the **CellAlgoSwitch**.***DlSchSwitch*** parameter	*Downlink Scheduling*	In inter-eNodeB CA based on relaxed backhaul, PCells but not SCells support downlink frequency selective scheduling.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Enhanced scheduling	None	None	This function affects only the UEs for which inter-eNodeB CA based on relaxed backhaul has taken effect. To mitigate the impact of inter-eNodeB delay, eNodeBs must estimate scheduling opportunities for CA UEs in advance. This may have a slight short-term impact on fairness among UEs but does not affect scheduling-related performance indicators.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Smart carrier selection based on virtual grids	**SMART_CARRIER_SELECTION_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter	*Multi-carrier Unified Scheduling*	• When both inter-eNodeB cell filtering (controlled by the **SccMeasRptLocalCell-PriorSw** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter) and smart carrier selection based on virtual grids (controlled by the **SMART_CARRIER_SELECTION_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter) are enabled, the number of handovers triggered for PCC anchoring may increase. • When both traffic model-aware smart carrier selection (controlled by the **TRAFFIC_AWARENESS_SCS_SW** and **INTER_ENB_UL_UU_CAPB_EVAL_SW** options of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter) and inter-eNodeB uplink 2CC aggregation based on relaxed backhaul are enabled, the eNodeB acts as follows during carrier evaluation based on uplink air interface capabilities: If two inter-eNodeB uplink CCs have been configured for a UE, the eNodeB considers the uplink air interface capability of the PCell only, when it calculates the capability of the serving cell combination for the UE.

RAT	Function Name	Function Switch	Reference	Description
				As a result, an incorrect handover occurs because the calculated air interface capability of the serving cell combination is less than the actual capability. You are not advised to enable both of the two functions.
FDD	Increasing the CRS transmission disabling rate	• **NCellDlRsrpMeasPara**.***DlRsrpAutoNCellMeasSwitch*** set to **ON** • **CellDlschAlgo**.***PilotOffRateIncreaseSw*** set to **BASED_ON_MEASUREMENT**	*eMIMO (FDD)*	After the neighboring cell measurement bandwidths are configured, PCell and SCell RSRP measurements by CA UEs will be affected, leading to a slight change in the number of CA UEs in the cells. After the function of increasing the CRS transmission disabling rate is enabled, data volume distribution between the PCell and SCells will be affected and the downlink PRB usages in the PCell and SCells will change. The function of increasing the CRS transmission disabling rate will offer lower gains when inter-eNodeB CA based on relaxed backhaul takes effect.
FDD	UMTS and LTE Zero Bufferzone	**UMTS_LTE_ZERO_BUFFER_ZONE_SW** option of the **ULZeroBufferZone**.***ZeroBufZoneSwitch*** parameter	*UMTS and LTE Zero Bufferzone*	There are fewer PUSCH and SRS resources in a cell in the bufferzone than in a common cell. Therefore, when the LTE bandwidth is 5 MHz or 10 MHz, using a cell in the bufferzone as a PCell for CA is not recommended. If the cell is used as a PCell, CA performance deteriorates.

RAT	Function Name	Function Switch	Reference	Description
FDD	GSM and LTE Spectrum Concurrency	**SpectrumCloud**.***SpectrumCloudSwitch*** set to **GL_SPECTRUM_CONCURRENCY**	*GSM and LTE Spectrum Concurrency*	For inter-eNodeB CA based on relaxed backhaul, pre-scheduling can be performed up to 35 ms earlier than actual scheduling. If LTE cells with GSM and LTE Spectrum Concurrency enabled act as SCells, the interference information may have changed at the moment of actual scheduling. As a result, the RB estimation for pre-scheduling is inaccurate, affecting the LTE network throughput.
FDD TDD	Terminal Awareness Differentiation	**RELAXED_BACKHAUL_CA_SW** option of the **UeCompat**.***BlacklistControlExtSwitch1*** parameter	*Terminal Awareness Differentiation*	Due to software and hardware exceptions of some CA UEs, UE compatibility issues (such as service drops) arise after inter-eNodeB SCells are configured for the UEs in relaxed backhaul scenarios. These issues affect network performance. To prevent inter-eNodeB SCell configuration for these UEs in relaxed backhaul scenarios, you can enable Terminal Awareness Differentiation with these UEs blacklisted and **RELAXED_BACKHAUL_CA_SW** of the **UeCompat**.***BlacklistControlExtSwitch1*** parameter selected.
FDD	Downlink intelligent joint scheduling	**DlCaUnifiedSchSwitch** option of the **CellDlschAlgo**.***DlCaUserSchSwitch*** parameter	*Smart 8T8R (FDD)*	Downlink intelligent joint scheduling changes the proportion of CA UE data packets distributed to each CC. Therefore, the transport load changes.

RAT	Function Name	Function Switch	Reference	Description
FDD	Intelligent unified scheduling enhancement	**DlCaUnifiedSchEnhSwitch** option of the **CellDlschAlgo**.***DlCaUserSchSwitch*** parameter	• *Smart 8T8R (FDD)* • *Massive MIMO Enhancements (FDD)*	Intelligent unified scheduling enhancement changes the proportion of CA UE data packets distributed to each CC. Therefore, the transport load changes.
TDD	Precise AMC	**PreciseAmcSwitch** option of the **CellAlgoSwitch**.***EmimoSwitch*** parameter	*Downlink Scheduling*	Precise AMC does not take effect for UEs on SCCs in inter-eNodeB CA based on relaxed backhaul.

16.3.4 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

The requirements described in **Boards** of **5.3.4 Hardware** must be fulfilled. In addition:

- BBU3910A does not support this function.
- For 5CC aggregation to 100 MHz, the UBBP and UMPT boards are recommended. If LBBPd boards are used, the peak data rate may not reach the expected value, due to the lower hardware capabilities.
- If the data rate of transmission from the EPC to the eNodeB exceeds the transmission capability of the BBP or main control board, the PCell cannot transfer data to SCells. As a result, the SCells may be activated but their scheduling is suspended.
- In adaptive configuration mode, the BBP specifications are as follows:
  – If the UBBPg/UBBPh is used:

    - If the **RELAXED_BH_CA_BBP_LINK_ENH_SW** option of the **EnodebAlgoExtSwitch**.***CaAlgoSwitch*** parameter is deselected, this BBP can work with a maximum of 26 other LBBPs or UBBPs for CA.
    - If the **RELAXED_BH_CA_BBP_LINK_ENH_SW** option of the **EnodebAlgoExtSwitch**.***CaAlgoSwitch*** parameter is selected, this BBP can work with a maximum of 108 other LBBPs or UBBPs for CA.
  - If the LBBPd/UBBPe/UBBPd/UBBPf is used, it can work with a maximum of 26 other LBBPs or UBBPs for CA.

  If the number of related BBPs reaches the preceding upper limit, the values of related indicators, such as the user-perceived rate, spectral efficiency, number of SCell additions, and number of CA UEs, may fluctuate in version upgrade and reset scenarios.

RF Modules

For details, see **RF Modules** in **5.3.4 Hardware**.

16.3.5 Networking

The requirements described in **5.3.5 Networking** must be fulfilled. In addition, CA in relaxed backhaul scenarios has the following requirements:

- It is recommended that this function be deployed in urban areas to prevent CA failures caused by large inter-site distances.

  In accordance with section J.1 "Deployment Scenarios" in 3GPP TS 36.300 V10.11.0, the delay spread among the inter-eNodeB CCs monitored at the UE cannot exceed 30.26 μs. A delay spread of 30.26 μs corresponds to a signal transmission distance difference of about 9 kilometers among the CCs.
- The eNodeBs must use the same software version.
- The eNodeBs involved in CA must use the same configuration mode for CA.
- The eNodeBs must be time-synchronized to within 3 μs. For the principles for and operations related to time synchronization, see *Synchronization*.

  The **DSP CLKTST** command can be used to query clock quality test monitoring results.
- Routes between inter-eNodeB cells must be established and working properly. eX2 or X2 interfaces must be set up between the eNodeBs. For details, see *eX2 Self-Management* or *S1 and X2 Self-Management*.
  - If the eNodeBs belong to the same operator and are managed by the same MAE-Access:

    eX2 or X2 interfaces between the eNodeBs support self-setup.
  - If the eNodeBs belong to different operators or are managed by different MAE-Access systems:

    SCTP links and endpoint groups for eX2 or X2 interfaces must be manually configured.

  Both the control plane and user plane must be configured for X2 interfaces. Otherwise, the routes between inter-eNodeB cells are unreachable, causing CA failures. The **DSP X2UPINFO** command (in endpoint mode) or **DSP IPPATH** command (in link mode) can be executed to check whether user-plane information has been configured for an X2 interface.

eNodeBs will automatically create IP PM sessions to detect link status of eX2 or X2 interfaces. If an IP PM session of bidirectional activation type is configured at either end of an eX2 or X2 interface, it may conflict with an automatically created session. If that is the case, the eX2 or X2 interface fails to work properly.

If three consecutive attempts of a cell to set up an eX2 or X2 route to an inter-eNodeB cell fail in a relaxed backhaul scenario, the subsequent setup attempts on this route will be prohibited within the next 5 minutes.

The maximum number of inter-eNodeB cells that can be associated varies depending on the type of main control board, as listed in **Table 16-7**.

Table 16-7 Maximum number of inter-eNodeB cells for each type of main control board

UMPTb	UMPTe	UMPTg	UMPTha
288	1152	1728	1728

NOTE

- If the **InterEnodebCellConfigOptSw** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is selected, inter-eNodeB cell management is performed on a per cell basis. This prevents cells that actually do not participate in inter-eNodeB CA from consuming the capacity in terms of connected inter-eNodeB cells.
- When the **InterEnodebCellConfigOptSw** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is changed from selected to deselected, the connected inter-eNodeB cell information maintained by the eNodeB is not updated immediately. This causes loss of the information about the cells that are served by a neighboring eNodeB with an eX2 or X2 interface to the local eNodeB but did not participate in inter-eNodeB CA before this option is deselected. As a result, redundant measurement configuration and PCC anchoring may occur. To restore the inter-eNodeB cell information that used to be stored on the local eNodeB, take the following action after deselecting this option: Remove and then add back all eX2 or X2 interfaces involved in inter-eNodeB CA. Alternatively, run the **RST APP** command.

- The inter-eNodeB transmission quality must meet requirements.
  - For transmission delay requirements, see **Usage Scenarios** in **16.1 Principles**.
  - Jitter and packet loss rate requirements are described in **Table 16-8**.

Table 16-8 Jitter and packet loss rate requirements

Scenario	Jitter (ms)	Packet Loss Rate (%)
Best	0	0.0001
Recommended	1	0.001
Tolerable	2	0.5

In tolerable scenarios described in **Table 16-8**, services can be successfully set up, but the QoS requirements of services other than VoLTE and push to talk (PTT) are not necessarily fulfilled.

NOTE

- If the transmission delay (maximum one-way delay or maximum RTT), jitter, or packet loss rate deteriorates, the data rates of CA UEs in inter-eNodeB CA based on relaxed backhaul decrease.
- If the downlink traffic to be carried on the serving board or eNodeB of the PCell exceeds the maximum data rate supported by the hardware, the end-to-end delay between the PCell and SCells may increase, so that the data rates in the SCells decrease or even drop to zero.
- If TCP is used for application layer data transmission and there is packet loss on the transmission link between the eNodeB and the EPC, the data rates of CA UEs may be adversely affected because of insufficient incoming application layer data packets. This insufficiency is caused by the limited size of the TCP sliding window.

Inter-eNodeB transmission quality information can be obtained using the following methods:

– Run the **DSP EX2UPINFO** command to query the maximum one-way delay and maximum RTT between eNodeBs with eX2 interfaces set up between them, or run the **DSP X2UPINFO** command to query the maximum one-way delay and maximum RTT between eNodeBs with X2 interfaces set up between them. If the status of clock synchronization between two eNodeBs does not meet accuracy requirements, the maximum one-way delay cannot be precisely measured. It can be estimated by taking half of the maximum RTT.

– Monitor performance counters related to IP Performance Monitor (PM). For details, see *IP Performance Monitor*.

- The bandwidths of the links between the eNodeBs must meet requirements.

  The required link bandwidth can be planned based on network conditions. It must be within the transmission bandwidth capacities of the eNodeBs.

  – Link bandwidth

    Required link bandwidth = Bandwidth of the SCC x Spectral efficiency of the SCC x Percentage of CA UEs on the SCC x Number of SCells to be associated

    Take a 20 MHz SCC as an example. Each time the serving eNodeB of the PCell associates an inter-eNodeB SCell with the PCell:

    ■ A link bandwidth of 35 Mbit/s is required for the uplink and also for the downlink, assuming the average SCC spectral efficiency of 3.5 and a CA UE percentage of 50% for the SCC. The required link bandwidth is calculated as follows:

      20 x 3.5 x 50% x 1 = 35 Mbit/s

      In consideration of transport protocol overheads, a bandwidth margin of about 12% must be reserved. As a result, the bandwidth to be configured is calculated as follows: 35 x (1 + 12%) = 39 Mbit/s.

    ■ A link bandwidth of 150 Mbit/s is required for the uplink and also for the downlink, assuming the maximum SCC spectral efficiency of 7.5 and a CA UE percentage of 100% for the SCC.

With the bandwidth margin of 12% considered, the bandwidth to be configured is calculated as follows: 150 x (1 + 12%) = 168 Mbit/s.

Spectral efficiency = (**L.Traffic.DL.SCH.QPSK.TB.bits** + **L.Traffic.DL.SCH.16QAM.TB.bits** + **L.Traffic.DL.SCH.64QAM.TB.bits** + **L.Traffic.DL.SCH.256QAM.TB.bits**)/(**L.ChMeas.PRB.DL.Used.Avg** x 3600)/180000

– Transmission bandwidth capacities of eNodeBs

The CA-related transmission bandwidth capacities of eNodeBs depend on the base station type and main control board type.

For macro and LampSite eNodeBs, the capacities depend on the main control board type and the setting of the **GlobalProcSwitch**.***ProcTypeForNonIdealServData*** parameter, as described in **Table 16-9**.

Table 16-9 CA-related transmission bandwidth capacities of macro and LampSite eNodeBs

Value of GlobalProcSwitch.*ProcTypeForNonIdealServData*	UMPTb	UMPTe	UMPTg/ UMPTga	UMPTha
SOFT	300 Mbit/s	600 Mbit/s	600 Mbit/s	600 Mbit/s
HARD	600 Mbit/s	1.2 Gbit/s	3 Gbit/s	3 Gbit/s

If the **GlobalProcSwitch**.***ProcTypeForNonIdealServData*** parameter is set to **HARD**, BBPs (except LBBPc) will process data to facilitate data forwarding at main control boards for inter-eNodeB CA based on relaxed backhaul. This processing helps improve the CA-related transmission bandwidth capacities of the eNodeBs. As a result, the **L.CA.Traffic.bits.RelaxedBackhaulCAUsed.DL.Scell** counter will produce a larger value, and the CPU usage of BBPs rises when the amount of processed data increases.

Any change to the setting of the **GlobalProcSwitch**.***ProcTypeForNonIdealServData*** parameter will cause the links between PCells and inter-eNodeB SCells to be reestablished. The reestablishment has the following impacts on existing CA UEs in inter-eNodeB CA based on relaxed backhaul:

- SCells are removed, so the instantaneous data rates of the UEs will decrease even temporarily to zero.
- In CA-group-based configuration mode, after the SCell removal, SCells will not be configured again for the UEs when the specific traffic volume conditions are fulfilled. SCells can be configured for the UEs only after they access the network again.

NOTE

In SCells participating in inter-eNodeB CA, the data rates on the radio interface are determined by the transport block sizes selected by eNodeBs for UEs. The sizes must not exceed the maximum allowed transmission bandwidth, and the size type is subject to protocol-stipulated restrictions. In case a selected transport block size does not precisely match the transmission bandwidth, the actual data throughput fails to reach the maximum allowed bandwidth even though spare radio resources are available in the SCells.

In inter-eNodeB CA based on relaxed backhaul, the PCell for a UE can allocate a maximum of 50 KB of data to each SCell in each TTI. As a result, the air interface data rate of the UE in the SCells cannot reach the peak value in some FDD scenarios.

16.3.6 Others

- UEs

  UEs must comply with 3GPP Release 12 or later and support the frequency bands of the carriers to be aggregated and their bandwidths. UEs must also support the peak data rates that CA can achieve.

  6CC–8CC aggregation requires UEs to comply with 3GPP Release 13 or later.

- EPC

  For this function to reach a theoretical peak data rate, the maximum bit rate that each UE subscribes to in the EPC cannot be lower than this theoretical value. For the theoretical values, see **16.2.1 Benefits**.

16.4 Operation and Maintenance

16.4.1 Data Configuration

16.4.1.1 Data Preparation

This function works in either CA-group-based or adaptive configuration mode. Prepare basic data as described in **5.4.1.1 Data Preparation**.

- In CA-group-based configuration mode

  In addition to **5.4.1.1.1 CA-Group-based Configuration Mode**, configure **CaGroup**, **CaGroupCell**, and **CaGroupSCellCfg** MOs on each eNodeB and ensure that the **CaGroup** and **CaGroupCell** configurations of the eNodeBs involved in CA are consistent. If the configurations are inconsistent, CA will not work. Moreover, set the parameters listed in **Table 16-10**.

Table 16-10 Parameters for activating the function in CA-group-based configuration mode

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	RelaxedBackhaulCaSwitch	Select this option for all the eNodeBs involved.

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	**ENodeBAlgoSwitch**.*CaAlgoSwitch*	TddRelaxedBackhaulCaSwitch	Select this option for all the eNodeBs involved.

- In adaptive configuration mode

  Moreover, set the parameters listed in **Table 16-11**.

Table 16-11 Parameters for activating the function in adaptive configuration mode

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	**ENodeBAlgoSwitch**.*CaAlgoSwitch*	RelaxedBackhaulCaSwitch	Select this option for all the eNodeBs involved.
CA Algorithm Switch	**ENodeBAlgoSwitch**.*CaAlgoSwitch*	TddRelaxedBackhaulCaSwitch	Select this option for all the eNodeBs involved.
CA Algorithm Extend Switch	**ENodeBAlgoSwitch**.*CaAlgoExtSwitch*	FreqBaseInterNBSccSwitch	Select this option for the serving eNodeBs of SCells.

In addition to the preceding parameters for function activation, prepare data as described in the "Data Preparation" sections for downlink 2CC, 3CC, 4CC, 5CC, or 6CC–8CC aggregation or for uplink 2CC aggregation if the corresponding number of inter-eNodeB CCs need to be aggregated in relaxed backhaul scenarios.

The parameters used for optimization of inter-eNodeB CA based on relaxed backhaul are described in **Table 16-12**.

Table 16-12 Parameters for optimizing inter-eNodeB CA based on relaxed backhaul

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	**ENodeBAlgoSwitch**.*CaAlgoSwitch*	RelaxedBHCaEnhanceSwitch	On a network where the inter-eNodeB one-way delay is between 4 ms and 8 ms (inclusive) and the RTT is between 8 ms and 16 ms (inclusive), to activate downlink CA, select this option.

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Extend Switch	**ENodeBAlgoSwitch**.***CaAlgoExtSwitch***	RelaxedBHCaEnh2Switch	On a network where the inter-eNodeB one-way delay is between 4 ms and 16 ms (inclusive) and the RTT is between 8 ms and 32 ms (inclusive), to activate downlink FDD CA, select this option.
CA Algorithm Extend Switch	**ENodeBAlgoSwitch**.***CaAlgoExtSwitch***	SccMeasRptLocalCellPriorSw	In normal scenarios, it is recommended that this option be deselected. In SCTP link congestion scenarios, it is recommended that this option be selected.
Relaxed Backhaul Ca Max Component Carrier Number	**CaMgtCfg**.***RelaxedBackhaulCaMaxCcNum***	None	Set this parameter based on site conditions.
Relaxed Backhaul CA UL Max CC Num	**CaMgtCfg**.***RelaxedBHCaUlMaxCcNum***	None	Set this parameter based on site conditions.
FDD TDD Relaxed BH CA DL Max CC Num	**CaMgtCfg**.***FTRelaxedBHCaDLMaxCcNum***	None	Set this parameter based on site conditions. This parameter cannot be set to **DL0CC** when the **CaMgtCfg**.***FTRelaxedBHCaUlMaxCcNum*** parameter is set to **2CC**.
FDD and TDD Relaxed Backhaul CA UL Max CC Num	**CaMgtCfg**.***FTRelaxedBHCaUlMaxCcNum***	None	Set this parameter based on site conditions.
Cell Level CA Algorithm Switch	**CaMgtCfg**.***CellCaAlgoSwitch***	RelaxedBhDlRlcRetransOptSw	Select this option.
Relaxed Backhaul SCC DL Target IBLER	**CaMgtCfg**.***RelaxedBhSccDlTargetIbler***	None	Set this parameter to **1**.

16.4.1.2 Using MML Commands (FDD)

Before using MML commands, refer to **16.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Before activating this function, configure cells or frequencies according to **5.4.1.2.1 CA-Group-based Configuration Mode (FDD)** or **5.4.1.2.3 Adaptive Configuration Mode (FDD)**.

The activation command examples for this function are as follows:

```
//Setting the clock synchronization mode of eNodeBs to time synchronization
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Turning on RelaxedBackhaulCaSwitch for each eNodeB
MOD ENODEBALGOSWITCH: CaAlgoSwitch=RelaxedBackhaulCaSwitch-1;
//Turning on FreqBaseIntereNBSccSwitch for the serving eNodeB of each possible SCell (required only in
adaptive configuration mode)
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=FreqBaseIntereNBSccSwitch-1;
//For downlink 2CC aggregation based on relaxed backhaul: setting RelaxedBackhaulCaMaxCcNum to 2CC
for the serving eNodeB of each possible PCell
MOD CAMGTCFG: LocalCellId=0, RelaxedBackhaulCaMaxCcNum=2CC;
//(Optional) For downlink 3CC aggregation based on relaxed backhaul: turning on CaDl3CCSwitch and
setting RelaxedBackhaulCaMaxCcNum to 3CC for the serving eNodeB of each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1, RelaxedBackhaulCaMaxCcNum=3CC;
//(Optional) For downlink 4CC aggregation based on relaxed backhaul: turning on CaDl4CCSwitch and
setting RelaxedBackhaulCaMaxCcNum to 4CC for the serving eNodeB of each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl4CCSwitch-1, RelaxedBackhaulCaMaxCcNum=4CC;
//(Optional) For downlink 5CC aggregation based on relaxed backhaul: turning on CaDl5CCSwitch and
setting RelaxedBackhaulCaMaxCcNum to 5CC for the serving eNodeB of each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl5CCSwitch-1, RelaxedBackhaulCaMaxCcNum=5CC;
//(Optional) For downlink 6CC–8CC aggregation based on relaxed backhaul: turning on DlMassiveCaSwitch
and setting RelaxedBackhaulCaMaxCcNum to DL_MASSIVE_CA for the serving eNodeB of each possible
PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=DlMassiveCaSwitch-1,
RelaxedBackhaulCaMaxCcNum=DL_MASSIVE_CA;
//(Optional) For uplink 2CC aggregation based on relaxed backhaul: turning on CaUl2CCSwitch and setting
RelaxedBHCaUlMaxCcNum to 2CC for the serving eNodeB of each possible PCell; moreover, turning on
MtaAlgSwitch for each eNodeB
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaUl2CCSwitch-1, RelaxedBHCaUlMaxCcNum=2CC;
MOD ENODEBALGOSWITCH: CaAlgoSwitch=MtaAlgSwitch-1;
//(Optional) Turning on RelaxedBHCaEnhanceSwitch for the serving eNodeB of each possible PCell to use
downlink CA on a relaxed backhaul network with an inter-eNodeB one-way delay between 4 ms and 8 ms
(inclusive) and an RTT between 8 ms and 16 ms (inclusive)
MOD ENODEBALGOSWITCH: CaAlgoSwitch=RelaxedBHCaEnhanceSwitch-1;
//(Optional) Turning on RelaxedBHCaEnh2Switch for the serving eNodeB of each possible PCell to use
downlink CA on a relaxed backhaul network with an inter-eNodeB one-way delay between 4 ms and 16 ms
(inclusive) and an RTT between 8 ms and 32 ms (inclusive)
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=RelaxedBHCaEnh2Switch-1;
```

Optimization Command Examples

```
//Turning on RelaxedBhDlRlcRetransOptSw for the serving eNodeB of each possible PCell to lift the MAC-
layer scheduling priority of downlink RLC retransmission packets for UEs in the relaxed-backhaul-based CA
state
```



```
MOD CAMGTCFG: LocalCellId=0,CellCaAlgoSwitch=RelaxedBhDlRlcRetransOptSw-1;
//Configuring RelaxedBhSccDlTargetIbler for the serving eNodeB of each possible SCell
MOD CAMGTCFG: LocalCellId=1,RelaxedBhSccDlTargetIbler=1;
//Prohibiting the serving eNodeB of the PCell for a CA UE from configuring any inter-eNodeB cell that is
working on the same frequency as an activated cell on this local eNodeB as an SCell for the UE
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=SccMeasRptLocalCellPriorSw-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off RelaxedBackhaulCaSwitch
MOD ENODEBALGOSWITCH: CaAlgoSwitch=RelaxedBackhaulCaSwitch-0;
```

16.4.1.3 Using MML Commands (TDD)

Before using MML commands, refer to **16.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Before activating this function, configure cells or frequencies according to **5.4.1.2.2 CA-Group-based Configuration Mode (TDD)** or **5.4.1.2.4 Adaptive Configuration Mode (TDD)**.

The activation command examples for this function are as follows:

```
//Setting the clock synchronization mode of eNodeBs to time synchronization
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Turning on TddRelaxedBackhaulCaSwitch for each eNodeB
MOD ENODEBALGOSWITCH: CaAlgoSwitch=TddRelaxedBackhaulCaSwitch-1;
//Turning on FreqBaseIntereNBSccSwitch for the serving eNodeB of each possible SCell (required only in
adaptive configuration mode)
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=FreqBaseIntereNBSccSwitch-1;
//For downlink 2CC aggregation based on relaxed backhaul: setting RelaxedBackhaulCaMaxCcNum to 2CC
for the serving eNodeB of each possible PCell
MOD CAMGTCFG: LocalCellId=0, RelaxedBackhaulCaMaxCcNum=2CC;
//(Optional) For downlink 3CC aggregation based on relaxed backhaul: turning on CaDl3CCSwitch and
setting RelaxedBackhaulCaMaxCcNum to 3CC for the serving eNodeB of each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1, RelaxedBackhaulCaMaxCcNum=3CC;
//(Optional) For downlink 4CC aggregation based on relaxed backhaul: turning on CaDl4CCSwitch and
setting RelaxedBackhaulCaMaxCcNum to 4CC for the serving eNodeB of each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl4CCSwitch-1, RelaxedBackhaulCaMaxCcNum=4CC;
//(Optional) For downlink 5CC aggregation based on relaxed backhaul: turning on CaDl5CCSwitch and
setting RelaxedBackhaulCaMaxCcNum to 5CC for the serving eNodeB of each possible PCell
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl5CCSwitch-1, RelaxedBackhaulCaMaxCcNum=5CC;
//(Optional) For uplink 2CC aggregation based on relaxed backhaul: turning on CaUl2CCSwitch and setting
RelaxedBHCaUlMaxCcNum to 2CC for the serving eNodeB of each possible PCell; moreover, turning on
MtaAlgSwitch for each eNodeB
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaUl2CCSwitch-1, RelaxedBHCaUlMaxCcNum=2CC;
MOD ENODEBALGOSWITCH: CaAlgoSwitch=MtaAlgSwitch-1;
//(Optional) Turning on RelaxedBHCaEnhanceSwitch for the serving eNodeB of each possible PCell to use
downlink CA on a relaxed backhaul network with an inter-eNodeB one-way delay between 4 ms and 8 ms
```

```
(inclusive) and an RTT between 8 ms and 16 ms (inclusive)
MOD ENODEBALGOSWITCH: CaAlgoSwitch=RelaxedBHCaEnhanceSwitch-1;
```

Optimization Command Examples

```
//Turning on RelaxedBhDlRlcRetransOptSw for the serving eNodeB of each possible PCell to lift the MAC-layer scheduling priority of downlink RLC retransmission packets for UEs in the relaxed-backhaul-based CA state
MOD CAMGTCFG: LocalCellId=0,CellCaAlgoSwitch=RelaxedBhDlRlcRetransOptSw-1;
//Configuring RelaxedBhSccDlTargetIbler for the serving eNodeB of each possible SCell
MOD CAMGTCFG: LocalCellId=1,RelaxedBhSccDlTargetIbler=1;
//Prohibiting the serving eNodeB of the PCell for a CA UE from configuring any inter-eNodeB cell that is working on the same frequency as an activated cell on this local eNodeB as an SCell for the UE
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=SccMeasRptLocalCellPriorSw-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off TddRelaxedBackhaulCaSwitch
MOD ENODEBALGOSWITCH: CaAlgoSwitch=TddRelaxedBackhaulCaSwitch-0;
```

16.4.1.4 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

16.4.2 Activation Verification

Counter Observation

If any of the counters listed in **Table 16-13** produces a non-zero value on a network that is serving CA UEs, inter-eNodeB CA based on relaxed backhaul has taken effect in the network.

Table 16-13 Counters used to verify activation of inter-eNodeB CA based on relaxed backhaul

Counter ID	Counter Name
1526732909	**L.Traffic.User.PCell.DL.RelaxedBackhaul-CA.Avg**
1526732955	**L.Traffic.User.SCell.DL.RelaxedBackhaul-CA.Avg**
1526732911	**L.ChMeas.PRB.DL.PCell.RelaxedBackhaul-CAUsed.Avg**
1526732912	**L.ChMeas.PRB.DL.SCell.RelaxedBackhaul-CAUsed.Avg**

Counter ID	Counter Name
1526743685	**L.Traffic.User.PCell.UL.RelaxedBackhaul-CA.Avg**
1526743658	**L.Traffic.User.SCell.UL.RelaxedBackhaul-CA.Avg**
1526743661	**L.Traffic.User.RelaxedBackhaul-CA.SCell.Active.UL.Avg**
1526769393	**L.Traffic.User.RelaxedBackhaul-CA.PCell.Active.DL.Avg**
1526769394	**L.Traffic.User.RelaxedBackhaul-CA.SCell.Active.DL.Avg**

For the **L.Traffic.User.RelaxedBackhaulCA.PCell.Active.DL.Avg** counter:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, this counter is stepped as long as an SCell is activated for relaxed-backhaul-based CA for a UE.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, this counter is stepped only when all SCells are activated for relaxed-backhaul-based CA for a UE.

For example, three carriers are configured but only two of them are active for relaxed-backhaul-based CA for a UE. That is, not all SCells are activated for the UE. In the presence of such a UE:

- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is selected, the **L.Traffic.User.RelaxedBackhaulCA.PCell.Active.DL.Avg** counter is stepped with this UE counted in.
- If the **CA_UE_MULTI_CC_STAT_OPT_SW** option of the **EnodebCounterParaGrp**.***EnodebCounterAlgoSwitch*** parameter is deselected, the **L.Traffic.User.RelaxedBackhaulCA.PCell.Active.DL.Avg** counter does not count this UE in.

Message Tracing

For the tools and the IEs to observe, see **Message Tracing** in **5.4.2 Activation Verification** and **Message Tracing** in **10.4.3 Activation Verification**.

16.4.3 Network Monitoring

For the counters used in scenarios of downlink 2CC, 3CC, 4CC or 5CC aggregation, downlink massive CA, and uplink 2CC aggregation, see the corresponding "Network Monitoring" sections.

In addition, monitor the counters in **Table 16-14** and compare the results with the network plan to evaluate network performance.

Table 16-14 Counters used to monitor performance of inter-eNodeB CA based on relaxed backhaul

Counter ID	Counter Name
1526732909	**L.Traffic.User.PCell.DL.RelaxedBackhaul-CA.Avg**
1526732910	**L.Traffic.User.PCell.DL.RelaxedBackhaul-CA.Max**
1526732955	**L.Traffic.User.SCell.DL.RelaxedBackhaul-CA.Avg**
1526732954	**L.Traffic.User.SCell.DL.RelaxedBackhaul-CA.Max**
1526732911	**L.ChMeas.PRB.DL.PCell.RelaxedBackhaul-CAUsed.Avg**
1526732912	**L.ChMeas.PRB.DL.SCell.RelaxedBackhaul-CAUsed.Avg**
1526733184	**L.Thrp.bits.DL.RelaxedBackhaulCAUser**
1526733185	**L.Thrp.Time.DL.RelaxedBackhaulCAUser**
1526733198	**L.Traffic.User.PCell.RelaxedBackhaulCA.OFF**
1526737803	**L.CA.Traffic.bits.RelaxedBackhaul-CAUsed.DL.Pcell**
1526737804	**L.CA.Traffic.bits.RelaxedBackhaul-CAUsed.DL.Scell**
1526743685	**L.Traffic.User.PCell.UL.RelaxedBackhaul-CA.Avg**
1526743658	**L.Traffic.User.SCell.UL.RelaxedBackhaul-CA.Avg**
1526743661	**L.Traffic.User.RelaxedBackhaul-CA.SCell.Active.UL.Avg**
1526769393	**L.Traffic.User.RelaxedBackhaul-CA.PCell.Active.DL.Avg**
1526769394	**L.Traffic.User.RelaxedBackhaul-CA.SCell.Active.DL.Avg**
1526759044	**L.CA.SCell.DLSCell.RelaxedBackhaul-CA.Mod.Att**
1526759045	**L.CA.SCell.DLSCell.RelaxedBackhaul-CA.Rmv.Att**
1526759046	**L.CA.SCell.DLSCell.RelaxedBackhaul-CA.Add.Att**

17 Inter-eNodeB CA Based on eNodeB Coordination

17.1 Principles

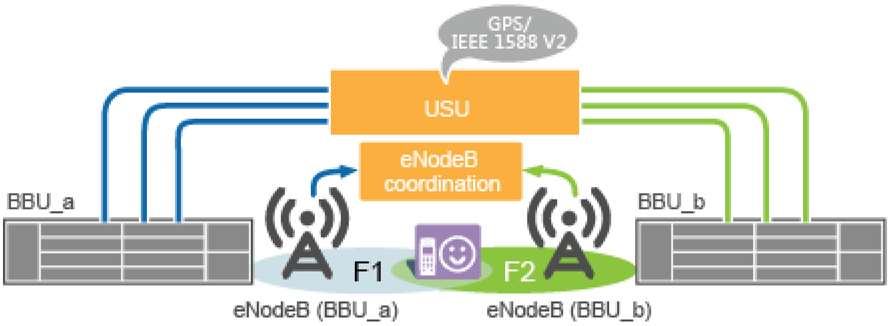
This function allows for downlink 2CC to 8CC aggregation and uplink 2CC aggregation between inter-eNodeB cells in eNodeB coordination scenarios.

For FDD, in eNodeB coordination scenarios, the eNodeBs are deployed in a centralized or distributed manner.

For TDD, in eNodeB coordination scenarios, the eNodeBs are deployed in a centralized manner.

Figure 17-1 illustrates an example of the scenarios.

Figure 17-1 Inter-eNodeB CA based on eNodeB coordination



In CA-group-based configuration mode, this function is not under switch control.

In adaptive configuration mode, this function is controlled by the switches listed in **Table 17-1**.

Table 17-1 Switch control over inter-eNodeB CA based on eNodeB coordination

RAT	Option	Description
FDD TDD	**FreqCfgCaOverBBUsSwitch** option of the **ENodeBAlgoSwitch**.***OverBBUsSwitch*** parameter	Select this option for all the eNodeBs involved.
FDD TDD	**FreqBaseIntereNBSccSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter	Select this option for the serving eNodeBs of SCells.

This function employs the same PCC anchoring and SCell management procedures as for intra-eNodeB CA. For details, see **4.6 Carrier Management for RRC_CONNECTED UEs** and **4.7 Carrier Management for RRC_IDLE UEs**.

Centralized Deployment

In the centralized architecture of eNodeB coordination, the eNodeBs exchange signaling messages and transmit service data through one or two levels of universal switching units (USUs). For details about this architecture, see *USU3910-based Multi-BBU Interconnection*.

CA works in this centralized architecture only when the inter-eNodeB round trip time (RTT) is less than 44 μs.

Distributed Deployment (FDD)

FDD CA works in the distributed architecture of eNodeB coordination.

CA works in this distributed architecture only when the inter-eNodeB RTT is less than 260 μs. For details about this architecture, see *USU3910-based Multi-BBU Interconnection*.

This function is controlled by the **DistributeCloudbbCaSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter, in addition to the switches listed in **Table 17-1**.

Hybrid Deployment (FDD)

FDD CA works in the hybrid architecture of eNodeB coordination.

In this architecture, a USU is working simultaneously in centralized and distributed deployment modes. However, a single eNodeB can be connected to the USU in only one mode: centralized or distributed. For details about this architecture, see *USU3910-based Multi-BBU Interconnection*.

17.2 Network Analysis

17.2.1 Benefits

This function enables inter-eNodeB carriers to be aggregated to increase user-perceived data rates for CA UEs.

17.2.2 Impacts

The impacts between this function and other functions are described in **17.3.3.3 Function Impacts**.

This function has the following impacts on the network, because of the delay:

- In the centralized architecture

  This function has a slight negative impact on the peak data rates of CA UEs but does not affect user experience. The reason is that the data to be sent does not arrive at SCells precisely at the moment of the SCells' scheduling occasion and the UEs cannot be scheduled for that moment.

- In the distributed or hybrid architecture (FDD only)
  - CA UEs cannot promptly acquire the real-time CQI changes about inter-eNodeB SCells. This causes slight deterioration in frequency-selective scheduling performance and an increase in IBLER.
  - Reports of HARQ retransmission demodulation results are delayed, affecting the single-UE peak data rate and resulting in an increase in RBLER.
  - CQI and IBLER feedback is delayed, causing variations in the data rates of UEs located medium or long distances from the center of their SCells.
  - Due to the differences in the RLC data arrival time for the aggregated carriers, CA UEs have to combine and organize the received data. This process places an additional burden on the UEs' CPUs. If the CPU capacity is insufficient, the data rates of the UEs fluctuate.
  - To minimize the impact of inter-eNodeB delay, RLC retransmissions occur only in PCells. If the Uu bandwidth of a PCell is used up by GBR services, RLC retransmissions for the CA UE are often blocked and the data rate of the UE fluctuates.
  - Due to possible errors in the estimated scheduling priorities of CA UEs, the PRBs in SCells for the UEs may not be fully utilized when the SCells are each serving a small number of non-CA UEs and the non-CA UE traffic is light.
  - When route setup is triggered for the first time between the PCell and an inter-eNodeB SCell for a CA UE, this route is set up assuming the eNodeB is in the centralized eNodeB coordination architecture by default, before the inter-eNodeB transmission delay is detected. If the detected inter-eNodeB transmission delay indicates the distributed eNodeB coordination architecture, the eNodeB removes this SCell for the UE. The eNodeB will attempt to configure this SCell for the UE again when the uplink or downlink traffic volume of the UE meets the SCell activation conditions described in **4.6.3.4 SCell Activation**. For subsequent CA UEs, the eNodeB configures this SCell considering that it is in the distributed eNodeB coordination architecture.

- Inter-eNodeB SCell configuration and removal in eNodeB coordination scenarios are affected by the CPU usage of the main control board (indicated by the **VS.BBUBoard.CPULoad.Max** counter). The impacts are as follows:
  - When the CPU usage of the main control board in the local eNodeB serving the PCell or in the peer eNodeB serving an SCell exceeds 60%, the local eNodeB imposes the restriction that SCell configuration can be performed only in a blind manner. When both the CPU usage of the main control board in the local eNodeB serving the PCell and that in the peer eNodeB serving an SCell become less than or equal to 50%, the local eNodeB removes the restriction on SCell configuration.
  - When the CPU usage of the main control board in the local eNodeB serving the PCell exceeds 70%, the local eNodeB removes all the SCells that have been configured in a non-blind manner.

17.3 Requirements

17.3.1 Licenses (FDD)

- In CA-group-based configuration mode

  If a CA group includes cells that are served by different eNodeBs in an eNodeB coordination scenario, each of the eNodeBs serving these cells requires one sales unit of the license for Inter-eNodeB CA based on Coordinated eNodeB.
- In adaptive configuration mode

  If any cell served by the local eNodeB is to be accompanied by any inter-eNodeB SCell in an eNodeB coordination scenario, each of the eNodeBs requires one sales unit of the license for Inter-eNodeB CA based on Coordinated eNodeB.

Table 17-2 lists the license model and sales unit for the feature.

Table 17-2 License model and sales unit

Feature ID	Feature Name	Model	Sales Unit
LAOFD-070202	Inter-eNodeB CA based on Coordinated eNodeB	LT1SCAFBCB00	per eNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

17.3.2 Licenses (TDD)

- In CA-group-based configuration mode

If a CA group includes cells that are served by different eNodeBs in an eNodeB coordination scenario, each of the eNodeBs serving these cells requires one sales unit of the license for Inter-eNodeB CA based on Coordinated eNodeB.

- In adaptive configuration mode

  If any cell served by the local eNodeB is to be accompanied by any inter-eNodeB SCell in an eNodeB coordination scenario, each of the eNodeBs requires one sales unit of the license for Inter-eNodeB CA based on Coordinated eNodeB.

Table 17-3 lists the license model and sales unit for the feature.

Table 17-3 License model and sales unit

Feature ID	Feature Name	Model	Sales Unit
TDLAOFD-110401	Inter-eNodeB CA based on Coordinated eNodeB	LT1SIRCBCB00	per eNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

17.3.3 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

17.3.3.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Downlink 2CC aggregation	None	**5 Downlink 2CC Aggregation**	Inter-eNodeB CA based on eNodeB coordination requires this function to be activated.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Downlink 3CC aggregation	**CaDl3CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	6 Downlink 3CC Aggregation	(Optional) This function must be activated if three carriers need to be aggregated in the downlink for inter-eNodeB CA based on eNodeB coordination.
FDD TDD	Downlink 4CC aggregation	**CaDl4CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	7 Downlink 4CC Aggregation	(Optional) This function must be activated if four carriers need to be aggregated in the downlink for inter-eNodeB CA based on eNodeB coordination.
FDD TDD	Downlink 5CC aggregation	**CaDl5CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	8 Downlink 5CC Aggregation	(Optional) This function must be activated if five carriers need to be aggregated in the downlink for inter-eNodeB CA based on eNodeB coordination.
FDD	Downlink massive CA	**DlMassiveCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	9 Downlink Massive CA (FDD)	(Optional) If six or more carriers need to be aggregated in the downlink, only FDD carriers can be used as the PCCs and downlink massive CA must be enabled.
FDD TDD	Downlink FDD+TDD CA	**InterFddTddCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	13 Downlink FDD+TDD CA	(Optional) This function must be activated if downlink inter-eNodeB FDD +TDD CA based on eNodeB coordination is required.
FDD TDD	Intelligent selection of serving cell combinations	**CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter	12 Intelligent Selection of Serving Cell Combinations	(Optional) This function must be activated if it is to be used for inter-eNodeB CA based on eNodeB coordination.

RAT	Function Name	Function Switch	Reference	Description
FDD TDD	Uplink 2CC aggregation	**CaUl2CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	10 Uplink 2CC Aggregation	(Optional) This function must be activated if two carriers need to be aggregated in the uplink for inter-eNodeB CA based on eNodeB coordination.
FDD TDD	Uplink FDD +TDD CA	**InterFddTddCaSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter	14 Uplink FDD+TDD CA	(Optional) This function must be activated if uplink inter-eNodeB FDD +TDD CA based on eNodeB coordination is required.

17.3.3.2 Mutually Exclusive Functions

None

17.3.3.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
FDD	Downlink MU-MIMO in TM9	**4TxTM9MuMimoSwitch** option of the **CellAlgoSwitch**.***EmimoSwitch*** parameter	*eMIMO (FDD)*	Downlink MU-MIMO in TM9 does not work in SCells of CA UEs in the distributed architecture of eNodeB coordination.
FDD	Downlink intelligent joint scheduling	**DlCaUnifiedSchSwitch** option of the **CellDlschAlgo**.***DlCaUserSchSwitch*** parameter	*Smart 8T8R (FDD)*	Downlink intelligent joint scheduling changes the proportion of CA UE data packets distributed to each CC. Therefore, the transport load changes.

RAT	Function Name	Function Switch	Reference	Description
FDD	Intelligent unified scheduling enhancement	**DlCaUnifiedSchEnhSwitch** option of the **CellDlschAlgo**.***DlCaUserSchSwitch*** parameter	• *Smart 8T8R (FDD)* • *Massive MIMO Enhancements (FDD)*	Intelligent unified scheduling enhancement changes the proportion of CA UE data packets distributed to each CC. Therefore, the transport load changes.
FDD TDD	UL CoMP based on eNodeB coordination	**UlJointReceptionOverBBUsSwitch** option of the **ENodeBAlgoSwitch**.***OverBBUsSwitch*** parameter	*UL CoMP*	UL CoMP based on eNodeB coordination and inter-eNodeB CA based on eNodeB coordination share the inter-BBU bandwidth. When the resources are limited, the first-come first-served principles are applied. Any function that fails to obtain resources cannot take effect.

17.3.4 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

The requirements described in **Boards** of **5.3.4 Hardware** must be fulfilled. In addition:

- For FDD, do not use BBU3910A or BBU3910C, which does not support this function.
- For TDD, do not use BBU3910A or UBBPfw1, which does not support this function.
- Do not use BBU5900A, which does not support this function.
- Before deploying this function, contact Huawei engineers for a resource audit. This function shares system resources with SFN, UL CoMP, and coordinated scheduling based power control (CSPC).

RF Modules

For details, see **RF Modules** in **5.3.4 Hardware**.

17.3.5 Networking

The requirements described in **5.3.5 Networking** must be fulfilled. In addition, CA in eNodeB coordination scenarios has the following requirements:

- The eNodeBs must use the same software version.
- The eNodeBs must be time-synchronized.
- The eNodeBs involved in CA must use the same configuration mode for CA.
- Routes between inter-eNodeB cells must be established and working properly.
  - When USU3910 is used, eX2 interfaces must be set up between the eNodeBs.
    - If the eNodeBs belong to the same operator and are managed by the same MAE-Access, eX2 self-setup works.
    - If the eNodeBs belong to different operators or are managed by different MAE-Access systems, Stream Control Transmission Protocol (SCTP) links and endpoint groups must be manually configured for eX2 interfaces. For details about the configuration, see *eX2 Self-Management*.

    If three consecutive attempts of a cell to set up an eX2 route to an inter-eNodeB cell fail in an eNodeB coordination scenario, the subsequent setup attempts on this route will be prohibited within the next 5 minutes.

For details about eNodeB coordination, see *USU3910-based Multi-BBU Interconnection*.

17.3.6 Others

- UEs

  UEs must comply with 3GPP Release 12 or later and support the frequency bands of the carriers to be aggregated and their bandwidths. UEs must also support the peak data rates that CA can achieve.

  6CC–8CC aggregation requires UEs to comply with 3GPP Release 13 or later.
- EPC

  For this function to reach a theoretical peak data rate, the maximum bit rate that each UE subscribes to in the EPC cannot be lower than this theoretical value. For the theoretical values, see **17.2.1 Benefits**.

17.4 Operation and Maintenance

17.4.1 Data Configuration

17.4.1.1 Data Preparation

This function works in either CA-group-based or adaptive configuration mode. Prepare basic data as described in **5.4.1.1 Data Preparation**.

- In CA-group-based configuration mode

  In addition to **5.4.1.1.1 CA-Group-based Configuration Mode**, configure **CaGroup**, **CaGroupCell**, and **CaGroupSCellCfg** MOs on each eNodeB and ensure that the **CaGroup** and **CaGroupCell** configurations of the eNodeBs involved in CA are consistent. If the configurations are inconsistent, CA will not work.
- In adaptive configuration mode

  In addition to **5.4.1.1.2 Adaptive Configuration Mode**, prepare data as described in **Table 17-4** for function activation.

Table 17-4 Parameters for activating adaptive inter-eNodeB CA based on eNodeB coordination

Parameter Name	Parameter ID	Option	Setting Notes
OverBBUsSwitch	**ENodeBAlgoSwitch**.***OverBBUsSwitch***	FreqCfgCaOverBBUsSwitch	Select this option for all the eNodeBs involved.
CA Algorithm Extend Switch	**ENodeBAlgoSwitch**.***CaAlgoExtSwitch***	FreqBaseInterNBSccSwitch	Select this option for the serving eNodeBs of SCells.

Moreover, in either configuration mode:

- To activate this function in the distributed architecture of eNodeB coordination (only for FDD), prepare data as described in **Table 17-5**.

Table 17-5 Parameters for activating inter-eNodeB CA based on distributed eNodeB coordination

Parameter Name	Parameter ID	Setting Notes
CA Algorithm Switch	**ENodeBAlgoSwitch**.***CaAlgoSwitch***	Select the **DistributeCloudbbCaSwitch** option.
Distribute Cloud BB CA Max Component Carrier Number	**CaMgtCfg**.***DisCloudBBCaMaxCcNum***	Set this parameter based on site conditions.

- Prepare data as described in the "Data Preparation" sections for downlink 2CC, 3CC, 4CC, 5CC, or 6CC–8CC aggregation or for uplink 2CC aggregation if the corresponding number of inter-eNodeB CCs need to be aggregated in eNodeB coordination scenarios.

17.4.1.2 Using MML Commands (FDD)

Before using MML commands, refer to **17.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before activating this function, perform the following configurations:

1. Configure a USU.

   For details, see *USU3910-based Multi-BBU Interconnection*.

2. Configure cells or frequencies.

   Perform the steps described in **5.4.1.2.1 CA-Group-based Configuration Mode (FDD)** or **5.4.1.2.3 Adaptive Configuration Mode (FDD)** on each eNodeB.

17.4.1.2.1 Centralized Architecture

Activation Command Examples

- CA-group-based configuration mode

```
//(Optional) Turning on CaDl3CCSwitch for the serving eNodeB of each possible PCell to enable
downlink 3CC aggregation for inter-eNodeB CA based on eNodeB coordination in the centralized
architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1;
//(Optional) Turning on both CaDl3CCSwitch and CaDl4CCSwitch for the serving eNodeB of each
possible PCell to enable downlink 4CC aggregation for inter-eNodeB CA based on eNodeB
coordination in the centralized architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1;
//(Optional) Turning on CaDl3CCSwitch, CaDl4CCSwitch, and CaDl5CCSwitch for the serving eNodeB
of each possible PCell to enable downlink 5CC aggregation for inter-eNodeB CA based on eNodeB
coordination in the centralized architecture
MOD CAMGTCFG: LocalCellId=0,
CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1&CaDl5CCSwitch-1;
//(Optional) Turning on CaDl3CCSwitch, CaDl4CCSwitch, CaDl5CCSwitch, and DlMassiveCaSwitch for
the serving eNodeB of each possible PCell to enable downlink 6CC–8CC aggregation for inter-eNodeB
CA based on eNodeB coordination in the centralized architecture
MOD CAMGTCFG: LocalCellId=0,
CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1&CaDl5CCSwitch-1&DlMassiveCaSwitch-1;
//(Optional) Turning on CaUl2CCSwitch for the serving eNodeB of each possible PCell to enable
uplink 2CC aggregation for inter-eNodeB CA based on eNodeB coordination in the centralized
architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaUl2CCSwitch-1;
```

- Adaptive configuration mode

```
//Turning on FreqCfgCaOverBBUsSwitch for each eNodeB
MOD ENODEBALGOSWITCH: OverBBUsSwitch=FreqCfgCaOverBBUsSwitch-1;
//Turning on FreqBaseIntereNBSccSwitch for the serving eNodeB of each possible SCell
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=FreqBaseIntereNBSccSwitch-1;
//(Optional) Turning on CaDl2CCExtSwitch for both the serving eNodeB of each possible PCell and
the serving eNodeB of each possible SCell to enable downlink aggregation of two CCs with total
bandwidth between 20 MHz and 40 MHz (inclusive) for inter-eNodeB CA based on eNodeB
```

```
coordination in the centralized architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl2CCExtSwitch-1;
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=CaDl2CCExtSwitch-1;
//(Optional) Turning on CaDl3CCSwitch for the serving eNodeB of each possible PCell to enable
downlink 3CC aggregation for inter-eNodeB CA based on eNodeB coordination in the centralized
architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1;
//(Optional) Turning on both CaDl3CCSwitch and CaDl4CCSwitch for the serving eNodeB of each
possible PCell to enable downlink 4CC aggregation for inter-eNodeB CA based on eNodeB
coordination in the centralized architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1;
//(Optional) Turning on CaDl3CCSwitch, CaDl4CCSwitch, and CaDl5CCSwitch for the serving eNodeB
of each possible PCell to enable downlink 5CC aggregation for inter-eNodeB CA based on eNodeB
coordination in the centralized architecture
MOD CAMGTCFG: LocalCellId=0,
CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1&CaDl5CCSwitch-1;
//(Optional) Turning on CaDl3CCSwitch, CaDl4CCSwitch, CaDl5CCSwitch, and DlMassiveCaSwitch for
the serving eNodeB of each possible PCell to enable downlink 6CC–8CC aggregation for inter-eNodeB
CA based on eNodeB coordination in the centralized architecture
MOD CAMGTCFG: LocalCellId=0,
CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1&CaDl5CCSwitch-1&DlMassiveCaSwitch-1;
//(Optional) Turning on CaUl2CCSwitch for the serving eNodeB of each possible PCell to enable
uplink 2CC aggregation for inter-eNodeB CA based on eNodeB coordination in the centralized
architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaUl2CCSwitch-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- CA-group-based configuration mode

```
//Removing all inter-eNodeB cells from the CA group at each eNodeB. Inter-eNodeB cells have
different eNodeB IDs from the local eNodeB ID.
RMV CAGROUPCELL: CaGroupId=0, LocalCellId=0, eNodeBId=1234;
```

- Adaptive configuration mode

```
//Turning off FreqCfgCaOverBBUsSwitch for each eNodeB
MOD ENODEBALGOSWITCH: OverBBUsSwitch=FreqCfgCaOverBBUsSwitch-0;
```

17.4.1.2.2 Distributed Architecture

Activation Command Examples

- CA-group-based configuration mode

```
//Turning on DistributeCloudbbCaSwitch for the serving eNodeB of each possible PCell
MOD ENODEBALGOSWITCH: CaAlgoSwitch=DistributeCloudbbCaSwitch-1;
//(Optional) Turning on CaDl3CCSwitch and setting DisCloudBBCaMaxCcNum to 3CC for the serving
eNodeB of each possible PCell to enable downlink 3CC aggregation for inter-eNodeB CA based on
eNodeB coordination in the distributed architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1, DisCloudBBCaMaxCcNum=3CC;
//(Optional) Turning on CaDl3CCSwitch and CaDl4CCSwitch, and setting DisCloudBBCaMaxCcNum to
4CC for the serving eNodeB of each possible PCell to enable downlink 4CC aggregation for inter-
eNodeB CA based on eNodeB coordination in the distributed architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1,
DisCloudBBCaMaxCcNum=4CC;
//(Optional) Turning on CaDl3CCSwitch, CaDl4CCSwitch, and CaDl5CCSwitch, and setting
DisCloudBBCaMaxCcNum to 5CC for the serving eNodeB of each possible PCell to enable downlink
5CC aggregation for inter-eNodeB CA based on eNodeB coordination in the distributed architecture
MOD CAMGTCFG: LocalCellId=0,
CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1&CaDl5CCSwitch-1,
DisCloudBBCaMaxCcNum=5CC;
//(Optional) Turning on CaDl3CCSwitch, CaDl4CCSwitch, CaDl5CCSwitch, and DlMassiveCaSwitch, and
setting DisCloudBBCaMaxCcNum to DL_MASSIVE_CA for the serving eNodeB of each possible PCell to
enable downlink 6CC–8CC aggregation for inter-eNodeB CA based on eNodeB coordination in the
```



```
distributed architecture
MOD CAMGTCFG: LocalCellId=0,
CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1&CaDl5CCSwitch-1&DlMassiveCaSwitch-1,
DisCloudBBCaMaxCcNum=DL_MASSIVE_CA;
//(Optional) Turning on CaUl2CCSwitch and setting DisCloudBBCaMaxCcNum to 2CC for the serving
eNodeB of each possible PCell to enable uplink 2CC aggregation for inter-eNodeB CA based on
eNodeB coordination in the distributed architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaUl2CCSwitch-1, DisCloudBBCaMaxCcNum=2CC;
```

- Adaptive configuration mode

```
//Turning on DistributeCloudbbCaSwitch for the serving eNodeB of each possible PCell
MOD ENODEBALGOSWITCH: CaAlgoSwitch=DistributeCloudbbCaSwitch-1;
//Turning on FreqCfgCaOverBBUsSwitch for each eNodeB
MOD ENODEBALGOSWITCH: OverBBUsSwitch=FreqCfgCaOverBBUsSwitch-1;
//Turning on FreqBaseIntereNBSccSwitch for the serving eNodeB of each possible SCell
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=FreqBaseIntereNBSccSwitch-1;
//(Optional) Turning on CaDl2CCExtSwitch for both the serving eNodeB of each possible PCell and
the serving eNodeB of each possible SCell to enable downlink aggregation of two CCs with total
bandwidth between 20 MHz and 40 MHz (inclusive) for inter-eNodeB CA based on eNodeB
coordination in the distributed architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl2CCExtSwitch-1;
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=CaDl2CCExtSwitch-1;
//(Optional) Turning on CaDl3CCSwitch and setting DisCloudBBCaMaxCcNum to 3CC for the serving
eNodeB of each possible PCell to enable downlink 3CC aggregation for inter-eNodeB CA based on
eNodeB coordination in the distributed architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1, DisCloudBBCaMaxCcNum=3CC;
//(Optional) Turning on CaDl3CCSwitch and CaDl4CCSwitch, and setting DisCloudBBCaMaxCcNum to
4CC for the serving eNodeB of each possible PCell to enable downlink 4CC aggregation for inter-
eNodeB CA based on eNodeB coordination in the distributed architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1,
DisCloudBBCaMaxCcNum=4CC;
//(Optional) Turning on CaDl3CCSwitch, CaDl4CCSwitch, and CaDl5CCSwitch, and setting
DisCloudBBCaMaxCcNum to 5CC for the serving eNodeB of each possible PCell to enable downlink
5CC aggregation for inter-eNodeB CA based on eNodeB coordination in the distributed architecture
MOD CAMGTCFG: LocalCellId=0,
CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1&CaDl5CCSwitch-1,
DisCloudBBCaMaxCcNum=5CC;
//(Optional) Turning on CaDl3CCSwitch, CaDl4CCSwitch, CaDl5CCSwitch, and DlMassiveCaSwitch, and
setting DisCloudBBCaMaxCcNum to DL_MASSIVE_CA for the serving eNodeB of each possible PCell to
enable downlink 6CC–8CC aggregation for inter-eNodeB CA based on eNodeB coordination in the
distributed architecture
MOD CAMGTCFG: LocalCellId=0,
CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1&CaDl5CCSwitch-1&DlMassiveCaSwitch-1,
DisCloudBBCaMaxCcNum=DL_MASSIVE_CA;
//(Optional) Turning on CaUl2CCSwitch and setting DisCloudBBCaMaxCcNum to 2CC for the serving
eNodeB of each possible PCell to enable uplink 2CC aggregation for inter-eNodeB CA based on
eNodeB coordination in the distributed architecture
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaUl2CCSwitch-1, DisCloudBBCaMaxCcNum=2CC;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off DistributeCloudbbCaSwitch for the serving eNodeB of each possible PCell
MOD ENODEBALGOSWITCH: CaAlgoSwitch=DistributeCloudbbCaSwitch-0;
```

17.4.1.2.3 Hybrid Architecture

For details, see **17.4.1.2.1 Centralized Architecture** and **17.4.1.2.2 Distributed Architecture**.

17.4.1.3 Using MML Commands (TDD)

Before using MML commands, refer to **17.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Before activating this function, perform the following configurations:

1. Configure a USU.

   For details, see *USU3910-based Multi-BBU Interconnection*.

2. Configure cells or frequencies.

   Perform the steps described in **5.4.1.2.2 CA-Group-based Configuration Mode (TDD)** or **5.4.1.2.4 Adaptive Configuration Mode (TDD)** on each eNodeB.

The activation command examples for this function are as follows:

- CA-group-based configuration mode

```
//(Optional) Turning on CaDl3CCSwitch for the serving eNodeB of each possible PCell to enable downlink 3CC aggregation
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1;
//(Optional) Turning on both CaDl3CCSwitch and CaDl4CCSwitch for the serving eNodeB of each possible PCell to enable downlink 4CC aggregation
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1;
//(Optional) Turning on CaDl3CCSwitch, CaDl4CCSwitch, and CaDl5CCSwitch for the serving eNodeB of each possible PCell to enable downlink 5CC aggregation
MOD CAMGTCFG: LocalCellId=0,
CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1&CaDl5CCSwitch-1;
//(Optional) Turning on CaUl2CCSwitch and TddCloudbbUlCaSwitch for the serving eNodeB of each possible PCell to enable uplink 2CC aggregation
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaUl2CCSwitch-1;
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=TddCloudbbUlCaSwitch-1;
```

- Adaptive configuration mode

```
//Turning on FreqCfgCaOverBBUsSwitch for each eNodeB
MOD ENODEBALGOSWITCH: OverBBUsSwitch=FreqCfgCaOverBBUsSwitch-1;
//Turning on FreqBaseIntereNBSccSwitch for the serving eNodeB of each possible SCell
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=FreqBaseIntereNBSccSwitch-1;
//Turning on CaDl2CCExtSwitch for both the serving eNodeB of each possible PCell and the serving eNodeB of each possible SCell to enable downlink aggregation of two CCs with total bandwidth between 30 MHz and 40 MHz (inclusive)
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl2CCExtSwitch-1;
MOD CAMGTCFG: LocalCellId=1, CellCaAlgoSwitch=CaDl2CCExtSwitch-1;
//(Optional) Turning on CaDl3CCSwitch for the serving eNodeB of each possible PCell to enable downlink 3CC aggregation
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1;
//(Optional) Turning on both CaDl3CCSwitch and CaDl4CCSwitch for the serving eNodeB of each possible PCell to enable downlink 4CC aggregation
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1;
//(Optional) Turning on CaDl3CCSwitch, CaDl4CCSwitch, and CaDl5CCSwitch for the serving eNodeB of each possible PCell to enable downlink 5CC aggregation
MOD CAMGTCFG: LocalCellId=0,
CellCaAlgoSwitch=CaDl3CCSwitch-1&CaDl4CCSwitch-1&CaDl5CCSwitch-1;
//(Optional) Turning on CaUl2CCSwitch and TddCloudbbUlCaSwitch for the serving eNodeB of each possible PCell to enable uplink 2CC aggregation
MOD CAMGTCFG: LocalCellId=0, CellCaAlgoSwitch=CaUl2CCSwitch-1;
MOD ENODEBALGOSWITCH: CaAlgoExtSwitch=TddCloudbbUlCaSwitch-1;
```


Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- CA-group-based configuration mode
```
//Removing all inter-eNodeB cells from the CA group at each eNodeB. Inter-eNodeB cells have different eNodeB IDs from the local eNodeB ID.
RMV CAGROUPCELL: CaGroupId=0, LocalCellId=0, eNodeBId=1234;
```
- Adaptive configuration mode
```
//Turning off FreqCfgCaOverBBUsSwitch for each eNodeB
MOD ENODEBALGOSWITCH: OverBBUsSwitch=FreqCfgCaOverBBUsSwitch-0;
```

17.4.1.4 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

17.4.2 Activation Verification

Counter Observation

If the counters listed in **Table 17-6** produce non-zero values on a network that is serving CA UEs, inter-eNodeB CA based on eNodeB coordination has taken effect in the network.

Table 17-6 Counters used to verify activation of inter-eNodeB CA based on eNodeB coordination

Counter ID	Counter Name
1526739785	**L.Traffic.User.PCell.USUCA.Avg**
1526739786	**L.Traffic.User.SCell.USUCA.Avg**

Message Tracing

For the tools and IEs to observe, see **Message Tracing** in **5.4.2 Activation Verification** and **Message Tracing** in **10.4.3 Activation Verification**.

17.4.3 Network Monitoring

For the counters used in scenarios of downlink 2CC, 3CC, 4CC or 5CC aggregation, downlink massive CA, and uplink 2CC aggregation, see the corresponding "Network Monitoring" sections.

18 SCC Buffer Optimization

18.1 Principles

Under a heavy CA load, service data is not allocated in a timely manner and therefore is concentrated in the PCell, with less-than-expected data distributed to SCells. This reduces the spectral efficiency of the SCells in case there are available spectrum resources in the SCells, affecting downlink user experience. To address this issue, SCC buffer optimization is introduced. This function is controlled by the **SCC_BUFFER_OPT_SW** option of the **CellDlSchExtParam**.***DlSchExtSwitch*** parameter. It takes effect only if it is enabled in SCells of a CA UE.

SCC buffer optimization enables the PCell to inform SCells of the latest buffered data volume of the UE in a timely manner, increasing the spectral efficiency of the SCells and improving downlink user experience.

18.2 Network Analysis

18.2.1 Benefits

SCC buffer optimization may increase the average downlink throughput of CA UEs.

18.2.2 Impacts

The impacts between this function and other functions are described in **18.3.2.3 Function Impacts**.

This function may have the following impacts on the network:

- The traffic distribution across frequency bands changes. Therefore, the values of all band-specific counters and indicators also change.
- The volume of data distributed to SCells increases.
- The average downlink throughput of CA UEs on the LTE side increases. As a result, in EN-DC scenarios, more data is distributed from the PDCP layer to the LTE side, causing changes in the average downlink UE throughput on the NR side.

18.3 Requirements

18.3.1 Licenses

None

18.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

18.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
FDD	Downlink 2CC aggregation	None	**5 Downlink 2CC Aggregation**	SCC buffer optimization can take effect only if downlink 2CC aggregation is enabled.

18.3.2.2 Mutually Exclusive Functions

None

18.3.2.3 Function Impacts

None

18.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations)
- DBS3900 LampSite and DBS5900 LampSite

Boards

No requirements

RF Modules

No requirements

18.3.4 Networking

None

18.3.5 Others

None

18.4 Operation and Maintenance

18.4.1 Data Configuration

18.4.1.1 Data Preparation

Table 18-1 describes the parameters used for function activation.

Table 18-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
DL Scheduling Extension Switch	**CellDlSchExtParam.*DlSchExtSwitch***	SCC_BUFFER_OPT_SW	Select this option.

18.4.1.2 Using MML Commands

Before using MML commands, refer to **18.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling SCC buffer optimization
MOD CELLDLSCHEXTPARAM: LocalCellId=0, DlSchExtSwitch=SCC_BUFFER_OPT_SW-1;
```

Deactivation Command Examples

```
//Disabling SCC buffer optimization
MOD CELLDLSCHEXTPARAM: LocalCellId=0, DlSchExtSwitch=SCC_BUFFER_OPT_SW-0;
```

18.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

18.4.2 Activation Verification

SCC buffer optimization has taken effect if the average downlink UE throughput (indicated by **User Downlink Average Throughput**) or average downlink throughput of CA UEs (calculated by using (**L.Thrp.bits.DL.CAUser – L.Thrp.bits.DL.LastTTI.CAUser**)/**L.Thrp.Time.DL.RmvLastTTI.CAUser**) increases.

18.4.3 Network Monitoring

Observe the following performance indicators to verify the increase in the overall UE data rate on the entire network and the CA UE data rate after this function is enabled:

- Average downlink UE throughput: **User Downlink Average Throughput**
- Average downlink throughput of CA UEs: (**L.Thrp.bits.DL.CAUser – L.Thrp.bits.DL.LastTTI.CAUser**)/**L.Thrp.Time.DL.RmvLastTTI.CAUser**

19 Collaboration Between CA and Other Key Technologies

19.1 Multi-Band Compatibility Enhancement

If multi-band compatibility enhancement is enabled for a cell, the eNodeB checks whether the primary and secondary frequency bands of the cell are present in the CA band combinations supported by individual CA UEs.

- The SCC band selection rules for CA UEs depend on whether flexible CA is enabled, as described in **Table 19-1**.

Table 19-1 SCC band selection rules for CA UEs

Flexible CA	UE Support for Adding the Primary Frequency Band as an SCC	UE Support for Adding the Secondary Frequency Band as an SCC	Frequency Band Selected as the SCC for the CA UE[a]
Enabled	Yes	No	Primary frequency band
	No	Yes	Secondary frequency band
	Yes	Yes	The frequency band with a higher SCC priority (specified by the **SccFreqCfg**.***SccPriority*** parameter) is selected.

<table>
<tr><th>Flexible CA</th><th>UE Support for Adding the Primary Frequency Band as an SCC</th><th>UE Support for Adding the Secondary Frequency Band as an SCC</th><th>Frequency Band Selected as the SCC for the CA UE[a]</th></tr>
<tr><td rowspan="3">Disabled</td><td>Yes</td><td>No</td><td>The primary frequency band is selected as the SCC if it has a higher priority[b].
If the secondary frequency band has a higher priority, CA cannot be performed for the UE.
If the primary and secondary frequency bands have the same priority, the UE randomly selects a frequency band as the SCC. In this case, CA may not be performed for the UE.</td></tr>
<tr><td>No</td><td>Yes</td><td>If the primary frequency band has a higher priority, CA cannot be performed for the UE.
The secondary frequency band is selected as the SCC if it has a higher priority.
If the primary and secondary frequency bands have the same priority, the UE randomly selects a frequency band as the SCC. In this case, CA may not be performed for the UE.</td></tr>
<tr><td>Yes</td><td>Yes</td><td>The frequency band with a higher SCC priority is selected.</td></tr>
<tr><td colspan="4">a: When flexible CA is disabled, UEs cannot perform CA in certain scenarios. This results in fewer CA UEs (indicated by L.Traffic.User.PCell.DL.Avg) and less traffic volume (indicated by L.Thrp.bits.DL.CAUser) than when flexible CA is enabled.
b: In this table, the priority of a frequency band refers to the priority for it to be configured as an SCC. It is specified by the SccFreqCfg.SccPriority parameter.</td></tr>
</table>

- The PCC band selection rules for CA UEs are described in **Table 19-2**.

Table 19-2 PCC band selection rules for CA UEs

Current Frequency Band of the PCC	SCC Addition Support by the Primary Frequency Band	SCC Addition Support by the Secondary Frequency Band	Frequency Band Selected as the PCC for the CA UE
Primary	Yes	Yes	Primary frequency band
Primary	Yes	No	Primary frequency band
Primary	No	Yes	Secondary frequency band: ● If the network side and the CA UE support priority adjustment of secondary MFBI frequency bands, the CA UE accesses the secondary frequency band. ● Otherwise, the CA UE is handed over to the secondary frequency band through an intra-cell handover.
Primary	No	No	The UE cannot perform CA.
Secondary	Yes	Yes	Secondary frequency band
Secondary	Yes	No	The CA UE is handed over to the primary frequency band through an intra-cell handover.
Secondary	No	Yes	Secondary frequency band
Secondary	No	No	The UE cannot perform CA.

- If both the primary and secondary frequency bands can be selected as the PCC and SCC, SCC band selection for the CA UE first depends on **Table 19-1**.

If such selection fails, **Table 19-2** is used to determine whether to trigger an intra-cell handover to add an SCC for the CA UE.

If a carrier involved in CA is used by multiple frequency bands, the primary and secondary bands must be all configured in the **PccFreqCfg**, **SccFreqCfg**, and **EutranInterNFreq** MOs. You are advised to set the parameters consistently for the primary and secondary bands. For details about multi-band compatibility enhancement, see *Multi-Band Compatibility Enhancement*.

19.2 Multiple-Antenna Technologies

19.2.1 Adaptive Handling of CA for MU Beamforming

Adaptive handling of CA for MU beamforming can be enabled when all the following conditions are met (the LBBPc does not support adaptive handling of CA for MU beamforming):

- The **FreqCfgSwitch** and **AdpCaSwitch** options of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter are selected.
- The **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch.*CaAlgoSwitch*** parameter is deselected.
- The **CaMubfPairingAdaptOptSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is selected.

When all the above conditions are met:

- **CaMgtCfg**.***HighLoadCellTypeNotAsScell*** set to **MASSIVE_MIMO_CELL**

  For CA UEs that cannot participate in pairing, high-load massive MIMO cells cannot be configured as SCells in the downlink at initial access, incoming RRC connection reestablishments, or incoming necessary handovers. As a result, the downlink user-perceived data rates and downlink PCell throughput of these UEs may decrease. If multiple TDD high-load massive MIMO cells are allowed to be added as SCells for CA UEs that can participate in pairing, some UEs in these SCells may not be paired for MU beamforming, affecting the downlink SCell throughput.

  In this case, if the "optimized selection of TDD high-load massive MIMO cells as SCells" function is enabled, a maximum of one TDD high-load massive MIMO SCell can be added for CA UEs that can participate in pairing. This increases the downlink SCell throughput and may decrease the number of carriers for CA UEs. This function is controlled by the **RSVD_SW_PARAM0_BIT28** option of the **EnodebRsvdParamExt**.***RsvdSwParam0*** parameter.

  A high-load massive MIMO cell is defined as follows:

  – With the **CaMgtCfg**.***HLUeCntThldForScellConfig*** parameter set to a value other than **65535**

    The eNodeB determines the load status of the massive MIMO cell based on the number of UEs that can be paired in the cell.

    ▪ Condition for entering the high-load state: Number of UEs that can be paired in the massive MIMO cell > **CaMgtCfg**.***HLUeCntThldForScellConfig***

    - Condition for exiting the high-load state: Number of UEs that can be paired in the massive MIMO cell ≤ max{0, **CaMgtCfg**.***HLUeCntThldForScellConfig*** – 20}
  - With the **CaMgtCfg**.***HLUeCntThldForScellConfig*** parameter set to **65535**
    - **MmLoadSteeringSw** option of the **CellAlgoSwitch**.***MlbAlgoSwitch*** parameter selected

      The eNodeB determines the load status of the massive MIMO cell based on the number of layers paired in the cell. A massive MIMO cell in which the average number of layers paired exceeds the value of **CellMLB**.***MmLoadSteeringLayerThld*** is defined as a high-load massive MIMO cell.
      - Condition for entering the high-load state: Average number of layers paired in the cell ≥ **CellMLB**.***MmLoadSteeringLayerThld*** x Step, where the step is always equal to 0.01
      - Condition for exiting the high-load state: Average number of layers paired in the cell < **CellMLB**.***MmLoadSteeringLayerThld*** x Step x 0.9, where the step is always equal to 0.01
    - **MmLoadSteeringSw** option of the **CellAlgoSwitch**.***MlbAlgoSwitch*** parameter deselected

      The method of determining high-load massive MIMO cells does not take effect.
- **CaMgtCfg**.***HighLoadCellTypeNotAsScell*** set to **NORMAL_CELL**

  High-load normal cells cannot be configured as SCells in the downlink for CA UEs at initial access, incoming RRC connection reestablishments, or incoming necessary handovers.

  A high-load normal cell is defined as follows:
  - With the **CaMgtCfg**.***HLUeCntThldForScellConfig*** parameter set to a value other than **65535**

    The eNodeB determines the load status of the cell based on the number of UEs in the cell. Number of UEs in a cell = Number of CA UEs that treat the cell as their PCell + Number of CA UEs that treat the cell as their SCell + Number of non-CA UEs that treat the cell as their serving cell
    - Condition for entering the high-load state: Number of UEs in the cell > **CaMgtCfg**.***HLUeCntThldForScellConfig***
    - Condition for exiting the high-load state: Number of UEs in the cell ≤ max{0, **CaMgtCfg**.***HLUeCntThldForScellConfig*** – 20}
  - With the **CaMgtCfg**.***HLUeCntThldForScellConfig*** parameter set to **65535**

    The eNodeB does not determine the load status of the cell.
- **CaMgtCfg**.***HighLoadCellTypeNotAsScell*** set to **ALL_CELL**

  During downlink SCell configuration for a CA UE at initial access, an incoming RRC connection reestablishment, or an incoming necessary handover:
  - If the CA UE cannot participate in pairing in high-load massive MIMO cells, these cells cannot be configured as SCells for the UE.
  - If a candidate SCell is a high-load normal cell, the cell cannot be configured as an SCell for the CA UE.

When the **SMART_CARRIER_SELECTION_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter is selected, whether a massive MIMO cell can be configured as an SCell depends on the carrier combination selected using smart carrier selection based on virtual grids. For details about smart carrier selection based on virtual grids, see *Multi-carrier Unified Scheduling*.

If the value of the **L.Traffic.User.SCell.DL.Avg** counter decreases after adaptive handling of CA for MU beamforming is enabled, this function has taken effect.

Adaptive handling of CA for MU beamforming has the following impacts on the network:

- The number of CA UEs that are in the downlink 3CC–8CC aggregation state decreases. The related counters are listed in the following table.

Counter ID	Counter Name	Counter Description
1526732907	**L.Traffic.User.PCell.DL.3CC.Avg**	Average number of CA UEs that treat the local cell as their PCell in the downlink 3CC CA state
1526737780	**L.Traffic.User.PCell.DL.4CC.Avg**	Average number of CA UEs that treat the local cell as their PCell in the downlink 4CC CA state
1526739805	**L.Traffic.User.PCell.DL.5CC.Avg**	Average number of CA UEs that treat the local cell as their PCell in the downlink 5CC CA state
1526749452	**L.Traffic.User.PCell.DL.6CC.Avg**	Average number of UEs that treat the local cell as their PCell in the downlink 6CC CA state
1526749453	**L.Traffic.User.PCell.DL.7CC.Avg**	Average number of UEs that treat the local cell as their PCell in the downlink 7CC CA state
1526749454	**L.Traffic.User.PCell.DL.8CC.Avg**	Average number of UEs that treat the local cell as their PCell in the downlink 8CC CA state
1526728427	**L.Traffic.User.SCell.DL.Avg**	Average number of downlink CA UEs that treat the local cell as their SCell
1526728425	**L.ChMeas.PRB.DL.SCell.Used.Avg**	Number of downlink PRBs occupied by CA UEs that treat the local cell as their SCell

Counter ID	Counter Name	Counter Description
1526729045	**L.CA.DLSCell.Add.Att**	Number of SCell addition attempts for CA UEs
1526729046	**L.CA.DLSCell.Add.Succ**	Number of successful SCell additions for CA UEs
1526728999	**L.CA.DLSCell.Act.Att**	Number of SCell activation executions for CA UEs
1526729000	**L.CA.DLSCell.Act.Succ**	Number of successful SCell activations for CA UEs

- The downlink throughput of CA UEs on the entire network decreases.

  Downlink throughput of CA UEs on the entire network = **L.Thrp.bits.DL.CAUser**/**L.Thrp.Time.DL.CAUser**

19.2.2 MIMO (FDD)

MIMO Schemes

In CA scenarios, each CC supports 2T or 4T MIMO (the number of antennas may vary with carriers). The MIMO scheme is separately configured for each CC.

- If the **CellMimoParaCfg.*MimoAdaptiveSwitch*** parameter is set to **NO_ADAPTIVE** for a cell, the transmission mode specified by the **CellMimoParaCfg.*FixedMimoMode*** parameter takes effect. The transmission mode can be TM2, TM3, TM4, or TM6.
- If the **CellMimoParaCfg.*MimoAdaptiveSwitch*** parameter is set to **OL_ADAPTIVE**, **CL_ADAPTIVE**, or **OC_ADAPTIVE** for a cell, the specified transmission mode adaptation takes effect in the cell.

If the **TM9Switch** option of the **CellAlgoSwitch.*EnhMIMOSwitch*** parameter is selected and the **CellCsiRsParaCfg.*CsiRsSwitch*** parameter is set to **FIXED_CFG**, TM9-capable CA UEs work in TM9 and other CA UEs work in a MIMO scheme depending on the setting of the **CellMimoParaCfg.*MimoAdaptiveSwitch*** parameter.

On a 4T MIMO network, CA works in downlink 4x2 MIMO scenarios if CA UEs support 2R and in downlink 4x4 MIMO scenarios if CA UEs support 4R.

When FDD and TDD cells serve as the PCell and an SCell for uplink FDD+TDD CA, respectively, the SRS subframe configurations of the two cells must overlap if the TDD SCell needs to support TM7, TM8, or TM9 without PMI.

For more details about MIMO, see *MIMO*.

UEs

The way the terminal industry chain has developed, some UEs do not support aggregation of the maximum number of CCs while using the highest-order MIMO configurations. For example, some UEs support aggregation of three 2x2 MIMO CCs or aggregation of one 4x4 MIMO CC and one 2x2 MIMO CC, but not aggregation of three 4x4 MIMO CCs.

For these UEs, the **CaMgtCfg**.***CaMimoPriorityStrategySw*** parameter can be used to determine the policy for adaptive selection of CA or high-order MIMO when the UEs have been configured with a maximum number of CCs.

- If this parameter is set to **CA_PRIOR**, CA takes precedence.
- If this parameter is set to **MIMO_PRIOR**, the eNodeB attempts to configure as many 4x4 MIMO-capable CCs as possible.

  For this purpose, the eNodeB removes SCells for a CA UE when two conditions are fulfilled. The UE falls back to be served by fewer CCs and enables rank-4-based transmission on the 4x4 MIMO-capable CCs. These conditions are:

  – The UE determines that SCell removal will allow a larger number of CCs to be capable of 4x4 MIMO.

  – The eNodeB determines that, after the fallback, any one of the 4x4 MIMO-capable CCs will meet the conditions described in **Table 19-3**.

Table 19-3 Conditions to be met by 4x4 MIMO-capable CCs

Modulation Scheme	Conditions to Meet
64QAM	CQI adjustment value used in rank-2-based transmission > **CellDlschAlgo**.***RankChangeAttemptMcsThld***
64QAM	IBLER < **CellDlschAlgo**.***RBDamageNearPointIblerTh***
256QAM	CQI adjustment value used in rank-2-based transmission > **CellDlschAlgo**.***RankChangeAttemptMcsThld*** + 7
256QAM	IBLER < **CellDlschAlgo**.***RBDamageNearPointIblerTh***

- If this parameter is set to **PEAK_RATE_PRIOR**, the eNodeB attempts to render CA UEs to reach highest theoretical peak data rates.

  For this purpose, the eNodeB removes SCells for a CA UE when two conditions are fulfilled. The UE falls back to be served by fewer CCs and enables rank-4-based transmission on the 4x4 MIMO-capable CCs. These conditions are:

  – The UE determines that SCell removal will help reach a higher theoretical peak data rate.

  – The eNodeB determines that, after the fallback, all of the 4x4 MIMO-capable CCs will meet the conditions described in **Table 19-3**.

When all 4x4 MIMO-capable CCs meet the conditions described in **Table 19-4**, the eNodeB configures SCells again so that the number of CCs for the UE reaches the maximum.

Table 19-4 Conditions to be met by all 4x4 MIMO-capable CCs

Modulation Scheme	Conditions to Meet
64QAM	Sum of the CQI adjustment values for all codewords < (**CellDlschAlgo**.***RankChangeAttemptMcsThld*** + 3) x 2
256QAM	Sum of the CQI adjustment values for all codewords < (**CellDlschAlgo**.***RankChangeAttemptMcsThld*** + 11) x 2

Sum of the CQI adjustment values for all codewords in rank-3-based transmission = CQI adjustment value for codeword 0 + CQI adjustment value for codeword 1 x 2

Sum of the CQI adjustment values for all codewords in rank-4-based transmission = CQI adjustment value for codeword 0 x 2 + CQI adjustment value for codeword 1 x 2

NOTE

If this parameter is set to **MIMO_PRIOR** or **PEAK_RATE_PRIOR** and uplink CA is enabled, uplink SCell configuration takes priority over 4x4 MIMO configuration for UEs capable of uplink CA after their fallback. The CCs with the uplink SCells may not be capable of 4x4 MIMO.

19.2.3 MIMO (TDD)

MIMO Schemes for PCells

When the eNodeB assigns MIMO schemes to the PCell of a CA UE:

- If the **CellBfMimoParaCfg.*BfMimoAdaptiveSwitch*** parameter is set to **NO_ADAPTIVE**, the value of the **CellBfMimoParaCfg.*FixedBfMimoMode*** parameter takes effect. The transmission mode can be TM2, TM3, TM4, TM7, TM8, TM9 without PMI, or TM9 with PMI.
- If the **CellBfMimoParaCfg.*BfMimoAdaptiveSwitch*** parameter is set to **TxD_BF_ADAPTIVE** or **MIMO_BF_ADAPTIVE**, adaptive switching between transmission modes for beamforming and MIMO takes effect.

For TM9 to take effect, the **TM9Switch** option of the **CellAlgoSwitch.*EnhMIMOSwitch*** parameter must be selected, and the **CellCsiRsParaCfg.*CsiRsSwitch*** parameter must be set to **FIXED_CFG** or **ADAPTIVE_CFG**. In such a case, either TM9 or any other transmission mode can be used for TM9-capable CA UEs in their PCells. TM9 cannot be used for other CA UEs, regardless of whether a fixed transmission mode or adaptive selection of a transmission mode is used.

MIMO Schemes for SCells

The transmission mode used in the SCells of CA UEs is controlled by the **CellBfMimoParaCfg.*SccBfMimoAdaptiveSwitch*** parameter:

- Set to **ON**

  The transmission mode is dependent on the **CellBfMimoParaCfg.*BfMimoAdaptiveSwitch*** parameter.

  – Set to **NO_ADAPTIVE**

    The value of the **CellBfMimoParaCfg.*FixedBfMimoMode*** parameter takes effect. The transmission mode can be TM2, TM3, TM4, TM7, TM8, TM9 without PMI, or TM9 with PMI. In this situation, the **NoSrsSccBfSwitch** option of the **CellAlgoSwitch**.***NoSrsSccBfAlgoSwitch*** parameter does not take effect.

  – If this parameter is set to **TxD_BF_ADAPTIVE** or **MIMO_BF_ADAPTIVE**, adaptive switching between transmission modes for beamforming and MIMO takes effect.

    ■ If there are SRSs in the SCell, see *Beamforming (TDD)*.

    ■ If there are no SRSs in the SCell, the **NoSrsSccBfSwitch** option of the **CellAlgoSwitch**.***NoSrsSccBfAlgoSwitch*** parameter affects which transmission mode will take effect in the SCell:

      ○ If this option is selected, see *Beamforming (TDD)*. Note that CA UEs complying with 3GPP Releases earlier than Release 12 do not support TM8.

      ○ If this option is deselected, TM7, TM8, and TM9 without PMI do not take effect.

  For TM7, TM8, TM9 without PMI, or TM9 with PMI to take effect, certain conditions must be fulfilled.

  – For TM9 with PMI to take effect

    The **TM9Switch** option of the **CellAlgoSwitch.*EnhMIMOSwitch*** parameter must be selected, and the **CellCsiRsParaCfg.*CsiRsSwitch*** parameter must be set to **FIXED_CFG** or **ADAPTIVE_CFG**. In such a case, either TM9 or any other transmission mode can be used for TM9-capable CA UEs in their SCells. TM9 cannot be used for other CA UEs, regardless of whether a fixed transmission mode or adaptive selection of a transmission mode is used.

  – For TM7 or TM8 to take effect in a TDD SCell with SRSs or in a TDD SCell without SRSs but with a weight value calculated based on the SRSs in another TDD cell

    At least one of the following conditions must be fulfilled:

    ■ Uplink 2CC aggregation has been enabled.

    ■ Beamforming in SCells has been enabled.

      With beamforming in SCells enabled, a CA UE in the SCell without SRSs can use the weight value calculated based on the SRSs in another TDD cell. The TDD SCell without SRSs and the TDD cell with SRSs must meet all of the following conditions:

    - The channel spacing between the center frequencies of the cells is less than or equal to 20 MHz.
    - The cells are served by the same RRU or AAU.
    - The cells have the same antenna configuration: 8T8R or 64T64R.
    - The **NoSrsSccBfSwitch** option of the **CellAlgoSwitch**.***NoSrsSccBfAlgoSwitch*** parameter has been selected for each of the cells.

    This function does not take effect for CA UEs for which there are SRSs in each TDD SCell. Beamforming is performed in a TDD SCell using the weight information for the cell itself.
  - For TM9 without PMI to take effect

    The preceding conditions for TM9 with PMI and the conditions for TM7 and TM8 must be all fulfilled.
- Set to **OFF**
  - For a non-SFN cell that acts as an SCell

    TM3 is always used in this SCell. An exception is that, if the serving RRUs for the cell are all 1T1R RRUs, only TM1 takes effect in the cell.
  - For an SFN cell that acts as an SCell
    - If its serving RRUs are all 1T1R RRUs, TM1 or TM2 is used, depending on the number of CRS ports configured for the cell.
    - If its serving RRUs are of any other combinations, TM3 is used in the cell.

When FDD and TDD cells serve as the PCell and an SCell for uplink FDD+TDD CA, respectively, the SRS subframe configurations of the two cells must overlap if the TDD SCell needs to support TM7, TM8, or TM9 without PMI. For example, the **SRSCfg**.***SrsSubframeCfg*** parameter is set to **SC4(4)** for the FDD cell.

For more details about MIMO, see *MIMO*.

UEs

The way the terminal industry chain has developed, some UEs do not support aggregation of the maximum number of CCs while using the highest-order MIMO configurations. For example, a UE with a maximum capability of 4x4 MIMO supports aggregation of three 2x2 MIMO CCs or aggregation of one 4x4 MIMO CC and one 2x2 MIMO CC, but not aggregation of three 4x4 MIMO CCs. A UE with a maximum capability of 8x8 MIMO supports aggregation of four 4x4 MIMO CCs or aggregation of two 8x8 MIMO CCs, but not aggregation of four 8x8 MIMO CCs.

For these UEs, Huawei has provided the **CaMgtCfg.*CaMimoPriorityStrategySw*** parameter for operators to determine the policy for adaptive selection of CA or high-order MIMO. For the policy to take effect for UEs with a maximum capability of 8x8 MIMO, operators must additionally set the **CellBfMimoParaCfg**.***CaTm9Rank8HighSpctEffThld*** and **CellBfMimoParaCfg**.***CaTm9Rank8LowSpctEffThld*** parameters.

- If the policy parameter is set to **CA_PRIOR**, CA takes precedence over high-order MIMO after UEs access the network. The number of CCs can reach the maximum value specified on the network.

- If this parameter is set to **MIMO_PRIOR**, high-order (4x4 or 8x8) MIMO takes precedence over CA after UEs access the network.

  For this purpose, the eNodeB removes SCells for a CA UE when two conditions are fulfilled. The UE falls back to be served by fewer CCs and enables rank-4-based transmission on the 4x4 MIMO-capable CCs or rank-8-based transmission on the 8x8 MIMO-capable CCs. These conditions are:

  – The UE determines that SCell removal will allow a larger number of CCs to be capable of high-order MIMO.
  – The eNodeB determines that, after the fallback, the channel quality of any one of the high-order MIMO-capable CCs will be higher than or equal to the high spectral efficiency threshold plus **CellBfMimoParaCfg**.***CaSccDelThldOffset***. This threshold is not configurable for 4x4 MIMO-capable CCs and is specified by the **CellBfMimoParaCfg**.***CaTm9Rank8HighSpctEffThld*** parameter for 8x8 MIMO-capable CCs.

- If the policy parameter is set to **PEAK_RATE_PRIOR**, the eNodeB attempts to enable CA UEs to reach highest theoretical peak data rates.

  For this purpose, the eNodeB removes SCells for a CA UE when two conditions are fulfilled. The UE falls back to be served by fewer CCs and enables rank-4-based transmission on the 4x4 MIMO-capable CCs or rank-8-based transmission on the 8x8 MIMO-capable CCs. These conditions are:

  – The UE determines that SCell removal will help reach a higher theoretical peak data rate.
  – The eNodeB determines that, after the fallback, the channel quality of all the high-order MIMO-capable CCs will be higher than or equal to the high spectral efficiency threshold plus **CellBfMimoParaCfg**.***CaSccDelThldOffset***. This threshold is not configurable for 4x4 MIMO-capable CCs and is specified by the **CellBfMimoParaCfg**.***CaTm9Rank8HighSpctEffThld*** parameter for 8x8 MIMO-capable CCs.

  When the channel quality of all the high-order MIMO-capable CCs becomes lower than the low spectral efficiency threshold plus **CellBfMimoParaCfg**.***CaSccAddThldOffset***, the eNodeB configures SCells again so that the number of CCs for the UE reaches the maximum. This threshold is not configurable for 4x4 MIMO-capable CCs and is specified by the **CellBfMimoParaCfg**.***CaTm9Rank8LowSpctEffThld*** parameter for 8x8 MIMO-capable CCs.

NOTE

If the policy parameter is set to **MIMO_PRIOR** or **PEAK_RATE_PRIOR** and uplink CA is enabled, uplink SCell configuration takes priority over high-order MIMO configuration for UEs capable of uplink CA after their fallback. The CCs with the uplink SCells may not be capable of 4x4 or 8x8 MIMO.

For more details about MIMO, see *MIMO*.

19.2.4 Massive MIMO (TDD)

Load-based Adaptive Activation of TDD Massive MIMO SCells

TDD massive MIMO uses a large number of antenna arrays to perform 3D beamforming and multi-user multiplexing, significantly lifting system capacity.

One of its key techniques is that the eNodeB measures the SRSs sent by UEs, uses them to estimate the weight values for downlink beamforming, and then applies the weights to the beams with which the UEs participate in pairing.

When a TDD massive MIMO cell acts as an SCell for a CA UE that does not support uplink CA, the UE does not send SRSs in this cell, so the UE cannot participate in pairing in this cell. This causes capacity loss in this SCell.

To resolve this issue, operators can use load-based adaptive activation of TDD massive MIMO SCells. This function determines the load status of TDD massive MIMO cells based on the number of UEs that can be paired in the cells.

This function takes effect when the **CaMgtCfg**.***ScellHeavyLoadUeNumThld*** parameter is set to a value other than **65535**.

- Situation: The number of UEs that can be paired in a cell exceeds the threshold specified by the **CaMgtCfg**.***ScellHeavyLoadUeNumThld*** parameter.

  This cell is considered to be heavily loaded. This massive MIMO cell cannot be activated as an SCell for any CA UE for which this cell has not been configured as an uplink SCell.
- Situation: The number of UEs that can be paired in a cell is less than or equal to max{0, **CaMgtCfg**.***ScellHeavyLoadUeNumThld*** – 20}.

  This cell is considered to be lightly loaded. This massive MIMO cell can be activated as an SCell for CA UEs for which this cell has not been configured as an uplink SCell.

When a TDD massive MIMO cell is heavily loaded, this function:

- Increases the user-perceived downlink data rate and downlink cell throughput in this cell.
- Decreases the user-perceived downlink data rates of CA UEs in their PCells, since this cell cannot be activated as an SCell for the CA UEs. The user-perceived downlink data rate and downlink cell throughput may decrease in the PCells of these CA UEs.

If the **CaMgtCfg**.***ScellHeavyLoadUeNumThld*** parameter is set to **65535**, this function does not take effect.

Load-based Selection of Uplink TDD Massive MIMO SCells

When a TDD massive MIMO cell acts as an SCell in the uplink for a CA UE that supports uplink CA, the UE can send SRSs and participate in pairing in this cell. This increases the capacity of this cell. When the cell is in the high load state, the CA UE is more likely to be paired in the cell.

When uplink 2CC aggregation is enabled in three or more cells, load-based selection of uplink TDD massive MIMO SCells can be enabled so that the eNodeB preferentially configures a high-load massive MIMO cell as the uplink SCell for CA UEs. This function is controlled by the **CaUlSmartSelectionSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter.

To be specific, during the initial access, an incoming RRC connection reestablishment, or an incoming handover of a CA UE, the eNodeB preferentially selects a high-load massive MIMO cell among candidate same-bandwidth TDD massive MIMO cells as an uplink SCell for the UE. This way, the capacity of the massive MIMO cell increases. This function can be verified by observing the

L.Traffic.User.SCell.UL.Avg counter. If the value of this counter increases in high-load cells, this function has taken effect.

This function does not take effect for inter-eNodeB CA based on relaxed backhaul.

UE Pairing During CA-oriented Pre-scheduling in Massive MIMO Scenarios

During CA-oriented pre-scheduling, PUCCH resources are preallocated to UEs in their SCells in a single-UE manner. Single-UE handling does not allow for UE pairing. Only the preallocated PUCCH resources can be used during actual scheduling, and only a limited number of UEs in their SCells can be allocated resources. However, multi-user (MU) beamforming allows for UE pairing, so that multiple UEs can use the same frequency-domain resources. It is possible that more UEs meet scheduling conditions in a cell acting as their SCell than UEs preallocated resources in this SCell. As a result, some UEs are not scheduled because of resource allocation failure. Due to the smaller number of UEs actually scheduled in the SCell, the downlink throughput of this SCell would decrease.

To resolve this issue, UE pairing during CA-oriented pre-scheduling can be enabled in massive MIMO scenarios. With this function, more UEs are actually scheduled in SCells and the downlink throughput of the SCells rises.

This function is enabled for an SCell when the **CaMgtCfg**.***CaPreallocationSccUeCount*** parameter is set to a value other than **0** for this SCell.

A larger value of the **CaMgtCfg**.***CaPreallocationSccUeCount*** parameter results in a larger number of UEs scheduled in the SCell. However, if the value is too large, the number of UEs scheduled in their PCell and the downlink throughput of the PCell decrease.

19.2.5 R13-compliant Carrier-Level Antenna Selection (TDD)

The UE transmit antenna selection capability defined by 3GPP Release 8 is UE-specific. If a UE reports that it supports UE-level antenna selection, it is assumed in 3GPP specifications by default that the UE supports antenna selection at all of its supported carriers. In addition, when transmit antenna switching is required at any CC of a CA UE, the other CCs of the UE must switch to the same target antenna at the same time.

For example, a UE supports carriers C1 and C2. If the UE supports UE-level antenna selection, the eNodeB assumes by default that the UE supports antenna selection at both C1 and C2. If the UE does not support UE-level antenna selection, the eNodeB assumes that the UE does not support antenna selection at either C1 or C2.

3GPP Release 13 has introduced reporting of carrier-level antenna selection capabilities of UEs. UEs report their antenna selection capabilities regarding each supported carrier and the antenna switching relationships between the carriers to the eNodeB. When UEs can report their carrier-level antenna selection capabilities and the eNodeB can identify the capabilities, dual-stream beamforming based on antenna selection can then be enabled. For details about this beamforming function, see *Beamforming (TDD)* and *Massive MIMO Basics (TDD)*.

This carrier-level antenna selection function is controlled by the **CARRIER_BASED_ANT_SEL_SWITCH** option of the

CellAlgoExtSwitch.***AntennaSelectionAlgoSwitch*** parameter. Before this function can be enabled, the **UeSRSAntSelectCtrlSwitch** option of the **ENodeBAlgoSwitch**.***CompatibilityCtrlSwitch*** parameter must be deselected to enable antenna selection for SRS transmission.

It is recommended that the function switch (the **CARRIER_BASED_ANT_SEL_SWITCH** option) be turned on for a cell when there are UEs capable of carrier-level antenna selection, as indicated by the CHR monitoring item **ueTxAntennaSelectionr13**, in the cell.

For a UE in the uplink CA state (for example, with carriers C1 and C2 aggregated), **Table 19-5** lists the possible scenarios identified by the eNodeB and the conditions for carrier-level antenna selection to take effect in the corresponding scenarios.

Table 19-5 Possible scenarios and conditions for carrier-level antenna selection

Scenario	Conditions for Carrier-Level Antenna Selection
The UE does not support carrier-level antenna selection. The eNodeB determines that neither C1 nor C2 supports antenna selection.	This function does not work.
The UE supports carrier-level antenna selection. The eNodeB determines that both C1 and C2 support antenna selection and the carriers must switch to the same target antenna at antenna switching.	Both C1 and C2 are TDD carriers, and the function switch has been turned on for both of them.
The UE supports carrier-level antenna selection. The eNodeB determines that both C1 and C2 support antenna selection and the carriers may switch to different target antennas at antenna switching.	The function switch has been turned on for one of the carriers, and this carrier is a TDD carrier. Alternatively, the function switch has been turned on for both of the carriers, and the two carriers are TDD carriers.
The UE supports carrier-level antenna selection. The eNodeB determines that C1 supports antenna selection but C2 does not.	C1 is a TDD carrier, and the function switch has been turned on for C1.
The UE supports carrier-level antenna selection. The eNodeB determines that C2 supports antenna selection but C1 does not.	C2 is a TDD carrier, and the function switch has been turned on for C2.

19.3 Connection Management

When CA is enabled, connection management has the following characteristics:

- After an RRC connection is set up between a CA UE and a cell, the cell acts as the PCell of the UE. The PCell transmits non-access stratum (NAS) messages for the UE.
- An eNodeB configures SCells for a CA UE by sending messages over the RRC connection. After the SCells are configured, the CA UE still has only one RRC connection with the network and is allocated only one cell radio network temporary identifier (C-RNTI).
- The PCell and SCells of a CA UE each have a complete set of channels with one exception: the PUCCH. This channel carries layer 1 uplink control information, such as ACKs or NACKs to downlink data, scheduling requests, and periodic CSI reports. The PUCCH exists only in the PCell.
- The PCell is set up for a CA UE during initial access or an RRC connection reestablishment. A handover is required to change the PCell for the UE. An RRC connection reestablishment procedure is triggered if a radio link failure (RLF) occurs in the PCell.
- SCells can be deactivated, but the PCell cannot. Only the serving eNodeB of the PCell can deactivate and remove SCells.

For more details about connection management, see *Connection Management*.

19.4 Mobility and Load Management

19.4.1 Mobility Management

This section describes how mobility management works when CA is enabled.

Handover Messages

The handover procedures for CA UEs have the following characteristics:

- The handover procedure for PCell changes is the same as a common handover procedure, in which the eNodeB sends an RRC Connection Reconfiguration message that contains the mobilityControlInfo IE to the UE.
- Unlike a common handover procedure, a handover procedure that involves an SCell update has the following characteristics:
  - If SCell configuration is required, the eNodeB sends an RRC Connection Reconfiguration message that contains the sCellToAddModList IE to the UE.
  - If SCell removal is required, the eNodeB sends an RRC Connection Reconfiguration message that contains the sCellToReleaseList IE to the UE.

The difference in the number of CCs between source and target eNodeBs has no impact on the handover procedure.

The RRC Connection Reconfiguration message and the preceding IEs can be traced as described in **5.4.2 Activation Verification**.

Measurement Configuration

If an eNodeB performs CA for a CA UE whose signal quality is so poor that an inter-frequency handover may occur, the spectral efficiency of the network decreases and the BLER increases. To prevent this, set thresholds as described in **Table 19-6**.

Table 19-6 Threshold setting rules

Threshold	Parameter	Setting Rules
Threshold for CA event A4 that triggers SCell configuration	• In CA-group-based configuration mode: A4 threshold = **CaMgtCfg**.***CarrAggrA4ThdRsrp*** + **CaGroupSCellCfg**.***SCellA4Offset*** (These parameters are configured on the PCell side.) • In adaptive configuration mode, A4 threshold varies as follows: For a CA UE for which SCell configuration is performed during initial access, an incoming handover, or an incoming RRC connection reestablishment: A4 threshold = **CaMgtCfg**.***CarrAggrA4ThdRsrp*** + **SccFreqCfg**.***SccA4Offset*** + **SccFreqCfg**.***SccA2RsrpThldExtendedOfs*** (These parameters are configured on the PCell side.) For a CA UE for which SCell configuration is periodically triggered based on the traffic volume: A4 threshold = **CaMgtCfg**.***CarrAggrA4ThdRsrp*** + **SccFreqCfg**.***SccA4Offset*** (These parameters are configured on the PCell side.)	This threshold must be greater than or equal to the threshold for inter-frequency handover event A4, which is specified by the **InterFreqHoGroup**.***InterFreqHoA4ThdRsrp*** parameter.

Threshold	Parameter	Setting Rules
Threshold for CA event A2 that triggers SCell removal	● In CA-group-based configuration mode: A2 threshold = **CaMgtCfg**.***CarrAggrA2ThdRsrp*** + **CaGroupSCellCfg**.***SCellA2Offset*** (These parameters are configured on the PCell side.) ● In adaptive configuration mode: A2 threshold = **CaMgtCfg**.***CarrAggrA2ThdRsrp*** + **SccFreqCfg**.***SccA2Offset*** + **SccFreqCfg**.***SccA2RsrpThldExtendedOfs*** (These parameters are configured on the PCell side.)	This threshold must be greater than or equal to the threshold for inter-frequency handover event A2, which is specified by the **InterFreqHoGroup**.***InterFreqHoA2ThdRsrp*** parameter.

Blacklist Control over Autonomous Gap

Certain UEs with the autonomous gap function enabled will encounter measurement errors or even service drops. These UEs can be blacklisted so that the autonomous gap function does not take effect for them. For details about this function, see *Terminal Awareness Differentiation*.

Handover Events

CA UEs report events for eNodeBs to evaluate the following handovers:

- Intra-frequency handover (Related parameters are configured in the **IntraFreqHoGroup** MO.)

  An eNodeB performs an intra-frequency handover for a CA UE when it receives an A3 measurement report from the UE in the PCell.

- Inter-frequency handover (Related parameters are configured in the **InterFreqHoGroup** MO.)

  After receiving an A2 measurement report from a CA UE in the PCell, the serving eNodeB of the PCell delivers the inter-frequency measurement configuration to the UE. The measurement configuration varies depending on whether SCells have been configured for the UE:

  – If SCells have been configured, the eNodeB delivers an A3 or A5 measurement configuration: A3 in case the **EutranInterNFreq**.***InterFreqHoEventType*** parameter is set to **EventA3** and A5 in case this parameter is set to any other value.

  – If no SCells have been configured, the eNodeB delivers a measurement configuration related to the event specified by the **EutranInterNFreq**.***InterFreqHoEventType*** parameter.

  The eNodeB performs an inter-frequency handover for the UE when it receives a measurement report of the specified event from the UE.

NOTE

- For the triggering conditions for these events, see *Mobility Management in Connected Mode*. Event A5 is triggered if the signal quality in the PCell is lower than the threshold for handover event A2 and the signal quality in at least one neighboring cell is higher than the threshold for handover event A4.
- It is recommended that threshold 1 for event A5 be the same as the threshold for inter-frequency handover event A2. If threshold 1 is lower than or equal to the threshold for blind handover event A2, the eNodeB will not deliver A5 measurement configurations to the UEs.

Frequency-Priority-based Handover

If the operator-specified frequency with the highest priority for frequency-priority-based handovers is different from the frequency with the highest priority for PCC anchoring, you are advised to select the **FreqPriBasedHoCaFiltSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter. In such a case, the eNodeB does not deliver the measurement configurations related to frequency-priority-based handovers to CA UEs, in order to prevent outgoing frequency-priority-based handovers of the CA UEs soon after these UEs are handed over to a frequency with a high PCC anchoring priority.

If a frequency has both the highest priority for frequency-priority-based handovers and the highest priority for PCC anchoring, you can deselect the **FreqPriBasedHoCaFiltSwitch** and **PccAnchorSwitch** options of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter. In such a case, mobility of both CA UEs and non-CA UEs is subject to the frequency priorities defined for frequency-priority-based handovers.

Load-based Handover

If the PCell of a CA UE meets the triggering conditions for load balancing and the UE meets the UE selection conditions, the eNodeB performs a load-based inter-frequency handover on the UE. For details about this type of handover, see *Intra-RAT Mobility Load Balancing*.

SCell Configuration During Handovers

To reduce the SCell configuration delay, enable SCell configuration during handovers by selecting the **HoWithSccCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter. The procedure for SCell configuration during handovers varies depending on the setting of **HoWithSccCfgSwitch**.

- If the **HoWithSccCfgSwitch** option is selected for both the source and target eNodeBs, this function is enabled. The procedure is as follows:
  a. The source eNodeB includes the reportAddNeighMeas IE in the handover-related A3, A4, and A5 measurement configurations delivered to the CA UE.
  b. The CA UE sends the source eNodeB the measurement reports that contain the measurement result of the best cell on each serving frequency (in the measResultBestNeighCell IE) and the PCell and SCell measurement results (in the measResultPCell and measResultSCell IEs).

c. The source eNodeB includes the best cell information in the CandidateCellInfoList IE of the handover request message and sends the message to the target eNodeB.

   The source eNodeB also includes the blind-configurable candidate SCells of the source cell in the CandidateCellInfoList IE of the handover request message if the **ENodeBAlgoSwitch**.**HoWithSccCfgAddBlindSwitch** parameter is set to **ON**.

   If **BIT26** of the **eNBRsvdPara**.***RsvdSwPara6*** parameter is on, relaxed-backhaul-based inter-eNodeB SCells are not included in the CandidateCellInfoList IE.

d. The target eNodeB acquires measurement results from the CandidateCellInfoList IE, determines the new SCell to be configured to accompany the target cell (acting as the PCell) of the handover, and updates the sCellToAddModList IE with the new SCell information. The target eNodeB then sends the updated information in the handover command to the source eNodeB.

   The eNodeB preferentially selects blind-configurable candidate SCells as new SCells. This takes effect in adaptive configuration mode and only if the **HoWithSccCfgBlindFirstSw** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is selected. It takes effect in CA-group-based configuration mode without switch control.

e. The source eNodeB sends an RRC Connection Reconfiguration message containing the mobilityControlInfo, sCellToReleaseList, and sCellToAddModList IEs to remove the original SCells and configure the new SCells during the handover.

After the **HoWithSccCfgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter is selected, the SCell configuration success rate changes as affected by the handover success rate. (The SCell configuration success rate can be observed using the **L.CA.DLSCell.Add.Succ** and **L.CA.DLSCell.Add.Att** counters.)

The **CaTrafficTriggerSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter does not take effect in this procedure. The source and target eNodeBs perform the procedure without considering traffic volume for configuring SCells, even if the **CaTrafficTriggerSwitch** option has been selected.

If this procedure fails, no SCell is configured for the UE that has been handed over to the target cell. The target eNodeB then performs an SCell configuration procedure. For details, see **4.6.3.1.2 CA-Group-based SCell Configuration Procedure** and **4.6.3.1.3 Adaptive SCell Configuration Procedure**.

NOTE

- The preceding procedure results in fewer blind or A4-based SCell configurations.
- SCell configuration during a handover requires that a routing relationship be set up between the SCell and the target cell of the handover.
- If SCell configuration during handovers is enabled, the eNodeB attempts to configure SCells without considering the load status of the candidate SCells, when PCC anchoring is triggered. This allows the SCells to be configured more quickly.
- During a blind handover of a CA UE with heavy uplink traffic load for MLB, the source eNodeB includes the PCell and SCell information for the UE in the handover request sent to the target eNodeB. Based on the received information, the target eNodeB selects the SCells to be configured for the CA UE after the handover.

- If the **HoWithSccCfgSwitch** option is deselected for both the source and target eNodeBs, this function is disabled. The procedure is as follows:

  The source eNodeB delivers an RRC Connection Reconfiguration message to remove the SCells of the CA UE and performs an intra- or inter-frequency handover for the UE. After the UE is handed over to the target cell, the target eNodeB configures SCells for the UE.

NOTE

SCell configuration, SCell change, and PCC anchoring may or may not be affected by event A2 reporting, depending on the setting of the **RcvA2CfgSccSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter:

- If this option is deselected, SCell configuration, SCell change, and PCC anchoring are affected by event A2 reporting.

  If the serving eNodeB of the PCell for a CA UE receives a handover-related event A2 report from the UE, the eNodeB stops configuring SCells, changing SCells, and performing PCC anchoring for the UE. If the eNodeB receives a handover-related event A1 report, the eNodeB can configure SCells in a blind manner (if blind SCell configuration is enabled) or after receiving an event A4 report related to carrier management from the UE, can change SCells, and can perform PCC anchoring for the UE.
- If this option is selected, SCell configuration, SCell change, and PCC anchoring are not affected by event A2 reporting. The serving eNodeB of the PCell for a CA UE can still configure SCells, change SCells, and perform PCC anchoring for the UE after receiving a handover-related event A2 report from the UE.

For more details about mobility management, see *Mobility Management in Connected Mode*.

19.4.2 Load Balancing (TDD)

Inter-CC Load Transfer Triggered by Cell Load

Each eNodeB measures and evaluates the load of each of the served cells involved in CA (referred to as CA cells in this section). If a CA cell enters the high load state, the serving eNodeB triggers load balancing between the CCs of UEs in the downlink CA state. With the load balancing algorithm, the eNodeB selects high- and low-load CA cells and dynamically adjusts the ratio of CA UE data volumes in the cells, balancing the load between the cells. This algorithm helps reduce the number of handovers of non-CA UEs triggered for MLB.

NOTE

- This function does not take effect for SFN cells.
- If three or more CCs are aggregated in the downlink for a UE, the ratio adjustment works between the PCell and SCells, not between SCells.

This function is controlled by the **DlCaLbAlgoSwitch** option of the **ENodeBAlgoSwitch.*CaLbAlgoSwitch*** parameter. When this option is selected, the **CaLoadBalancePreAllocSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter must be deselected.

For this load balancing algorithm to take effect, operators must also set the following parameters:

- **eNodeBMlb.*DlCaLbMaxCCNum***

  This parameter specifies the maximum number of downlink CCs aggregated for each CA UE whose load is to be transferred for load balancing.

  – If this parameter is set to **2CC**, the eNodeB transfers the load of CA UEs in the downlink 2CC aggregation state.

  – If this parameter is set to **3CC**, the eNodeB transfers the load of CA UEs in the downlink 2CC aggregation state and those in the downlink 3CC aggregation state. Currently, this parameter does not provide an option for 4CC or 5CC. The value **3CC** is recommended.

- **eNodeBMlb.*CaUeMinDataVolRatio***

  This parameter specifies the minimum required ratio (expressed as a percentage) of the data volume on a high-load CC to that on a low-load CC for each CA UE. For example:

  – If this parameter is set to **20**, the minimum ratio for a single CA UE is 1:5.

  – If this parameter is set to **0**, the minimum ratio is 0:1, which means that all data of the CA UE on the high-load CC can be transferred to the low-load CC. The value **0** is recommended.

19.4.3 Uplink-Traffic-based MLB (TDD)

The ratio of uplink-to-downlink subframes for a TDD cell is usually small (typically 1:3). It is possible that the number of uplink subframes is sufficient in an FDD cell but insufficient in a TDD cell. When uplink traffic is heavy, MLB can be enabled for a TDD cell that will act as PCells for CA UEs in FDD+TDD CA. This MLB function is controlled by the **UlHeavyTrafficMlbTFCASw** option of the **CellAlgoSwitch.*TrafficMlbSwitch*** parameter. When the load of the TDD cell exceeds a specified threshold, the eNodeB hands over the CA UEs with heavy uplink traffic from the TDD cell to a specified FDD frequency or changes the PCell to an FDD cell. This function improves user experience with uplink data rates for CA UEs in the TDD cell.

If the target FDD cell of the handover is included in the existing serving cell combination for FDD+TDD CA, the function of blind handover to FDD cells can be used to shorten the handover delay. This function is enabled by setting the **CaMgtCfg**.***UlHeavyTrafficMlbTFCAOptSw*** parameter to **ON**.

Uplink-traffic-based MLB under CA requires load information exchange. The eNodeB does not hand over CA UEs to a neighboring FDD cell whose load information has not been obtained.

For details about the MLB procedure, see *Intra-RAT Mobility Load Balancing*.

NOTE

If SCell configuration is required during the preceding handover procedure, see **SCell Configuration During Handovers** in **19.4.1 Mobility Management** for details.

19.4.4 Admission and Congestion Control

This section describes how admission control and congestion control work when CA is enabled.

Admission Control

Admission control under CA differs from admission control without CA in the following ways:

- Admission control based on the number of UEs

  When an eNodeB receives an access request from a CA UE (for RRC connection setup or reestablishment, an incoming handover, or a transition from out-of-synchronization to in-synchronization), the eNodeB performs admission control in both the PCell and SCells based on the number of UEs in the individual cells. If the eNodeB accepts the access request, it treats the UE as an RRC_CONNECTED UE in each serving cell. However, only the RRC_CONNECTED UEs in the PCell consume UE count license units. Take 2CC aggregation as an example. If all UEs on the entire network have their SCells activated and are working in the 2CC aggregation state, the maximum number of UEs that can access the network decreases by half, and the number of consumed UE count license units is equal to the actual number of UEs that have accessed the network.

  The UE count threshold used to determine whether to admit a CA UE to an SCell is equal to 90% of the **CellRacThd.*AcUserNumber*** parameter value specified for this cell.

  When a UE attempts to access a cell, the eNodeB preferentially releases other CA UEs that treat the cell as their SCell, if the number of UEs has already reached the maximum number allowed. The release maximizes the total number of UEs on the entire network.

- Admission control based on QoS satisfaction rates

  This type of admission control applies to GBR services of CA UEs only in their PCells.

  - If a CA UE is admitted to its PCell, the UE accesses the network.
  - If the UE is not admitted to its PCell but is capable of preemption, the UE performs preemption in this PCell. If the preemption is successful, the UE accesses the network.
  - If the UE is not admitted to its PCell and if it is incapable of preemption or the preemption fails, the UE cannot access the network.

For more details about admission control, see *Admission and Congestion Control*.

Congestion Control

Huawei eNodeBs relieve traffic congestion mainly by releasing GBR services. When a cell is overloaded, GBR services in the cell do not meet their QoS requirements.

If the **DlLdcSwitch** and **UlLdcSwitch** options of the **CellAlgoSwitch.RacAlgoSwitch** parameter are selected for the cell, the eNodeB releases low-priority GBR services in the cell to ensure satisfaction rates for high-priority GBR services. This congestion control mechanism works in CA scenarios.

If congestion control over GBR services is triggered in a cell because there are insufficient radio resources, the eNodeB releases GBR services of CA UEs that treat this cell as their PCell and of non-CA UEs in this cell. (Non-CA UEs are terminals that do not support CA.) When the eNodeB selects the UEs whose services are to be released, it removes CA UEs that treat the cell as their SCell from the list of prioritized candidates. If a service to be released is the only GBR service of a UE that meets conditions for redirection, the eNodeB redirects the UE. If the service is not the only GBR service or the conditions for redirection have not been met, the eNodeB releases this GBR service of the UE.

For more details about congestion control, see *Admission and Congestion Control*.

19.4.5 High-Speed UE Return

High-speed UE return transfers high-speed UEs from public-network cells back to overlapping inter-frequency cells on networks dedicated to high-speed railways.

High-speed UEs are frequently handed over within the public network because the public-network cells cover relatively small areas. The eNodeBs on the public network identify high-speed CA UEs based on how often they are handed over. If an eNodeB identifies a CA UE as a high-speed UE and detects that one of its SCells belongs to the dedicated network, the eNodeB redirects the UE to that SCell without instructing the UE to perform inter-frequency measurements.

This function is controlled by the **HSCABlindRedirectSw** option of the **CellAlgoSwitch**.***HighSpeedSchOptSwitch*** parameter.

For the mechanism and engineering guidelines for this function, see *High Speed Mobility*.

19.5 Resource Management

19.5.1 Scheduling

This section describes how scheduling works when CA is enabled. For more details about scheduling, see *Uplink Scheduling* and *Downlink Scheduling*.

Mechanisms of Scheduling Under CA

This section describes the characteristics of scheduling in CA scenarios.

For GBR services, scheduling with CA enabled is almost the same as scheduling with CA disabled. The goal of scheduling is still to meet the QoS requirements of GBR services.

For non-GBR services, Huawei has designed two scheduling methods for the enhanced proportional fair (EPF) policy: basic and differentiated. The method used is specified by the **CellDlschAlgo**.***CaSchStrategy*** parameter. The default method is basic scheduling. The scheduling method must be consistent between serving cells to prevent data transmission exceptions. The following presents the characteristics of the two methods:

- Basic scheduling

  When PRBs on a network are abundant, basic scheduling works in the same way as differentiated scheduling. CA UEs can use sufficient PRBs to meet their service requirements.

  When there is PRB congestion on a network, basic scheduling works to maintain fairness. The data rate of a CA UE is almost the same as the data rate of a non-CA UE that is running services with the same QCI.

  Due to inconsistent spectral efficiency for the CA UE between CCs, basic scheduling does not achieve the same data rates for the CA UE and the non-CA UE when planning to allocate the same number of PRBs to the two UEs.

  - If the difference in spectral efficiency between CCs is large (for example, when the CA UE is located at the edge of its PCell or SCell), there is a large difference in the number of allocated PRBs between the CA UE and the non-CA UE.
  - If the spectral efficiency is almost the same between CCs, the number of PRBs allocated to the CA UE is close to that of PRBs allocated to the non-CA UE.

  The **CellDlschAlgo.*CaSchWeight*** parameter specifies a differentiated scheduling factor for CA UEs when the scheduling method is basic scheduling. This parameter can be set to values of **0** to **10**, which correspond to actual values ranging from 0 to 1. It takes effect only for downlink scheduling. A larger parameter value results in more scheduling differentiation between CA UEs and non-CA UEs. A smaller parameter value results in fairer scheduling. This is a parameter to be set only for PCells. If this parameter is set to its default value **0**, basic scheduling works in the same way as before. If this parameter is set to a non-zero value, the number of PRBs allocated to a CA UE is (1 + n x m) times the number of PRBs allocated to a non-CA UE, where n is the number of activated SCells and m is the actual value of the parameter that specifies the differentiated scheduling factor.

- Differentiated scheduling

  The data rate of a CA UE for priority calculation is defined as the data rate only on the current CC of the UE. On each CC, the CA UE is allocated the same number of RBs as a non-CA UE on the same CC would be. Therefore, the number of RBs allocated to the CA UE is the sum of the average number of RBs per non-CA UE in each of the serving cells. The data rate of the CA UE served by n CCs is almost n times the data rate of a non-CA UE, assuming that the spectral efficiencies of the CCs are close to each other, UEs are evenly distributed on the CCs, and channel conditions are the same between the CCs.

  In differentiated scheduling, a CA UE is treated as a common UE on each CC. It is scheduled separately in the CCs. Therefore, a CA UE can be allocated more PRBs and deliver a better user experience than a non-CA UE. However, radio resources for non-CA UEs decrease.

Currently, frequency selective scheduling is not used in SCells by default. To enable frequency selective scheduling in SCells, set the **CellDlschAlgo.*CaSccDopMeas*** parameter to **FROMPCC**.

NOTE

If an SCell is configured in the uplink for a CA UE, uplink scheduling uses the method configured for downlink scheduling: basic or differentiated scheduling.

PCC/SCC Scheduling Principles

UE scheduling on their PCCs and SCCs varies depending on the CA scenario:

- When RLC unacknowledged mode (UM) is used:
  - In inter-eNodeB CA based on centralized eNodeB coordination and in intra-eNodeB CA

    If the service type is VoLTE or emergency call, the serving eNodeB of the PCell schedules the UE only on the PCC. (If a UE reverts from the uplink CA state to the single-carrier state due to a lack of power, one of the original SCCs may become the only carrier and scheduling on this SCC occurs.) If the service type is neither VoLTE nor emergency call, the eNodeB can schedule the UE only on the PCC and its intra-eNodeB SCCs.
  - In inter-eNodeB CA based on relaxed backhaul

    If the service type is VoLTE or PTT, the serving eNodeB of the PCell schedules the UE only on the PCC.

    If the service type is neither VoLTE nor PTT, the eNodeB schedules the UE only on the PCC and its intra-eNodeB SCCs.

  For details about RLC UM, see section 4.2.1 "RLC entities" in 3GPP TS 36.322 V10.0.0.
- In TDD CA scenarios, scheduling optimization based on cell capability for target-rate-guaranteed UEs can be enabled by selecting the **CELL_CAPB_BASED_GBR_OPT_SW** option of the **CellAlgoExtSwitch**.***WTTxAlgoSwitch*** parameter. This function prevents UEs in the cell with weak capabilities from occupying excessive resources.

  Scheduling optimization based on cell capability for target-rate-guaranteed UEs takes effect only for specific CA UEs identified based on QCIs or SPIDs. For details, see *Rate Control Based on User Types*.

  The **BEST_EFFORT_GBR_SW** option of the **CellAlgoSwitch**.***DlSchExtSwitch*** parameter must be selected before enabling scheduling optimization based on cell capability for target-rate-guaranteed UEs.

  Scheduling optimization based on cell capability for target-rate-guaranteed UEs lifts the scheduling priorities of UEs with strong cell capabilities on their SCCs, based on the cell capabilities of UEs on different carriers. The cell capability of a UE is directly proportional to the spectral efficiency of the UE and the cell bandwidth, and is inversely proportional to the cell load. The higher the spectral efficiency of the UE and the larger the cell bandwidth, the stronger the cell capability of the UE. The higher the cell load, the weaker the cell capability of the UE.

Precise Scheduling for CA UEs

With precise scheduling, the eNodeB dynamically distributes traffic to CCs based on the real-time traffic volumes and scheduling capabilities of the CCs. This function is enabled if the **CaEnhancedPreAllocSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is selected.

The following constraints apply to the **CaEnhancedPreAllocSwitch** option:

- In FDD, this option does not take effect for inter-eNodeB CA based on relaxed backhaul.
- In TDD, if this option is selected in inter-eNodeB CA based on relaxed backhaul, traffic distribution to CCs is not based on the real-time traffic volume. Instead, the amount of traffic distributed from the PCell to SCells is adjusted only in preallocation situations.
- In TDD, before this option is selected, the **DlCaLbAlgoSwitch** option of the **ENodeBAlgoSwitch**.***CaLbAlgoSwitch*** parameter must be deselected.
- Before this option is selected, the **CaLoadBalancePreAllocSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter must be deselected.

If the **EpfEnhancedSwitch** option of the **CellAlgoSwitch**.***DlSchSwitch*** parameter is selected, it is recommended that the **CellDlschAlgo**.***EpfEnhancedSchOptSwitch*** parameter be set to **ON**. With this setting, UEs in their SCells for inter-BBP CA can also use the priority calculation method for enhanced proportional fair (EPF) enhancement, resulting in shorter initial-packet delay in the SCells.

CCE-Usage-based Downlink Data Split for CA UEs (FDD)

Low-frequency LTE cells can serve a relatively large number of UEs at the cell edge due to their large coverage. When these cells are congested, the experience of non-CA UEs in the cells deteriorate. To address this issue, CCE-usage-based downlink data split for CA UEs can be enabled in their PCells. This function relieves cell congestion and improves the experience of non-CA UEs by lowering user experience of CA UEs in the cells. It takes effect only in cells with a bandwidth less than or equal to 5 MHz. This function is controlled by the **CellDlschAlgo**.***DataSplitCceUsageHighThld*** parameter. When this function takes effect for a cell, the value of the **L.CA.Traffic.bits.DL.PCell** counter decreases for the cell.

The relationships between the configured values and effective values of the related threshold parameters are described in **Table 19-7**.

Table 19-7 Relationships between the configured values and effective values of the related threshold parameters

Parameter	Configured Value	Effective Value
CellDlschAlgo.***DataSplitCceUsageHighThld***	**0** to **100**	Same as the configured value

Parameter	Configured Value	Effective Value
CellDlschAlgo.***DataSplitCceUsageLowThld***	**0** to **100**	• The effective value of this parameter is **0** if the configured value is greater than or equal to the effective value of the **CellDlschAlgo**.***DataSplitCceUsageHighThld*** parameter. • The effective value of this parameter is the same as its configured value if the configured value is less than the effective value of the **CellDlschAlgo**.***DataSplitCceUsageHighThld*** parameter.

If an SCell has been activated for a CA UE, the CCE usage of the PCell for the UE is calculated as follows:

CCE usage = max((**L.ChMeas.CCE.ULUsed.Equivalent** + **L.ChMeas.CCE.DLUsed.Equivalent**)/**L.ChMeas.CCE.AvailPower.Equivalent** x 100%, (**L.ChMeas.CCE.CommUsed** + **L.ChMeas.CCE.ULUsed** + **L.ChMeas.CCE.DLUsed**)/**L.ChMeas.CCE.Avail** x 100%)

The calculation result will be filtered. The value obtained after filtering will be compared with the values of **CellDlschAlgo**.***DataSplitCceUsageHighThld*** and **CellDlschAlgo**.***DataSplitCceUsageLowThld*** to determine whether the PCell is in the congested state.

Table 19-8 provides the definitions of the PCell congested state and non-congested state and the corresponding handling mechanisms.

Table 19-8 Definitions of the PCell congested state and non-congested state and the corresponding handling mechanisms

PCell Status	Definition	Handling Mechanism
Congested	The CCE usage in the PCell for the CA UE is greater than the effective value of the **CellDlschAlgo**.***DataSplitCceUsageHighThld*** parameter.	The downlink data of the UE is transmitted only in the LTE SCells. However, in the case of relaxed-backhaul-based inter-eNodeB CA, the downlink RLC retransmission data of inter-eNodeB SCells is still transmitted in the PCell.

PCell Status	Definition	Handling Mechanism
Non-congested	The CCE usage in the PCell for the CA UE is less than or equal to the effective value of the **CellDlschAlgo**.***DataSplitCceUsage-LowThld*** parameter.	Downlink data of the UE is transmitted as usual.

If the CCE usage in the PCell for the CA UE is less than or equal to the effective value of **CellDlschAlgo**.***DataSplitCceUsageHighThld*** and greater than the effective value of **CellDlschAlgo**.***DataSplitCceUsageLowThld***, the congestion status of the PCell is the same as that in the last second.

If the downlink data of a CA UE cannot be transmitted in the PCell:

- The user-perceived data rate decreases. In this situation, if the SCells are unavailable due to coverage or air interface quality deterioration or because the transmission rate of downlink data from the core network to the eNodeB exceeds the transmission specification of the BBP or main control board, the user-perceived data rate will notably decrease.
- The downlink retransmission rate and downlink scheduling delay will increase for the CA UE but decrease for non-CA UEs. Therefore, the downlink retransmission rate and downlink packet delay of the cell will change.
- More resources in the PCell can be spared to non-CA UEs or VoLTE services. When there are a large number of small-packet services, the number of scheduling times in the PCell increases. As a result, the CCE usage in the PCell increases.

If both CCE-usage-based downlink data split for CA UEs and multi-carrier unified scheduling for CA (controlled by the **CA_MULTI_CARR_UNIFIED_SCH_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchEnSw*** parameter) are enabled, the former function preferentially takes effect in terms of downlink data split for CA UEs. When this is the case, the gains offered by the latter function decrease. For details about the gains, see *Multi-carrier Unified Scheduling*.

Basic and Enhanced Flexible Control of CA Performance Optimization

Ultra-low-latency scheduling (controlled by the **ULTRA_LOW_LATENCY_SCH_SW** option of the **MultiCarrUnifiedSch**.***MultiCarrierUnifiedSchSw*** parameter) causes a large increase in the CPU usage of baseband processing units (BBPs). To reduce the BBP CPU usage in heavy-load scenarios, the basic and enhanced functions of flexible control of CA performance optimization have been introduced. These functions are controlled by the **EnodebAlgoExtSwitch**.***CaFlexCtrlOptThldOffset*** parameter.

- If this parameter is set to **255**, neither basic nor enhanced flexible control of CA performance optimization takes effect.
- If this parameter is set to **0**, basic flexible control of CA performance optimization takes effect, but enhanced flexible control of CA performance optimization does not. In this case, the eNodeB adaptively optimizes the CA

scheduling-related processing performance based on service characteristics and CPU loads.

- If this parameter is set to a value in the range of 1 to 20, both basic and enhanced flexible control of CA performance optimization take effect and the eNodeB uses this parameter to control the threshold for the flexible control functions.

The setting of the **EnodebAlgoExtSwitch**.***CaFlexCtrlOptThldOffset*** parameter has the following network impacts:

- If this parameter is set to a value in the range of 0 to 20, the BBP CPU usage of the eNodeB decreases. In heavy-load scenarios, the volume of downlink data in cells increases, the downlink UE throughput slightly decreases, and the downlink IBLER fluctuates.
- If this parameter is set to a value in the range of 1 to 20, a smaller value of this parameter has a greater impact on the user-perceived data rates of CA UEs and results in higher system reliability; a larger value of this parameter has a smaller impact on the user-perceived data rates of CA UEs but a greater impact on system reliability. The more the SCell-activated UEs on the network, the greater the impact of these functions on real-time CA data split.

When ultra-low-latency scheduling is enabled, it is recommended that the **EnodebAlgoExtSwitch**.***CaFlexCtrlOptThldOffset*** parameter be set to **1**.

19.5.2 Power Control

If an SCell has been configured and activated for a CA UE in the uplink, uplink power control works separately on the two CCs. The UE sends CC-specific extended power headroom reports (PHRs) to the eNodeB. Based on these reports, the eNodeB calculates the total required power for the two CCs. The required power for the PCC or an SCC is equal to the maximum transmit power supported by the UE minus the power headroom indicated in the PCC- or SCC-specific extended PHR, respectively. If the total required power exceeds the maximum transmit power of the UE while the transmit power per PRB remains unchanged, the eNodeB reduces the number of uplink PRBs for the UE till the UE enters the single carrier state.

NOTE

Extended PHR is an information element introduced by 3GPP for uplink CA UEs. It contains the PHR for each CC. For details, see section 6.1.3.6a "Extended Power Headroom MAC Control Element" in 3GPP TS 36.321 V11.0.0.

The **UlCaPuschPcOptSwitch** option of the **CellAlgoSwitch**.**UlPcAlgoSwitch** parameter can be selected to increase the number of UEs in the UL CA state. That is because when this option is selected, the transmit power per PRB decreases for CA UEs so that more PRBs can be allocated to the UEs. This function increases uplink throughput of these UEs when resources are sufficient. However, selecting this option does not produce gains for uplink CA UEs that experience poor uplink SINR in their SCells in relaxed backhaul scenarios.

19.5.3 MTA

For UEs on which uplink CA has taken effect, both PCells and SCells provide uplink resources. Using a single timing advance (TA) has a significant impact on demodulation performance in the following scenarios: (1) There is a noticeable difference (for example, about 78 meters, which is equivalent to a TA) between the distances of a CA UE to the receive antennas of the PCell and an SCell. (2) The PCell and an SCell are not served by the same pRRU of a LampSite eNodeB. To improve uplink performance, the multiple timing advances (MTA) function is recommended. For MTA to take effect, the MTA switch must be turned on and UEs must be capable of MTA. The MTA switch varies depending on the CA scenario:

- Single-duplex-mode uplink 2CC aggregation: the **MtaAlgSwitch** option of the **ENodeBAlgoSwitch**.**CaAlgoSwitch** parameter
- FDD+TDD uplink 2CC aggregation: the **FTMtaAlgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter

After an SCell is configured and activated in both downlink and uplink for a CA UE, the eNodeB sends a PDCCH order to the UE and, as instructed, the UE initiates a non-contention-based random access procedure in the SCell. The eNodeB then sends the UE a random access response (RAR) to configure a secondary timing advance group (sTAG) for the UE. The network maintains a TA for each serving cell of the UE.

NOTE

In this scenario, the UE sends a non-contention-based preamble in the uplink SCell, whereas the eNodeB sends an RAR to the UE in the PCell. The number of times UEs have sent non-contention-based preambles in a cell that acts as an uplink SCell is measured by the **L.RA.Dedicate.Att** and **L.RA.Dedicate.sTAG.Att** counters. The number of times the eNodeB has sent the corresponding RARs in a cell that acts as the PCell is measured by the **L.RA.Dedicate.Resp** and **L.RA.Dedicate.sTAG.Resp** counters.

If the number of preamble retransmissions in the SCell reaches the maximum number allowed, the random access procedure fails. The UE stops random access without instructing the PCell to perform an RRC connection reestablishment. As a result, the UE is out-of-synchronization in the uplink, and only downlink scheduling is allowed. If the SCell activation conditions are met and the resynchronization timer expires, a random access procedure is triggered again. The eNodeB removes the SCell if consecutive random access attempts fail.

NOTE

For details about PDCCH order, see section 8.0 "UE procedure for transmitting the physical uplink shared channel" in 3GPP TS 36.213 V11.4.0.

If a TDD cell and an FDD cell are acting as the PCell and SCell respectively in uplink 2CC aggregation for a UE and the **FTMtaAlgSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoExtSwitch*** parameter is selected:

- The eNodeB preferentially instructs the UE to send SRSs in the 13th orthogonal frequency division multiplexing (OFDM) symbol of an uplink subframe. Uplink scheduling of the UE in the FDD cell yields to the SRS transmission to ensure beamforming gains in the TDD cell.
- If the **CellPdcchAlgo**.**ComSigCongregLv** parameter is set to **CONGREG_LV8** for the TDD cell, the PDCCH control channel element (CCE) aggregation level for common control signaling is so high that the eNodeB transmit power might be insufficient for sending a Random Access Response message to the UE. This negatively affects configuration of the uplink SCell for the UE.

In a network with macro-micro or micro-micro CoMP enabled, the average measured TA value for a micro cell is 2 microseconds greater than the actual value when certain conditions are met. In this micro cell, the values of individual counters **L.RA.TA.UE.Index0** to **L.RA.TA.UE.Index11** are not smooth. In addition, if TA is used to estimate the distance between a UE in the cell and the antenna of the eNodeB serving the cell, the estimation result is inaccurate. The conditions are as follows:

- The **MtaAlgSwitch** option of the **ENodeBAlgoSwitch**.**CaAlgoSwitch** parameter is selected.
- The **CaUl2CCSwitch** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter is selected.
- Any of the following options is selected:
  - **UlJointReceptionSwitch** option of the **CellAlgoSwitch**.***UplinkCompSwitch*** parameter
  - **UlCompForVideoSwitch** option of the **CellAlgoSwitch**.***UplinkCompSwitch*** parameter
  - **UlVoiceJROverRelaxedBHSw** option of the **ENodeBAlgoSwitch**.***OverBBUsSwitch*** parameter
  - **UlNonVoiceJROverRelaxedBHSw** option of the **ENodeBAlgoSwitch**.***OverBBUsSwitch*** parameter

The **TaDistanceMappingSwitch** option of the **CellCounterParaGroup**.***CellCounterAlgoSwitch*** parameter can be selected to resolve this problem.

19.5.4 RAN Sharing

In RAN sharing scenarios, the operators involved allocate resources to their subscribers based on the predefined proportions of resources. To enhance fairness in resource usage among the operators that share a carrier, differentiated scheduling is recommended.

CA UEs can access only operator-specific carriers (either in PCells or in SCells). For example, there are three cells working on frequencies f1, f2, and f3, respectively:

- Cell working on f1 is shared by operators A and B
- Cell working on f2 is dedicated to operator A.
- Cell working on f3 is dedicated to operator B.

On this network, only the carriers of f1 and f2 can be aggregated for CA UEs of operator A, and only the carriers of f1 and f3 can be aggregated for CA UEs of operator B. Cells on f3 cannot be accessed by or configured as SCells for CA UEs of operator A. Cells on f2 cannot be accessed by or configured as SCells for CA UEs of operator B.

If equivalent PLMNs are configured or inter-PLMN handovers are enabled, carriers of frequencies owned by different operators can be aggregated. For example, cells on f2 and f3 can be aggregated. For details about the configuration, see *Mobility Management in Connected Mode*.

In RAN sharing scenarios, different eNodeBs must be assigned different eNodeB IDs. Otherwise, errors will occur in SCell configuration because target eNodeBs cannot be identified.

For more details about RAN sharing, see *RAN Sharing*.

19.6 RAN-Terminal Coordination

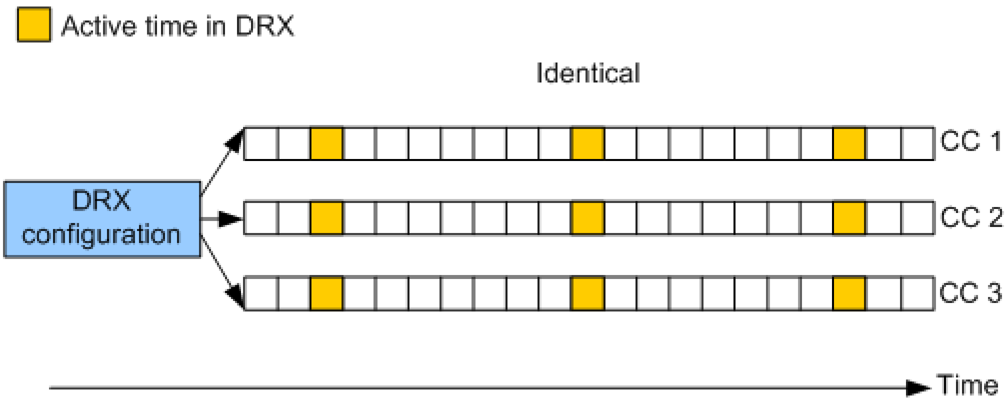
19.6.1 DRX Control

In discontinuous reception (DRX) mode, UEs save power by not monitoring the physical downlink control channel (PDCCH) for certain periods.

According to section 5.7 "Discontinuous Reception (DRX)" in 3GPP TS 36.321 V11.2.0, the same active time applies to all activated serving cells.

For this purpose, all the activated serving cells of a CA UE use the DRX parameters of the PCell, including On Duration Timer, DRX Inactivity Timer, and DRX Short Cycle Timer. **Figure 19-1** shows an example of the DRX configuration under CA.

Figure 19-1 Common DRX configuration



When CA is enabled, DRX control works as follows:

- If DRX is enabled in the PCell and all of the SCells for a CA UE, the UE states in the cells are handled as follows when the serving eNodeB of the PCell sends a MAC CE to activate the SCells:
  - If the UE has entered the DRX state in the PCell, the UE now also enters the DRX state in SCells, with the DRX parameters for the PCell applied to the SCells.
  - If the UE has not entered the DRX state in the PCell, the eNodeB determines whether the UE should enter the DRX state in all the PCell and SCells based on the traffic volume of the UE.
- If DRX is enabled in the PCell but disabled in an SCell, the UE exits and does not return to the DRX state after the SCell is configured for the UE.
- If DRX is disabled in the PCell, the UE will not enter the DRX state in either the PCell or SCells, regardless of whether DRX is enabled in the SCells.
- If DRX and uplink FDD+TDD CA are both enabled, the **DrxParaGroup.OnDurationTimer** parameter must be set to a value greater than four subframes for the FDD cell.
- QCI-specific DRX for the UE is controlled only by the PCell.

For more details about DRX control, see *DRX and Signaling Control*.

19.6.2 SPID- and IP-based SCell Management

CA policies can be defined for specific SPIDs and IP addresses. The **SpidCfg**.***MaximumNumberOfCarriers*** parameter controls the maximum number of CCs that can be aggregated for specific UEs. For details, see *Flexible User Steering*.

19.6.3 Blind SCell Configuration for Fast-Moving UEs

Experience of fast-moving UEs can now be improved by prohibiting blind SCell configuration for non-fast-moving UEs. This, therefore, has a negative impact on experience of non-fast-moving UEs. This function is controlled by the **FAST_MOVING_SCELL_BLIND_CFG_SW** option of the **CaMgtCfg**.***CellCaAlgoExtSwitch*** parameter and the **SCC_BLIND_CONFIG_SW** option of the **CellQciPara**.***QciAlgoSwitch*** parameter.

- If the **FAST_MOVING_SCELL_BLIND_CFG_SW** option of the **CaMgtCfg**.***CellCaAlgoExtSwitch*** parameter is deselected and the **SCC_BLIND_CONFIG_SW** option of the **CellQciPara**.***QciAlgoSwitch*** parameter is deselected for at least one QCI in a cell, the eNodeB identifies a UE with any bearer of a QCI with **SCC_BLIND_CONFIG_SW** selected as a fast-moving UE. SCells can be configured in a blind manner only for fast-moving UEs. For other UEs, SCells can be configured only based on measurements. If the **SCC_BLIND_CONFIG_SW** option of the **CellQciPara**.***QciAlgoSwitch*** parameter is selected for all QCIs in a cell, this function does not take effect, as blind SCell configuration is allowed for all UEs.

  NOTE

  The function controlled by the **SCC_BLIND_CONFIG_SW** option of the **CellQciPara**.***QciAlgoSwitch*** parameter does not take effect when intelligent selection of serving cell combinations is enabled (by setting the **CaSmartSelectionSwitch** option of the **ENodeBAlgoSwitch**.***CaAlgoSwitch*** parameter) in CA-group-based or adaptive configuration mode.

- If the **FAST_MOVING_SCELL_BLIND_CFG_SW** option of the **CaMgtCfg**.***CellCaAlgoExtSwitch*** parameter is selected, the eNodeB identifies a UE as a fast-moving UE when either of the following conditions is met: (1) The **FAST_MOVING_UE_FLAG** option of the **SpidCfg**.***AggregationAttribute*** parameter is selected for the SPID of the UE; (2) The **FAST_MOVING_UE_FLAG** option of the **QciPara**.***AggregationAttribute*** parameter is selected for at least one QCI of the bearers for the UE. SCells can be configured in a blind manner only for fast-moving UEs. For other UEs, SCells can be configured only based on measurements.

19.6.4 QCI-specific CA Policies

When adaptive CA is enabled, QCI-specific CA policies work, which allows the eNodeB to use different CA policies for UEs running services with different QCIs. For details about adaptive CA, see **4.6.1.3 Adaptive PCC Anchoring Procedure** and **4.6.3.1.3 Adaptive SCell Configuration Procedure**.

To configure QCI-specific CA policies, set the **QciPara**.***MultiCarrCfgGroupId*** parameter to the same value as the **CellMultiCarrCfgGrp**.***MultiCarrCfgGroupId*** parameter, which identifies the measurement configuration group for a frequency (**CellMultiCarrCfgGrp**.***DlArfcn***) relevant to a cell

(**CellMultiCarrCfgGrp**.***LocalCellId***) of an operator (**CellMultiCarrCfgGrp**.***CnOperatorId***).

QCI-specific CA policies include the following:

- QCI-specific CA PCC anchoring policy

  Assume that the **CellMultiCarrCfgGrp**.***RatType*** parameter is set to **EUTRAN**, and the frequency identified by the **CellMultiCarrCfgGrp**.***DlArfcn*** parameter is an E-UTRAN frequency and is already configured using the **PccFreqCfg**.***PccDlEarfcn*** parameter. For the specified QCI, the PCC priority of this frequency is specified by the **CellMultiCarrCfgGrp**.***LtePreferredPccPriority*** parameter. This priority takes precedence over the priority specified by the **PccFreqCfg**.***PreferredPccPriority*** parameter.

  UEs running services with the specified QCI may or may not measure the frequency identified by the **CellMultiCarrCfgGrp**.***DlArfcn*** parameter for PCC anchoring, depending on the **FORBID_MEAS_FLAG** option of the **CellMultiCarrCfgGrp**.***AggregationAttribute*** parameter.

  – If this option is selected, the measurement is prohibited.
  – If this option is deselected, the measurement is allowed.

- QCI-specific SCell management policy

  Assume that the **CellMultiCarrCfgGrp**.***RatType*** parameter is set to **EUTRAN**, and the frequency identified by the **CellMultiCarrCfgGrp**.***DlArfcn*** parameter is an E-UTRAN frequency and is already configured using the **PccFreqCfg**.***PccDlEarfcn*** or **SccFreqCfg**.***SccDlEarfcn*** parameter. For the specified QCI, the SCell measurement and blind SCell configuration attributes of this frequency are specified by the **CellMultiCarrCfgGrp**.***AggregationAttribute*** parameter.

  UEs running services with the specified QCI may or may not measure the frequency identified by the PCell parameter **CellMultiCarrCfgGrp**.***DlArfcn*** for SCell configuration, depending on the **FORBID_MEAS_FLAG** option of the PCell parameter **CellMultiCarrCfgGrp**.***AggregationAttribute***.

  – If this option is selected, the measurement is prohibited.
  – If this option is deselected, the measurement is allowed.

  For UEs running services with the specified QCI, the frequency identified by the PCell parameter **CellMultiCarrCfgGrp**.***DlArfcn*** may or may not be configured as an SCC in a blind manner, depending on the **FORBID_BLIND_CFG_FLAG** option of the PCell parameter **CellMultiCarrCfgGrp**.***AggregationAttribute***.

  – If this option is selected, the blind configuration is prohibited.
  – If this option is deselected, the blind configuration is allowed.

  The eNodeB delivers the A2 measurement configuration related to an SCell to a CA UE after configuring the SCell for the UE based on A4 measurements or in a blind manner. After receiving an A2 measurement report that contains the SCell, the eNodeB sends an RRC Connection Reconfiguration message to remove the SCell. If QCI-specific SCell management is enabled, the RSRP thresholds for events A4 and A2 are as follows (all parameters in the following formulas are configured on the PCell side):

  – RSRP threshold for event A4

- For SCell configuration during initial access, an incoming handover, or an incoming RRC connection reestablishment

  RSRP threshold for event A4 = **CaMgtCfg**.***CarrAggrA4ThdRsrp*** + **SccFreqCfg**.***SccA4Offset*** + **SccFreqCfg**.***SccA2RsrpThldExtendedOfs*** + **CellMultiCarrCfgGrp**.***LteSccA4RsrpThldOffset***

- For periodic SCell configuration based on traffic volume

  RSRP threshold for event A4 = **CaMgtCfg**.***CarrAggrA4ThdRsrp*** + **SccFreqCfg**.***SccA4Offset*** + **CellMultiCarrCfgGrp**.***LteSccA4RsrpThldOffset***

– RSRP threshold for event A2 related to SCell removal

  RSRP threshold for event A2 related to SCell removal = **CaMgtCfg**.***CarrAggrA2ThdRsrp*** + **SccFreqCfg**.***SccA2Offset*** + **SccFreqCfg**.***SccA2RsrpThldExtendedOfs*** + **CellMultiCarrCfgGrp**.***LteSccA2RsrpThldOffset***

NOTE

The actual value range of the RSRP threshold for event A4 and the RSRP threshold for event A2 is –140 dBm to –43 dBm. If the calculation result is less than –140 dBm, –140 dBm takes effect. If the calculation result is greater than –43 dBm, –43 dBm takes effect.

For a single value of **CellMultiCarrCfgGrp**.***MultiCarrCfgGroupId***, if different cells (identified by the **CellMultiCarrCfgGrp**.***LocalCellId*** parameter) have an identical combination of the **CellMultiCarrCfgGrp**.***DlArfcn***, **CellMultiCarrCfgGrp**.***CnOperatorId***, and **CellMultiCarrCfgGrp**.***RatType*** parameter settings, then these cells must also have an identical combination of the following parameter settings, to prevent ping-pong handovers:

- **CellMultiCarrCfgGrp**.***LtePreferredPccPriority***
- **CellMultiCarrCfgGrp**.***LteSccA2RsrpThldOffset***
- **CellMultiCarrCfgGrp**.***LteSccA4RsrpThldOffset***

19.6.5 SPID-specific CA Policies

When adaptive CA is enabled, CA PCC anchoring policies and SCell management policies can be defined on a per SPID basis on the eNodeB. For details about adaptive CA, see **4.6.1.3 Adaptive PCC Anchoring Procedure** and **4.6.3.1.3 Adaptive SCell Configuration Procedure**.

The **SpidCfg**.***MultiCarrCfgGroupId*** parameter identifies a measurement configuration group for the frequencies of multiple carriers. This parameter is associated with the **CellMultiCarrCfgGrp**.***MultiCarrCfgGroupId*** parameter, which identifies the measurement configuration group for a frequency (**CellMultiCarrCfgGrp**.***DlArfcn***) relevant to a cell (**CellMultiCarrCfgGrp**.***LocalCellId***) of an operator (**CellMultiCarrCfgGrp**.***CnOperatorId***). By setting the two ID parameters to an identical value, operators can apply CA PCC anchoring policies and SCell management policies to specific SPIDs. For more details, see *Flexible User Steering*.

19.7 VoLTE

Collaboration between CA and voice over LTE (VoLTE) is controlled by the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter, the **QCI_VOLTE_CA_COEXIST_SW** option of the **CellQciPara**.***QciAlgoSwitch*** parameter, and the **VolteSupportCaInterFreqMeasSw** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter.

There are two scenarios:

- CA for UEs with VoLTE services ongoing
- Initiation of VoLTE services after UEs enter the CA state

When QCI-specific VoLTE and CA coexistence is enabled (by selecting the **QCI_VOLTE_CA_COEXIST_SW** option of the **CellQciPara**.***QciAlgoSwitch*** parameter), data service experience of CA UEs can be guaranteed, but voice quality may fluctuate.

CA for UEs with VoLTE Services Ongoing

CA for UEs with VoLTE services ongoing is controlled by the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter, the **QCI_VOLTE_CA_COEXIST_SW** option of the **CellQciPara**.***QciAlgoSwitch*** parameter, and the **VolteSupportCaInterFreqMeasSw** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter.

- If the **QCI_VOLTE_CA_COEXIST_SW** option of the **CellQciPara**.***QciAlgoSwitch*** parameter is selected for any of the QCIs configured for such a CA UE, the processing for this UE is the same as that with the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter set to **255**.
- If the **QCI_VOLTE_CA_COEXIST_SW** option of the **CellQciPara**.***QciAlgoSwitch*** parameter is deselected for all QCIs configured for a CA UE, the processing for this UE depends on the value of the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter.
  - With the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter set to a value within the range of **-140** to **-43**
    - If the PCell RSRP of a VoLTE UE is less than or equal to this threshold, then:

      The VoLTE UE reports CA event A2. PCC anchoring and SCell configuration do not take effect for the VoLTE UE.

      If the VoLTE UE is located at the cell edge, the concurrent data services on the UE cannot obtain CA gains, and therefore the user-perceived data rates of these data services may decrease while the packet loss rates and block error rates of voice services may increase. However, KPIs such as the VoLTE call drop rate may improve.

      Upon release of the bearer with a QCI of 1, the eNodeB initiates an SCell configuration procedure for the UE with the current serving cell as its PCell.
    - If the PCell RSRP of a VoLTE UE is greater than this threshold, then:

      Whether PCC anchoring and SCell configuration take effect for the VoLTE UE depends on the setting of the

VolteSupportCaInterFreqMeasSw option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter.

- Option selected

  If the VoLTE UE has never reported CA event A2 because the PCell RSRP is less than or equal to the threshold when it is in the RRC_CONNECTED state, PCC anchoring or SCell configuration based on inter-frequency measurements or virtual grids, as well as blind SCell configuration, take effect for the VoLTE UE.

  If the VoLTE UE has ever reported CA event A2 because the PCell RSRP is less than or equal to the threshold when it is in the RRC_CONNECTED state, PCC anchoring or SCell configuration based on inter-frequency measurements or virtual grids, as well as blind SCell configuration, take effect for the VoLTE UE only after the QCI-1 bearer of the UE is released.

- Option deselected

  PCC anchoring and SCell configuration based on inter-frequency measurements, as well as PCC anchoring based on virtual grids, do not take effect for the VoLTE UE, but blind SCell configuration and SCell configuration based virtual grids can be triggered.

– With the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter set to **0**

  PCC anchoring and SCell configuration do not take effect for the VoLTE UE.

  If CA-related measurements are active for the VoLTE UE, all these measurements are immediately released.

  The concurrent data services on the UE cannot obtain CA gains, and therefore the user-perceived data rates of these data services may decrease while the packet loss rates and block error rates of voice services may increase. However, KPIs such as the VoLTE call drop rate may improve.

  Upon release of the bearer with a QCI of 1, the eNodeB initiates an SCell configuration procedure for the UE with the current serving cell as its PCell.

– With the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter set to **255**

  Whether PCC anchoring and SCell configuration take effect for the VoLTE UE depends only on the setting of the **VolteSupportCaInterFreqMeasSw** option of the **CaMgtCfg**.***CellCaAlgoSwitch*** parameter.

  - Option selected

    PCC anchoring or SCell configuration based on inter-frequency measurements or virtual grids, as well as blind SCell configuration, can be triggered for the VoLTE UE.

  - Option deselected

    PCC anchoring and SCell configuration based on inter-frequency measurements, as well as PCC anchoring based on virtual grids, do not take effect for the VoLTE UE, but blind SCell configuration and SCell configuration based virtual grids can be triggered.

SCell changes for CA increase the number of RRC connection reconfiguration messages, causing voice quality to deteriorate. Whitelist control over CA and voice decoupling can reduce the number of RRC connection reconfiguration messages for whitelisted voice service UEs. Its procedure varies depending on phases: before QCI-1 bearers are set up, when QCI-1 bearers are present, and after QCI-1 bearers are released. For details, see *VoLTE*.

Initiation of VoLTE Services After UEs Enter the CA State

When VoLTE services are initiated for UEs with carriers aggregated, the UEs may or may not stay in the CA state, depending on the settings of the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter and the **QCI_VOLTE_CA_COEXIST_SW** option of the **CellQciPara**.***QciAlgoSwitch*** parameter.

- If the **QCI_VOLTE_CA_COEXIST_SW** option of the **CellQciPara**.***QciAlgoSwitch*** parameter is selected for any of the QCIs configured for such a CA UE, the processing for this UE is the same as that with the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter set to **255**.
- If the **QCI_VOLTE_CA_COEXIST_SW** option of the **CellQciPara**.***QciAlgoSwitch*** parameter is deselected for all QCIs configured for a CA UE, the processing for this UE depends on the value of the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter.
  - With the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter set to a value within the range of **-140** to **-43**
    - If the PCell RSRP of a CA UE is less than or equal to the value of the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter, then:

      The CA UE exits the CA state.
    - If the PCell RSRP of the CA UE is greater than the value of the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter, then:

      The CA UE can stay in the CA state, and its VoLTE service is scheduled only on the PCC.
  - With the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter set to **0**

    The CA UE exits the CA state.
  - With the **CaMgtCfg**.***VolteCaA2RsrpThld*** parameter set to **255**

    The CA UE can stay in the CA state, and its VoLTE service is scheduled only on the PCC.

20 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, LOFD-001016 or TDLOFD-001016.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

21 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, LOFD-001016 or TDLOFD-001016.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

22 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

23 Reference Documents

1. *Mobility Management in Connected Mode*
2. *Intra-RAT Mobility Load Balancing*
3. *Idle Mode Management*
4. *Emergency Call*
5. *CS Fallback*
6. *Multi-Band Compatibility Enhancement*
7. *VoLTE*
8. *ANR Management*
9. *SFN*
10. *CSPC*
11. *eMBMS*
12. *LCS*
13. *Network Assisted Interference Cancellation*
14. *Terminal Awareness Differentiation*
15. *DL CoMP (TDD)*
16. *DL CoMP (FDD)*
17. *TDM eICIC (FDD)*
18. *GSM and LTE Spectrum Concurrency*
19. *D-MIMO (TDD)*
20. *Relay*
21. *Extended Cell Range*
22. *Super Combined Cell (FDD)*
23. *Energy Conservation and Emission Reduction*
24. *Auto Neighbor Group Configuration*
25. *UMTS and LTE Spectrum Sharing*
26. *UMTS and LTE Zero Bufferzone*
27. *MIMO*
28. *UL CoMP*

29. *Uplink Interference Cancellation (FDD)*
30. *Uplink Coordinated Scheduling*
31. *LTE FDD and NR Dynamic Spectrum Sharing*
32. *LTE FDD and NR Hybrid Dynamic Spectrum Sharing*
33. *LTE FDD and NR SUL Dynamic Spectrum Sharing*
34. *Beamforming (TDD)*
35. *Massive MIMO Basics (FDD)*
36. *Massive MIMO Enhancements (FDD)*
37. *Massive MIMO Basics (TDD)*
38. *Massive MIMO Enhancements (TDD)*
39. *Uplink Scheduling*
40. *Downlink Scheduling*
41. *Dedicated Carrier for TM9*
42. *USU3910-based Multi-BBU Interconnection*
43. *eMIMO (FDD)*
44. *eX2 Self-Management*
45. *S1 and X2 Self-Management*
46. *High Speed Mobility*
47. *IP Performance Monitor*
48. *Subframe Configuration (TDD)*
49. *Connection Management*
50. *Admission and Congestion Control*
51. *DRX and Signaling Control*
52. *RAN Sharing*
53. *Short TTI (FDD)*
54. *WBB*
55. *LTE Spectrum Coordination*
56. *Massive MIMO Optimization in WTTx Scenarios (TDD)*
57. *Flexible Bandwidth based on Overlap Carriers (FDD)*
58. *Compact Bandwidth (FDD)*
59. *Multi-carrier Unified Scheduling*
60. 3GPP TS 36.211: "Physical channels and modulation"
61. 3GPP TS 36.331: "Radio Resource Control (RRC)"
62. 3GPP TS 36.321: "Medium Access Control (MAC) protocol specification"
63. 3GPP TS 36.322: "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Link Control (RLC) protocol specification"
64. 3GPP TS 36.101: "Evolved Universal Terrestrial Radio Access (E-UTRA); User Equipment (UE) radio transmission and reception"
65. 3GPP TS 36.306: "User Equipment (UE) radio access capabilities"
66. 3GPP TS 36.213: "Physical layer procedures"
67. 3GPP TS 36.300: "Overall description"

68. *Multi-RAT Coordinated Energy Saving*
69. *Soft Split Resource Duplex (TDD)*
70. *EPC-based NSA Basics*
71. *EPC-based NSA Performance Enhancement*
72. *Licensed Assisted Access (TDD)*
73. *Dynamic Power Sharing Between LTE Carriers*
74. *Superior Uplink Coverage (FDD)*
75. *UL and DL Decoupling* in *5G RAN Feature Documentation*
76. *Seamless Intra-Band Carrier Joining (TDD)*
77. *Seamless Intra-Band Carrier Joining (FDD)*
78. *Dynamic Multi-Carrier Management (FDD)*
79. *Automatic Congestion Handling*
80. *Physical Resource Slicing (FDD)*
81. *Synchronization*
82. *Smart 8T8R (FDD)*
83. *Flexible User Steering*
84. *Rate Control Based on User Types*
85. *RuralLink Solution*
86. *Green Site*
87. *NSA-SA Selection Based on User Experience*
88. *LTE and NR Power Allocation*
89. *Interference Detection and Suppression*
90. *Low-Band Booster*
91. *License Control Item Lists (FDD)*
92. *License Control Item Lists (CIoT)*
93. *License Control Item Lists (TDD)*
94. *eMTC*
95. *PTT*
96. *TMA (TDD)*
97. *UMTS and LTE Spectrum Sharing Based on DC-HSDPA*
98. *LTE AHR (TDD)*
99. *LTE FDD and NR Inter-RAT Unified Scheduling*
100. *3900 & 5900 Series Base Station Product Documentation*

